



SMART COCKPIT.COM

A319/A320/A321

Indicating & Recording



INTRODUCTION

The Electronic Instrument System (EIS) presents data on six identical Display Units (DUs):

- The Electronic Flight Instrument System (EFIS) displays mostly flight parameters and navigation data on the Primary Flight Displays (PFDs) and Navigation Displays (NDs).
- The Electronic Centralized Aircraft Monitor (ECAM) presents data on the Engine/Warning Display (E/WD) and System Display (SD) :
 - Primary engine indications, fuel quantity, flap and slat position
 - Warning and caution alerts, or memos
 - Synoptic diagrams of aircraft systems, and status messages
 - Permanent flight data

COCKPIT ARRANGEMENT

R

CAPTAIN :

EFIS CONTROL PANEL

NAVIGATION DISPLAY

MASTER WARNING
AND CAUTION LIGHTS

PRIMARY FLIGHT
DISPLAY

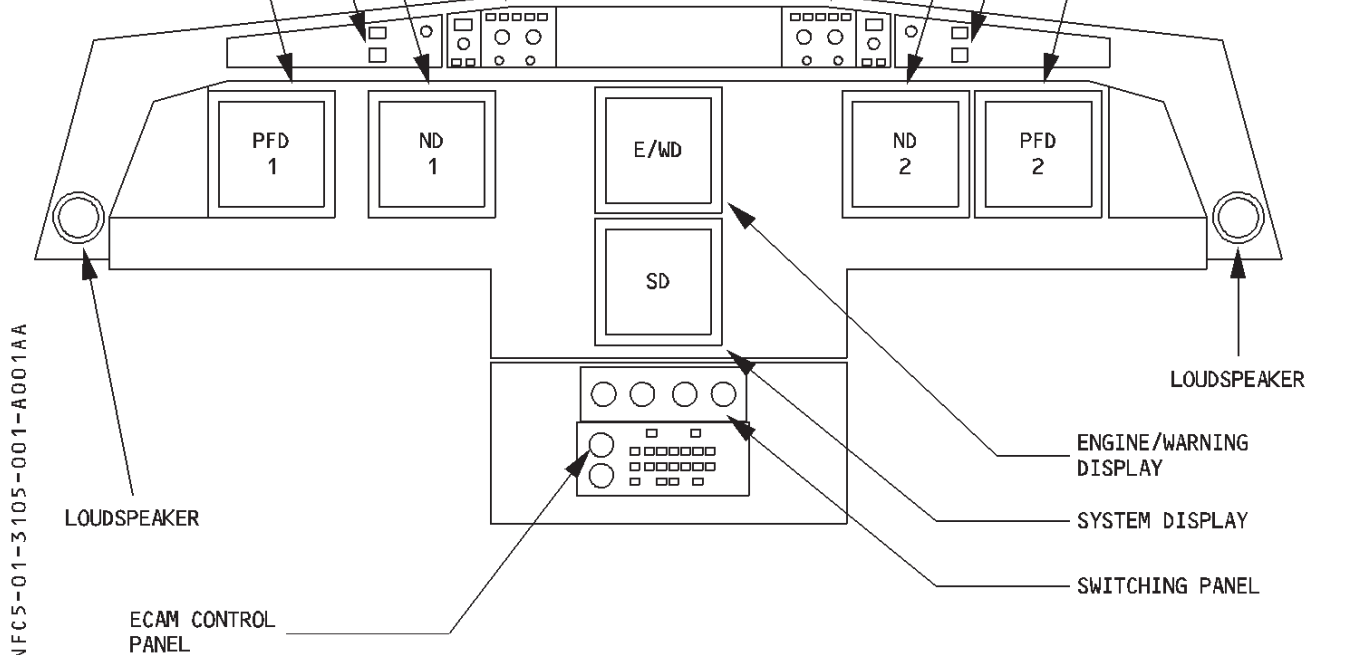
FIRST OFFICER :

EFIS CONTROL PANEL

NAVIGATION DISPLAY

MASTER WARNING
AND CAUTION LIGHTS

PRIMARY FLIGHT
DISPLAY



ARCHITECTURE

DISPLAY UNIT (DU)

The instrument panels have six identical units.
These DUs are full-color cathode ray tubes.

DISPLAY MANAGEMENT COMPUTER (DMC)

Three identical display management computers acquire and process all the signals received from sensors and other computers to generate the images to be displayed on the primary flight displays, navigation displays, engine/warning display, and system display.
Each DMC has two independent channels an EFIS channel and an ECAM channel and is able to drive simultaneously one PFD, one ND, and either of the ECAMs in its engine warning or system status task.

SYSTEM DATA ACQUISITION CONCENTRATOR (SDAC)

The two identical SDACs acquire data, then generate signals. Some of these signals go to the three DMCs, which use them to generate displays of system pages and engines parameters. Others go to the flight warning computers, which use them to generate ECAM messages and aural alerts.

FLIGHT WARNING COMPUTER (FWC)

The two identical FWCs generate alert messages, memos, aural alerts, and synthetic voice messages. For this purpose they acquire data :

- directly from aircraft sensors or systems to generate red warnings.
- through the SDACs to generate amber cautions.

The ECAM display units display the alert messages generated by the FWCs.

The FWCs also generate.

- radio altitude callouts.
- decision height callouts.
- landing distance and landing speed increments.

ATTENTION-GETTERS

The FWCs also drive the attention-getters. Each pilot has a set of these on the panel under the glareshield. They are :

- a master warning light that flashes “MASTER WARN” in red for red warnings
- a master caution light that illuminates “MASTER CAUT” in amber for amber cautions

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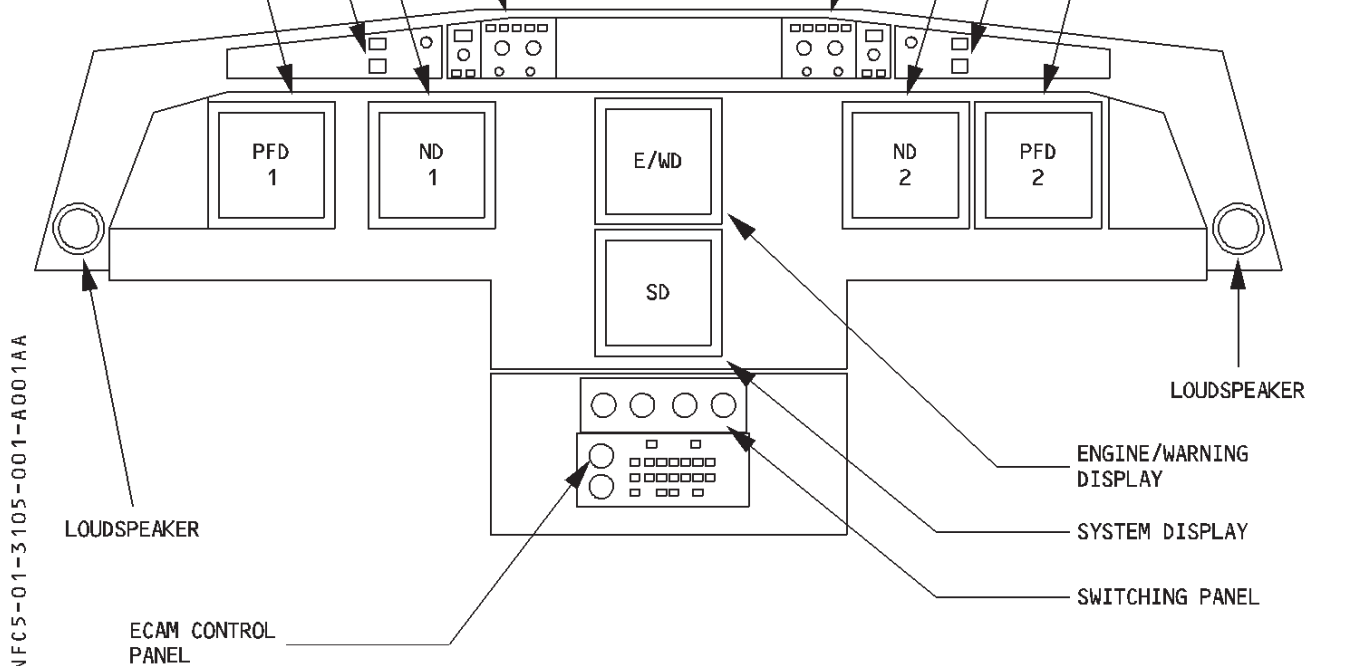
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NFC5-01-3105-001-A001AA

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These DUs are full-color Liquid Crystal Displays (LCD).

DISPLAY MANAGEMENT COMPUTER (DMC)

R Three identical Display Management Computers (DMCs) obtain data from the different
R sensors and computers, and send it to the display units. The display units then generate
R and display the applicable images. Each DMC has a single DMC channel, and can
R simultaneously supply one PFD, one ND and both ECAM display units (E/WD and SD).

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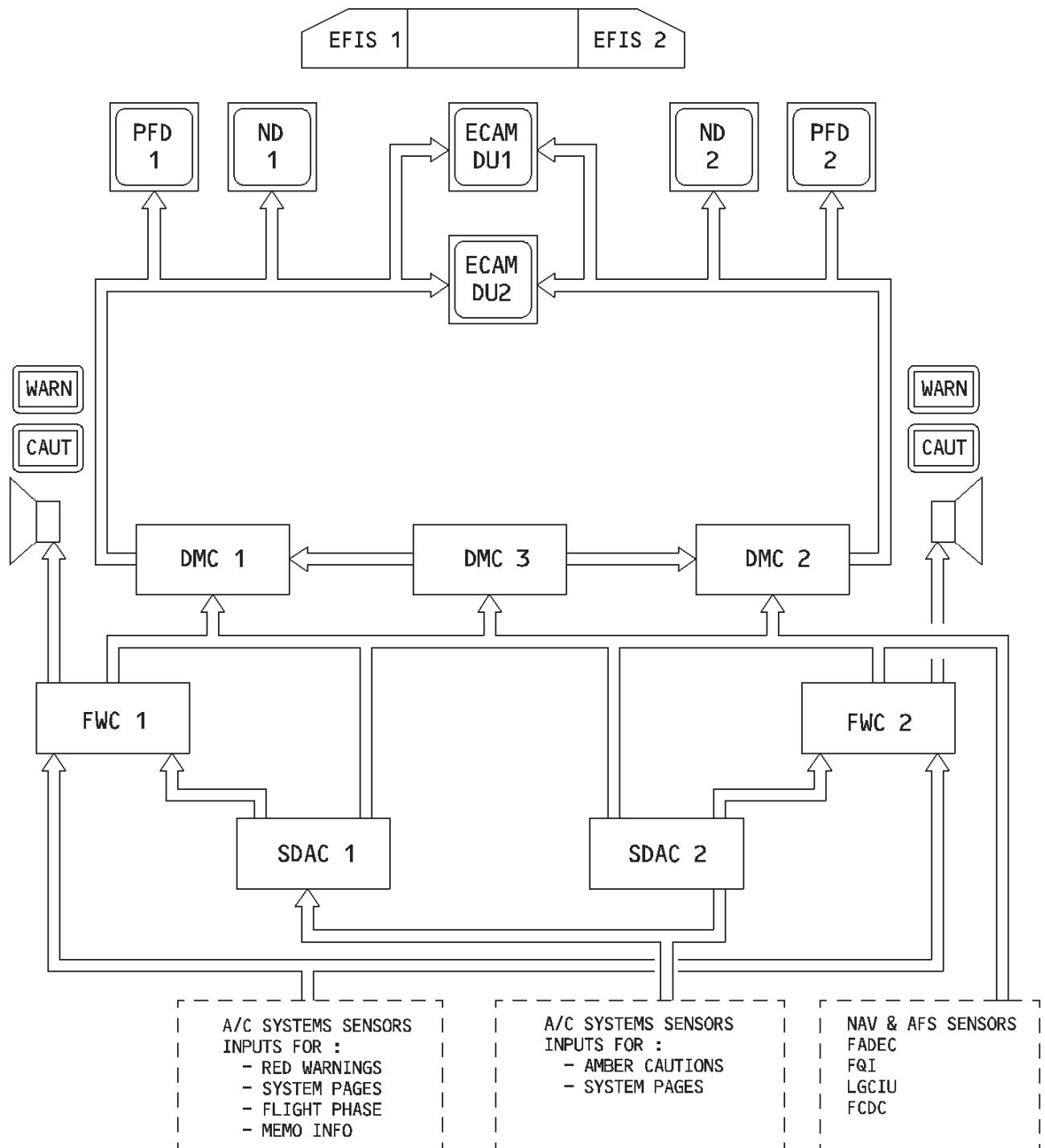
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LOUDSPEAKER

The communications loudspeakers announce aural alerts and voice messages, and do so even when they are turned off.

EIS BLOCK DIAGRAM

R



NFC5-01-3105-003-A001AA

CONTROLS AND SWITCHING

ECAM CONTROL PANEL (ECP)

- R The ECAM Control Panel, located on the pedestal, includes :
- R — Such E/WD controls, as CLR, STS, and the brightness control knob.
- R — Such SD controls, as ENG, BLEED, PRESS..., system page selector, and the brightness
- R control knob.

EIS DMC SWITCHING SELECTOR

- R A switch near the center of the SWITCHING panel which is located just above the ECAM
- R control panel, enables the flight crew to replace the Captain or First Officer's Display Management Computer (DMC 1, or DMC 2) by DMC 3.

ECAM/ND SWITCHING

A switch on the right-hand side of the SWITCHING panel enables the flight crew to transfer the ECAM System Display to either the Captain or First Officer's Navigation Display.

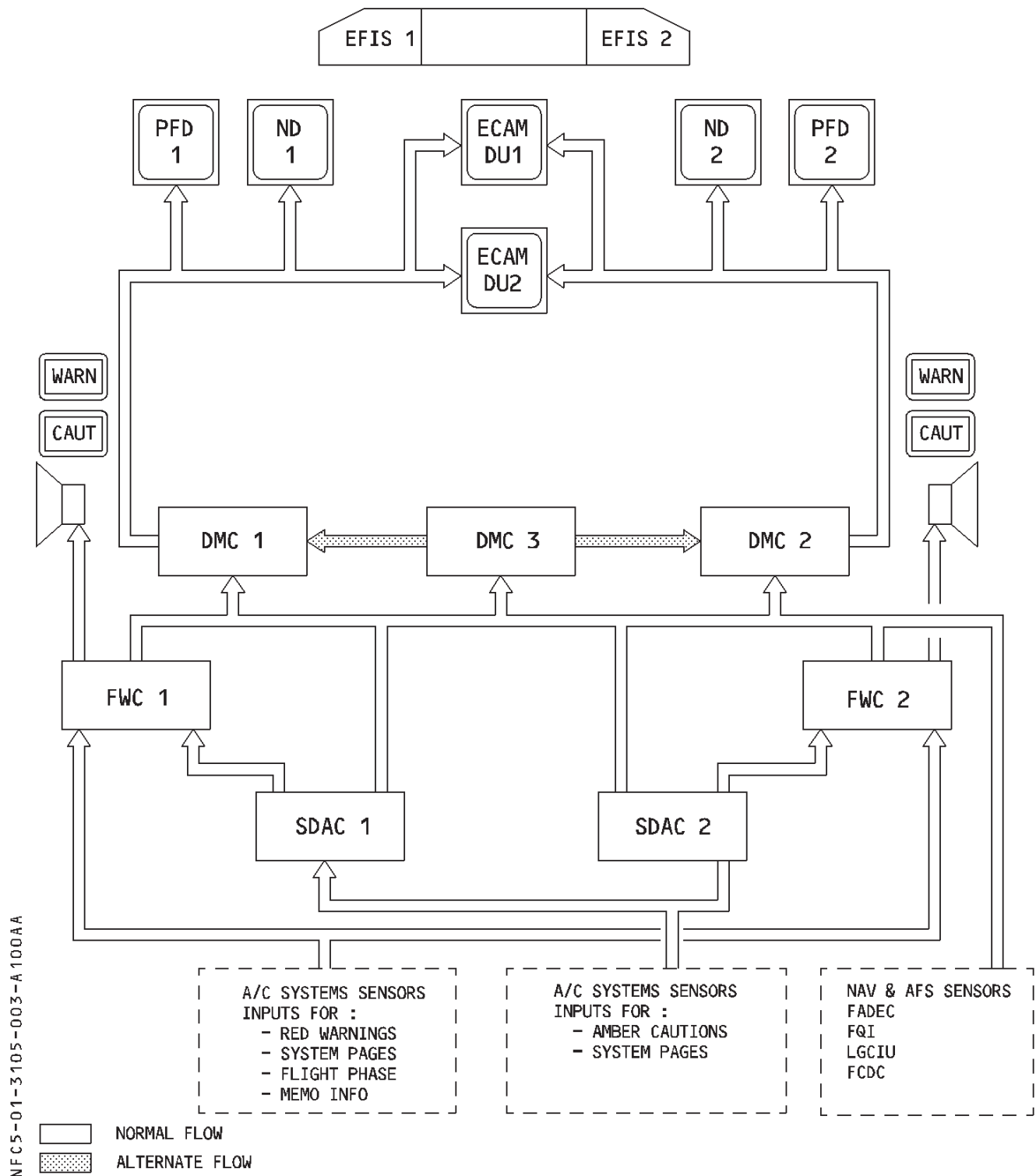
EFIS SWITCHING

A PFD/ND XFR pushbutton on each side console enables the pilot to swap displays to the respective outside DUs.

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EFIS SWITCHING

A PFD/ND XFR pushbutton on each side console enables the pilot to swap displays to the respective outside DUs.

RECONFIGURING THE DISPLAY MANAGEMENT COMPUTER (DMC)

In normal operation :

- DMC 1 supplies data to the Captain's PFD and ND, and the upper ECAM DU.
- DMC 2 supplies data to the First Officer's PFD and ND, and the lower ECAM DU.
- DMC 3 is on standby.

If a DMC fails (corresponding DU shows a diagonal line), the flight crew can replace DMC 1 or 2 with DMC 3 by turning the EIS DMC switch on the SWITCHING panel to "CAPT 3" or "F/O 3".

RECONFIGURING DISPLAY UNITS (DUs)**FAILURE OF UPPER ECAM DU (OR CTL/BRIGHTNESS KNOB TURNED TO OFF)**

If the upper ECAM display fails, or is switched off :

- The engine/warning page automatically replaces the system/status page on the lower ECAM DU.

The flight crew can have the system/status page displayed by :

- Using the "ECAM/ND XFR" switch, on the SWITCHING panel, to move it to a navigation display unit (NDU), or
- R – Pushing and holding (for a maximum of 3 minutes) the related system page pushbutton on the ECAM control panel to temporarily display it on the lower ECAM DU (in place of the engine/warning page).

FAILURE OF LOWER ECAM DU (OR CTL/BRIGHTNESS KNOB TURNED TO OFF)

If the lower ECAM display fails, or is switched off, the flight crew can display the system/status page by :

- Using the "ECAM/ND XFR" switch, on the SWITCHING panel, to display it on NDU, or
- R – Pushing and holding (for a maximum of 3 minutes) the related system page pushbutton, on the ECAM control panel to temporarily display it on the upper ECAM DU (in place of the engine/warning page).

FAILURE OF BOTH ECAM DUs

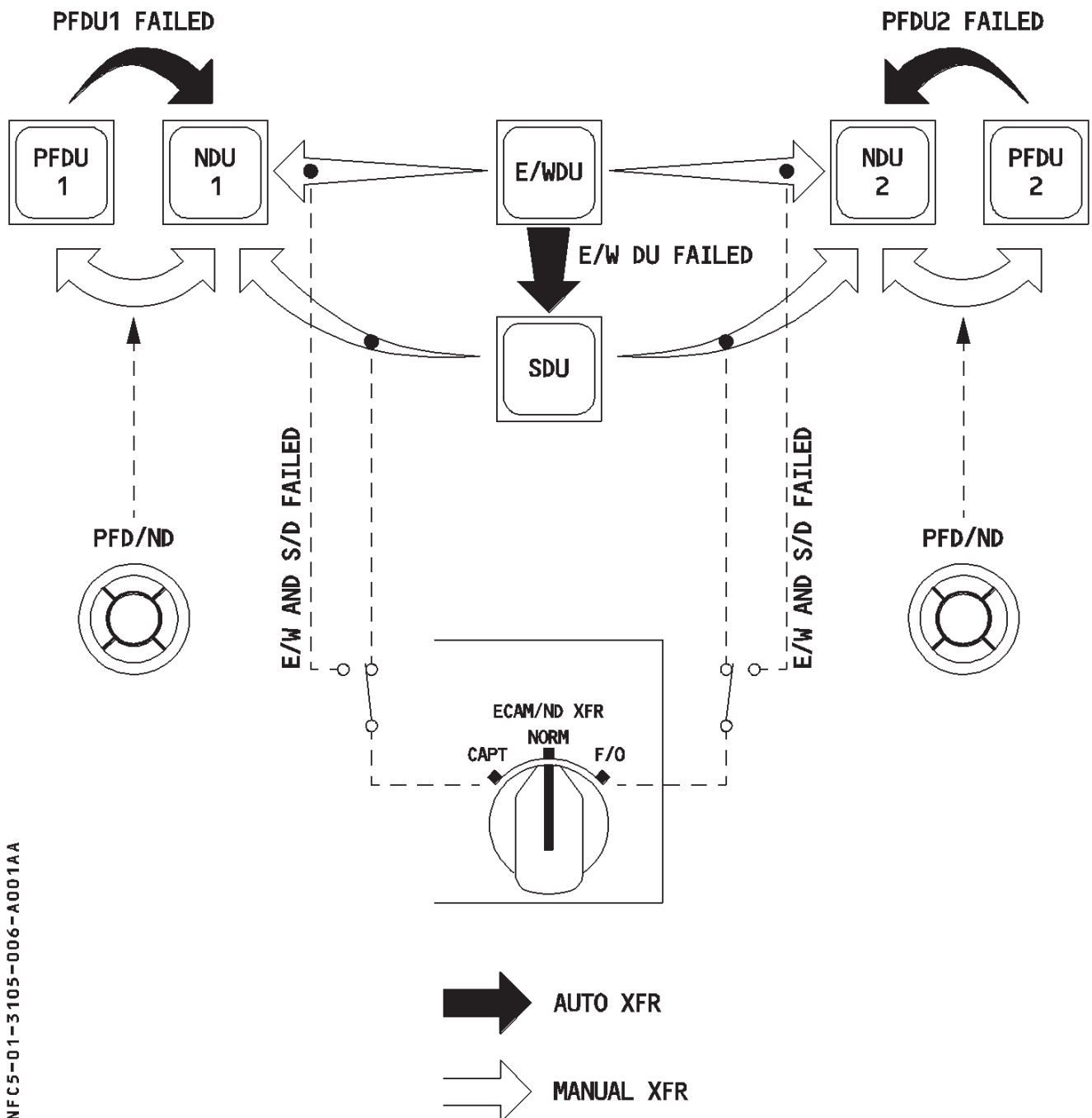
If both ECAM displays fail, the flight crew may :

- Use the "ECAM/ND XFR" on the SWITCHING panel to display the engine/warning page on a navigation display and, if needed,
- R – Push and hold (for a maximum of 3 minutes) the related system page pushbutton on the ECAM control panel, to temporarily display the system/status page on an ND.

PFDU/NDU RECONFIGURATION

- R If a PFDU fails, the system automatically transfers the PFD image to the NDU.
- The pilot can also make this transfer manually by :
- turning the PFD ON-OFF/brightness control OFF, or
 - pressing the PFD/ND/XFR pushbutton, which cross-changes the images between the PFDU and the NDU.
- If an NDU fails, the pilot can use the PFD/ND/XFR pushbutton to transfer the ND image to the PFDU.

DU RECONFIGURATION



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**RECONFIGURING THE DISPLAY MANAGEMENT COMPUTER (DMC)**

- R In normal operation, each DMC drives the following Display Units :
- R – DMC 1 drives the CAPT PFD, CAPT ND the ECAM DUs.
- R – DMC 2 drives the F/O PFD and F/O ND.
- R – DMC 3 is on standby, ready to drive any DU.
- R If DMC 1 or 2 fails (the “INVALID DATA” message is displayed on the DUs), the crew
- R manually selects the DMC 3 source (“CAPT 3” or “F/O 3”).
- R If DMC 1 fails (or DMC 3, if “CAPT 3” was selected), DMC 2 automatically drives the
- R ECAM.

RECONFIGURING DISPLAY UNITS (DUs)**FAILURE OF UPPER ECAM DU (OR CTL/BRIGHTNESS KNOB TURNED TO OFF)**

If the upper ECAM display fails, or is switched off :

- The engine/warning page automatically replaces the system/status page on the lower ECAM DU.

The flight crew can display the system/status page by :

- Using the “ECAM/ND XFR” switch, on the SWITCHING panel, to move it to a Navigation Display Unit (NDU), or
- Pushing and holding (for a maximum of 3 minutes) the related system page pushbutton, on the ECAM control panel, to temporarily display it on the lower ECAM DU (instead of the engine/warning page).

FAILURE OF LOWER ECAM DU (OR CTL/BRIGHTNESS KNOB TURNED TO OFF)

If the lower ECAM display fails, or is switched off, the flight crew can display the system/status page by :

- Using the “ECAM/ND XFR” switch, on the SWITCHING panel, to display it on the NDU, or
- Pushing and holding (for a maximum of 3 minutes) the related system page pushbutton, on the ECAM control panel, to temporarily display it on the upper ECAM DU (instead of the engine/warning page).

FAILURE OF BOTH ECAM DUs

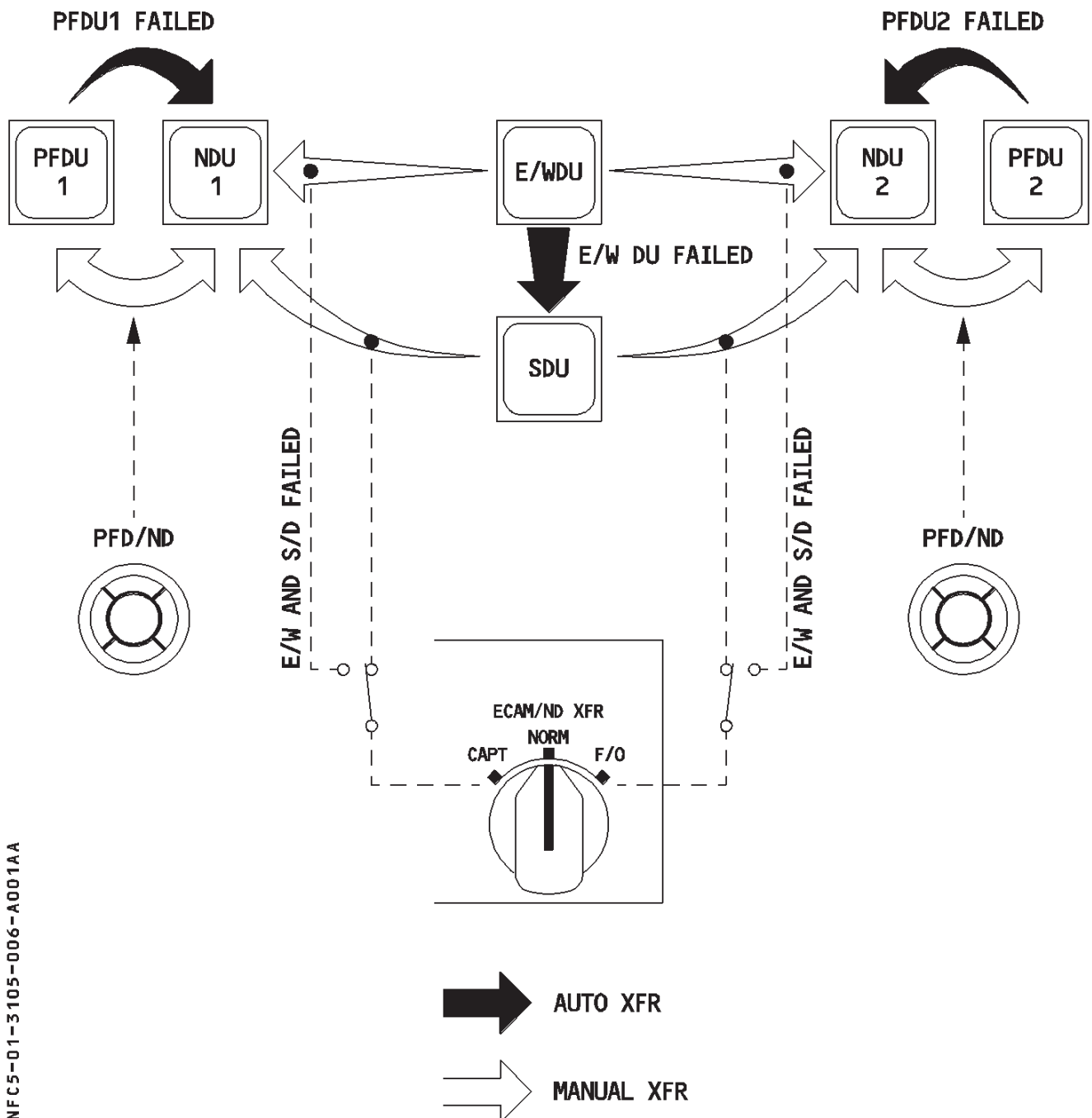
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- Use the “ECAM/ND XFR”, on the SWITCHING panel, to display the engine/warning page on a navigation display and, if needed,
- Push and hold (for a maximum of 3 minutes) the related system page pushbutton, on the ECAM control panel, to temporarily display the system/status page on an ND.

PFDU/NDU RECONFIGURATION

- R If a PFDU fails, the system automatically transfers the PFD image to the NDU.
- The pilot can also make this transfer manually by :
- turning the PFD ON-OFF/brightness control OFF, or
 - pressing the PFD/ND/XFR pushbutton, which cross-changes the images between the PFDU and the NDU.
- If an NDU fails, the pilot can use the PFD/ND/XFR pushbutton to transfer the ND image to the PFDU.

DU RECONFIGURATION



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MESSAGES ON THE DISPLAY UNITS (DUs)

These messages are displayed on either the EFIS or the ECAM Display Unit (DU) depending on the current EFIS or ECAM configuration.

FAILURE OF A DU

If a DU fails, the flight crew may find one of the following displays :

- A blank screen with an “F” letter in amber, or
- A distorted display, or
- A blank screen with the “INVALID DISPLAY UNIT” message in amber.

FEEDBACK MESSAGES

The DU displays the following messages in amber when the Display Management Computer (DMC) detects a discrepancy between the parameters obtained by the DMC and the operational parameters displayed on the DU :

- “CHECK CAPT PFD” (“CHECK F/O PFD”) if the discrepancy concerns the PFD parameters
- “CHECK CAPT ND” (“CHECK F/O ND”) if the discrepancy concerns the ND parameters
- “CHECK EWD” if the discrepancy concerns the E/WD parameters
- “CHECK SD” if the discrepancy concerns the SD parameters.

In addition, if the aircraft is on ground, the “DU NOT MONITORED” message is displayed in amber when there is only one DMC, instead of two DMCs, that provides the affected DU with feedback information.

This message means that there is either a DMC test in progress or that there is an EIS failure.

In the case of an EIS failure, a maintenance action is necessary.

SIDE1/SIDE2 DISCREPANCY MESSAGES

The DU displays the following messages along with a caution on the ECAM E/WD when there is a discrepancy between the parameters displayed on the Captain’s EFIS and the First Officer’s EFIS :

- Both PFDs display the message “CHECK ATT” if there is a discrepancy of at least 5 degrees between the attitude values, pitch and/or roll
- Both PFDs display the message “CHECK ALT” if there is a discrepancy between the altitude values greater than 250 ft when the flight crew selects a QNH different from STD, or 500 ft when the flight crew selects a QNH STD
- Both PFDs and NDs display the message “CHECK HDG” if there is a discrepancy of at least 5 degrees between the heading values.

Note : The message “CHECK HDG” flashes for a few seconds on the NDs, and then it remains steady. If the flight crew selects the PLAN mode on the NDs the message “CHECK HDG” does not appear.

DU RESET

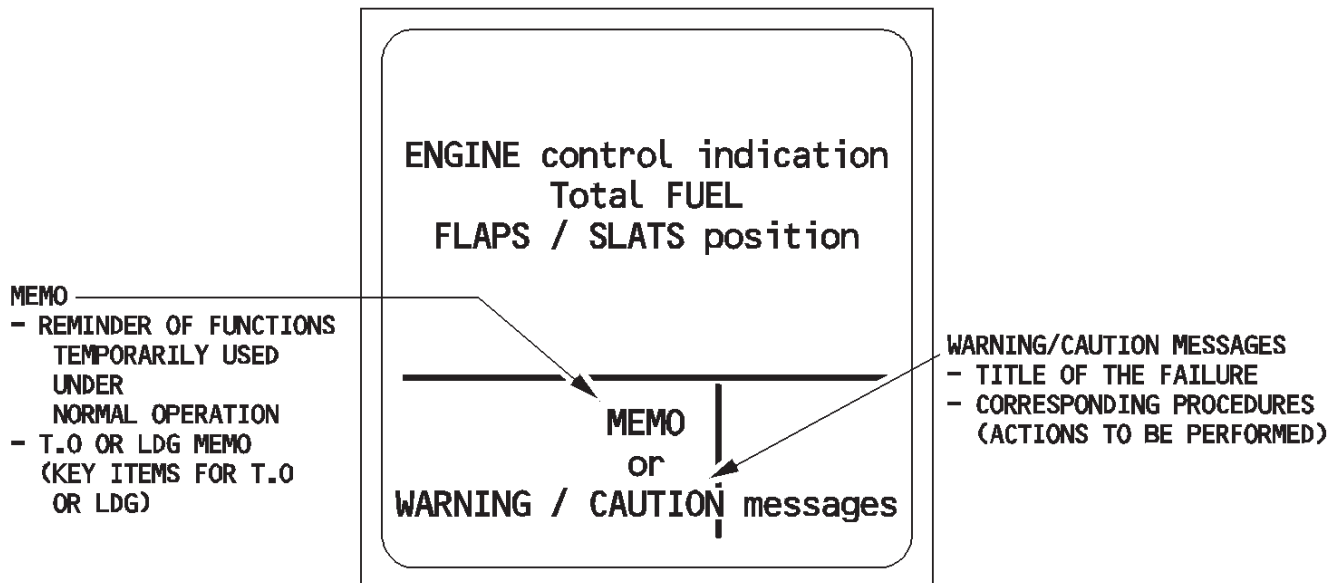
In the case of a DU reset, the message "SELF TEST IN PROGRESS" can be displayed in green and/or the message "WAITING FOR DATA" may be displayed in green during the EIS initialization.

ECAM DU ARRANGEMENT

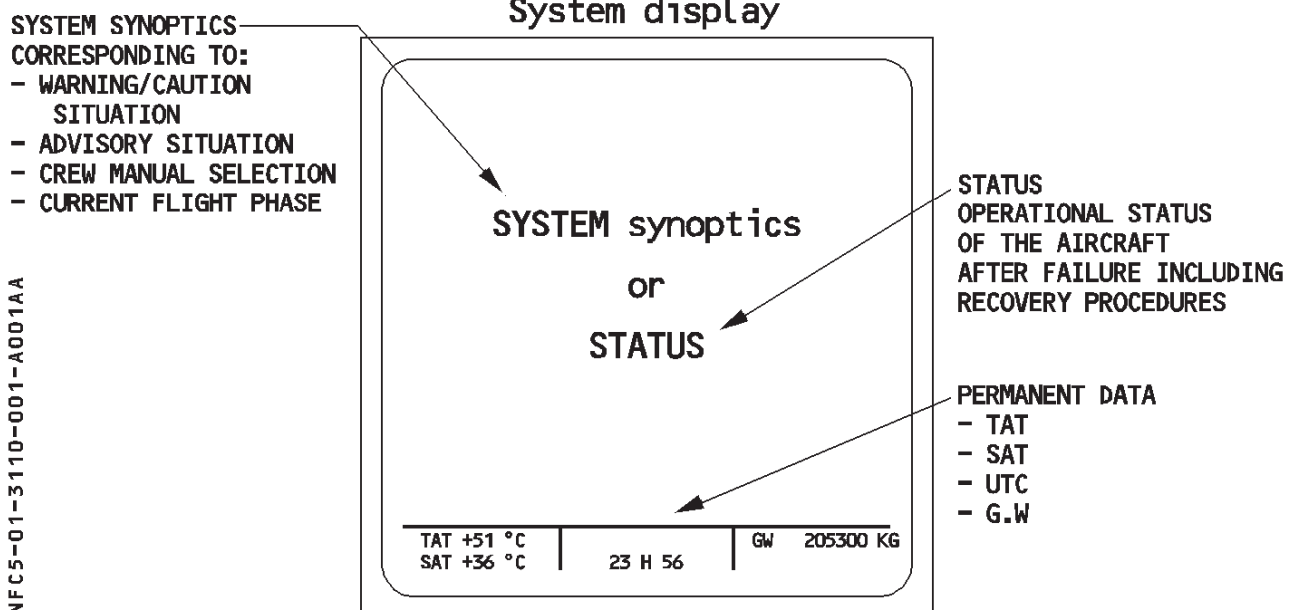
The ECAM has two display units :

- one for the engine/warning display (E/WD).
- one for the system/status display (SD).

Engine / warning display



System display



NFC5-01-3110-001-A001AA

COLOR CODE

The ECAM display uses a color code that indicates the importance of the failure or the indication.

RED : The configuration or failure requires immediate action.

AMBER : The flight crew should be aware of the configuration or failure, but need not take immediate action.

GREEN : The item is operating normally.

WHITE : These titles and remarks guide the flight crew, as they execute various procedures.

BLUE : These are actions to be carried out, or limitations.

MAGENTA : These are particular messages that apply to particular pieces of equipment or situations (inhibition messages, for example).

WARNING/CAUTION CLASSIFICATION

R

	LEVEL	SIGNIFICATION	AURAL	VISUAL
FAILURE MODE	Level 3	Red warning : The configuration, or failure requires immediate action : – Aircraft in dangerous configuration, or limit flight conditions (eg: stall, o/speed) – System failure altering flight safety (eg : Eng fire, excess cab alt)	Continuous Repetitive Chime (CRC) or specific sound or synthetic voice	– MASTER WARN light red flashing or specific red light – Warning message (red) on E/WD – Automatic call of the relevant system page on the S/D *
	Level 2	Amber caution : The flight crew should be aware of the configuration or failure, but does not need to take any immediate action. However, time and situation permitting, these cautions should be considered without delay to prevent any further degradation of the affected system : – System failure without any direct consequence on the flight safety (eg: HYD G SYS LO PR)	Single Chime (SC)	– MASTER CAUT light amber steady – Caution message (amber) on E/WD – Automatic call of the relevant system page on the S/D * .
	Level 1	Amber caution : Requires crew monitoring : – Failures leading to a loss of redundancy or system degradation (eg : FCDC fault)	NONE	– Caution message (amber) on E/WD generally without procedure.
INFORMATION	ADVISORY	System parameters monitoring	NONE	– Automatic call of the relevant system page on the S/D. The affected parameter pulses green.
	MEMO	Information : Recalls normal or automatic selection of functions which are temporarily used	NONE	– Green, Amber, or Magenta message on E/WD

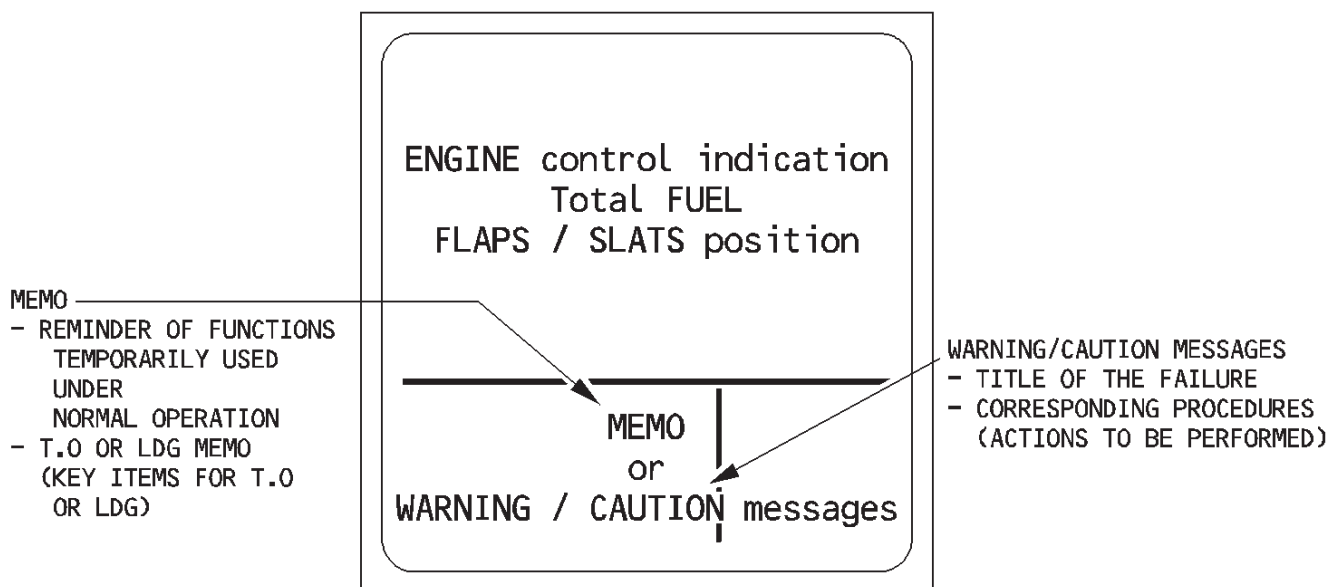
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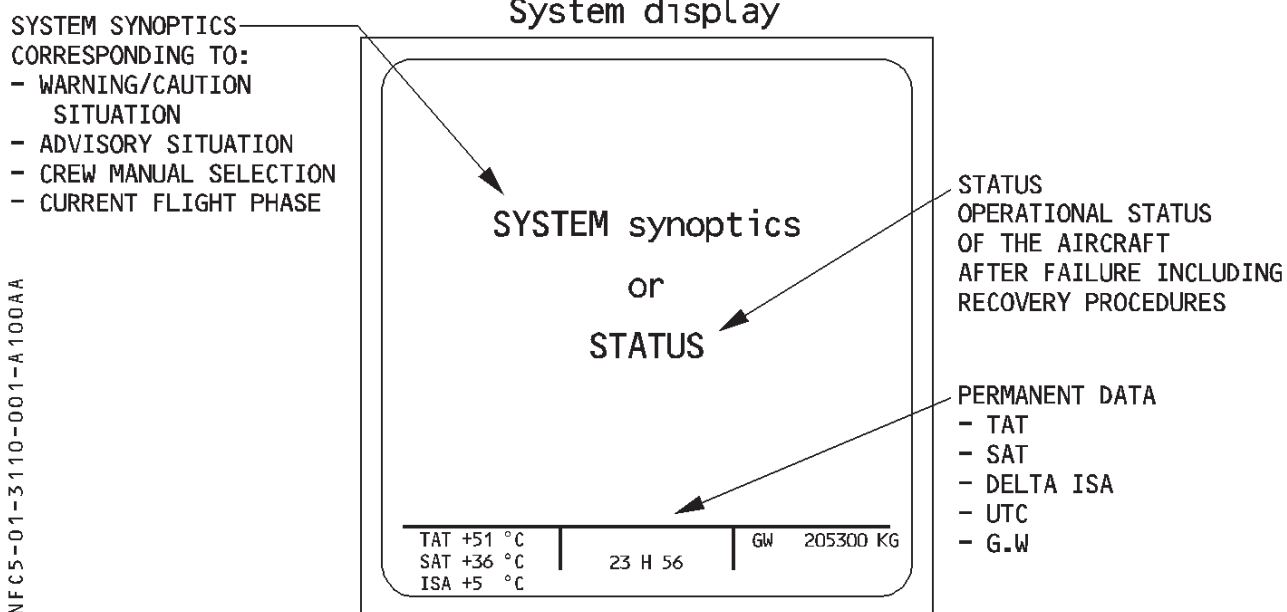
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PRIORITY RULES

There are three priority levels for warnings and cautions :

- A level 3 warning has priority over a level 2 caution which has priority over a level 1 caution.

The FWC observes these priorities.

TYPES OF FAILURES

Independent : a failure that affects an isolated system or item of equipment without degrading the performance of others in the aircraft.

Primary : a failure of a system or an item of equipment that costs the aircraft the use of other systems or items of equipment.

Secondary : the loss of a system or an item of equipment resulting from a primary failure.

R AUDIO INDICATORS

AUDIO INDICATOR	MEANING	DURATION	AUDIO INDICATOR CANCELLATION *
CONTINUOUS REPETITIVE CHIME	RED WARNINGS	PERMANENT	Press * MASTER WARN It
SINGLE CHIME	AMBER CAUTION	1/2 seconds	
CAVALRY CHARGE ***	A/P DISCONNECTION BY TAKE OVER pb	1.5 seconds	Second push on TAKE OVER pb
	A/P DISCONNECTION DUE TO FAILURE	PERMANENT	Press MASTER WARN It or TAKE OVER pb
TRIPLE CLICK	LANDING CAPABILITY CHANGE	1/2 sec. (3 pulses)	
CRICKET + "STALL" message (synthetic voice)	STALL	PERMANENT	NIL
BUZZER	CABIN CALL	3 seconds	NIL
	EMER CABIN CALL	3 seconds REPEATED 3 TIMES	NIL
	MECH CALL **	As long as outside pb pressed	NIL
CONTINUOUS BUZZER **	SELCAL CALL	PERMANENT	Press RESET key on ACP
"WINDSHEAR" (synthetic voice)	WINDSHEAR	REPEATED 3 TIMES	NIL

- R *
- The pilot can cancel any audio indicator, by pressing :
- The EMER CANC pushbutton on the ECAM control panel,
 - The MASTER WARN pushbutton, except for OVERSPEED or L/G NOT DOWN warnings.
- **
- The pilot can cancel the continuous buzzer, by pressing the MASTER CAUT pushbutton.
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- When the flight crew presses the MASTER WARN pushbutton (following a warning that leads to an AP disconnection), the Cavalry Charge sound is only inhibited when it has been emitted in order to avoid an unintended cancellation.

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	A/P DISCONNECTION DUE TO FAILURE	PERMANENT	Press MASTER WARN It or TAKE OVER pb
TRIPLE CLICK	LANDING CAPABILITY CHANGE or "GPS PRIMARY LOST" in approach	1/2 sec. (3 pulses)	
CRICKET + "STALL" message (synthetic voice)	STALL	PERMANENT	NIL
BUZZER	CABIN CALL	3 seconds	NIL
	EMER CABIN CALL	3 seconds REPEATED 3 TIMES	NIL
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CONTINUOUS BUZZER **	SELCAL CALL	PERMANENT	Press RESET key on ACP
"WINDSHEAR" (synthetic voice)	WINDSHEAR	REPEATED 3 TIMES	NIL
" GO AROUND WINDSHEAR AHEAD" (synthetic voice)	Windshear ahead detected during the landing phase	PERMANENT	NIL
"WINDSHEAR AHEAD" (twice) (synthetic voice)	Windshear ahead detected during the takeoff phase	PERMANENT	NIL
"MONITOR RADAR DISPLAY" (synthetic voice)	Windshear ahead detected caution message	PERMANENT	NIL

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CONTINUOUS REPETITIVE CHIME	RED WARNINGS	PERMANENT	Press * MASTER WARN It
SINGLE CHIME	AMBER CAUTION	0.5 seconds	
CAVALRY CHARGE ***	A/P DISCONNECTION BY TAKE OVER pb	1.5 seconds	Second push on TAKE OVER pb
	A/P DISCONNECTION DUE TO FAILURE	PERMANENT	Press MASTER WARN It or TAKE OVER pb
TRIPLE CLICK	Landing capability downgrade or "GPS PRIMARY LOST" in approach, or mode reversion	0.5 sec. (3 pulses)	
CRICKET + "STALL" message (synthetic voice)	STALL	PERMANENT	NIL
BUZZER	CABIN CALL	3 seconds	NIL
	EMER CABIN CALL	3 seconds REPEATED 3 TIMES	NIL
	MECH CALL **	As long as outside pb pressed	NIL
CONTINUOUS BUZZER **	SELCAL CALL	PERMANENT	Press RESET key on ACP
"WINDSHEAR" (synthetic voice)	WINDSHEAR	REPEATED 3 TIMES	NIL
"GO AROUND WINDSHEAR AHEAD" (synthetic voice)	Windshear ahead detected during the landing phase	PERMANENT	NIL
"WINDSHEAR AHEAD" (twice) (synthetic voice)	Windshear ahead detected during the takeoff phase	PERMANENT	NIL
"MONITOR RADAR DISPLAY" (synthetic voice)	Windshear ahead detected caution message	PERMANENT	NIL

- R *
- The pilot can cancel any audio indicator, by pressing the :
 - EMER CANC pushbutton on the ECAM control panel, or
 - MASTER WARN pushbutton, except for OVERSPEED or L/G NOT DOWN warnings.
- ** The pilot can cancel the continuous buzzer, by pressing the MASTER CAUT pushbutton.
- *** When the flight crew presses the MASTER WARN pushbutton (following a warning that leads to an AP disconnection), the Cavalry Charge sound is only inhibited when it has been emitted, in order to avoid an unintended cancellation.

R

AUDIO INDICATOR	MEANING	DURATION	AUDIO INDICATOR CANCELLATION *
C CHORD	ALTITUDE ALERT (Refer to 1.31.40)	1.5 seconds or PERMANENT	new ALTITUDE selection or press MASTER WARN pb
AUTO CALL OUT (synthetic voice)	HEIGHT ANNOUNCEMENT BELOW 2500 FT (Refer to 1.34.10)	PERMANENT	NIL
GROUND PROXIMITY WARNING (synthetic voice)	(Refer to 1.34.70)	PERMANENT	NIL
"PRIORITY LEFT" "PRIORITY RIGHT" (synthetic voice)	A/P TAKE OVER pb	1 second	NIL
"RETARD" (synthetic voice)	Thrust levers not in IDLE position for landing	ONE TIME at 20 ft (10 ft in autoland with A/THR ON), then PERMANENT until thrust levers are set to IDLE	NIL
"SPEED, SPEED, SPEED" (Synthetic voice)	Current thrust is not sufficient to recover a positive flight through pitch control	Every 5 seconds until thrust is increased	THRUST LEVER(s)
"DUAL INPUT" (synthetic voice)	Both sidesticks are moved simultaneously	Every 5 seconds	One sidestick deactivated

- R *
- The pilot can cancel any audio indicator by pressing :
 - The EMER CANC pushbutton on the ECAM control panel,
 - The MASTER WARN pushbutton, except for OVERSPEED or L/G NOT DOWN warnings.



R

AUDIO INDICATOR	MEANING	DURATION	AUDIO INDICATOR CANCELLATION *
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AUTO CALL OUT (synthetic voice)	HEIGHT ANNOUNCEMENT BELOW 2500 FT (Refer to 1.34.10)	PERMANENT	NIL
GROUND PROXIMITY WARNING (synthetic voice)	(Refer to 1.34.70)	PERMANENT	NIL
"PRIORITY LEFT" "PRIORITY RIGHT" (synthetic voice)	A/P TAKE OVER pb	1 second	NIL
"RETARD" (synthetic voice)	Thrust levers not in IDLE position for landing	ONE TIME at 20 ft (10 ft in autoland with A/THR ON), then PERMANENT until thrust levers are set to IDLE	NIL
TCAS (synthetic voice)	(Refer to 1.34.80)	PERMANENT	NIL
"SPEED, SPEED, SPEED" (synthetic voice)	Current thrust is not sufficient to recover a positive flight through pitch control	Every 5 seconds until thrust is increased	THRUST LEVER(s)

- R *
- The pilot can cancel any audio indicator by pressing :
 - The EMER CANC pushbutton on the ECAM control panel,
 - The MASTER WARN pushbutton, except for OVERSPEED or L/G NOT DOWN warnings.

R

AUDIO INDICATOR	MEANING	DURATION	AUDIO INDICATOR CANCELLATION *
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AUTO CALL OUT (synthetic voice)	HEIGHT ANNOUNCEMENT BELOW 2500 FT (Refer to 1.34.10)	PERMANENT	NIL
GROUND PROXIMITY WARNING (synthetic voice)	(Refer to 1.34.70)	PERMANENT	NIL
"PRIORITY LEFT" "PRIORITY RIGHT" (synthetic voice)	A/P TAKE OVER pb	1 second	NIL
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"SPEED, SPEED, SPEED" (Synthetic voice)	Current thrust is not sufficient to recover a positive flight through pitch control	Every 5 seconds until thrust is increased	THRUST LEVER(s)
"DUAL INPUT" (synthetic voice)	Both sidesticks are moved simultaneously	Every 5 seconds	One sidestick deactivated

- R *
- The pilot can cancel any audio indicator by pressing :
 - The EMER CANC pushbutton on the ECAM control panel,
 - The MASTER WARN pushbutton, except for OVERSPEED or L/G NOT DOWN warnings.

AUDIO INDICATOR	MEANING	DURATION	AUDIO INDICATOR CANCELLATION *
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AUTO CALL OUT (synthetic voice)	HEIGHT ANNOUNCEMENT BELOW 2500 FT (Refer to 1.34.10)	PERMANENT	NIL
GROUND PROXIMITY WARNING (synthetic voice)	(Refer to 1.34.70)	PERMANENT	NIL
"PRIORITY LEFT" "PRIORITY RIGHT" (synthetic voice)	A/P TAKE OVER pb	1 second	NIL
"RETARD" (synthetic voice)	Thrust levers not in IDLE position for landing	ONE TIME at 20 ft (10 ft in autoland with A/THR ON), then PERMANENT until thrust levers are set to IDLE	NIL
TCAS (synthetic voice)	(Refer to 1.34.80)	PERMANENT	NIL
"SPEED, SPEED, SPEED" (Synthetic voice)	Current thrust is not sufficient to recover a positive flight through pitch control	Every 5 seconds until thrust is increased	THRUST LEVER(s)
"DUAL INPUT" (synthetic voice)	Both sidesticks are moved simultaneously	Every 5 seconds	One sidestick deactivated
"PITCH, PITCH" (Synthetic voice)	Aircraft pitch attitude becomes excessive during flare and landing.	one time	NIL

- * The pilot can cancel any audio indicator by pressing :
- The EMER CANC pushbutton on the ECAM control panel,
 - The MASTER WARN pushbutton, except for OVERSPEED or L/G NOT DOWN warnings.

GENERAL

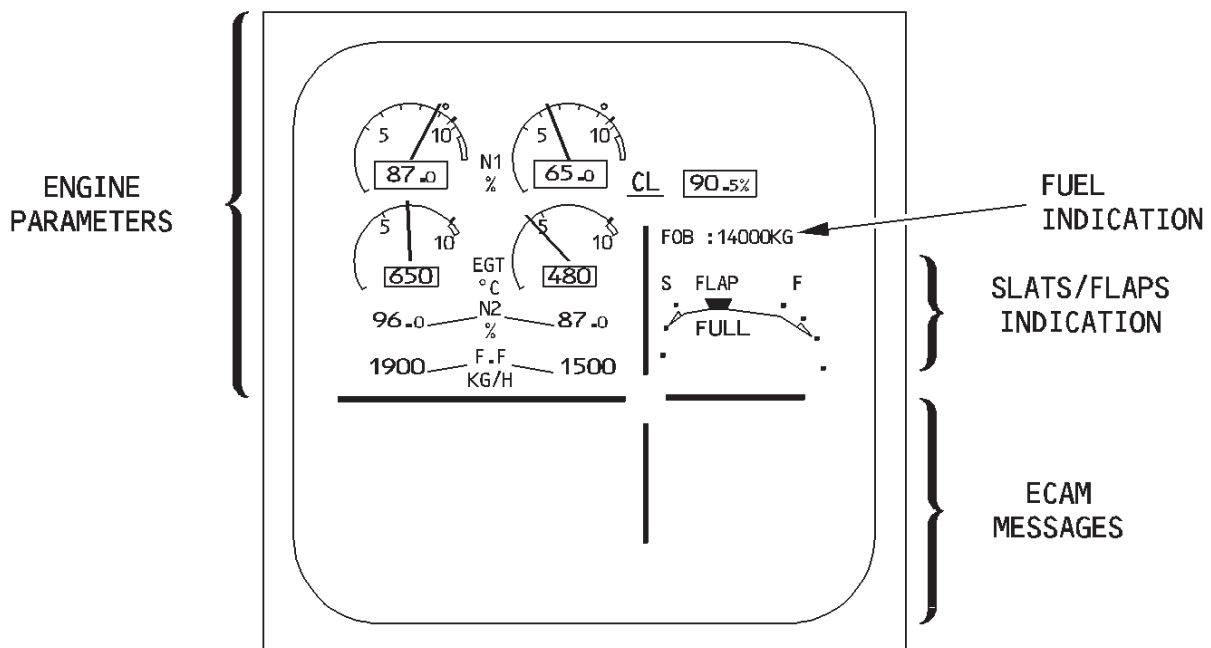
The E/WD appears on the upper ECAM display unit (DU).

— The upper part of this DU displays :

- R · Engine parameters (refer to 1.70.90)
- R · Fuel on board (FOB) (refer to 1.28.20)
- R · Position of slats and flaps (refer to 1.27.40)

— The lower part of this DU displays messages generated by the FWC :

- Warning and caution messages when a failure occurs
- Memos when there is no failure



The lower part, which is dedicated to ECAM messages, is divided into two parts of several lines each.

Left part : — Primary or independent warnings and cautions, or
— Memo information

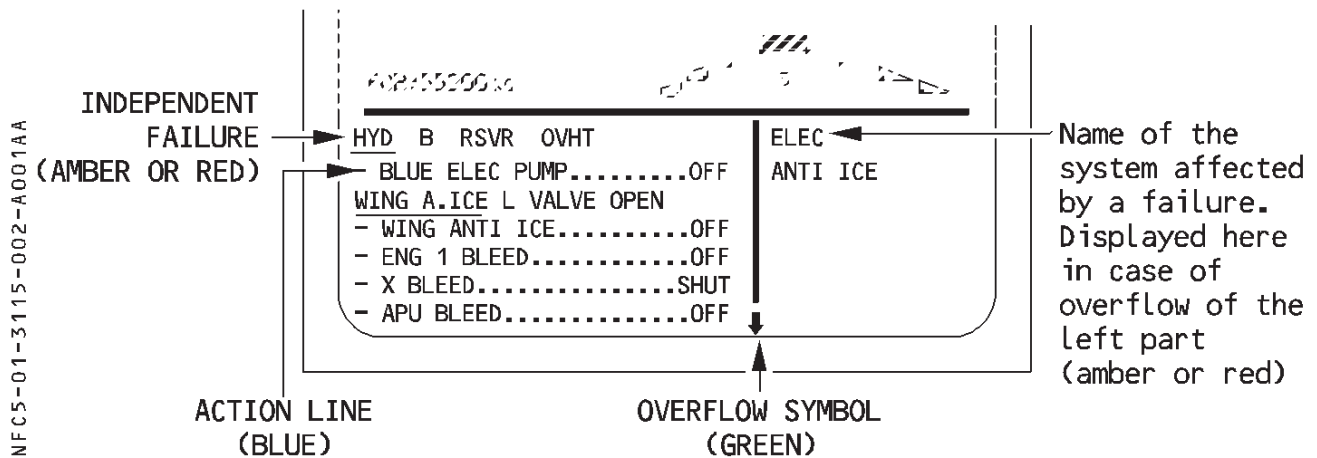
Right part : — Title of system affected by a primary or independent warning or caution in case of overflow on the left part, or
— Secondary failure, or
— Memo, or
— Special lines (such as "AP OFF", "LAND ASAP")

As soon as the FWC detects a failure, and if there is no flight phase inhibition active, the E/WD displays the title of the failure and actions to be taken.

The action line clears automatically when the flight crew has executed the required action.

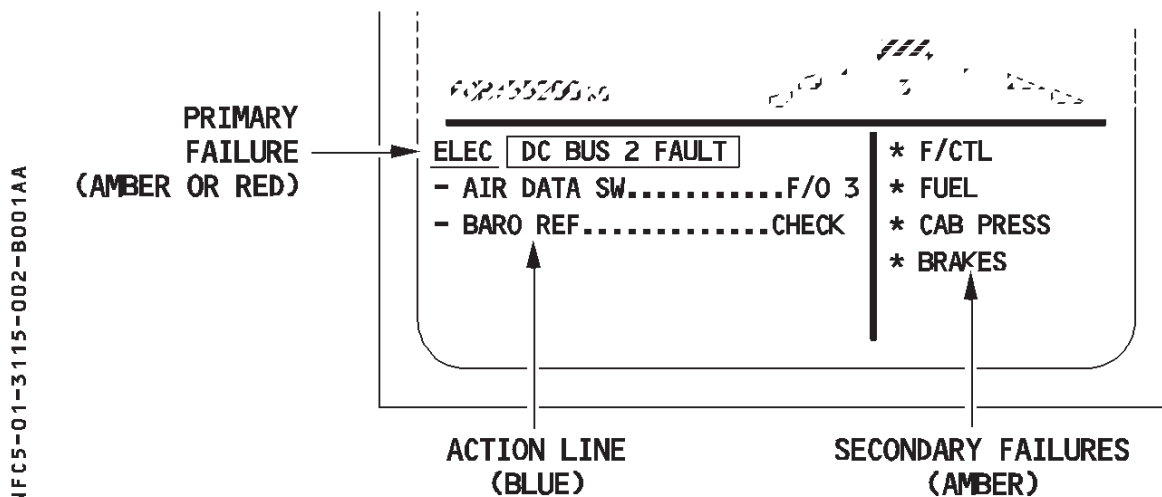
Note : Certain actions will not disappear after execution.

INDEPENDENT FAILURE



If there are too many ECAM messages for the amount of space available in the lower part of the E/WD, a green arrow appears at the bottom of the display, pointing down to show that the information has overflowed off the screen. The pilot can scroll down to view additional messages by pushing the CLR pushbutton on the ECAM control panel (on the pedestal, just below the lower ECAM DU).

PRIMARY and SECONDARY FAILURE



The ECAM DU displays a primary failure as a boxed title.

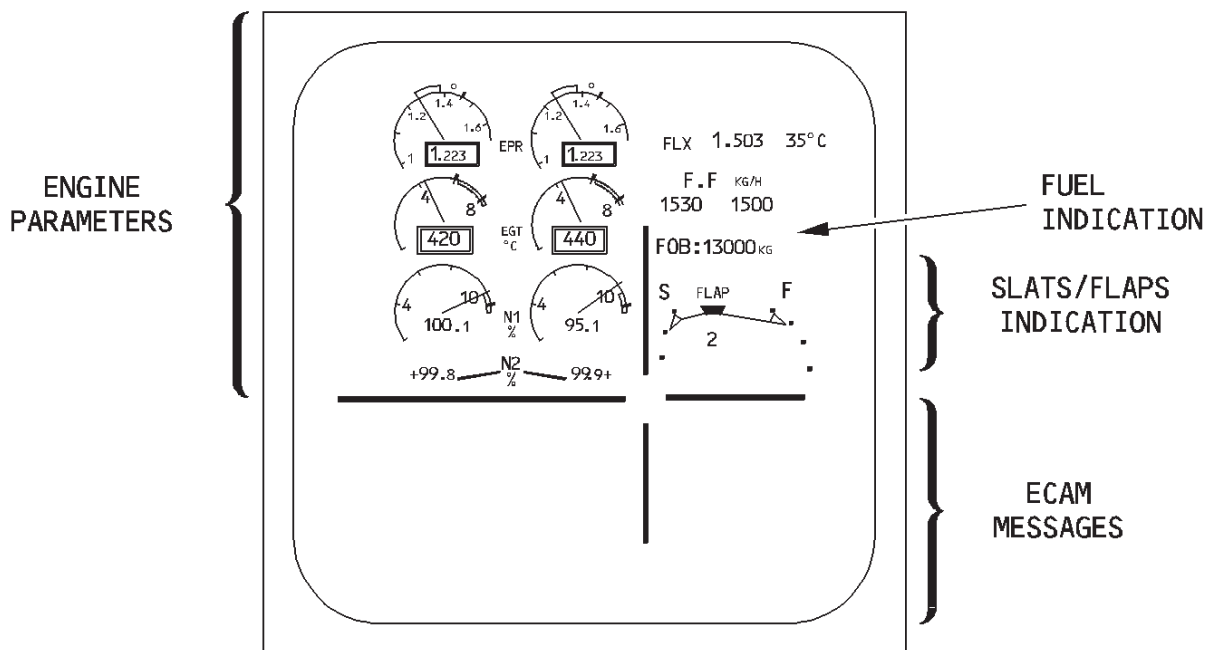
It identifies a secondary failure by putting a star in front of the title of the affected system.

R Note : The DU displays the overflow symbol, if primary or secondary failures overflow. In
R case of ELEC EMER CONFIG, the secondary failures are inhibited.

GENERAL

The E/WD appears on the upper ECAM display unit (DU).

- The upper part of this DU displays :
 - Engine parameters (refer to 1.70.95)
 - Fuel on board (FOB) (refer to 1.28.20)
 - Position of slats and flaps (refer to 1.27.40)
- The lower part of this DU displays messages generated by the FWC :
 - Warning and caution messages when a failure occurs
 - Memos when there is no failure



The lower part, which is dedicated to ECAM messages, is divided into two parts of several lines each.

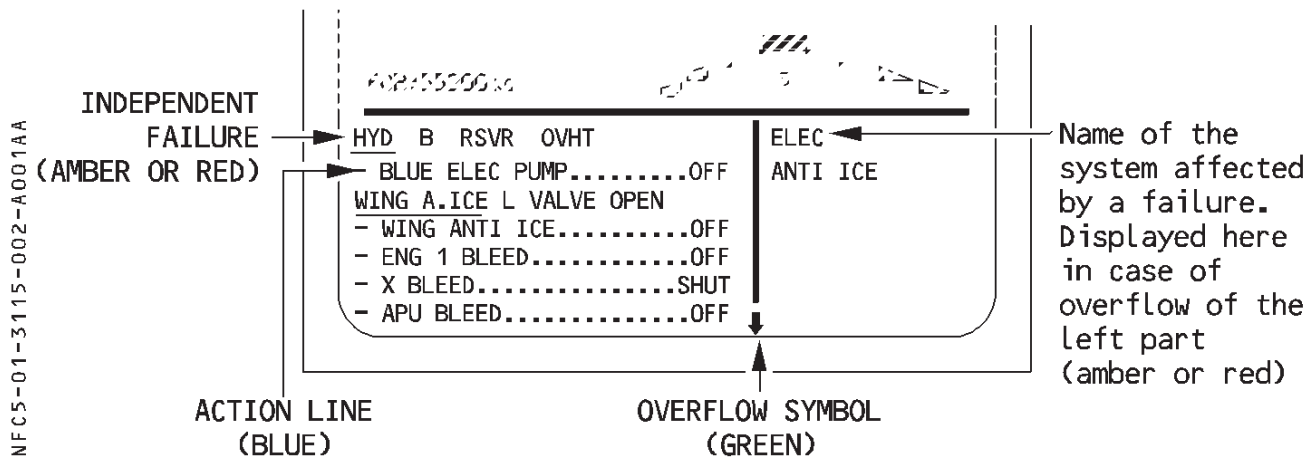
- Left part : – Primary or independent warnings and cautions, or
– Memo information
- Right part : – Title of system affected by a primary or independent warning or caution in case of overflow on the left part, or
– Secondary failure, or
– Memo, or
– Special lines (such as "AP OFF", "LAND ASAP")

As soon as the FWC detects a failure, and if there is no flight phase inhibition active, the E/WD displays the title of the failure and the actions to be taken.

The action line automatically clears when the flight crew has executed the required action.

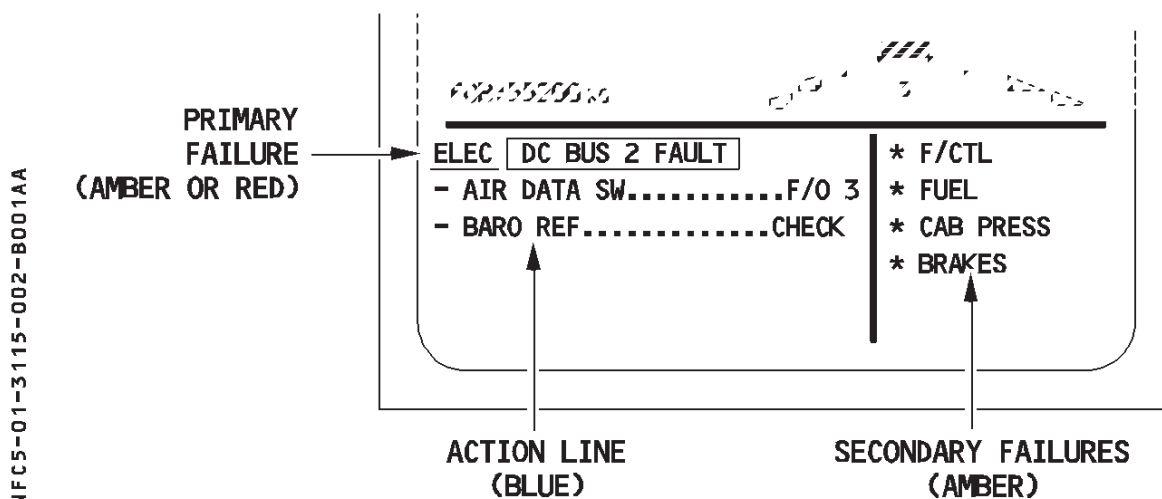
Note : Certain actions will not disappear after execution.

INDEPENDENT FAILURE



If there are too many ECAM messages for the amount of space available in the lower part of the E/WD, a green arrow appears at the bottom of the display, pointing down to show that the information has overflowed off the screen. The pilot can scroll down to view additional messages by pushing the CLR pushbutton on the ECAM control panel (on the pedestal, just below the lower ECAM DU).

PRIMARY and SECONDARY FAILURE



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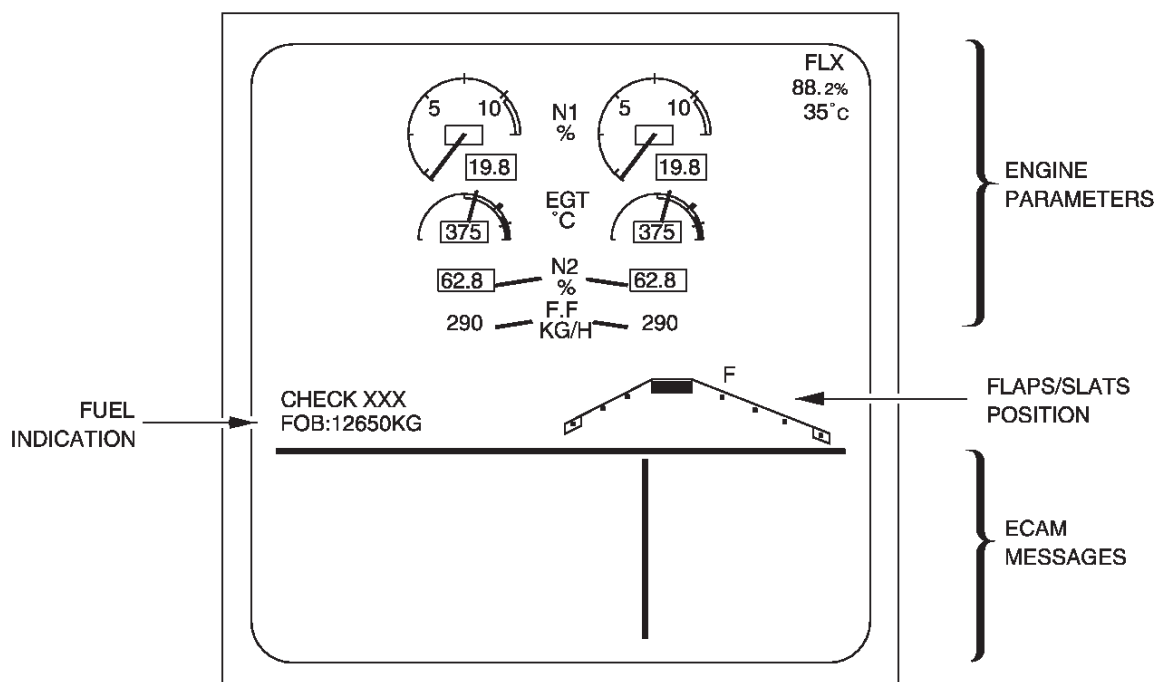
It identifies a secondary failure by putting a star in front of the title of the affected system.

R Note : The DU displays the overflow symbol, if primary or secondary failures overflow. In
R case of ELEC EMER CONFIG, the secondary failures are inhibited.

GENERAL

The Engine Warning Display (E/WD) appears on the ECAM upper Display Unit (DU).

- The upper part of this DU displays :
 - Engine parameters (Refer to 1.70.90)
 - Feedback messages (Refer to 1.31.05)
 - Fuel On Board (FOB) (Refer to 1.28.20)
 - Slats/Flaps' position (Refer to 1.27.40)
- The lower part of this DU displays messages generated by the FWC :
 - Warning and caution messages when a failure occurs
 - Memos when there is no failure.



The lower part of the DU, dedicated to ECAM messages, is divided into two sections that have several lines each.

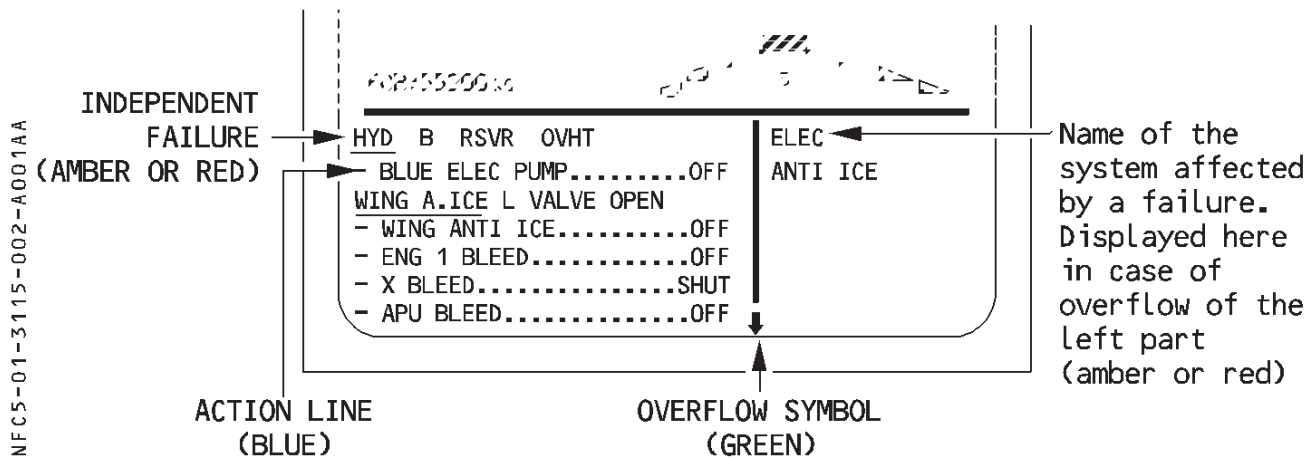
Bottom left : – Primary or independent warnings and cautions, or
– Memo information.

Bottom right : – Title of the system affected by a primary or independent warning or caution, in the case of overflow on the bottom left part, or
– Secondary failure, or
– Memo, or
– Special lines (e.g. "AP OFF", "LAND ASAP").

As soon as the FWC detects a failure, and if there is no flight phase inhibition active, the E/WD displays the title of the failure and the actions that the flight crew must perform. The action line automatically clears, when the flight crew performs the necessary action.

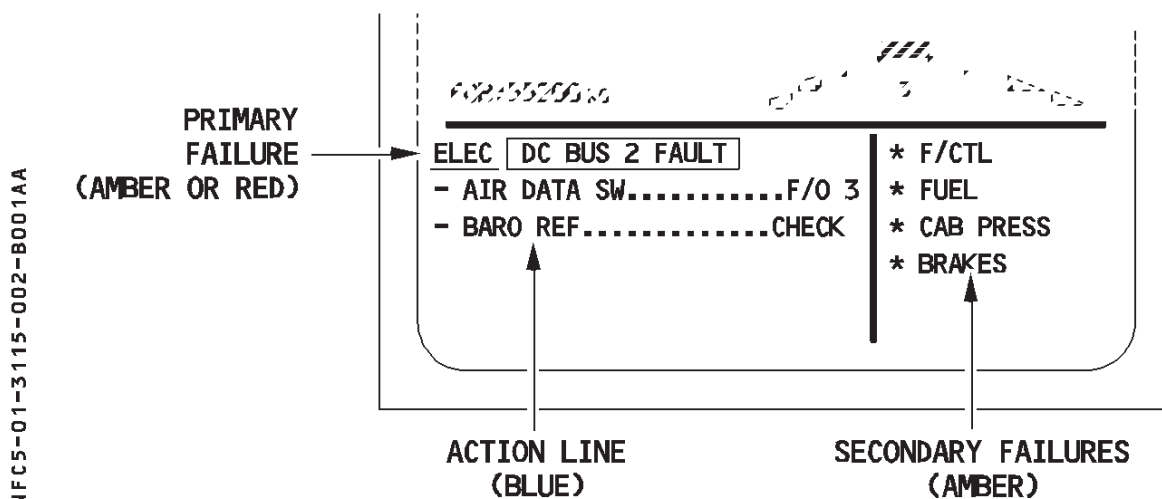
Note : Some action lines do not disappear from the E/WD even after the flight crew performs the necessary action.

INDEPENDENT FAILURE



If there are too many ECAM messages for the amount of space available in the lower part of the E/WD, a green arrow appears at the bottom of the display, pointing down to show that the information has overflowed off the screen. The pilot can scroll down to view additional messages by pushing the CLR pushbutton on the ECAM control panel (on the pedestal, just below the lower ECAM DU).

PRIMARY and SECONDARY FAILURE



The ECAM DU displays a primary failure as a boxed title.

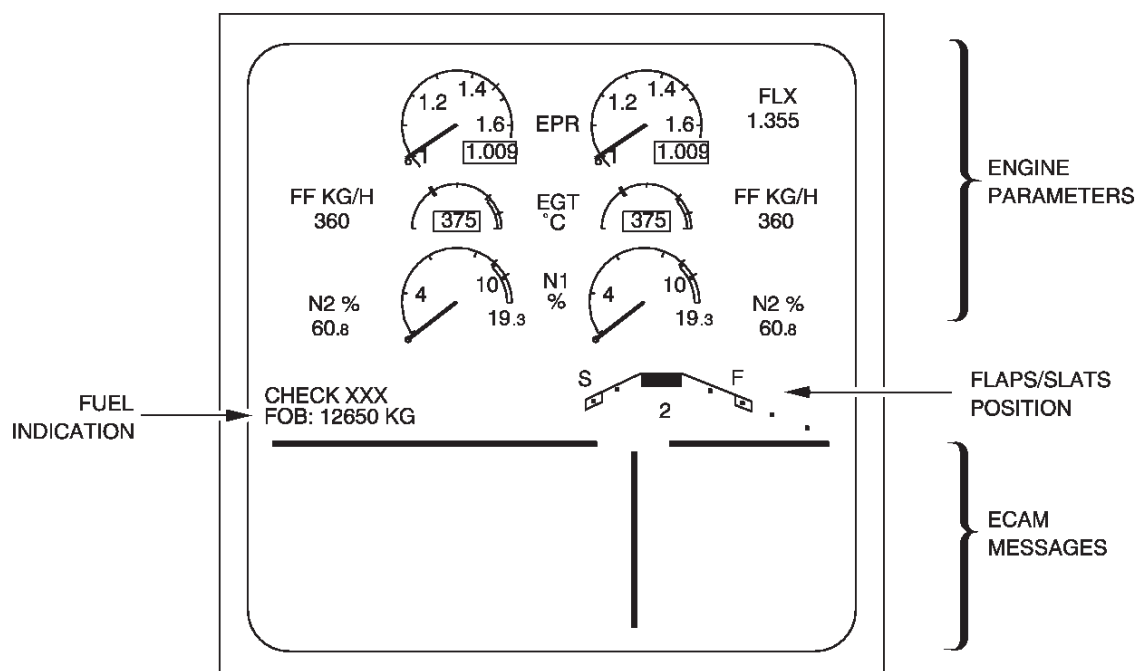
It identifies a secondary failure by putting a star in front of the title of the affected system.

R Note : The DU displays the overflow symbol, if primary or secondary failures overflow. In
R case of ELEC EMER CONFIG, the secondary failures are inhibited.

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 - Feedback messages (Refer to 1.31.05)
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- The lower part of this DU displays messages generated by the FWC :
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 - Memos when there is no failure.



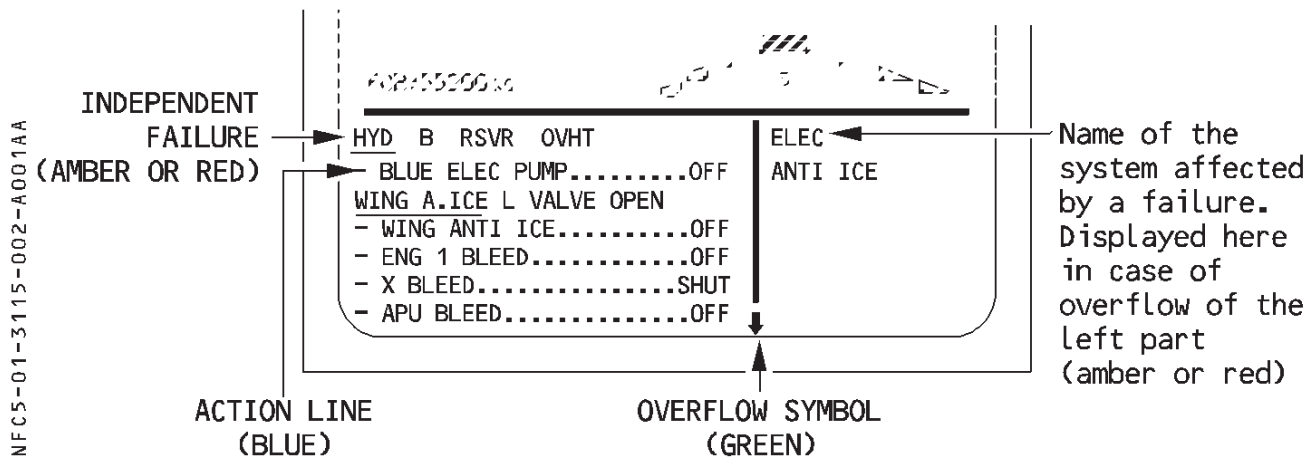
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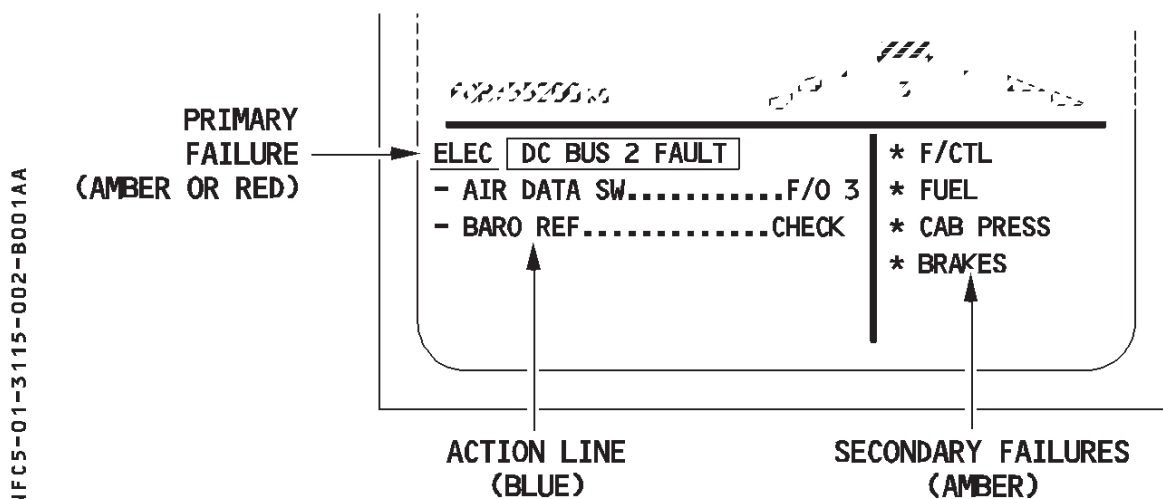
Note : Some action lines do not disappear from the E/WD even after the flight crew performs the necessary action.

INDEPENDENT FAILURE



If there are too many ECAM messages for the amount of space available in the lower part of the E/WD, a green arrow appears at the bottom of the display, pointing down to show that the information has overflowed off the screen. The pilot can scroll down to view additional messages by pushing the CLR pushbutton on the ECAM control panel (on the pedestal, just below the lower ECAM DU).

PRIMARY and SECONDARY FAILURE



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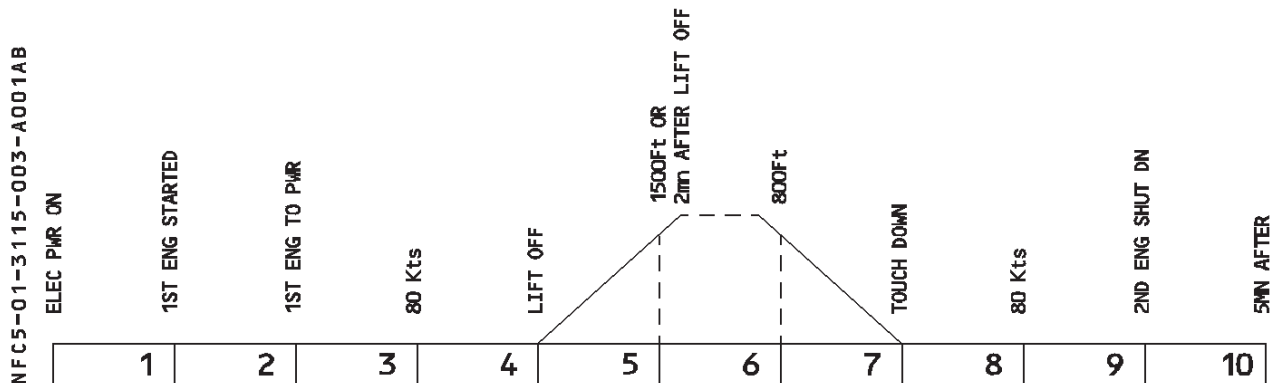
It identifies a secondary failure by putting a star in front of the title of the affected system.

R Note : The DU displays the overflow symbol, if primary or secondary failures overflow. In
R case of ELEC EMER CONFIG, the secondary failures are inhibited.

FLIGHT PHASES

GENERAL

The FWC divides its functions according to these ten flight phases :

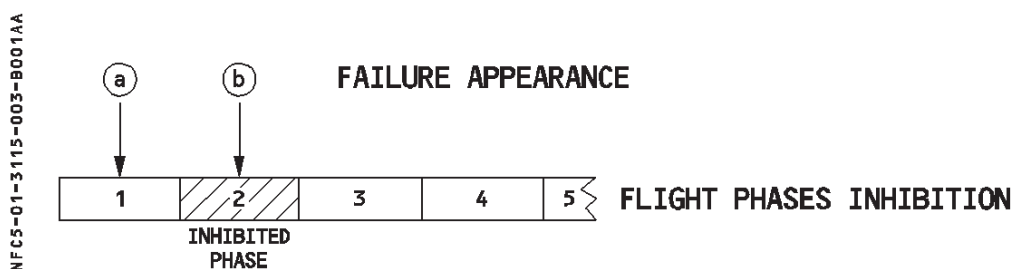


To improve its operational efficacy, the computer inhibits some warnings and cautions for certain flight phases. It does so to avoid alerting the pilots unnecessarily at times when they have high workloads, such as during takeoff or landing. In these two phases, the DU displays magenta memos : "T.O. INHIBIT" (flight phases 3, 4, and 5), and "LDG INHIBIT" (flight phases 7 and 8).

Note : These flight phases are different from and independent of the ones that the FMGC uses.

FLIGHT PHASE INHIBITION

Two cases are possible (for instance) :



Effect on E/WD :

- (a) The failure occurs during phase 1. The E/WD displays the warning immediately and continues to display it as long as the failure is present, even in phase 2.
- (b) The failure occurs during phase 2. The E/WD displays the warning only when the aircraft has entered phase 3, where it is not inhibited. Then the warning remains displayed as long as the failure is present.

MEMOS

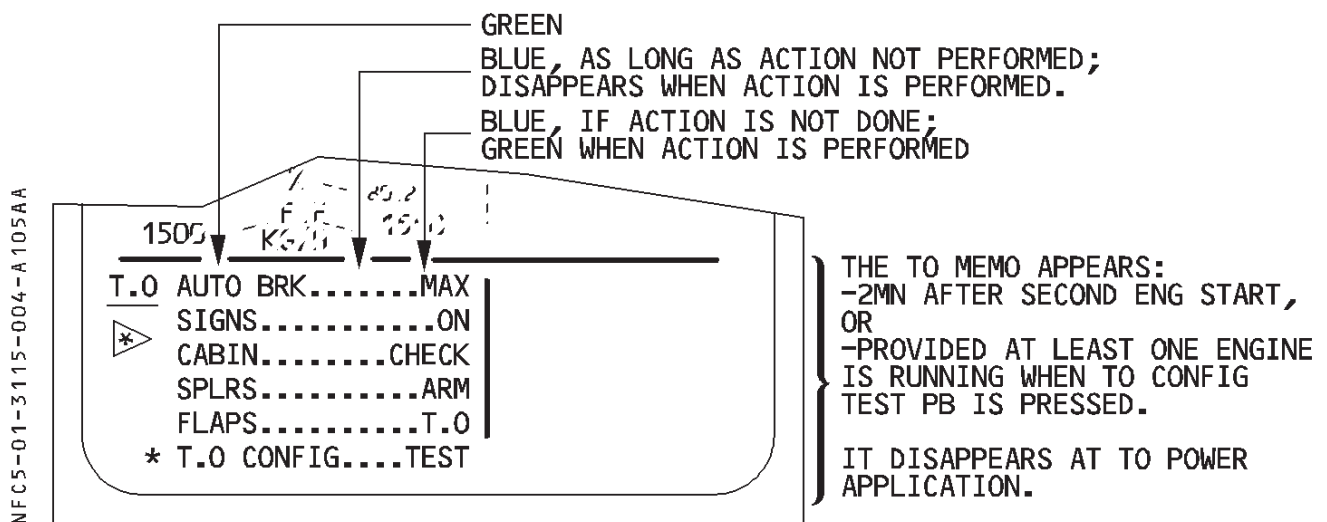
DISPLAY

- R Memos appear in the lower part of the E/WD. They are normally in green, but may be amber in abnormal situations.
 Memos list functions or systems that are temporarily used in normal operations.
 Each chapter of the "Warning and Cautions" section of this manual lists memo messages.

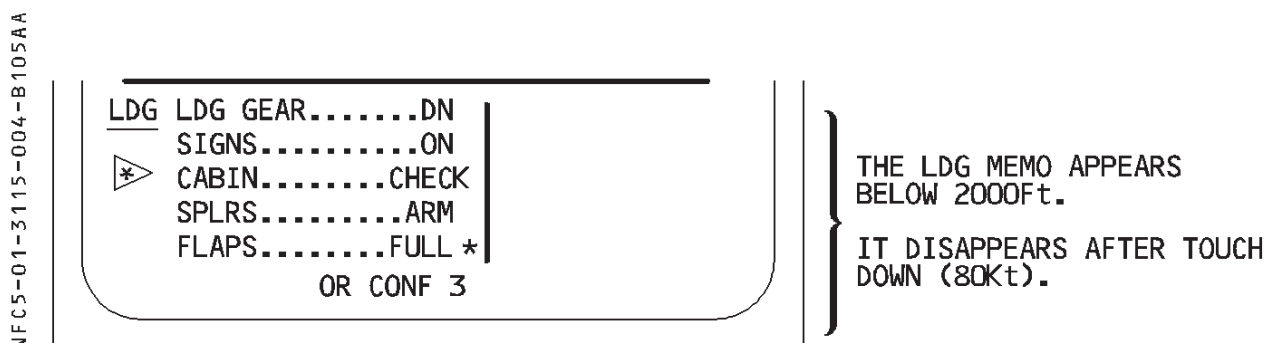
TO AND LDG MEMOS

During the takeoff and landing phases, the right side of the memo area displays specific T.O. INHIBIT or LDG INHIBIT (magenta) memos.

Takeoff and landing memos are displayed, as follows, during the related flight phases :



- * This line disappears when the test is completed. It is replaced by "TO CONFIG NORMAL", if aircraft configuration is correct.
 The test is requested again, if the configuration becomes abnormal.



- * "CONF 3" is displayed in alternate or direct law, or if the GPWS LDG FLAP 3 pushbutton is ON.

CONFIGURATION WARNINGS

The following warnings and cautions appear in the lower part of the E/WD if the aircraft is not in takeoff configuration when the flight crew presses the T.O. CONFIG pushbutton on the ECAM control panel or applies takeoff power.

WARNINGS / CAUTIONS	T.O. CONFIG TEST	T.O. POWER
<u>CONFIG</u> RUD TRIM NOT IN T.O RANGE (R)	TRIGGERED	TRIGGERED
<u>CONFIG</u> PITCH TRIM NOT IN T.O RANGE (R)		
<u>CONFIG</u> FLAPS NOT IN T.O CONFIG (R)		
<u>CONFIG</u> SPD BRK NOT RETRACTED (R)		
<u>CONFIG</u> SLATS NOT IN T.O CONFIG (R)		
<u>CONFIG</u> L SIDESTICK FAULT (R)		
<u>CONFIG</u> R SIDESTICK FAULT (R)		
<u>DOOR</u> (A)		
<u>FWS</u> OEB/FWC DISCREPANCY (A)		
<u>BRAKES</u> HOT (A)		
<u>ENG 1</u> OIL LO TEMP (A)		NOT TRIGGERED
<u>ENG 2</u> OIL LO TEMP (A)		
<u>CONFIG</u> PARK BRK ON (R)	NOT TRIGGERED	TRIGGERED
<u>ENG</u> FLEX TEMP NOT SET (A)		

(R) Red warning

(A) Amber caution

CONFIGURATION WARNINGS

The following warnings and cautions appear in the lower part of the E/WD if the aircraft is not in takeoff configuration when the flight crew presses the T.O. CONFIG pushbutton on the ECAM control panel or applies takeoff power.

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<u>CONFIG</u> FLAPS NOT IN T.O CONFIG (R)		
<u>CONFIG</u> SPD BRK NOT RETRACTED (R)		
<u>CONFIG</u> SLATS NOT IN T.O CONFIG (R)		
<u>CONFIG</u> L SIDESTICK FAULT (R)		
<u>CONFIG</u> R SIDESTICK FAULT (R)		
<u>DOOR</u> (A)		
<u>FWS</u> OEB/FWC DISCREPANCY (A)		
<u>BRAKES</u> HOT (A)		
<u>ENG 1</u> OIL LO TEMP (A)		NOT TRIGGERED
<u>ENG 2</u> OIL LO TEMP (A)		
<u>CONFIG</u> PARK BRK ON (R)	NOT TRIGGERED	TRIGGERED
<u>ENG</u> FLEX THR LEVERS NOT SET (A)		

(R) Red warning

(A) Amber caution

CONFIGURATION WARNINGS

The following warnings and cautions appear in the lower part of the E/WD if the aircraft is not in takeoff configuration when the flight crew presses the T.O. CONFIG pushbutton on the ECAM control panel or applies takeoff power.

WARNINGS / CAUTIONS	T.O. CONFIG TEST	T.O. POWER
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<u>CONFIG</u> PITCH TRIM NOT IN T.O RANGE (R)		
<u>CONFIG</u> FLAPS NOT IN T.O CONFIG (R)		
<u>CONFIG</u> SPD BRK NOT RETRACTED (R)		
<u>CONFIG</u> SLATS NOT IN T.O CONFIG (R)		
<u>CONFIG</u> L SIDESTICK FAULT (R)		
<u>CONFIG</u> R SIDESTICK FAULT (R)		
<u>DOOR</u> (A)		
<u>FWS</u> OEB/FWC DISCREPANCY (A)		
<u>BRAKES</u> HOT (A)		
<u>CONFIG</u> PARK BRK ON (R)	NOT TRIGGERED	
<u>ENG</u> THR LEVERS NOT SET (A)		

(R) Red warning

(A) Amber caution

GENERAL

The system/status display (SD) uses the lower ECAM DU to display :

- pages showing synoptic diagrams of the aircraft systems, or
- the status page.

SYSTEM PAGES

The lower ECAM DU can display 12 system pages :

(For description see relevant FCOM chapter).

- ENGINE (secondary engine parameters)
- BLEED (air bleed)
- CAB PRESS (cabin pressurization)
- ELEC (electric power)
- HYD (hydraulic)
- FUEL (fuel)
- APU (auxiliary power unit)
- COND (air conditioning)
- DOOR/OXY (doors/oxygen)
- WHEEL (landing gear, braking, ground spoilers, etc.)
- F/CTL (flight controls)
- CRUISE (cruise)

The pilot may manually call up a system page for display on the lower ECAM DU, or the system may automatically display a page.

– Manual :

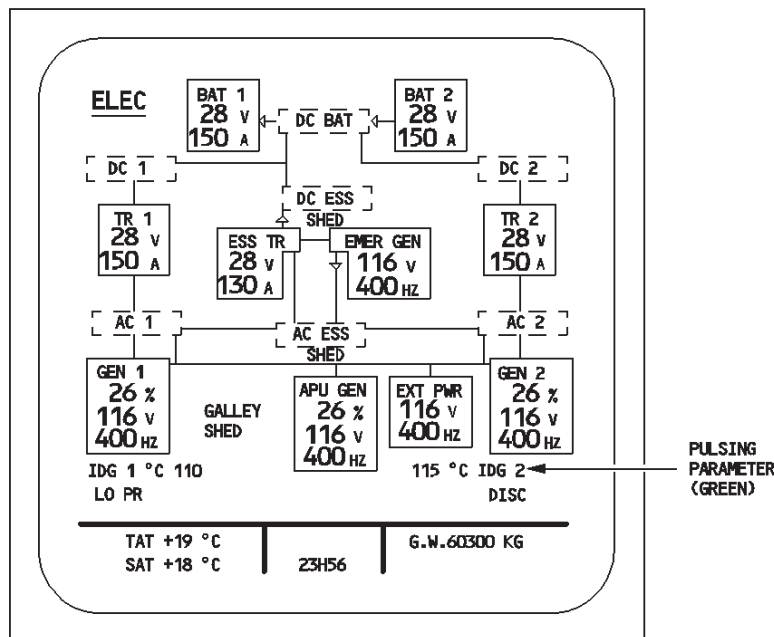
- The pilot can use the pushbutton on the ECAM control panel to call up any system page, except the CRUISE page, for display at any time.
- The corresponding pushbutton on the ECAM control panel lights up.
- A failure-related or advisory display automatically replaces a page the pilot has manually called up.

– Automatic, related to a failure :

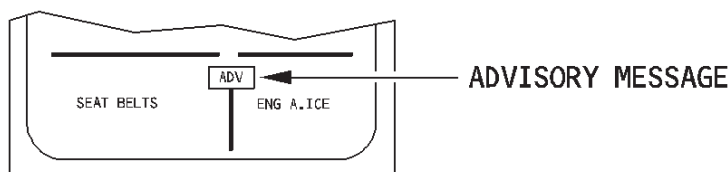
- The relevant system page automatically appears, as soon as any fault or malfunction triggers a caution or warning message.

– Automatic, advisory :

- The relevant system page automatically appears, when a parameter drifts out of its normal range.
- The value (shown in green) pulses, as long as it is outside its limits.
- The advisory mode is inhibited in some flight phases.



Note : If an advisory is triggered when the ECAM is in the single-display configuration, an advisory message appears on the upper part of the E/WD, and the associated key on the ECAM control panel flashes to identify the appropriate system page.



GENERAL

The system/status display (SD) uses the lower ECAM DU to display :

- pages showing synoptic diagrams of the aircraft systems, or
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SYSTEM PAGES

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- DOOR/OXY (doors/oxygen)
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The pilot may manually call up a system page for display on the lower ECAM DU, or the system may automatically display a page.

– Manual :

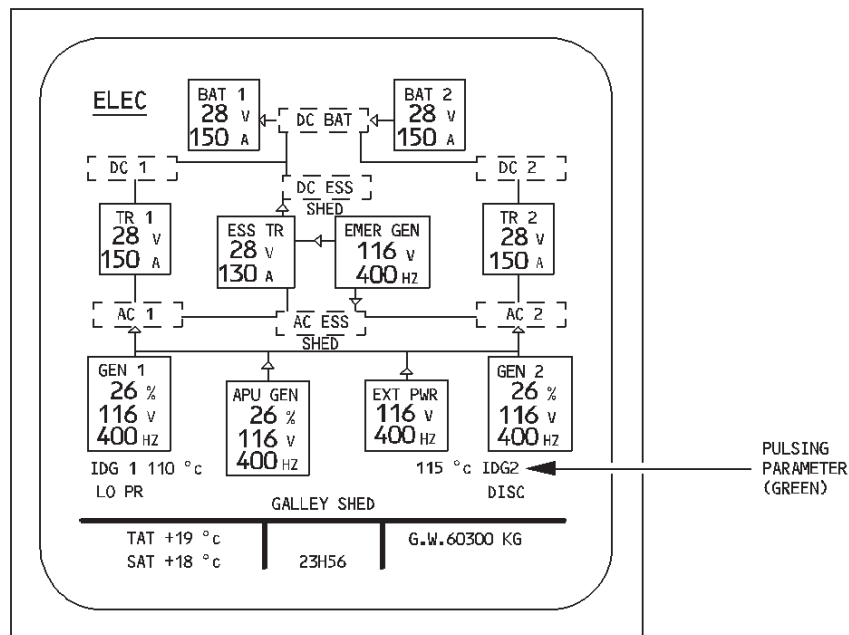
- The pilot can, at any time, use the pushbutton on the ECAM's control panel to call up and display any system page, except the CRUISE page.
- The corresponding pushbutton on the ECAM control panel lights up.
- A failure-related or advisory display automatically replaces a page that the pilot has manually called up.

– Automatic, related to a failure :

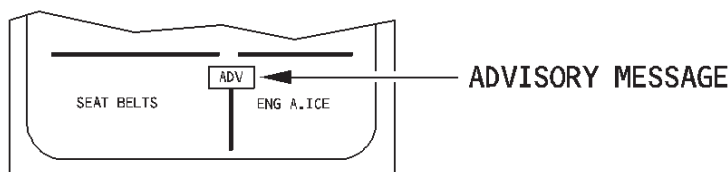
- The relevant system page automatically appears, as soon as any fault or malfunction triggers a caution or warning message.

– Automatic, advisory :

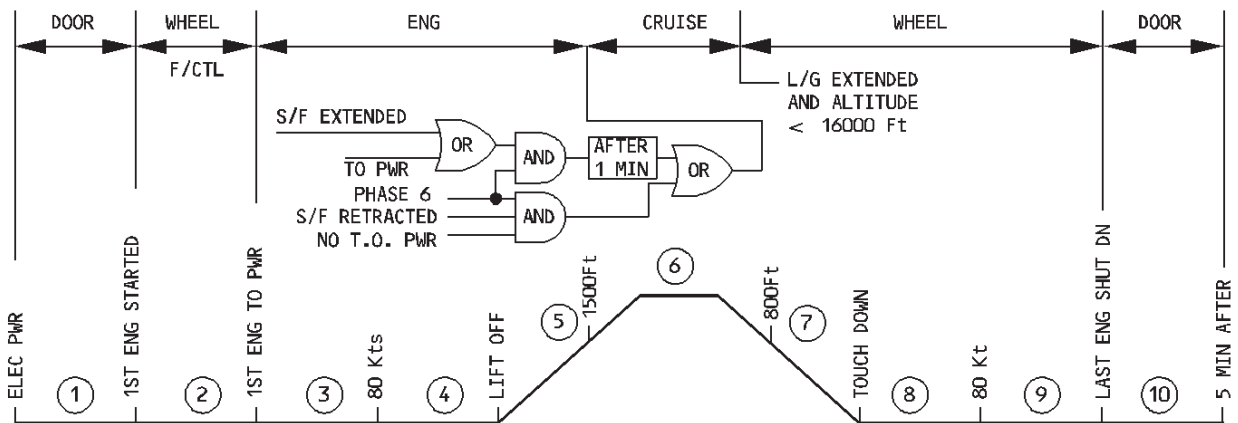
- The relevant system page automatically appears, when a parameter drifts out of its normal range.
- The value (shown in green) pulses, as long as it is outside its limits.
- The advisory mode is inhibited in some flight phases.



Note : If an advisory is triggered, when the ECAM is in the single-display configuration, an advisory message appears on the upper part of the E/WD, and the associated key on the ECAM control panel flashes to identify the appropriate system page.

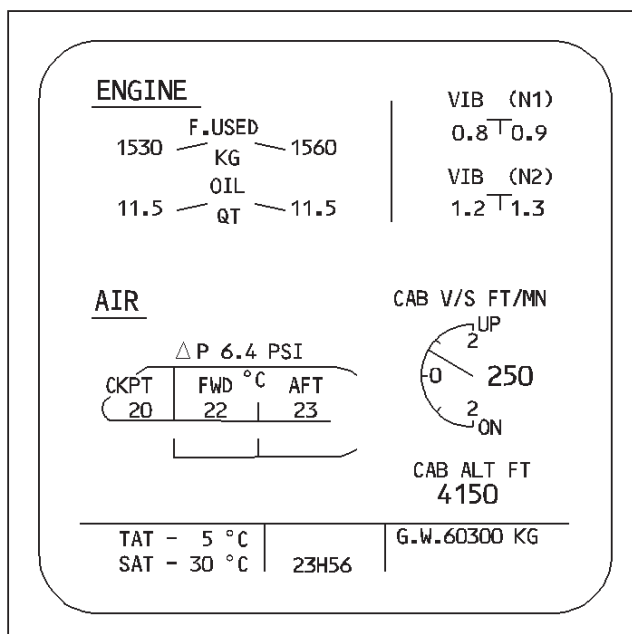


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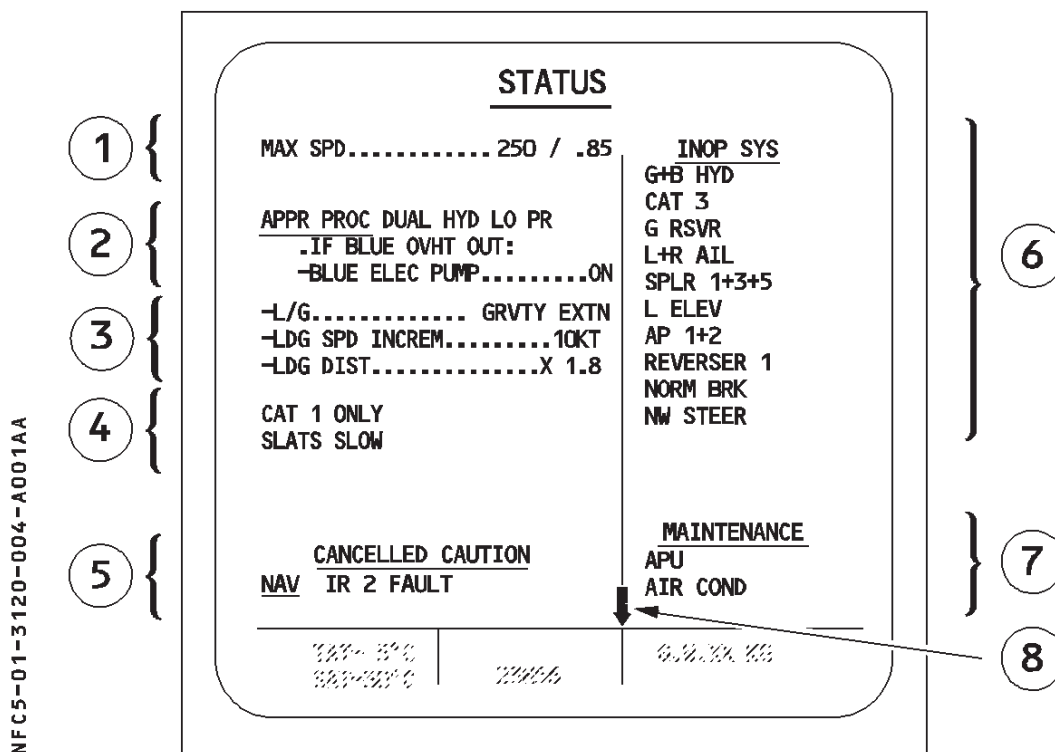


- For a description of the ENGINE and AIR indications that appear when the SD is displaying the CRUISE page, see the relevant FCOM chapter.

R



STATUS PAGE



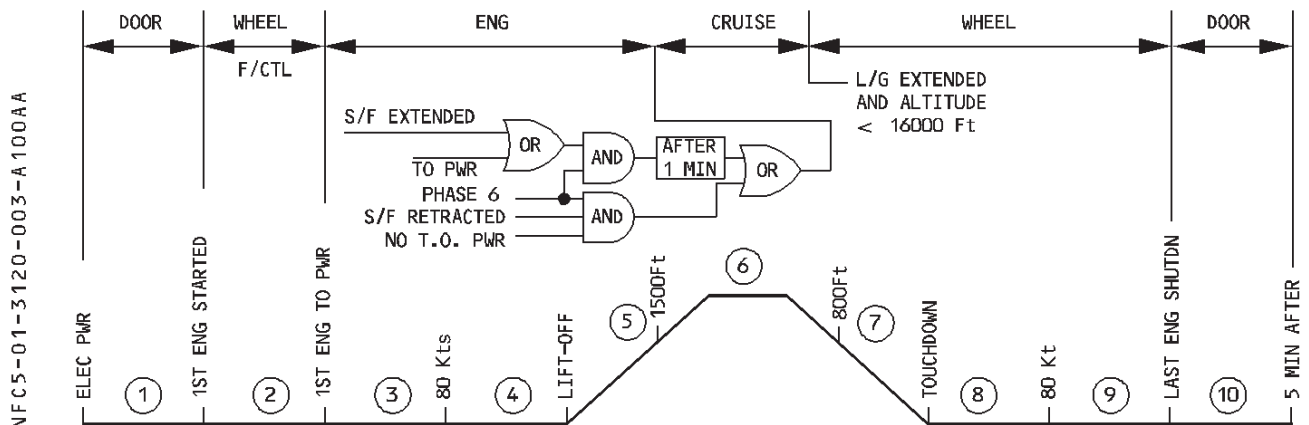
R The status page displays an operational summary of the aircraft status after the SD has displayed a failure. As shown in the illustration above, the summary includes :

- ① Limitations (speed, flight level) : Blue
- ② Approach procedures : White/Red or Amber
- ③ Procedures (corrections to apply for landing) : Blue
- ④ Information : Green
- ⑤ Cancelled caution : White
- ⑥ Inoperative system : Amber
- ⑦ Maintenance status : White
- ⑧ Symbol displayed if data overflows the left or right area.

The pilot scrolls the display to view overflow by pressing the CLR pushbutton.

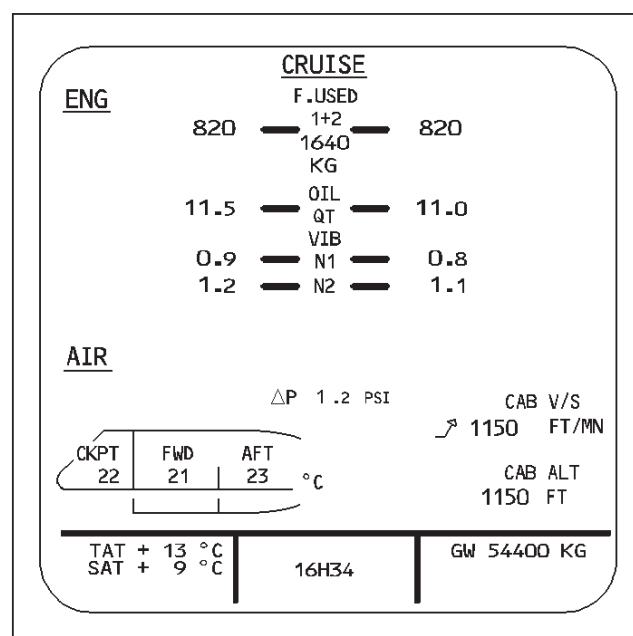
Note : The titles of the different parts of the display are white and underlined.

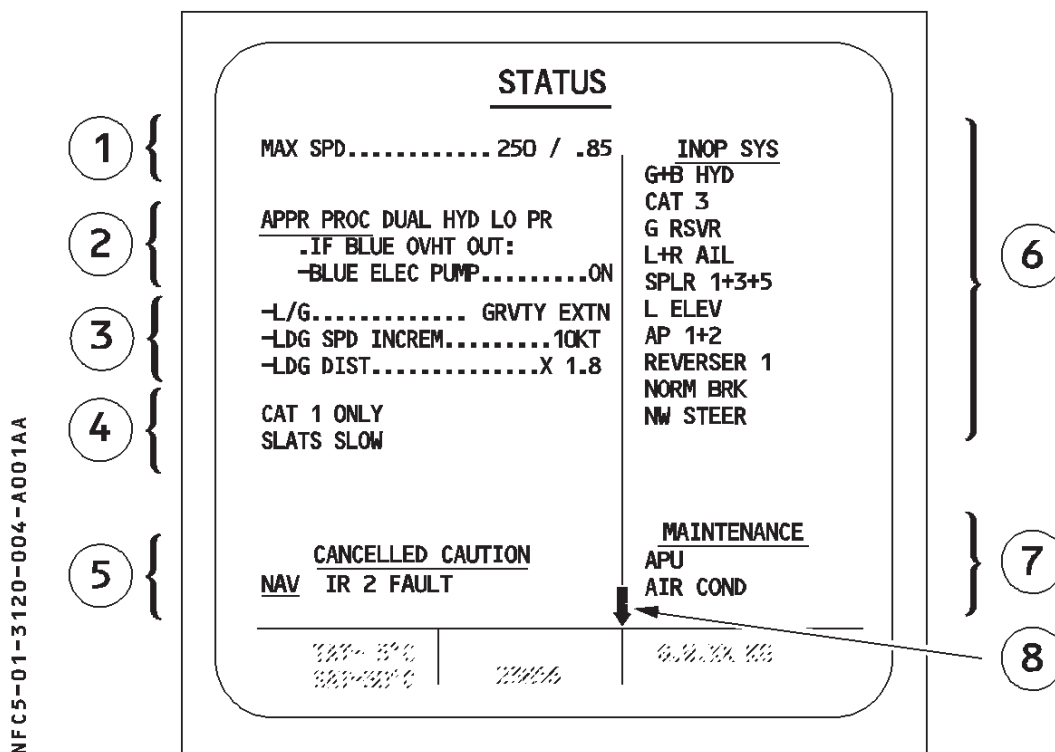
- Automatic, flight phase mode
 - If no other mode is engaged, the SD displays the system page related to the present flight phase, as shown in the following diagram.



- Phase 2 : The F/CTL page replaces the WHEEL page for 20 seconds, when either pilot moves his sidestick (more than 3° in pitch or roll), or when the rudder pedal deflection is more than 22°.
- The APU page appears when the APU MASTER switch is ON. It disappears when APU RPM has been above 95 % for 10 seconds, or when the APU MASTER switch is switched OFF.
- The ENGINE page appears at the beginning of the start sequence, or when a pilot selects "CRANK". It disappears 10 seconds after the end of the start sequence, when the ENG MODE sel is set to NORM.

For a description of the ENGINE and AIR indications that appear, when the SD is displaying the CRUISE page, see the relevant FCOM chapter.



STATUS PAGE


R The status page displays an operational summary of the aircraft status after the SD has displayed a failure. As shown in the illustration above, the summary includes :

- ① Limitations (speed, flight level) : Blue
- ② Approach procedures : White/Red or Amber
- ③ Procedures (corrections to apply for landing) : Blue
- ④ Information : Green
- ⑤ Cancelled caution : White
- ⑥ Inoperative system : Amber
- ⑦ Maintenance status : White
- ⑧ Symbol displayed if data overflows the left or right area.

The pilot scrolls the display to view overflow by pressing the CLR pushbutton.

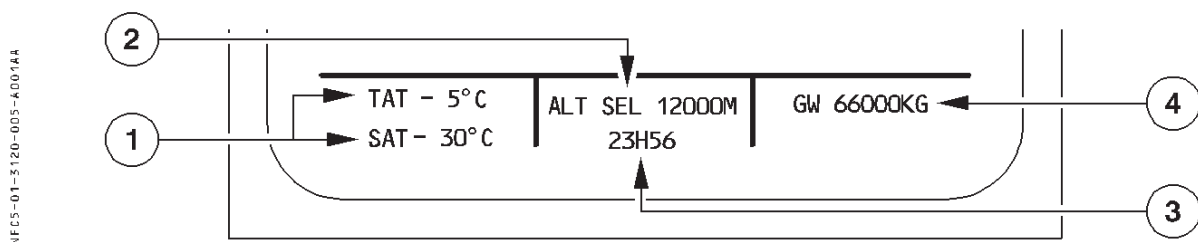
Note : The titles of the different parts of the display are white and underlined.

The STATUS page automatically appears, immediately after the flight crew clears all the pages related to the current failure. The STATUS page also appears automatically during descent, when the slats are extended, unless if the page is empty. The flight crew may manually call up the STATUS page if they press the STS key on the ECAM control panel. If the STATUS page holds messages other than "CANCELLED CAUTION", or the MAINTENANCE part, the E/WD screen displays "STS" (status reminder).

If the STATUS page holds messages in the MAINTENANCE part at engine shutdown, the "STS" (status reminder) flashes on the E/WD screen.

The screen displays the MAINTENANCE message, only when the aircraft is on ground, before engine start-up or after engine shutdown (Phases 1 and 10).

PERMANENT DATA



① Temperature

The screen displays the Total Air Temperature (TAT) and Static Air Temperature (SAT) in green.

② G LOAD - ALT SEL

The screen displays one of the following items :

- The Load factor (G LOAD) in amber, when the value is above 1.4 g or below 0.7 g. The display of the load factor is inhibited during flight phases 1, 2 and 3.
- Altitude in green selected via the Flight Control Unit (FCU), if the flight crew selects metric units, and provided the load factor is not displayed.

③ UTC

The screen displays the Universal Time Coordinated (UTC), synchronized with the cockpit clock in green.

④ GW

The screen displays the Gross Weight (GW) in green, as soon as the flight crew starts the first engine. The last two digits are dashed, if accuracy is degraded. On ground, blue dashes are displayed instead of the indication, if no computed data is available.

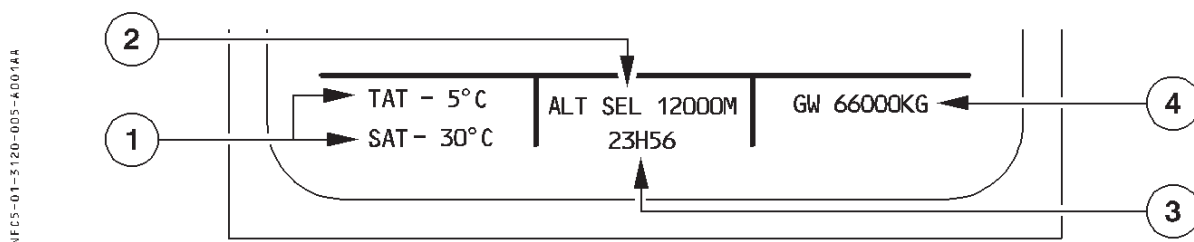
Note : *When the Fuel On Board (FOB) indicated on the ECAM exceeds 23 200 kg (51 147 lb), the GW indication is inaccurate. When the FOB on the ECAM goes below 23 200 kg (51 147 lb) GW accuracy is recovered.*

The STATUS page automatically appears, immediately after the flight crew clears all the pages related to the current failure. The STATUS page also appears automatically during descent, when the slats are extended, unless if the page is empty. The flight crew may manually call up the STATUS page if they press the STS key on the ECAM control panel. If the STATUS page holds messages other than "CANCELLED CAUTION", or the MAINTENANCE part, the E/WD screen displays "STS" (status reminder).

If the STATUS page holds messages in the MAINTENANCE part at engine shutdown, the "STS" (status reminder) flashes on the E/WD screen.

The screen displays the MAINTENANCE message, only when the aircraft is on ground, before engine start-up or after engine shutdown (Phases 1 and 10).

PERMANENT DATA



① Temperature

The screen displays the Total Air Temperature (TAT) and Static Air Temperature (SAT) in green.

② G LOAD - ALT SEL

The screen displays one of the following items :

- The Load factor (G LOAD) in amber, when the value is above 1.4 g or below 0.7 g. The display of the load factor is inhibited during flight phases 1, 2 and 3.
- Altitude in green selected via the Flight Control Unit (FCU), if the flight crew selects metric units, and provided the load factor is not displayed.

③ UTC

The screen displays the Universal Time Coordinated (UTC), synchronized with the cockpit clock in green.

④ GW

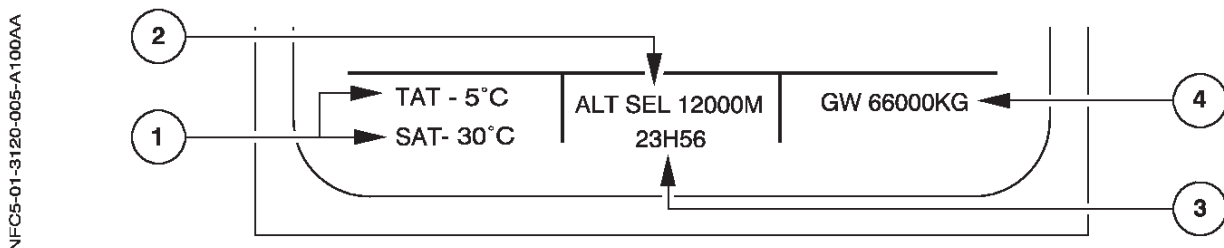
The screen displays the Gross Weight (GW) in green, as soon as the flight crew starts the first engine. The last two digits are dashed, if accuracy is degraded. On ground, blue dashes are displayed instead of the indication, if no computed data is available.

The STATUS page automatically appears, immediately after the flight crew clears all the pages related to the current failure. The STATUS page also appears automatically during descent, when the slats are extended, unless if the page is empty. The flight crew may manually call up the STATUS page if they press the STS key on the ECAM control panel. If the STATUS page holds messages other than "CANCELLED CAUTION", or the MAINTENANCE part, the E/WD screen displays "STS" (status reminder).

If the STATUS page holds messages in the MAINTENANCE part at engine shutdown, the "STS" (status reminder) flashes on the E/WD screen.

The screen displays the MAINTENANCE message, only when the aircraft is on ground, before engine start-up or after engine shutdown (Phases 1 and 10).

PERMANENT DATA



① Temperature

The screen displays Total Air Temperature (TAT) and Static Air Temperature (SAT) in green.

② G LOAD - ALT SEL

The screen displays one of the following items :

- Load factor (G LOAD) in amber, when the value is above 1.4 g or below 0.7 g. The display of the load factor is inhibited during flight phases 1, 2, 3, 9 and 10.
- Altitude in green selected via the Flight Control Unit (FCU), if the flight crew selects metric units, and provided the load factor is not displayed.
- "CHECK CAPT (F/O) PFD", "CHECK CAPT (F/O) ND" or "CHECK SD" all in amber (Refer to FCOM 1.31.05).

Note : The display of the feedback messages takes priority over the load factor indication.

③ UTC

The screen displays the Universal Time Coordinated (UTC), synchronized with the cockpit clock in green.

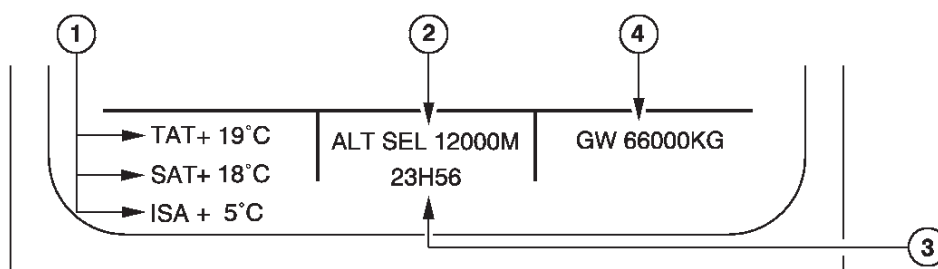
④ GW

The screen displays the Gross Weight (GW) in green, as soon as the flight crew starts the first engine. The last two digits are dashed, if accuracy is degraded. On ground, blue dashes are displayed instead of the indication, if no computed data is available.

The STATUS page automatically appears, immediately after the flight crew clears all the pages related to the current failure. The STATUS page also appears automatically during descent, when the slats are extended, unless if the page is empty. The flight crew may manually call up the STATUS page if they press the STS key on the ECAM control panel. If the STATUS page holds messages other than "CANCELLED CAUTION", or the MAINTENANCE part, the E/WD screen displays "STS" (status reminder). If the STATUS page holds messages in the MAINTENANCE part at engine shutdown, the "STS" (status reminder) flashes on the E/WD screen. The screen displays the MAINTENANCE message, only when the aircraft is on ground, before engine start-up or after engine shutdown (Phases 1 and 10).

PERMANENT DATA

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① Temperature

The screen displays Total Air Temperature (TAT) and Static Air Temperature (SAT) in green.

International Standard Atmosphere temperature (ISA) is displayed in green, in standard altitude mode and when the SAT is valid.

② G LOAD - ALT SEL

The screen displays one of the following items :

- Load factor (G LOAD) in amber, when the value is above 1.4 g or below 0.7 g. The display of the load factor is inhibited during flight phases 1, 2, 3, 9 and 10.
- Altitude in green, selected via the Flight Control Unit (FCU) if the flight crew selects metric units, and provided the load factor is not displayed.
- "CHECK CAPT (F/O) PFD", "CHECK CAPT (F/O) ND" or "CHECK SD" all in amber (Refer to FCOM 1.31.05).

Note : The display of the feedback messages takes priority over the load factor indication.

③ UTC

The screen displays the Universal Time Coordinated (UTC), synchronized with the cockpit clock in green.

④ GW

The screen displays the Gross Weight (GW) in green, as soon as the flight crew starts the first engine. The last two digits are dashed, if accuracy is degraded. On ground, blue dashes are displayed instead of the indication, if no computed data is available.

GENERAL

If ECAM detects a failure :

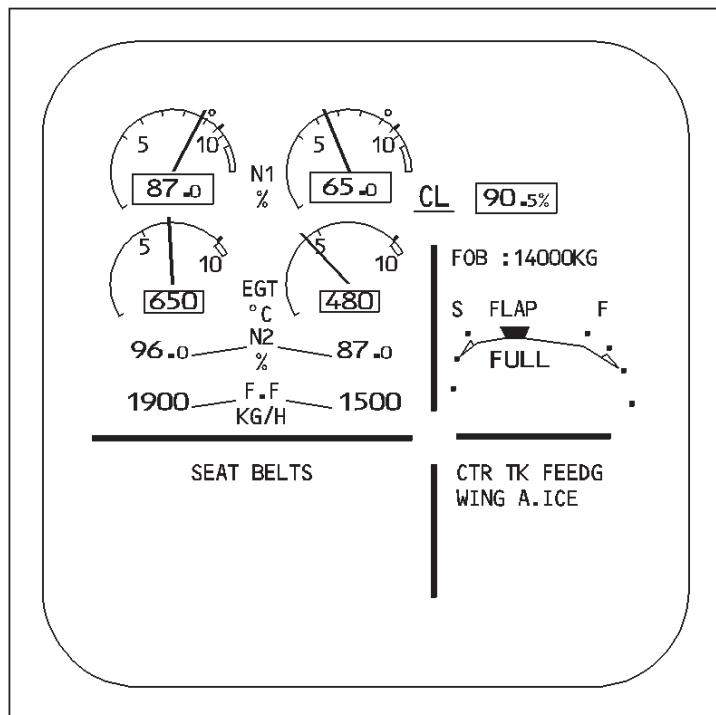
- The E/WD displays warning or caution messages.
- The master warning or master caution lights light up (except in the case of a level 1 caution).
- The system sounds an aural signal (except in the case of a level 1 caution).
- The system display (SD) shows the system page for the affected system.
- R – The CLR pushbutton on the ECAM control panel lights up.

In addition, a local warning light controlled directly by the affected system can light up. After completing remedial procedures, the flight crew must push the CLR pushbutton repeatedly until the displays return to their normal configurations :

- MEMO messages on the E/WD
- The system page related to the present flight phase on the SD.
- The CLR light on the ECAM control panel turned off.

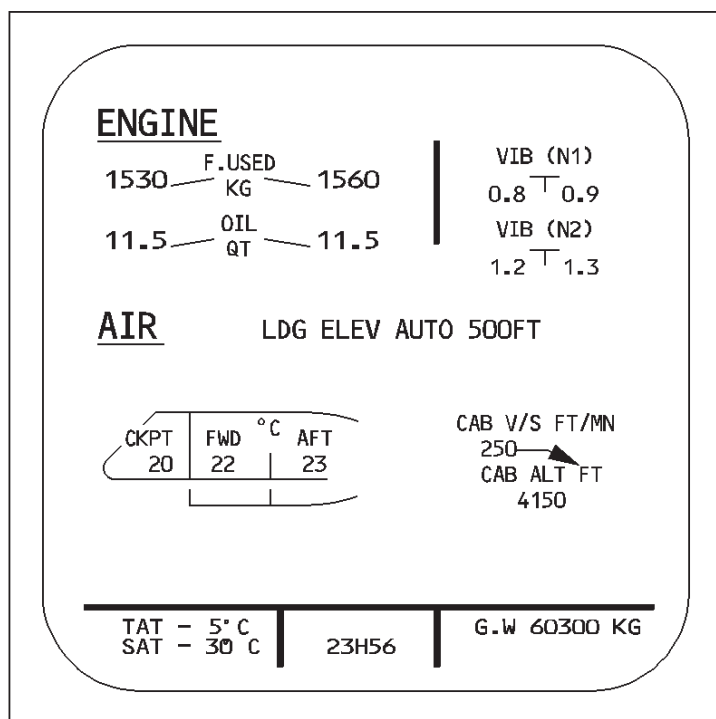
EXAMPLE**1 – THE ECAM DETECTS NO FAILURE**

R

ECAM UPPER DISPLAY (E/WD)

- ENGINE CONTROL PARAMETERS
- FUEL QUANTITY INDICATION
- FLAPS/SLATS POSITION

- MEMO INFORMATION

ECAM LOWER DISPLAY (SD)

- FLIGHT PHASE RELATED SYSTEM PAGE (CRUISE PAGE IN THIS EXAMPLE)

- PERMANENT DATA

NFC5-01-3125-002-A005AB

GENERAL

If ECAM detects a failure :

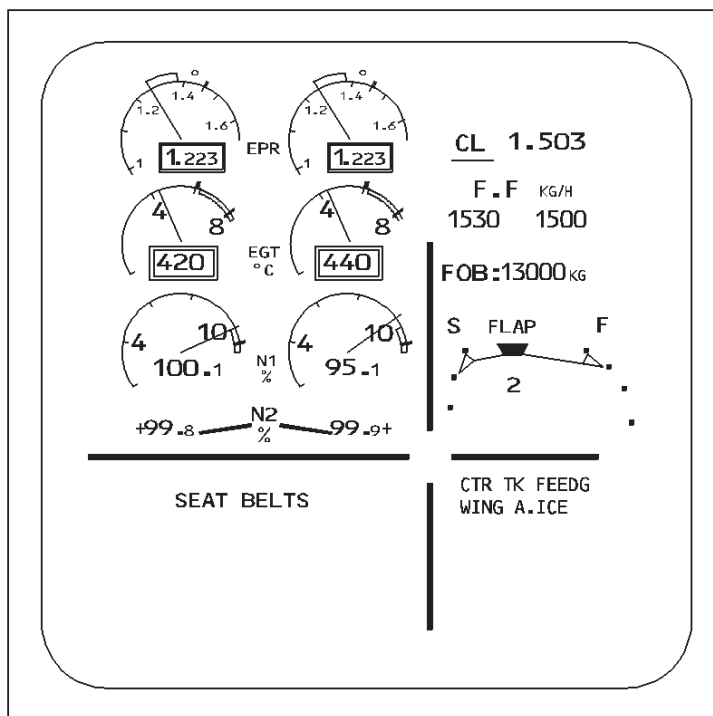
- The E/WD displays warning or caution messages.
- The master warning or master caution lights light up (except in the case of a level 1 caution).
- The system sounds an aural signal (except in the case of a level 1 caution).
- The system display (SD) shows the system page for the affected system.
- R – The CLR pushbutton on the ECAM control panel lights up.

In addition, a local warning light controlled directly by the affected system can light up. After completing remedial procedures, the flight crew must push the CLR pushbutton repeatedly until the displays return to their normal configurations :

- MEMO messages on the E/WD
- The system page related to the present flight phase on the SD.
- The CLR light on the ECAM control panel turned off.

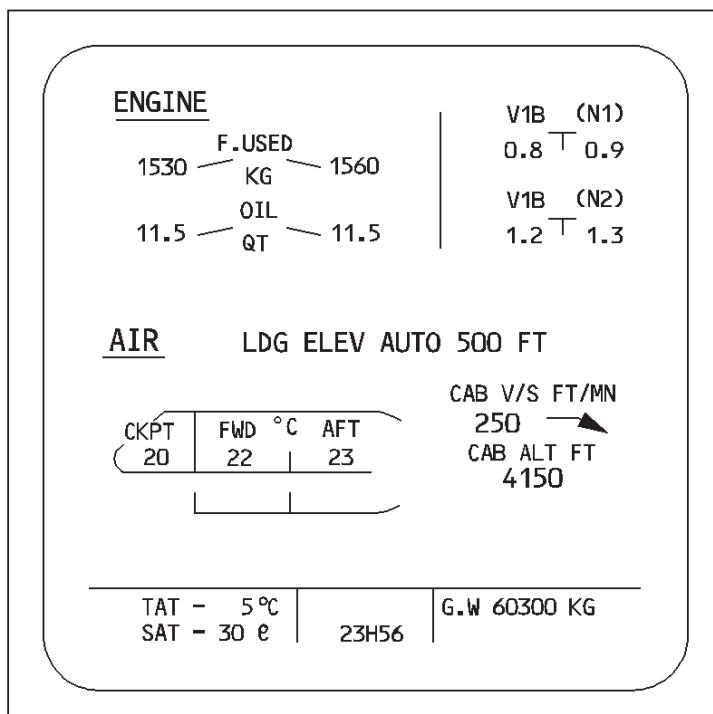
EXAMPLE**1 – THE ECAM DETECTS NO FAILURE**

R

**ECAM UPPER DISPLAY (E/WD)**

- ENGINE CONTROL PARAMETERS
- FUEL QUANTITY INDICATION
- FLAPS/SLATS POSITION

- MEMO INFORMATION

**ECAM LOWER DISPLAY**

- FLIGHT PHASE RELATED SYSTEM PAGE (CRUISE PAGE IN THIS EXAMPLE)

- PERMANENT DATA

NFC5-01-3125-002-A050AA

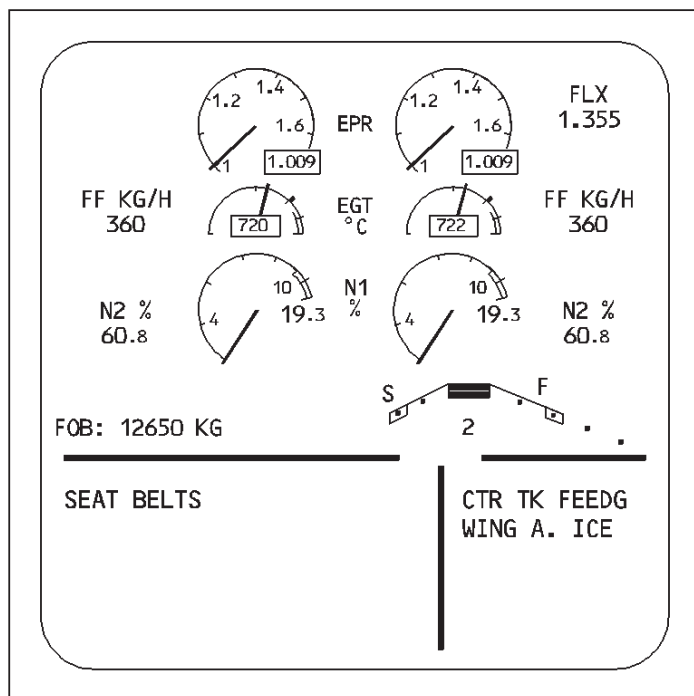
GENERAL

If ECAM detects a failure :

- The E/WD displays warning or caution messages.
- The master warning or master caution lights light up (except in the case of a level 1 caution).
- The system sounds an aural signal (except in the case of a level 1 caution).
- The system display (SD) shows the system page for the affected system.
- R – The CLR pushbutton on the ECAM control panel lights up.

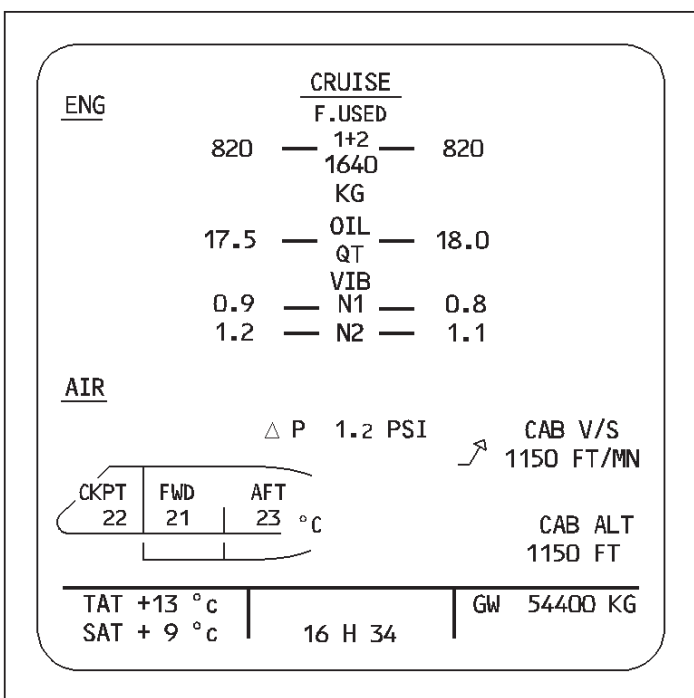
In addition, a local warning light controlled directly by the affected system can light up. After completing remedial procedures, the flight crew must push the CLR pushbutton repeatedly until the displays return to their normal configurations :

- MEMO messages on the E/WD
- The system page related to the present flight phase on the SD.
- The CLR light on the ECAM control panel turned off.

EXAMPLE**1 – THE ECAM DETECTS NO FAILURE****ECAM UPPER DISPLAY (E/WD)**

- ENGINE CONTROL PARAMETERS
- FUEL QUANTITY INDICATION
- FLAPS/SLATS' POSITION

- MEMO INFORMATION

**ECAM LOWER DISPLAY**

- FLIGHT PHASE RELATED SYSTEM PAGE (CRUISE PAGE IN THIS EXAMPLE)

- PERMANENT DATA

NFC5-01-3125-002-A110AA

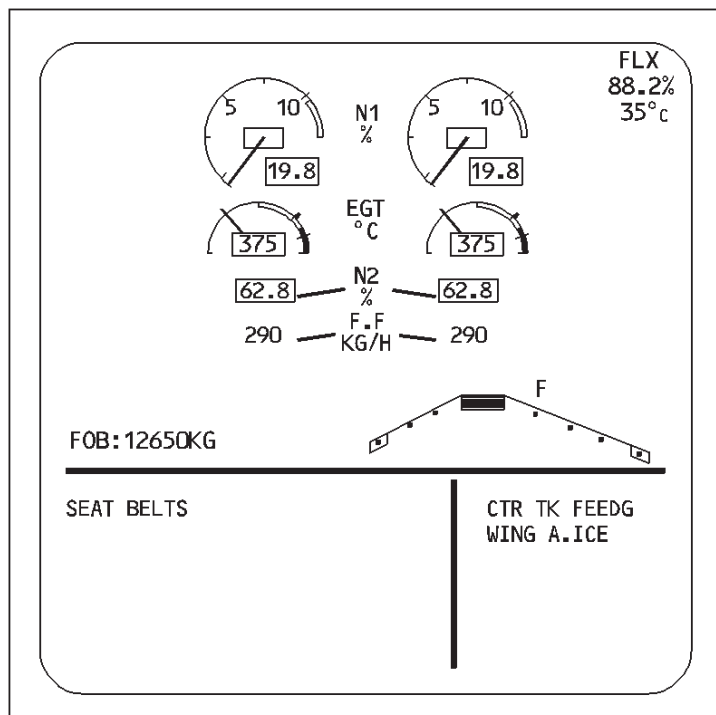
GENERAL

If ECAM detects a failure :

- The E/WD displays warning or caution messages.
- The master warning or master caution lights light up (except in the case of a level 1 caution).
- The system sounds an aural signal (except in the case of a level 1 caution).
- The system display (SD) shows the system page for the affected system.
- R – The CLR pushbutton on the ECAM control panel lights up.

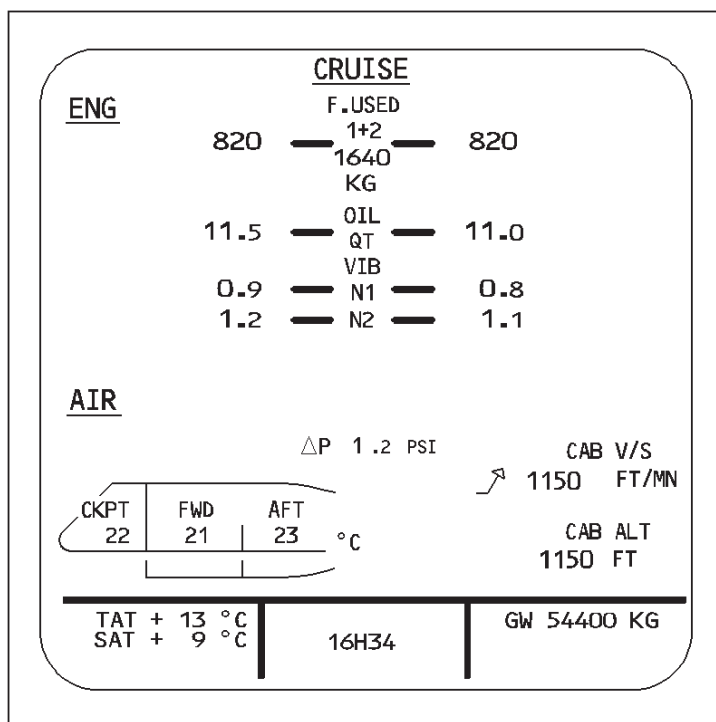
In addition, a local warning light controlled directly by the affected system can light up. After completing remedial procedures, the flight crew must push the CLR pushbutton repeatedly until the displays return to their normal configurations :

- MEMO messages on the E/WD
- The system page related to the present flight phase on the SD.
- The CLR light on the ECAM control panel turned off.

EXAMPLE**1 – THE ECAM DETECTS NO FAILURE**ECAM UPPER DISPLAY (E/WD)

- ENGINE CONTROL PARAMETERS
- FUEL QUANTITY INDICATION
- FLAPS/SLATS POSITION

- MEMO INFORMATION

ECAM LOWER DISPLAY (SD)

- FLIGHT PHASE RELATED SYSTEM PAGE (CRUISE PAGE IN THIS EXAMPLE)

- PERMANENT DATA

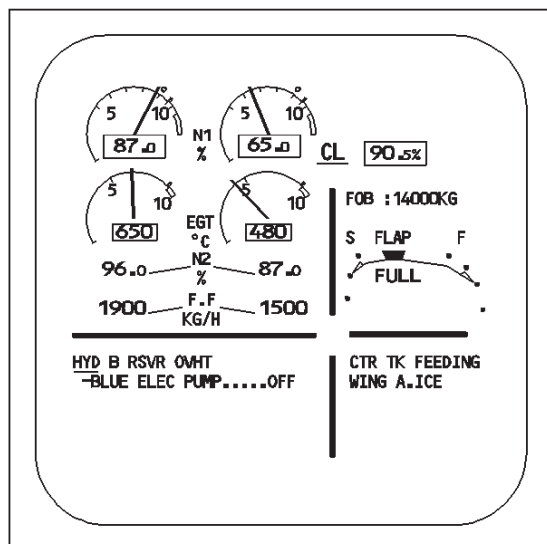
NFC5-01-3125-002-A115AA

R 2 – THE ECAM DETECTS A FAILURE

For example, a hydraulic reservoir is overheating.

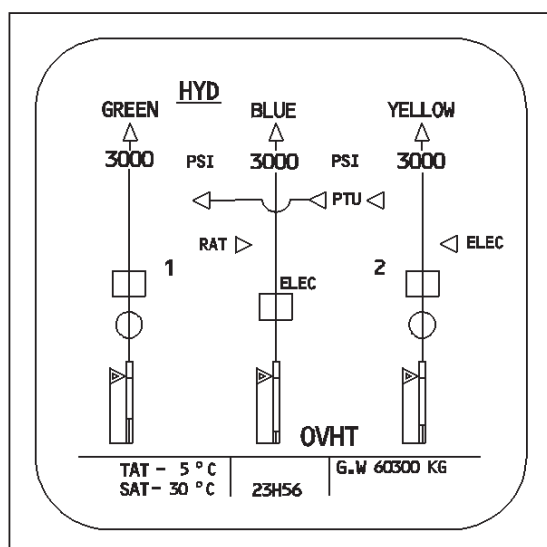
COCKPIT INDICATIONS

- A single chime sounds.
 - Both MASTER CAUTION lights come on and stay on.
 - A FAULT light on the overhead HYD panel comes on.
 - The memo space on the E/WD displays the message “HYD B RSVR OVHT” and the instruction “BLUE ELEC PUMP OFF”.
 - The lower ECAM display (SD) automatically calls up the diagram of the hydraulic systems and displays “OVHT” in amber by the blue system.
- R – The CLR pushbutton on the ECAM control panel lights up.



ECAM UPPER DISPLAY (E/WD)

- LEFT PART
 - . INDEPENDENT FAILURE
 - TITLE OF THE FAILURE
 - ACTIONS TO BE PERFORMED
- RIGHT PART
 - .MEMO INFORMATION



ECAM LOWER DISPLAY (SD)

- SYNOPSIS OF THE AFFECTED SYSTEM AUTOMATICALLY CALLED OVHT IS DISPLAYED IN AMBER

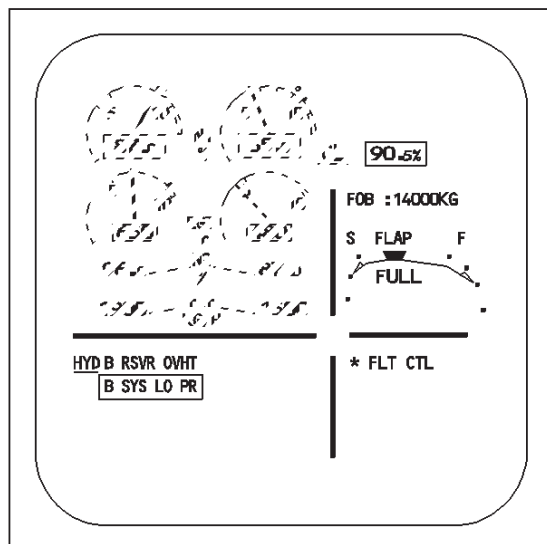
NFC5-01-3125-003-A005AA

3 – THE FLIGHT CREW FOLLOWS THE INSTRUCTION DISPLAYED ON THE E/WD

- R The crew switches off the BLUE ELEC PUMP pushbutton, depressurizing the blue hydraulic circuit.

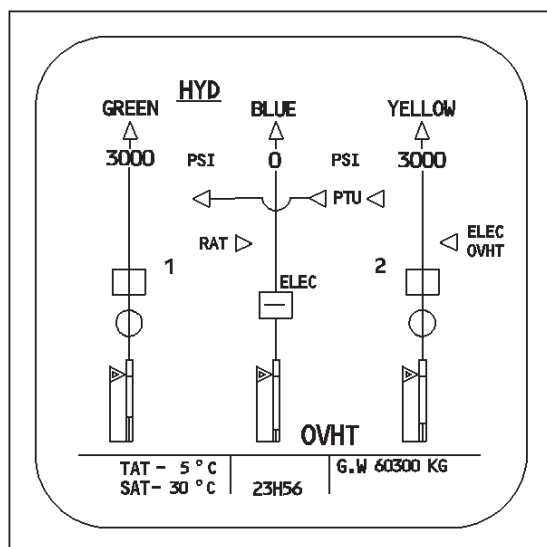
COCKPIT INDICATIONS

- A single chime sounds.
- Both MASTER CAUTION lights stay on.
- A FAULT/OFF light on the overhead panel comes on.
- The second part of the message on the E/WD changes to "B SYS LO PR".
- The system diagram on the SD shows an amber zero for the pressure in the blue system, along with the amber "OVHT".
- The right side of the memo area indicates a secondary failure in the flight control system.
- The CLR pushbutton on the ECAM control panel stays lighted.



ECAM UPPER DISPLAY (E/WD)

- LEFT PART
 - . INDEPENDENT FAILURE AND PRIMARY FAILURE
- RIGHT PART
 - . SECONDARY FAILURE



ECAM LOWER DISPLAY (SD)

- THE SYNOPTIC OF THE SYSTEM PAGE IS CHANGED ACCORDING TO THE NEW SYSTEM CONFIGURATION
- OVHT AND THE PRESSURE ARE DISPLAYED IN AMBER

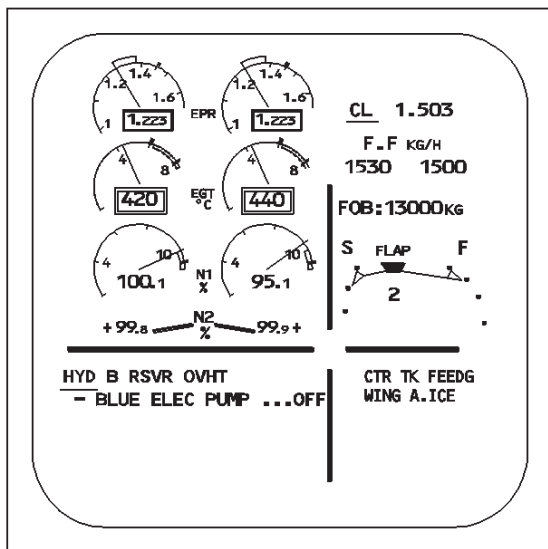
NFC5-01-3125-004-A001AA

R 2 – THE ECAM DETECTS A FAILURE

For example, a hydraulic reservoir is overheating.

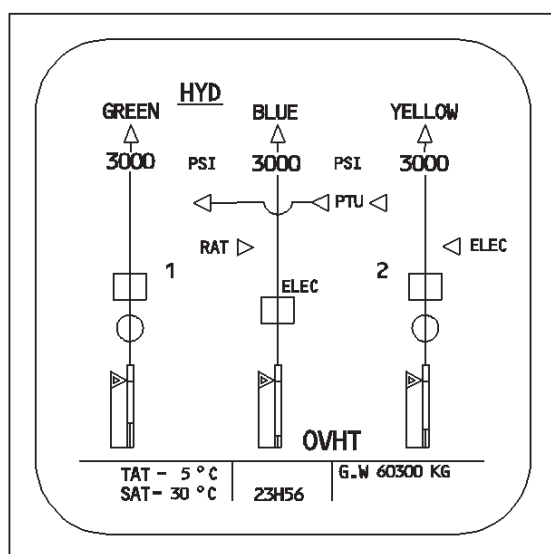
COCKPIT INDICATIONS

- A single chime sounds.
 - Both MASTER CAUTION lights come on and stay on.
 - A FAULT light on the overhead HYD panel comes on.
 - The memo space on the E/WD displays the message “HYD B RSVR OVHT” and the instruction “BLUE ELEC PUMP OFF”.
 - The lower ECAM display (SD) automatically calls up the diagram of the hydraulic systems and displays “OVHT” in amber by the blue system.
- R
- The CLR pushbutton on the ECAM control panel lights up.



ECAM UPPER DISPLAY (E/WD)

- LEFT PART
 - . INDEPENDENT FAILURE
 - TITLE OF THE FAILURE
 - ACTIONS TO BE PERFORMED
- RIGHT PART
 - . MEMO INFORMATION



ECAM LOWER DISPLAY (SD)

- SYNOPTIC OF THE AFFECTED SYSTEM AUTOMATICALLY CALLED OVHT IS DISPLAYED IN AMBER

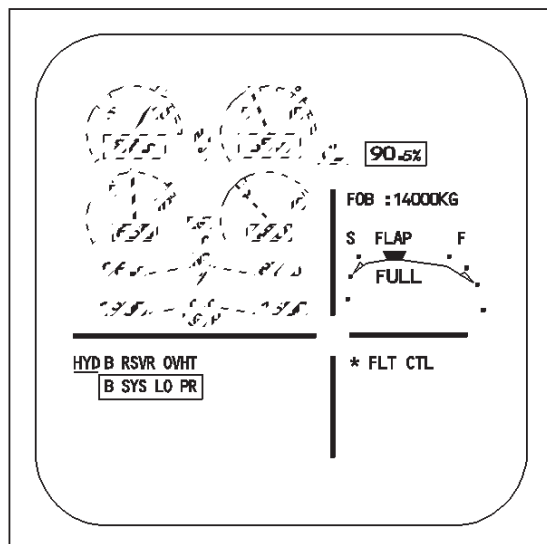
NFC5-01-3125-003-A050AA

3 – THE FLIGHT CREW FOLLOWS THE INSTRUCTION DISPLAYED ON THE E/WD

- R The crew switches off the BLUE ELEC PUMP pushbutton, depressurizing the blue hydraulic circuit.

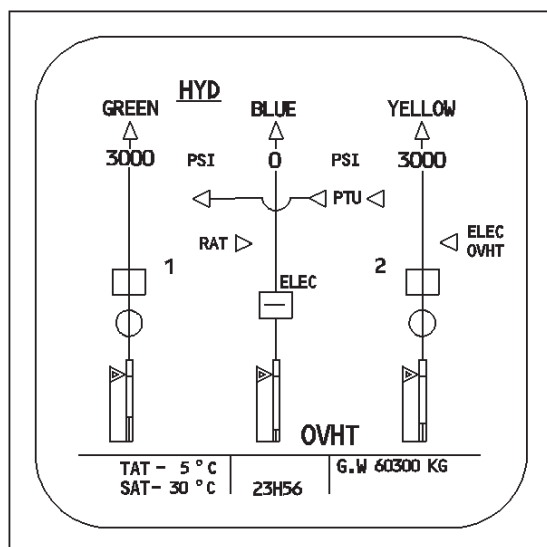
COCKPIT INDICATIONS

- A single chime sounds.
- Both MASTER CAUTION lights stay on.
- A FAULT/OFF light on the overhead panel comes on.
- The second part of the message on the E/WD changes to "B SYS LO PR".
- The system diagram on the SD shows an amber zero for the pressure in the blue system, along with the amber "OVHT".
- The right side of the memo area indicates a secondary failure in the flight control system.
- The CLR pushbutton on the ECAM control panel stays lighted.



ECAM UPPER DISPLAY (E/WD)

- LEFT PART
 - . INDEPENDENT FAILURE AND PRIMARY FAILURE
- RIGHT PART
 - . SECONDARY FAILURE



ECAM LOWER DISPLAY (SD)

- THE SYNOPTIC OF THE SYSTEM PAGE IS CHANGED ACCORDING TO THE NEW SYSTEM CONFIGURATION
- OVHT AND THE PRESSURE ARE DISPLAYED IN AMBER

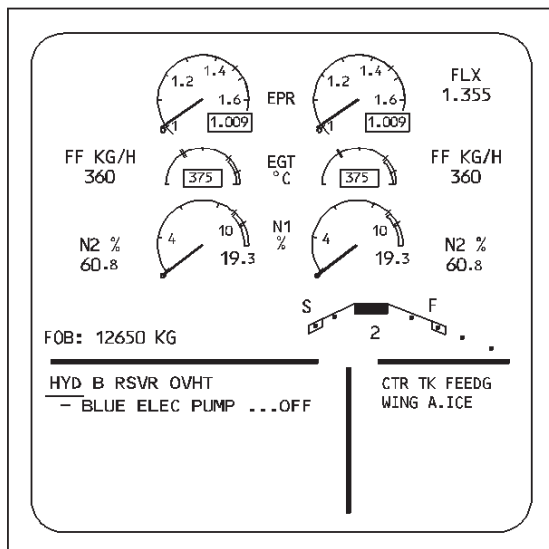
NFC5-01-3125-004-A001AA

2 – THE ECAM DETECTS A FAILURE

For example, a hydraulic reservoir is overheat.

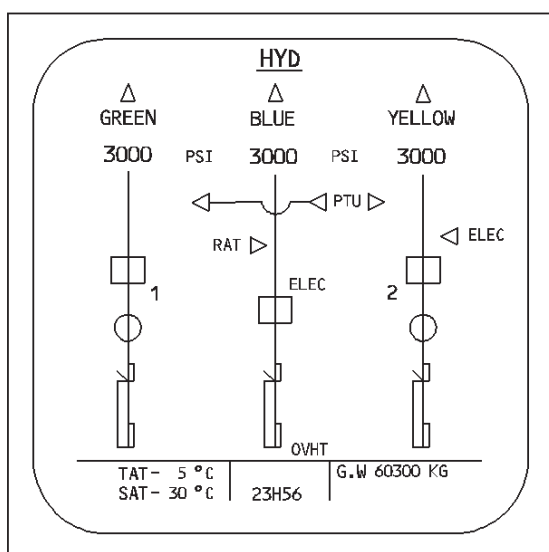
COCKPIT INDICATIONS

- A single chime sounds.
- Both MASTER CAUTION lights come on, and stay on.
- A FAULT light, on the overhead HYD panel, comes on.
- The memo space on the E/WD displays the “HYD B RSVR OVHT” message, and the “BLUE ELEC PUMP OFF” instruction.
- The lower ECAM display (SD) automatically calls up the hydraulic systems’ diagram, and displays “OVHT” in amber next to the blue system.
- The ECAM’s CLR pushbutton lights up.



ECAM UPPER DISPLAY (E/WD)

- BOTTOM LEFT
 - INDEPENDENT FAILURE
 - TITLE OF THE FAILURE
 - ACTIONS TO BE PERFORMED
- BOTTOM RIGHT
 - MEMO INFORMATION



ECAM LOWER DISPLAY (SD)

- SYNOPTIC OF THE AFFECTED SYSTEM AUTOMATICALLY CALLED
- OVHT IS DISPLAYED IN AMBER

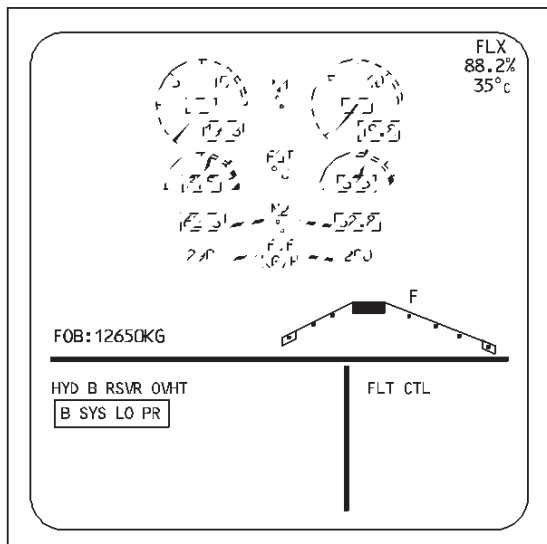
NFC5-01-3125-003-A110AA

3 – THE FLIGHT CREW FOLLOWS THE INSTRUCTIONS DISPLAYED ON THE E/WD

The flight crew switches off the BLUE ELEC PUMP pushbutton, depressurizing the blue hydraulic circuit.

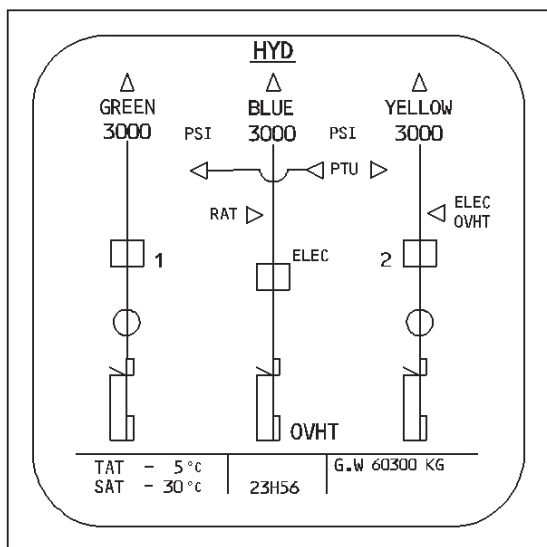
COCKPIT INDICATIONS

- A single chime sounds.
- Both MASTER CAUTION lights stay on.
- A FAULT/OFF light, on the overhead panel, comes on.
- The second part of the message on the E/WD changes to “B SYS LO PR”.
- The SD’s system diagram shows an amber zero for the pressure in the blue system, along with the amber “OVHT”.
- The right side of the memo area indicates a secondary failure in the flight control system.
- The ECAM control panel’s CLR pushbutton remains on.



ECAM UPPER DISPLAY (E/WD)

- BOTTOM LEFT
 - . INDEPENDENT FAILURE AND PRIMARY FAILURE
- BOTTOM RIGHT
 - . SECONDARY FAILURE



ECAM LOWER DISPLAY (SD)

- THE SYNOPTIC OF THE SYSTEM PAGE IS CHANGED ACCORDING TO THE NEW SYSTEM CONFIGURATION
- OVHT AND THE PRESSURE ARE DISPLAYED IN AMBER

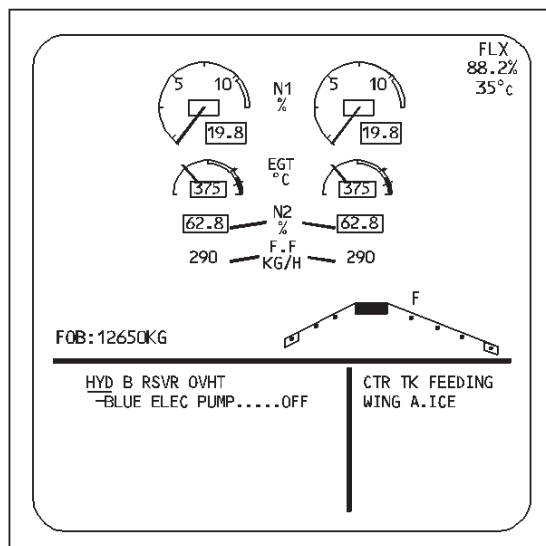
NFC5-01-3125-004-A100AA

2 – THE ECAM DETECTS A FAILURE

For example, a hydraulic reservoir is overheat.

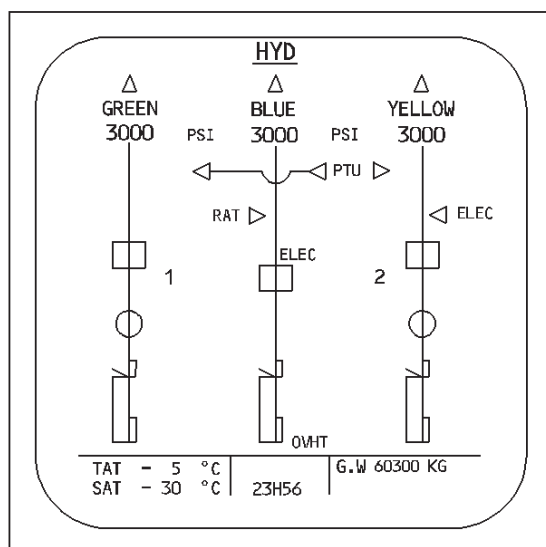
COCKPIT INDICATIONS

- A single chime sounds.
- Both MASTER CAUTION lights come on, and stay on.
- A FAULT light, on the overhead HYD panel, comes on.
- The memo space on the E/WD displays the “HYD B RSVR OVHT” message, and the “BLUE ELEC PUMP OFF” instruction.
- The lower ECAM display (SD) automatically calls up the hydraulic systems’ diagram, and displays “OVHT” in amber next to the blue system.
- The ECAM’s CLR pushbutton lights up.



ECAM UPPER DISPLAY (E/WD)

- BOTTOM LEFT
 - . INDEPENDENT FAILURE
 - TITLE OF THE FAILURE
 - ACTIONS TO BE PERFORMED
- BOTTOM RIGHT
 - .MEMO INFORMATION



ECAM LOWER DISPLAY (SD)

- SUMMARY OF THE AFFECTED SYSTEM AUTOMATICALLY DISPLAYED. OVHT IS DISPLAYED IN AMBER.

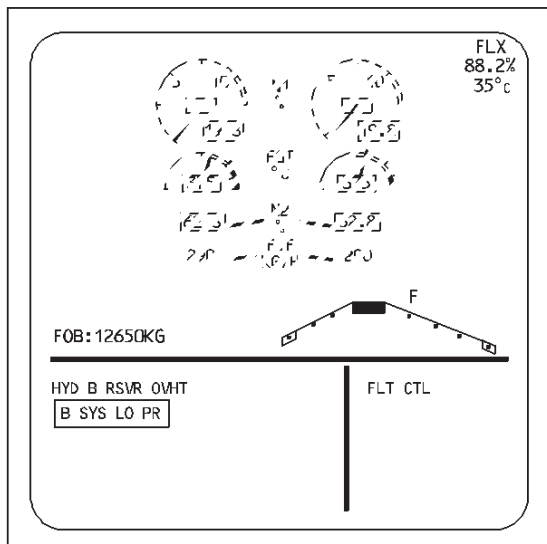
NFC5-01-3125-003-A120AA

3 – THE FLIGHT CREW FOLLOWS THE INSTRUCTIONS DISPLAYED ON THE E/WD

The flight crew switches off the BLUE ELEC PUMP pushbutton, depressurizing the blue hydraulic circuit.

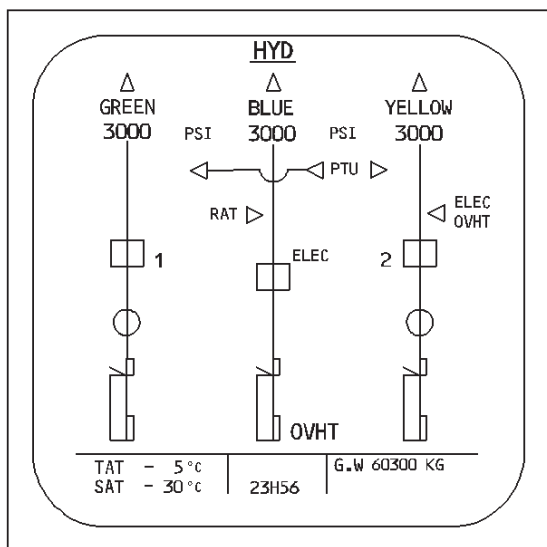
COCKPIT INDICATIONS

- A single chime sounds.
- Both MASTER CAUTION lights stay on.
- A FAULT/OFF light, on the overhead panel, comes on.
- The second part of the message on the E/WD changes to “B SYS LO PR”.
- The SD’s system diagram shows an amber zero for the pressure in the blue system, along with the amber “OVHT”.
- The right side of the memo area indicates a secondary failure in the flight control system.
- The ECAM control panel’s CLR pushbutton remains on.



ECAM UPPER DISPLAY (E/WD)

- BOTTOM LEFT
 - INDEPENDENT FAILURE AND PRIMARY FAILURE
- BOTTOM RIGHT
 - SECONDARY FAILURE



ECAM LOWER DISPLAY (SD)

- THE SYNOPTIC OF THE SYSTEM PAGE IS CHANGED ACCORDING TO THE NEW SYSTEM CONFIGURATION
- OVHT AND THE PRESSURE ARE DISPLAYED IN AMBER

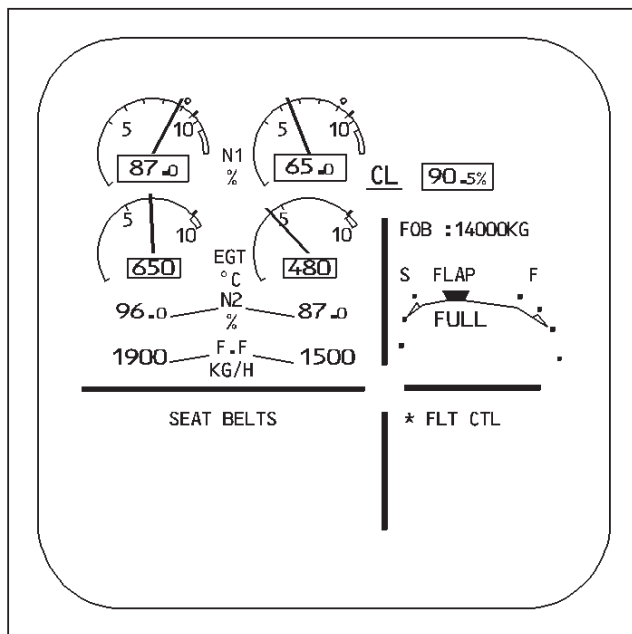
NFC5-01-3125-004-A100AA

4 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON ON THE ECP

COCKPIT INDICATIONS

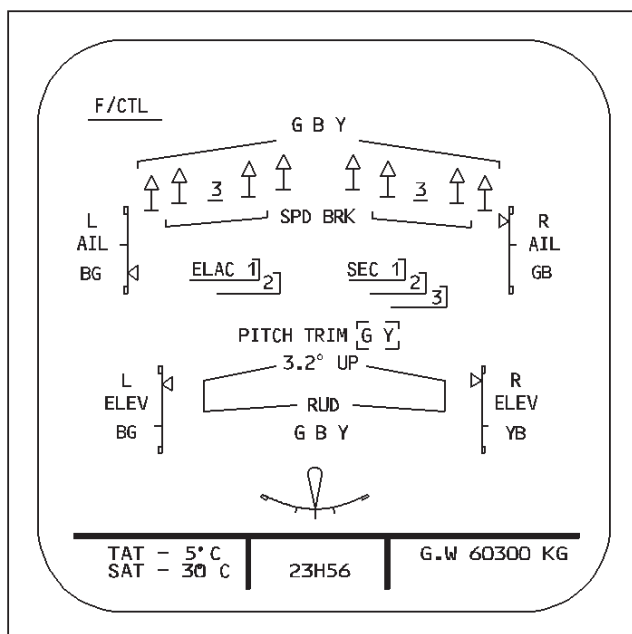
- The CLR pushbutton remains lit.
- The FAULT/OFF light stays on.
- The hydraulic system messages disappear from the E/WD, and the right side of the memo area indicates a secondary failure in the flight control system.
- The SD automatically calls up the flight control system page, with surface actuator indications associated with the blue hydraulic system shown in amber.

R



ECAM UPPER DISPLAY (E/WD)

- LEFT PART
MEMO INFORMATION
- RIGHT PART
SECONDARY FAILURE



ECAM LOWER DISPLAY (SD)

- F/CTL SYSTEM PAGE
AUTOMATICALLY DISPLAYED
FAULTY SPOILERS (n°3)
AND SURFACE ACTUATORS PRESSURE
INDICATIONS B ARE DISPLAYED
IN AMBER

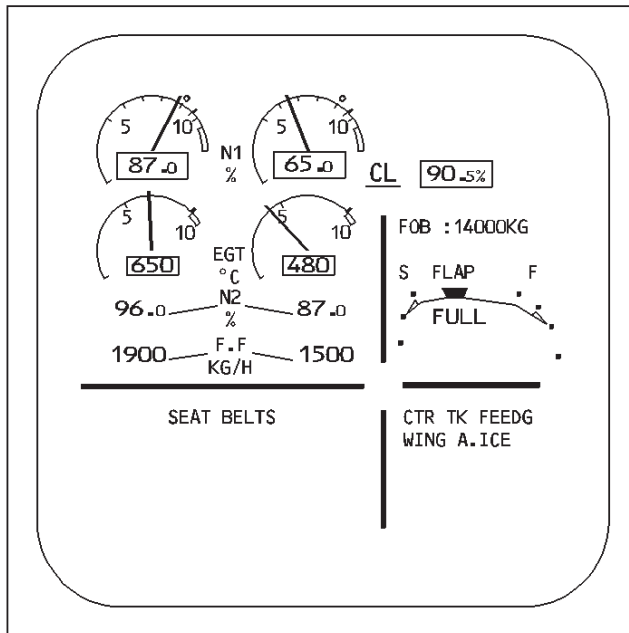
NFC5-01-3125-005-A005AA

5 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON A SECOND TIME

COCKPIT INDICATIONS

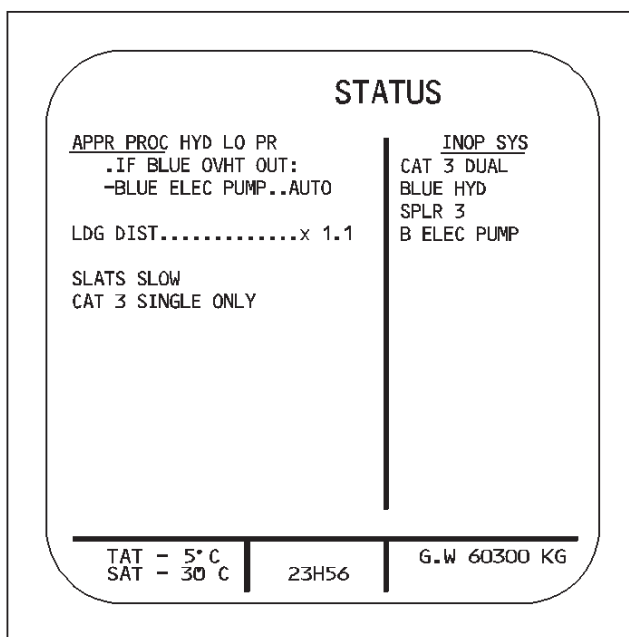
- The CLR and STS pushbuttons on the ECP light up.
- The FAULT/OFF lights stay on.
- The memo area on the E/WD returns to normal.
- The STATUS page automatically appears on the SD, displaying the procedures for completing the flight with a faulty blue system.

R



ECAM UPPER DISPLAY (E/WD)

- FULL MEMO DISPLAYED



ECAM LOWER DISPLAY (SD)

- THE STATUS PAGE IS AUTOMATICALLY DISPLAYED TO:
 - .PROVIDE THE PROCEDURE TO BE APPLIED FOR APPROACH.
 - .PROVIDE LANDING DISTANCE FACTORS AND INFORMATION.
 - .LIST THE INOPERATIVE SYSTEMS.

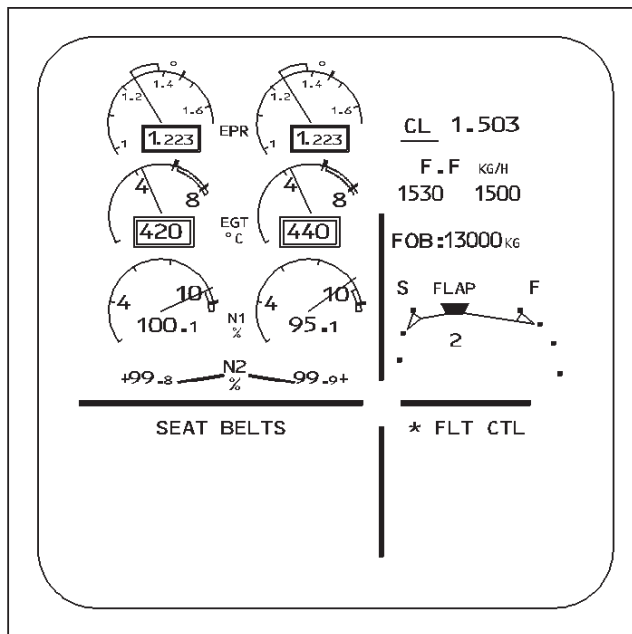
NFC5-01-3125-006-A005AA

4 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON ON THE ECP

COCKPIT INDICATIONS

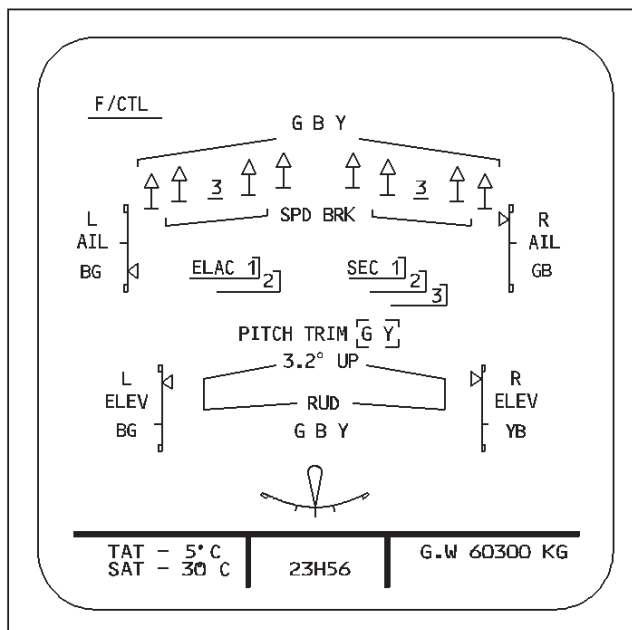
- The CLR pushbutton remains lit.
- The FAULT/OFF light stays on.
- The hydraulic system messages disappear from the E/WD, and the right side of the memo area indicates a secondary failure in the flight control system.
- The SD automatically calls up the flight control system page, with surface actuator indications associated with the blue hydraulic system shown in amber.

R



ECAM UPPER DISPLAY (E/WD)

- LEFT PART
MEMO INFORMATION
- RIGHT PART
SECONDARY FAILURE



ECAM LOWER DISPLAY (SD)

- F/CTL SYSTEM PAGE
AUTOMATICALLY DISPLAYED
FAULTY SPOILERS (n°3)
AND SURFACE ACTUATORS PRESSURE
INDICATIONS B ARE DISPLAYED
IN AMBER

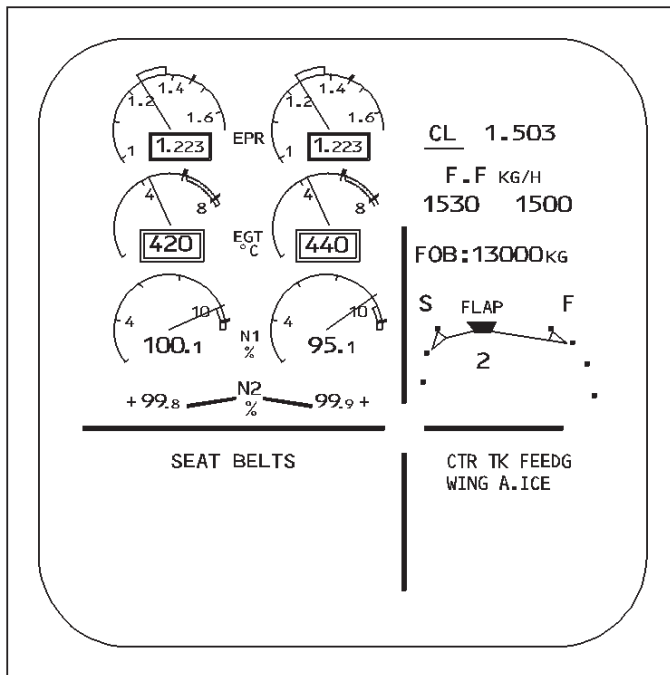
NFC5-01-3125-005-A050AA

5 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON A SECOND TIME

COCKPIT INDICATIONS

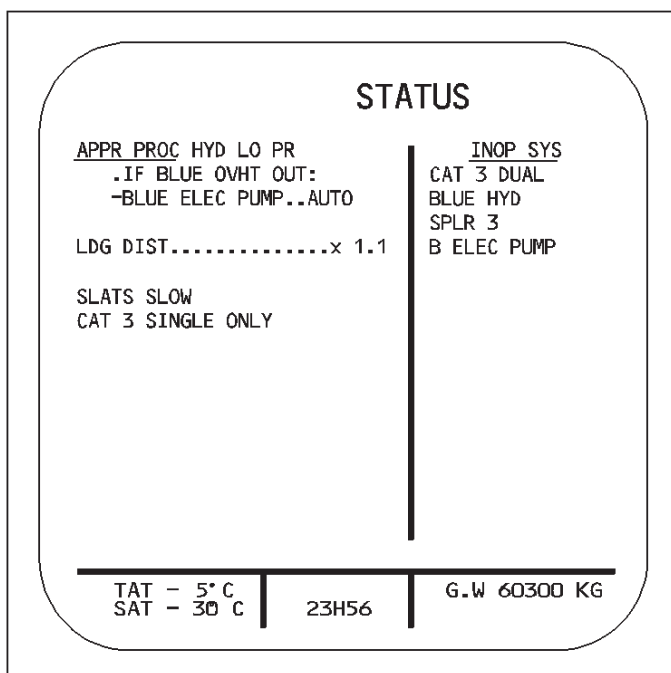
- The CLR and STS pushbuttons on the ECP light up.
- The FAULT/OFF lights stay on.
- The memo area on the E/WD returns to normal.
- The STATUS page automatically appears on the SD, displaying the procedures for completing the flight with a faulty blue system.

R



ECAM UPPER DISPLAY (E/WD)

- FULL MEMO DISPLAYED



ECAM LOWER DISPLAY (SD)

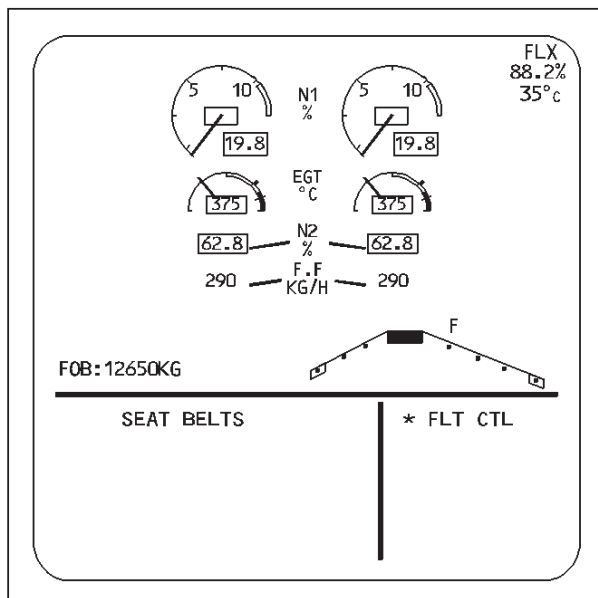
- THE STATUS PAGE IS AUTOMATICALLY DISPLAYED TO:
 - .PROVIDE THE PROCEDURE TO BE APPLIED FOR APPROACH.
 - .PROVIDE LANDING DISTANCE FACTORS AND INFORMATION.
 - .LIST THE INOPERATIVE SYSTEMS.

NFC5-01-3125-006-A050AA

4 – ONE OF THE PILOTS PUSHES THE ECP's CLR PUSHBUTTON

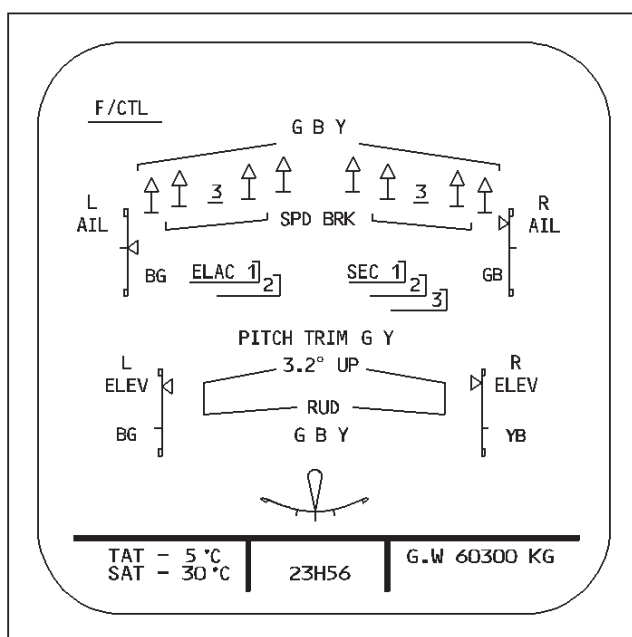
COCKPIT INDICATIONS

- The CLR pushbutton stays on.
- The FAULT/OFF light stays on.
- Hydraulic system messages disappear from the E/WD, and the right side of the memo area indicates a secondary failure in the flight control system.
- The SD automatically calls up the flight control system page, with surface actuator indications (associated with the blue hydraulic system) shown in amber.



ECAM UPPER DISPLAY (E/WD)

- BOTTOM LEFT
MEMO INFORMATION
- BOTTOM RIGHT
SECONDARY FAILURE



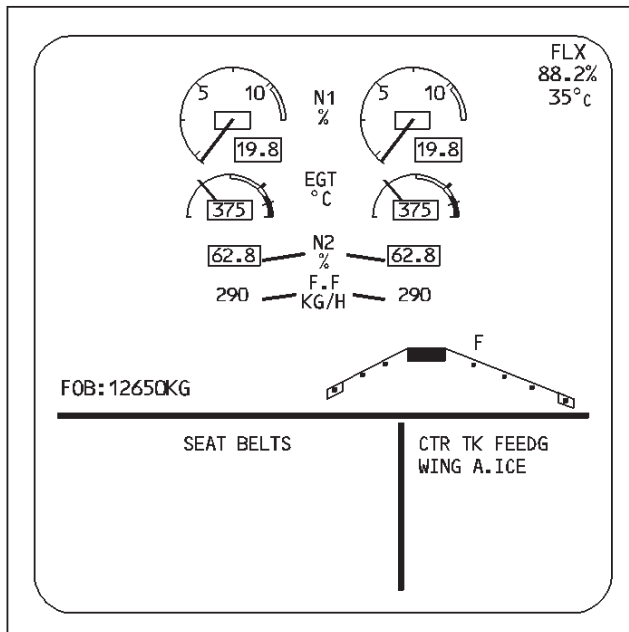
ECAM LOWER DISPLAY (SD)

- F/CTL SYSTEM PAGE
AUTOMATICALLY DISPLAYS
FAULTY SPOILERS (n°3)
AND SURFACE ACTUATOR B PRESSURE
INDICATIONS IN AMBER

5 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON A SECOND TIME

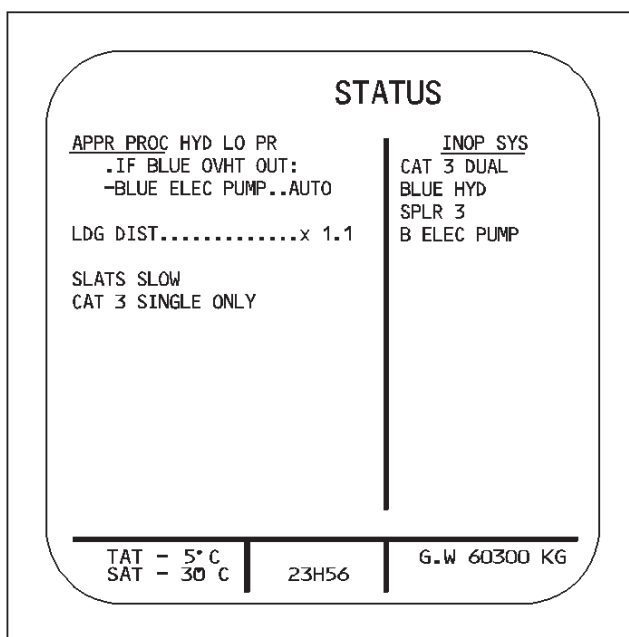
COCKPIT INDICATIONS

- The ECP's CLR and STS pushbuttons light up.
- The FAULT/OFF lights stay on.
- The E/WD's memo area returns to normal.
- The STATUS page automatically appears on the SD, displaying the procedures for completing the flight with a faulty blue system.



ECAM UPPER DISPLAY (E/WD)

- FULL MEMO DISPLAYED



ECAM LOWER DISPLAY (SD)

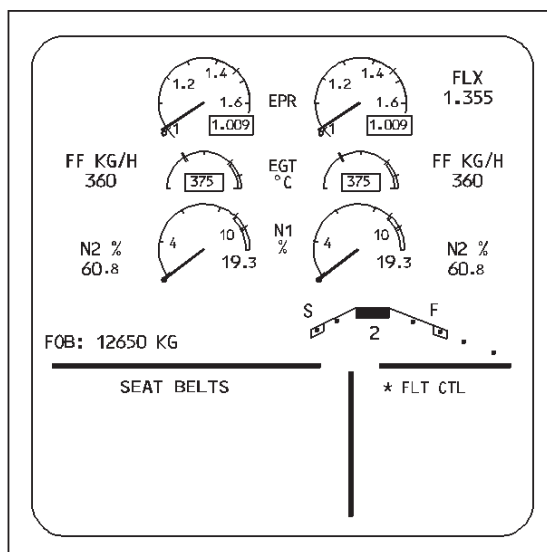
- THE STATUS PAGE IS AUTOMATICALLY DISPLAYED TO:
- .PROVIDE THE PROCEDURE TO BE APPLIED FOR APPROACH.
- .PROVIDE LANDING DISTANCE FACTORS AND INFORMATION.
- .LIST THE INOPERATIVE SYSTEMS.

NFC5-01-3125-006-A110AA

4 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON ON THE ECP

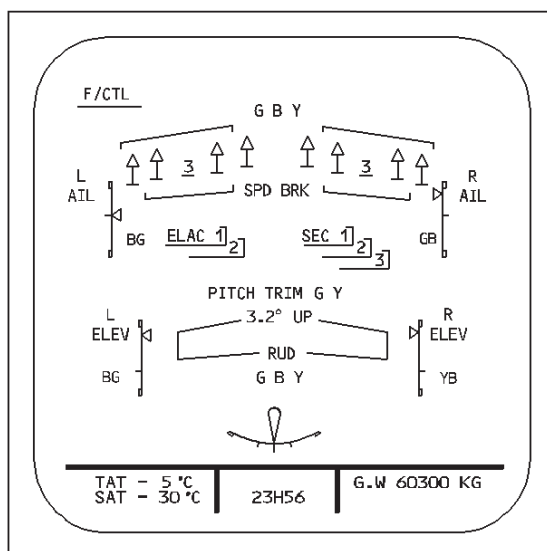
COCKPIT INDICATIONS

- The CLR pushbutton stays on.
- The FAULT/OFF light stays on.
- Hydraulic system messages disappear from the E/WD, and the right side of the memo area indicates a secondary failure in the flight control system.
- The SD automatically calls up the flight control system page, with surface actuator indications (associated with the blue hydraulic system) shown in amber.



ECAM UPPER DISPLAY (E/WD)

- BOTTOM LEFT
• MEMO INFORMATION
- BOTTOM RIGHT
• SECONDARY FAILURE



ECAM LOWER DISPLAY (SD)

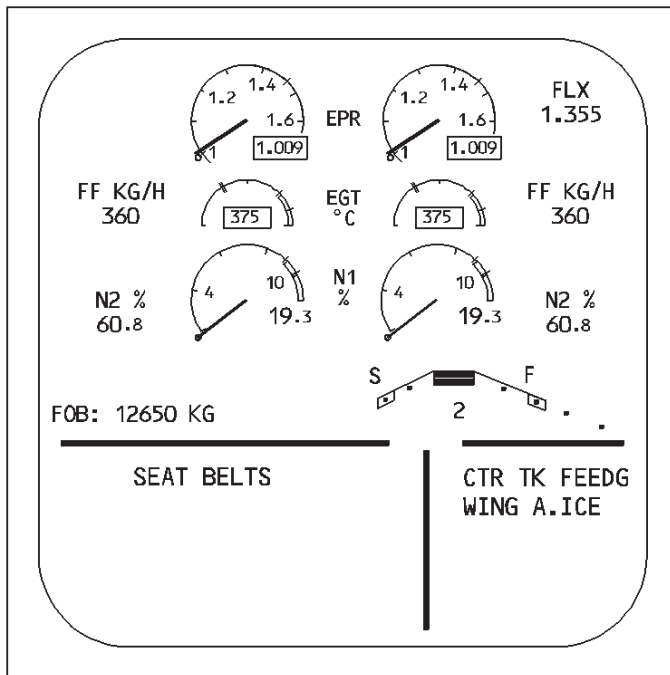
- F/CTL SYSTEM PAGE
AUTOMATICALLY DISPLAYS
FAULTY SPOILERS (n°3)
AND SURFACE ACTUATOR B PRESSURE
INDICATIONS IN AMBER.

NFC5-01-3125-005-A120AA

5 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON A SECOND TIME

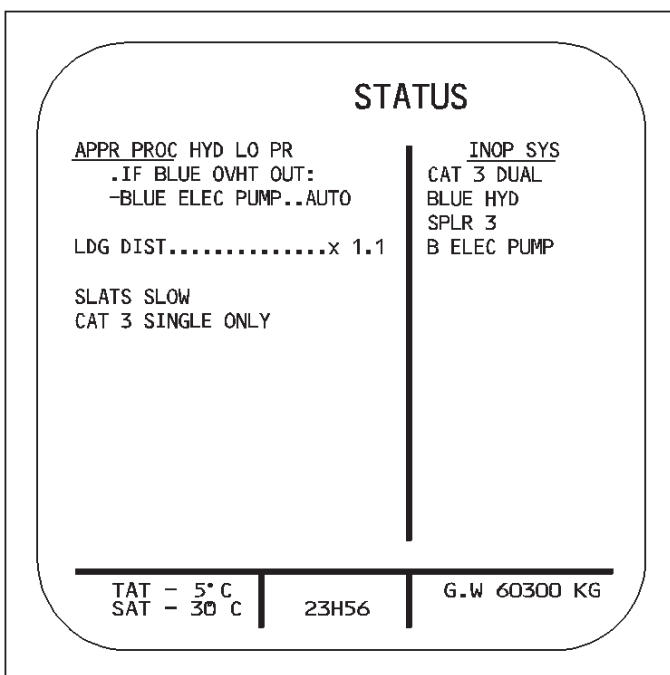
COCKPIT INDICATIONS

- The ECP's CLR and STS pushbuttons light up.
- The FAULT/OFF lights stay on.
- The E/WD's memo area returns to normal.
- The STATUS page automatically appears on the SD, displaying the procedures for completing the flight with a faulty blue system.



ECAM UPPER DISPLAY (E/WD)

- FULL MEMO DISPLAYED



ECAM LOWER DISPLAY (SD)

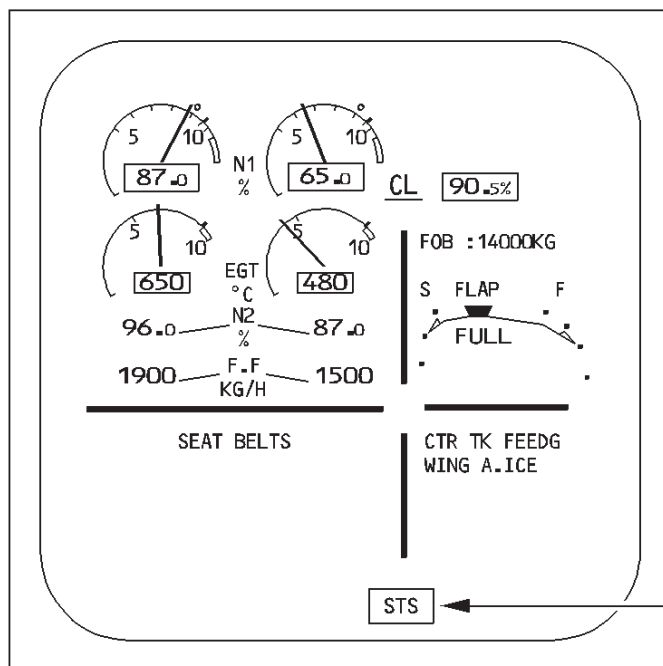
- THE STATUS PAGE IS AUTOMATICALLY DISPLAYED TO:
 - .PROVIDE THE PROCEDURE TO BE APPLIED FOR APPROACH.
 - .PROVIDE LANDING DISTANCE FACTORS AND INFORMATION.
 - .LIST THE INOPERATIVE SYSTEMS.

6 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON A THIRD TIME

COCKPIT INDICATIONS

- The CLR pushbutton light goes off.
- The FAULT/OFF lights stay on.
- A status reminder appears at the bottom of the E/WD.
- The SD automatically calls up the system page for the flight phase.

R

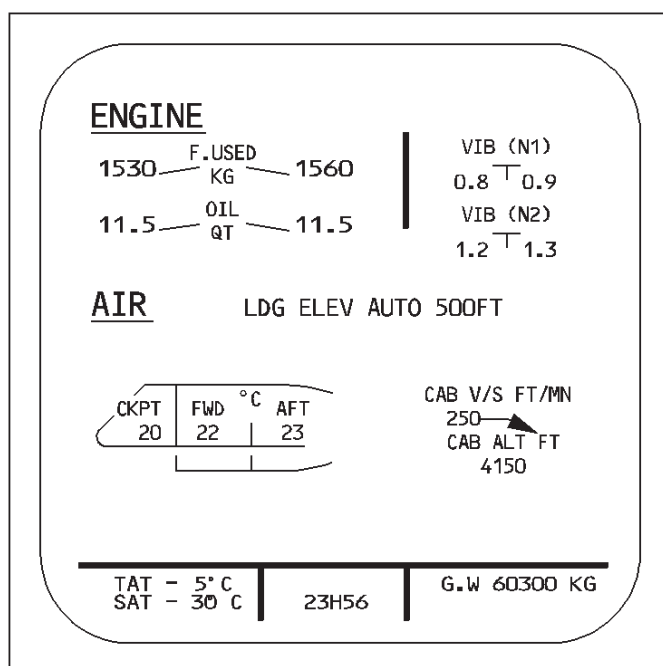


ECAM UPPER DISPLAY (E/WD)

– FULL MEMO DISPLAYED

STATUS REMINDER

ECAM LOWER DISPLAY (SD)



– RETURN TO THE FLIGHT PHASE
RELATED SYSTEM PAGE :
CRUISE PAGE

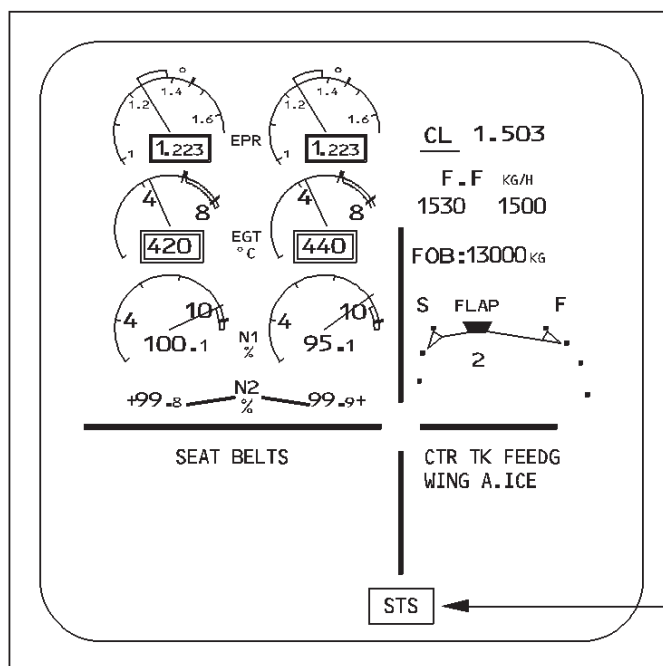
NFC5-01-3125-007-A005AA

6 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON A THIRD TIME

COCKPIT INDICATIONS

- The CLR pushbutton light goes off.
- The FAULT/OFF lights stay on.
- A status reminder appears at the bottom of the E/WD.
- The SD automatically calls up the system page for the flight phase.

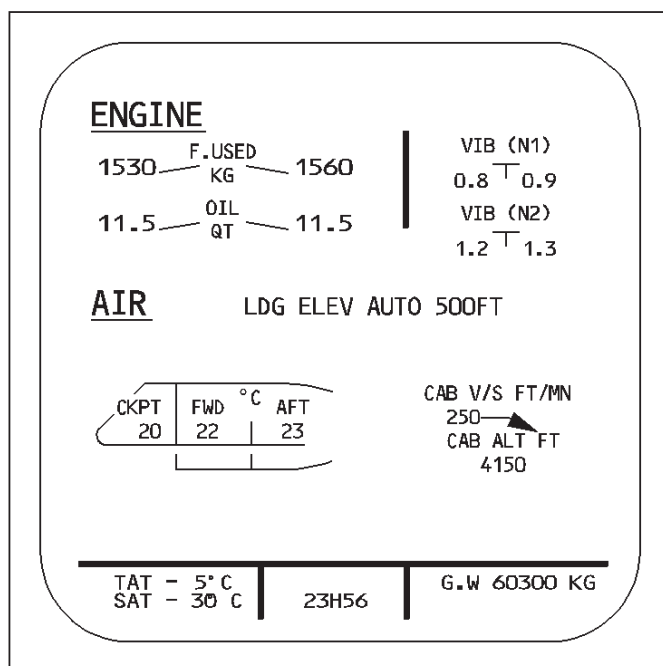
R



ECAM UPPER DISPLAY (E/WD)

– FULL MEMO DISPLAYED

ECAM LOWER DISPLAY (SD)



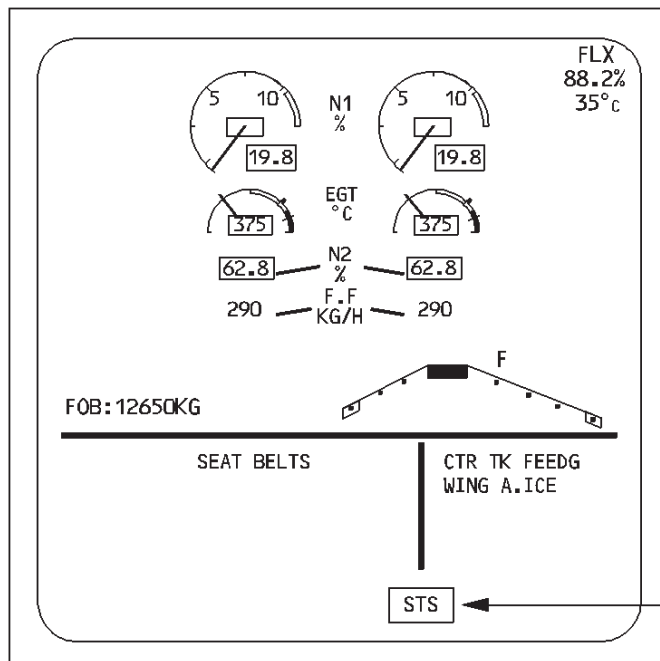
– RETURN TO THE FLIGHT PHASE
RELATED SYSTEM PAGE :
CRUISE PAGE

NFC5-01-3125-007-A050AA

6 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON A THIRD TIME

COCKPIT INDICATIONS

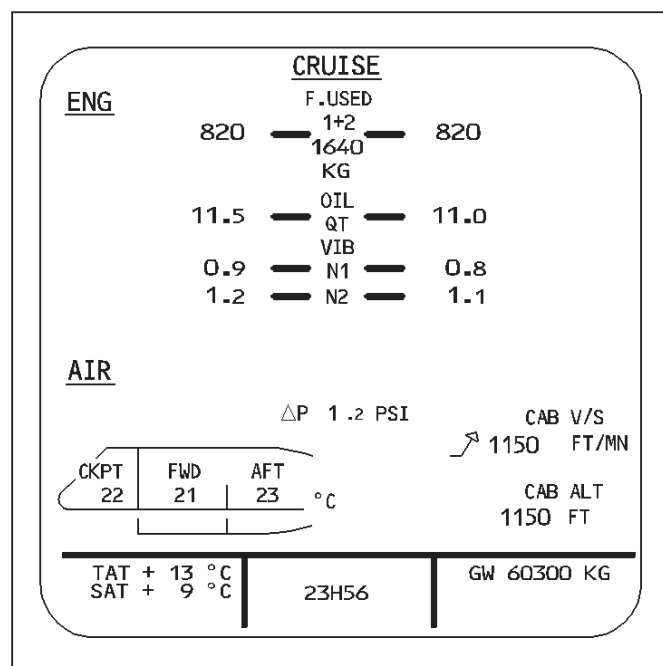
- The CLR pushbutton light goes off.
- The FAULT/OFF lights stay on.
- A status reminder appears at the bottom of the E/WD.
- The SD automatically displays the system page corresponding to the flight phase.



ECAM UPPER DISPLAY (E/WD)

– FULL MEMO DISPLAYED

STATUS REMINDER



ECAM LOWER DISPLAY (SD)

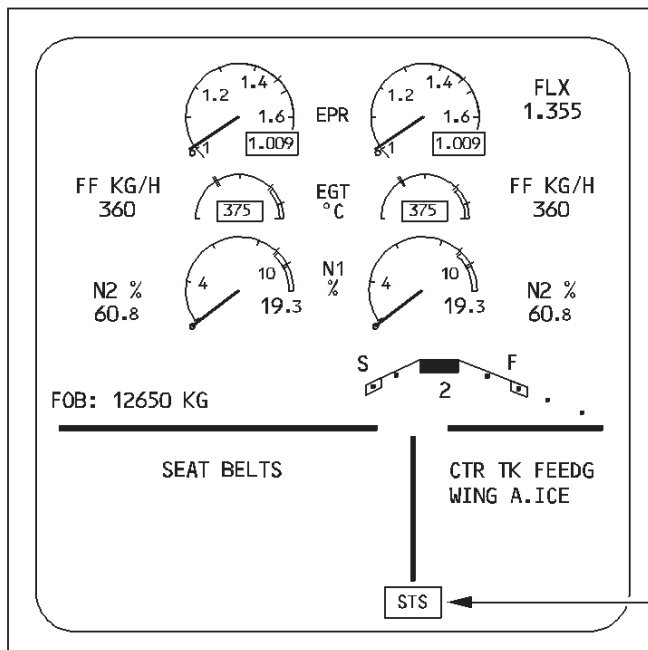
- RETURN TO THE FLIGHT PHASE
RELATED SYSTEM PAGE :
CRUISE PAGE

NFC5-01-3125-007-A110AA

6 – ONE OF THE PILOTS PUSHES THE CLR PUSHBUTTON A THIRD TIME

COCKPIT INDICATIONS

- The CLR pushbutton light goes off.
- The FAULT/OFF lights stay on.
- A status reminder appears at the bottom of the E/WD.
- The SD automatically displays the system page corresponding to the flight phase.

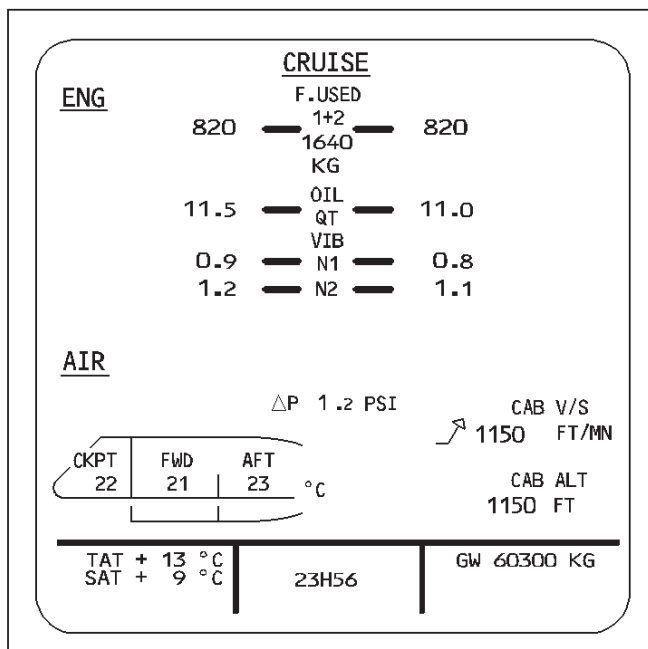


ECAM UPPER DISPLAY (E/WD)

– FULL MEMO DISPLAYED

STATUS REMINDER

ECAM LOWER DISPLAY (SD)



– RETURN TO THE FLIGHT PHASE
RELATED SYSTEM PAGE :
CRUISE PAGE

GENERAL

The OEB reminder function provides operational help to the crew by enabling them to clearly identify (on the ECAM) all procedures and status messages affected by an OEB. When a situation leading to a warning/caution occurs, a message informs the crew in real time that an OEB exists for the displayed warning and/or status and, consequently, that the procedure and/or status presented on the ECAM is not applicable.

Then the crew must refer to the QRH where the correct information is provided.

DESCRIPTION

The OEB reminder flag may apply to the :

- ECAM procedure only,
- ECAM procedure and corresponding status messages,
- Status message only.

PROCEDURE ONLY AFFECTED

- The ECAM warning title remains unaltered,
- All corresponding actions are suppressed and replaced by the “REFER TO QRH PROC” message,
- The related status messages on the ECAM system display remains unaltered.

COCKPIT INDICATION

<u>AIR</u> PACK1 OVHT	
–REFER TO QRH PROC	

ECAM UPPER DISPLAY (E/WD)

STATUS	
●WHEN PACK OVHT OUT:	<u>INOP SYS</u>
–PACK1.....ON	PACK1

ECAM LOWER DISPLAY (SD)

NFC5-01-3127-001-A100AA

PROCEDURE AND STATUS AFFECTED

- The ECAM warning title remains unaltered,
- All corresponding actions are suppressed and replaced by the “REFER TO QRH PROC” message,
- The related status messages on the ECAM system display remains unchanged, except for the additional “REFER TO QRH PROC” title.

COCKPIT INDICATION

<u>AIR</u> PACK1 OVHT	
–REFER TO QRH PROC	

ECAM UPPER DISPLAY (E/WD)

NFC5-01-3127-002-A100AA

STATUS	
<u>REFER TO QRH PROC</u>	<u>INOP SYS</u>
●WHEN PACK OVHT OUT:	
–PACK1.....ON	PACK1

ECAM LOWER DISPLAY (SD)

STATUS MESSAGE ONLY AFFECTED

- The ECAM warning title remains unaltered,
- The corresponding procedure remains unchanged, except for the additional “FOR STS REFER TO QRH” line.
- The related status messages on the ECAM system display remains unchanged, except for the additional “REFER TO QRH PROC” title.

COCKPIT INDICATION

<u>AIR PACK1 OVHT</u> –PACK1.....OFF ●WHEN PACK OVHT OUT: –PACK1.....ON –FOR STS REFER TO QRH	
---	--

ECAM UPPER DISPLAY (E/WD)

STATUS	
<u>REFER TO QRH PROC</u> ●WHEN PACK OVHT OUT: –PACK1.....ON	<u>INOP SYS</u> PACK1

ECAM LOWER DISPLAY (SD)

NFC5-01-3127-003-A100AB

OEB DATABASE

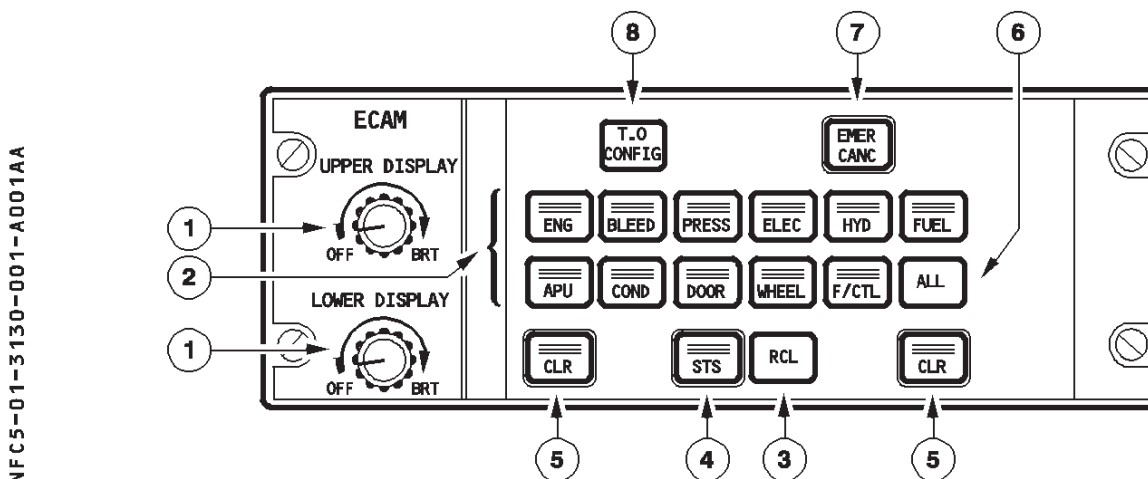
The OEB database lists the warnings and cautions affected by an OEB.

The OEB database can be :

- Loaded manually on the aircraft via the MCDU, and stored in both FWCs.
- Crossloaded from one FWC to the other FWC.
- Updated by entering a code via the MCDU.
- Checked via the MCDU.

Note : *The code provided on the OEB is designed to ensure that the OEB database is not updated before the OEB is available.*

ECAM CONTROL PANEL



① OFF / BRT knobs

Used to turn the ECAM DUs on and off, and to control their brightness (automatic adjustment of brightness for ambient light conditions is superimposed on this manual control).

Note : When the pilot turns the UPPER DISPLAY knob to OFF, the engine/warning (E/W) display appears on the lower display unit (automatic transfer).

② System page pushbuttons

- Call up the corresponding system pages on the SD.
- Light up, when pushed for manual selection, or when an advisory is detected.
- Call up the aircraft system page corresponding to the present flight phase or the current warning when pushed a second time.

When only one ECAM display is on, the pilot can display a system page for up to 3 minutes by pushing and holding the system page pushbutton.

- If an advisory condition arises, the relevant system page is not automatically displayed, but the pushbutton light pulses.
- If an ECAM warning is triggered, the relevant system page is not automatically displayed, and the system page pushbutton does not light up.

③ RCL pushbutton

The pilot pushes the RCL pushbutton to call up the warning messages, the caution messages, and the status page, that may have been suppressed by the activation of the CLR pushbutton or by flight-phase-related inhibition.

If there are no suppressed warnings or cautions, the E/WD shows "NORMAL" for five seconds.

If the pilot holds this pushbutton down for more than three seconds, the E/WD displays any caution messages that were suppressed by the EMER CANC pushbutton.

④ STS (status) pushbutton

The pilot pushes this pushbutton to display the STATUS page on the lower SD. The pushbutton remains lit, as long as the SD displays the STS page. If the system has no status messages, the status page displays "NORMAL" for five seconds.

The pilot can clear the STATUS page by pushing the CLR pushbutton, or by pushing the STS pushbutton a second time.

When only one ECAM display is on :

- It displays the STATUS page only when the pilot pushes the STATUS pushbutton and holds it. He can display the next STATUS page, if any, by releasing the pushbutton and pushing it again (before two seconds have elapsed). The new page then appears after a short delay.
- The pilot can keep the STS pushbutton pressed to display the STATUS page for a maximum of three minutes, after which the ECAM automatically displays the engine/warning page.

R

⑤ CLR pushbutton

This pushbutton remains lit as long as the E/WD is displaying a warning or caution message, or a status message on the SD.

If it is lit, pressing it changes the ECAM display.

R

⑥ ALL pushbutton

When this pushbutton is pressed and held down, the SD successively displays all the system pages at one-second intervals.

If the ECAM control panel fails, the pilot can use this pushbutton to page through the system pages until he comes to the one he wants to look at. He then releases the pushbutton to select that page.

⑦ EMER CANC pb

This pushbutton affects the following :

- Warnings :
 - Cancel (stop) an aural warning for as long as the failure condition continues.
 - Extinguishes the MASTER WARNINGS lights.
 - Does not affect the ECAM message display.
- Cautions :
 - Cancel any present caution (single chime, MASTER CAUTION lights, ECAM message) for the rest of the flight.
 - Automatically calls up the STATUS page, which displays “CANCELLED CAUTION” and the title of the failure that is inhibited.

The inhibition is automatically suppressed when Flight Phase 1 is initiated. The pilot may restore it manually by pressing the RECALL pushbutton for more than three seconds.

R Note : *This pushbutton should only be used to suppress spurious MASTER CAUTIONS.*

⑧ T.O. CONFIG pb

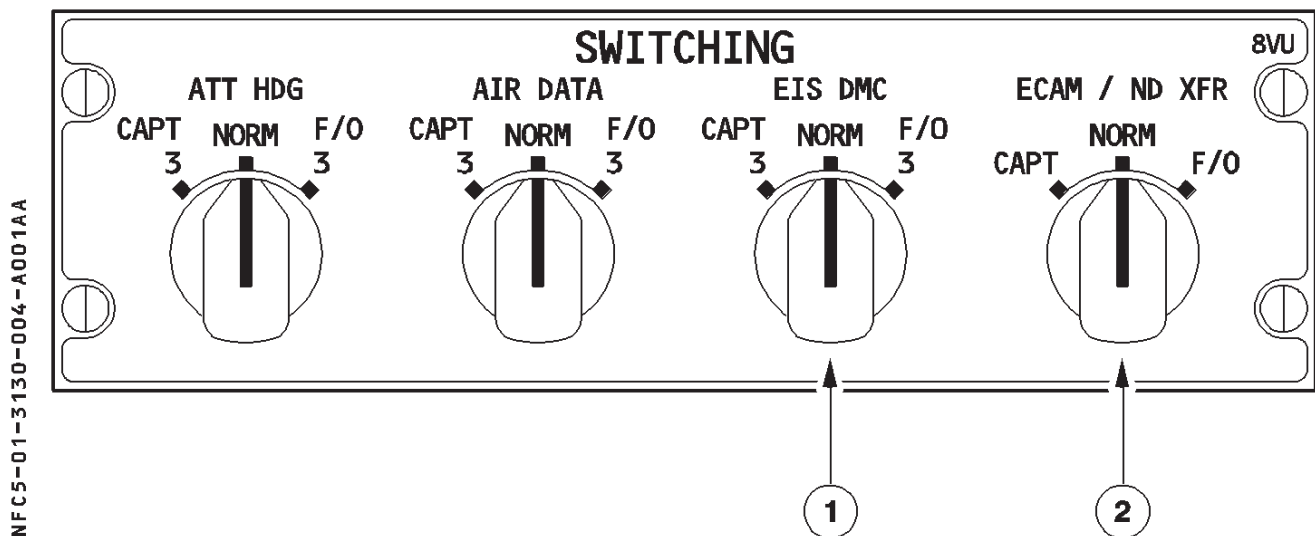
This pushbutton simulates the application of takeoff power. This is a test that triggers a warning, if the aircraft is not in takeoff configuration. (See 1.31.15).

If the configuration is correct, the E/WD displays the “TO CONFIG NORMAL” message in the TO MEMO section.

Note : *If the ECAM control panel fails, the CLR, RCL, STS, EMER CANC, and ALL pushbuttons remain operative, because their contacts are directly wired to the flight warning and display management computers.*

SWITCHING PANEL

ON PEDESTAL



① EIS DMC selector switch

NORM : DMC 1 supplies data to PFD 1, ND 1, and the upper ECAM DU.

DMC 2 supplies data to PFD 2, ND 2, and the lower ECAM DU.

CAPT 3: DMC 3 replaces DMC 1.

F/O 3 : DMC 3 replaces DMC 2.

Note : If a DMC fails, each of its associated DUs displays a diagonal line.

R ② ECAM/ND transfer selector switch

R Transfers the system/status display to either the Captain's or the First Officer's ND.

Note : If both ECAM DUs (E/WD and SD) fail, the flight crew can use this switch to transfer the E/W display to either navigation display.

⑦ EMER CANC pb

This pushbutton affects the following :

- Warnings :
 - Cancel (stop) an aural warning for as long as the failure condition continues.
 - Extinguishes the MASTER WARNINGS lights.
 - Does not affect the ECAM message display.
- Cautions :
 - Cancel any present caution (single chime, MASTER CAUTION lights, ECAM message) for the rest of the flight.
 - Automatically calls up the STATUS page, which displays “CANCELLED CAUTION” and the title of the failure that is inhibited.

The inhibition is automatically suppressed when Flight Phase 1 is initiated. The pilot may restore it manually by pressing the RECALL pushbutton for more than three seconds.

R Note : *This pushbutton should only be used to suppress spurious MASTER CAUTIONS.*

⑧ T.O. CONFIG pb

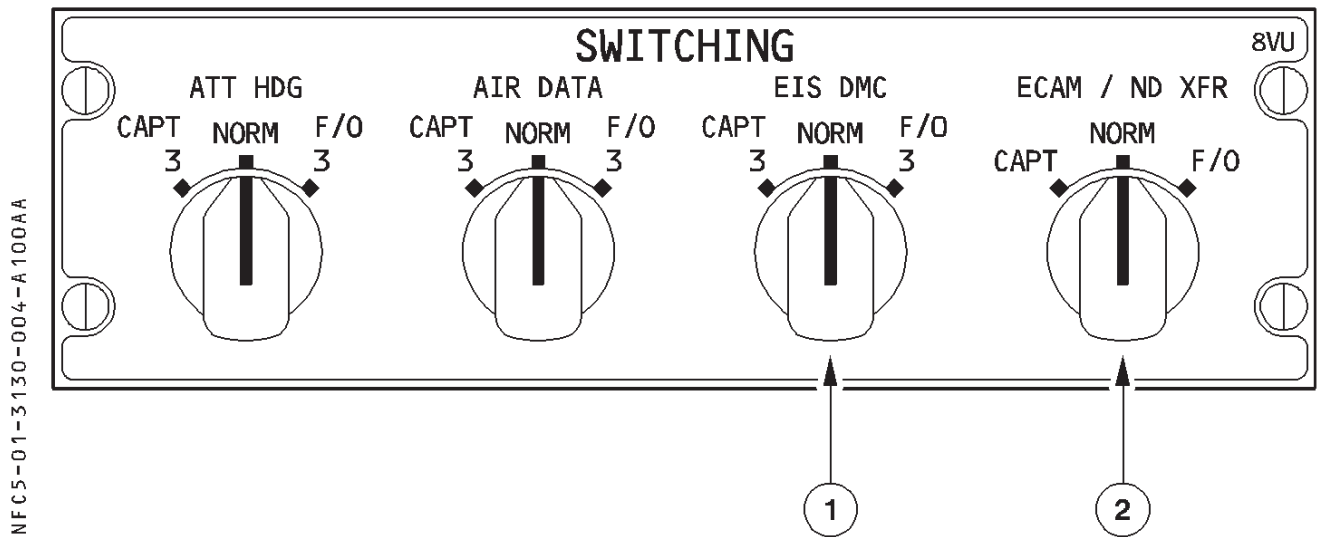
This pushbutton simulates the application of takeoff power. This is a test that triggers a warning, if the aircraft is not in takeoff configuration. (See 1.31.15).

If the configuration is correct, the E/WD displays the “TO CONFIG NORMAL” message in the TO MEMO section.

Note : *If the ECAM control panel fails, the CLR, RCL, STS, EMER CANC, and ALL pushbuttons remain operative, because their contacts are directly wired to the flight warning and display management computers.*

SWITCHING PANEL

ON PEDESTAL



① EIS Display Management Computer (DMC) selector

NORM : DMC 1 supplies the CAPT's PFD, the CAPT's ND, and the ECAM's DUs.

DMC 2 supplies the F/O's PFD and the F/O's ND.

CAPT 3: DMC 3 replaces DMC 1.

F/O 3 : DMC 3 replaces DMC 2.

Note : If a DMC fails, each of its associated DUs displays an "INVALID DATA" message.

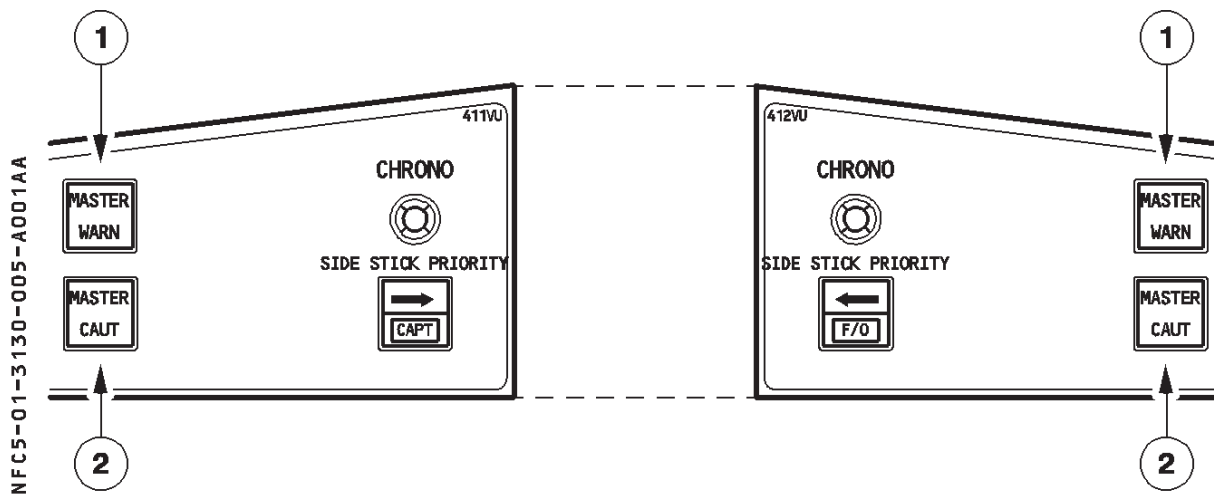
② ECAM/ND transfer selector

Transfers the system/status display to either the Captain or the First Officer's ND.

The "ECAM ON ND" message is displayed on the lower ECAM display.

Note : If both ECAM DUs (E/WD and SD) fail, the flight crew may use this switch to transfer the E/WD display to either navigation display. In this case, the "ECAM ON ND" message is not displayed.

ATTENTION GETTERS



① MASTER WARN lights

- Flash red for level 3 warning.
- Accompanied by an aural warning (continuous repetitive chime, specific sounds or synthetic voice).

② MASTER CAUT lights

- Light up steady amber for a level 2 caution.
- Accompanied by a single chime.

These lights go out when :

- One pilot presses the light (except for some red warnings, such as the overspeed and stall warnings).
- The warning/caution situation is over.
- The pilot presses the CLR pushbutton on the ECAM control panel (except for some red warnings, such as the overspeed and stall warnings).
- The pilot presses the EMER CANC pushbutton on the ECAM control panel.

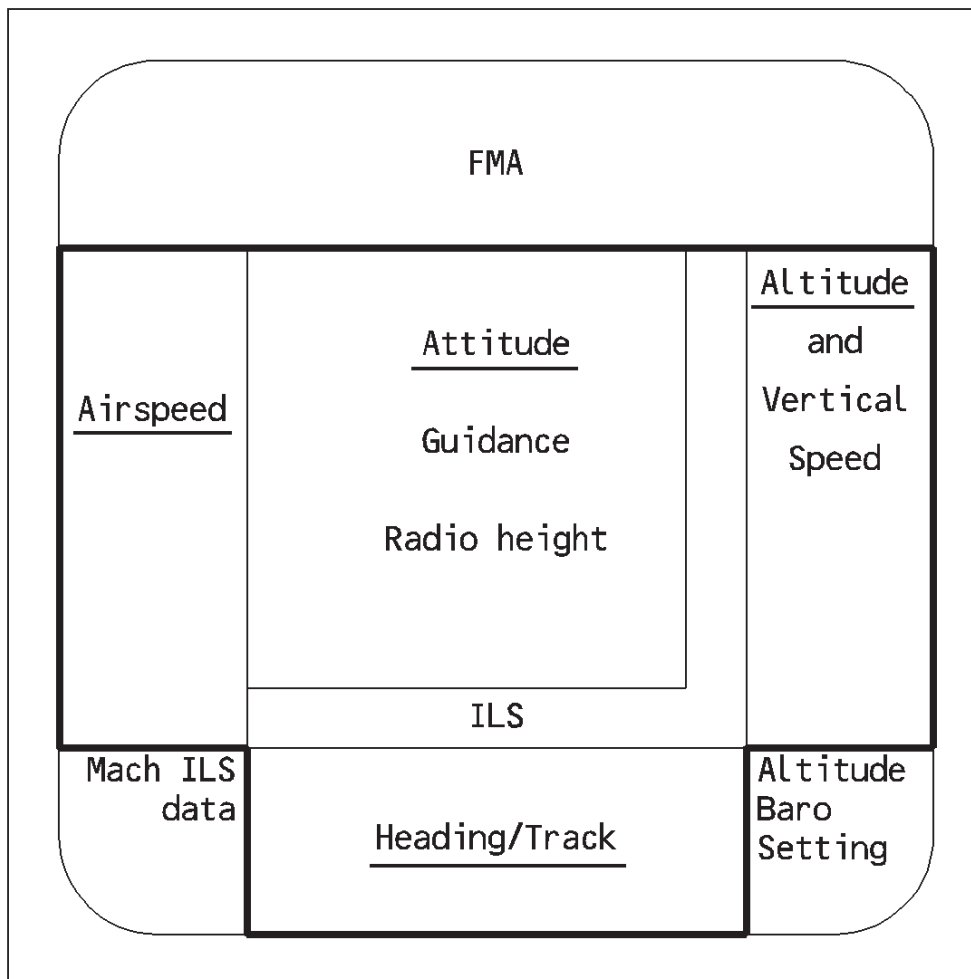
The aural warnings cease when :

- One pilot presses the MASTER WARN light (except for some red warnings, such as the overspeed and stall warnings).
- The warning situation is over.
- The pilot presses the EMER CANC pushbutton on the ECAM control panel.

GENERAL

The Primary Flight Display (PFD) provides the following information to the flight crew :

- Attitude and Guidance
- Airspeed
- Altitude (baro and radio) and vertical speed
- Heading and Track
- FMGS modes (Flight Mode Annunciator)
- Vertical and lateral deviations
- Radio navigation information (ILS, DME).

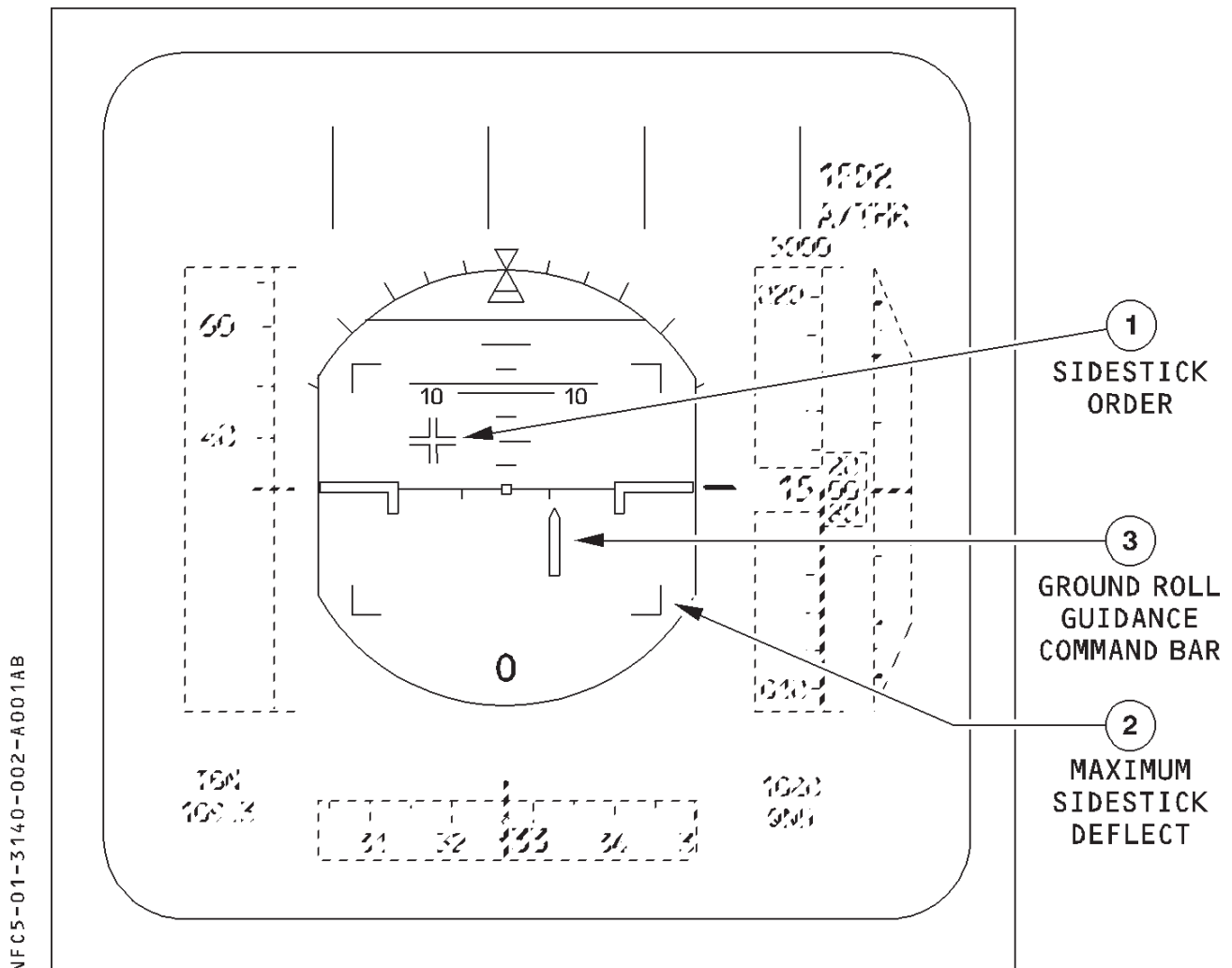


The FWC monitors such main parameters as : Attitude, heading, and altitude. Also refer to the "FLAGS AND MESSAGES DISPLAYED ON PFD" Chapter.

Note : A grey background appears on the speed, the heading vertical speed, and the altitude speed scales of the PFD. If the Primary Flight Display (PFD) Unit temperature exceeds a defined threshold, the grey background disappears, in order to limit power consumption and prevent a DU overheat. Any additional increase in temperature will lead to a complete cut off of the power supply to this display unit.

R
R

SPECIFIC GROUND INDICATIONS



① Sidestick order indication (white)

This is displayed, as soon as one engine is started.

It indicates the total of the pilot's and copilot's sidestick orders (shown here as left wing down, pitch up).

② Max Sidestick Deflection (white)

This is displayed, as soon as one engine is started.

③ Ground Roll Guidance Command Bar (green)

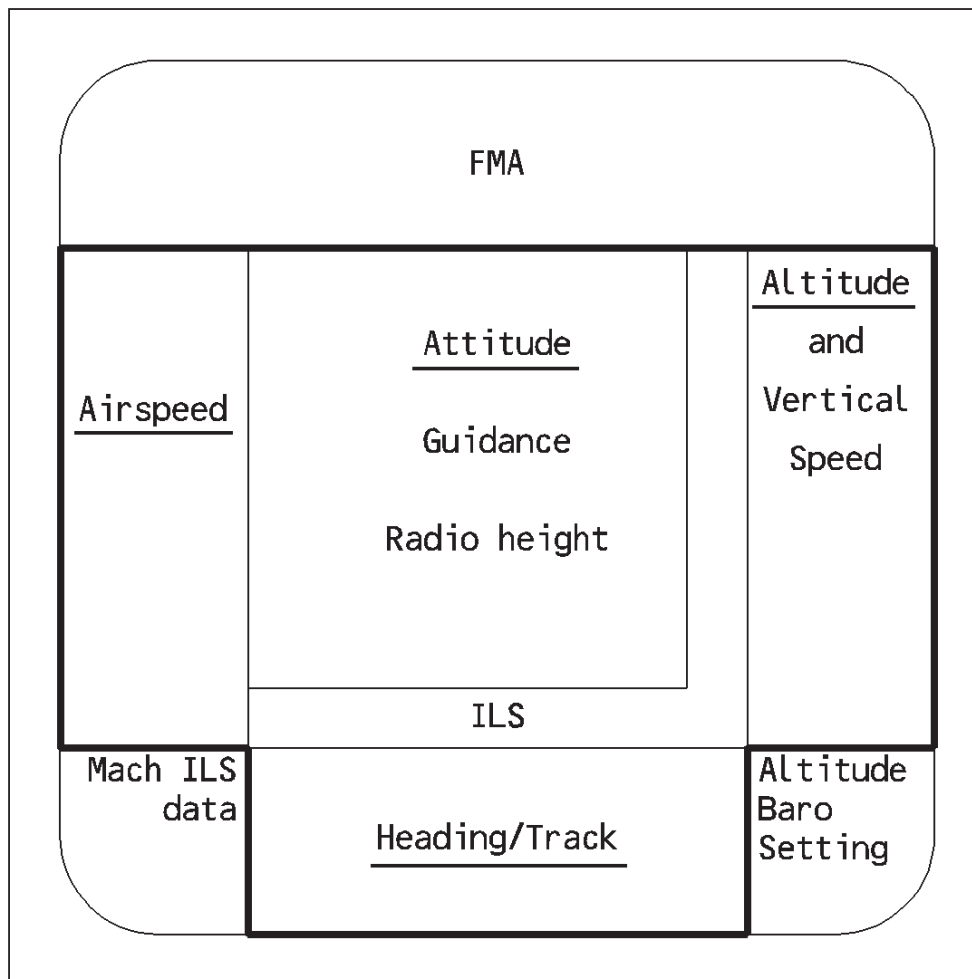
This symbol is displayed when the aircraft is on the ground, or below 30 feet radio altitude, provided a localizer signal is available. It shows the flight director yaw orders, to keep the runway centerline.

R
R

GENERAL

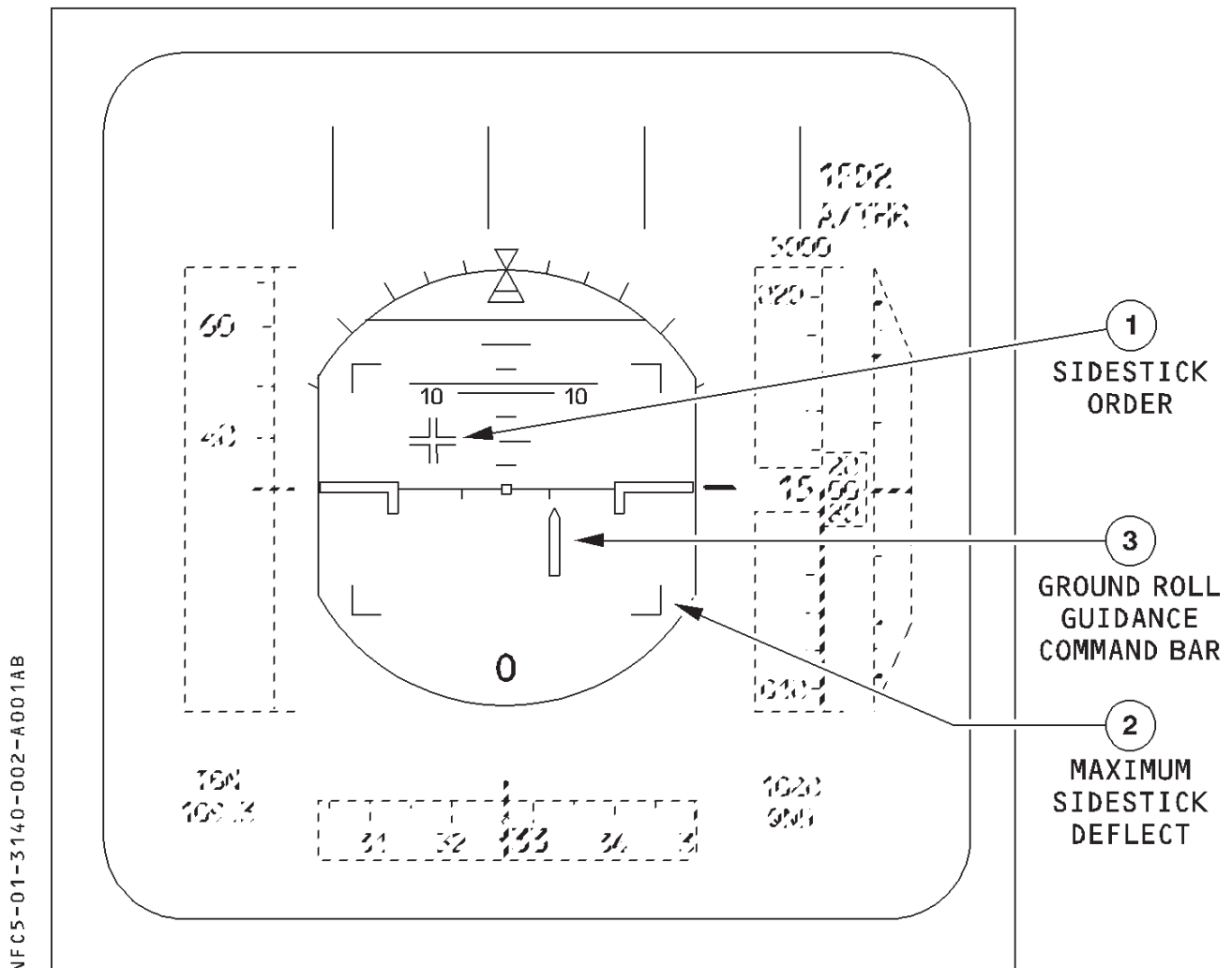
The Primary Flight Display (PFD) provides the following information :

- Attitude and Guidance
- Airspeed
- Altitude (Baro and Radio) and Vertical Speed
- Heading and Track
- FMGS modes (Flight Mode Annunciator)
- Vertical and Lateral Deviations
- Radio navigation information (ILS, DME).



The FWC monitors such main parameters as : Attitude, heading, and altitude. For more information, refer to the "FLAGS AND MESSAGES DISPLAYED ON PFD" Chapter.

SPECIFIC GROUND INDICATIONS



① Sidestick order indication (white)

This is displayed, as soon as one engine is started.

It indicates the total of the pilot's and copilot's sidestick orders (shown here as left wing down, pitch up).

② Max Sidestick Deflection (white)

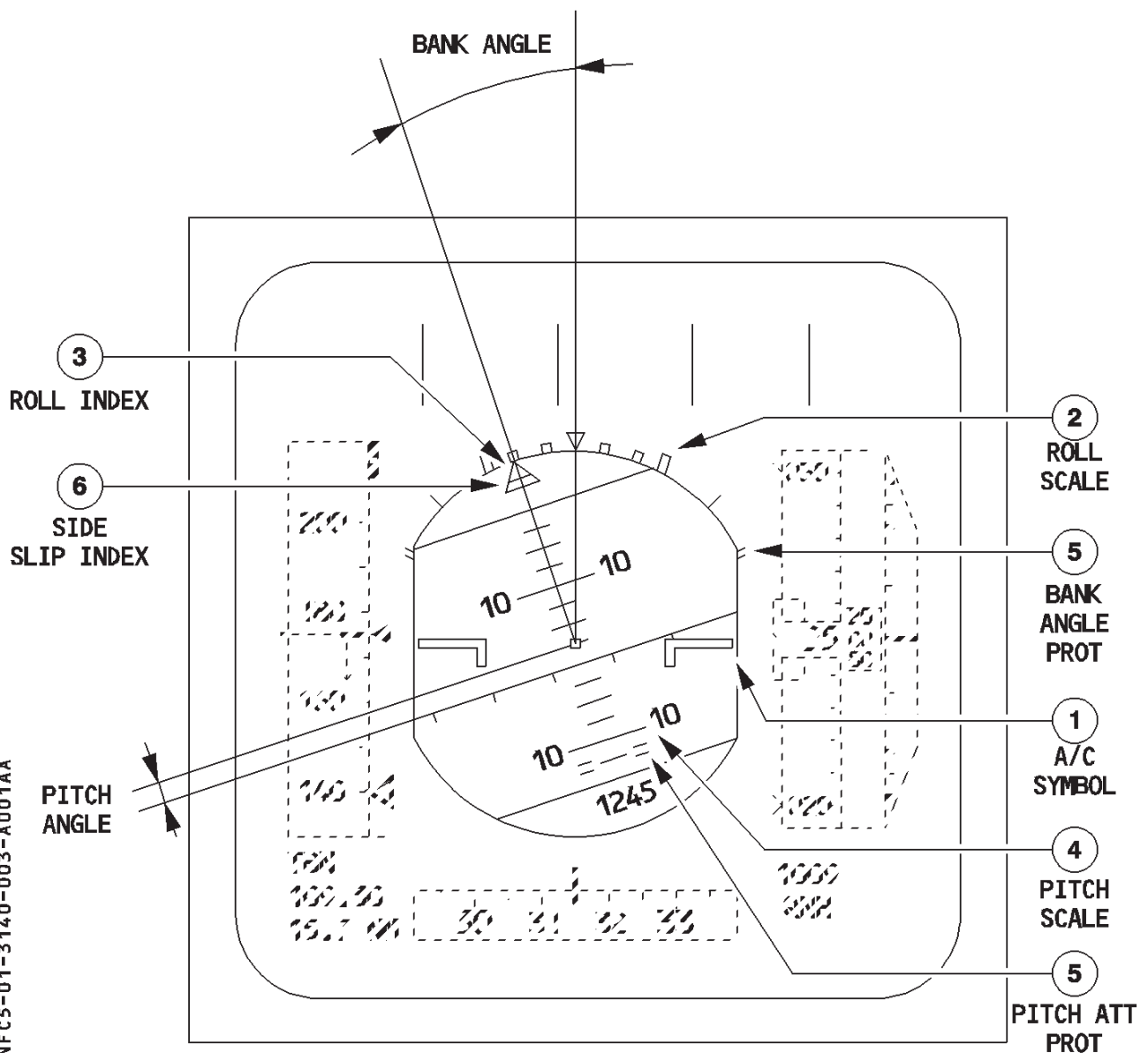
This is displayed, as soon as one engine is started.

③ Ground Roll Guidance Command Bar (green)

This symbol is displayed when the aircraft is on the ground, or below 30 feet radio altitude, provided a localizer signal is available. It shows the flight director yaw orders, to keep the runway centerline.

R
R

NFC5-01-3140-003-A001AA



This symbol is in black, and outlined in yellow. The yellow outline is dimmed if the crew selects TRK-FPA, unless the FMA is in the TOGA or FLX mode.

This scale is in white, and has markers at 0,10, 20, 30, and 45 degrees of bank.

This pointer indicates the bank angle. When the bank angle exceeds 45°, all the PFD symbols, except those for attitude, speed, heading, altitude, and vertical speed, disappear. The display returns to normal when the bank angle decreases below 40°.

④ Pitch Scale (white)

This scale has markers every ten degrees between 80° nose up and 80° nose down (every 2.5° between 10° nose down and 30° nose up). When pitch angle exceeds 25° nose up or 13° nose down, all the PFD displays except attitude, speed, speed trend, heading, altitude, and vertical speed disappear. Beyond 30°, large red arrowheads indicate that the attitude has become excessive and show the direction to move the nose in order to reduce it. The display returns to normal when pitch angle becomes less than 22° nose up or 10° nose down.

⑤ Flight Control Protection Symbols

The display shows these symbols in green :

- on the roll scale at $\pm 67^\circ$ to mark the bank angle limits
- on the pitch scale at 15° nose down or 30° nose up to mark the pitch limits.

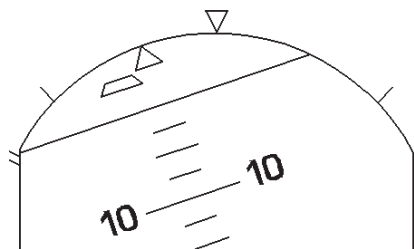
An amber $\times$ replaces these symbols if the corresponding protection is lost.

(Refer to 1.27.30)

⑥ Sideslip Index (yellow)

R

This trapezoidal index moves beneath the roll index. On the ground it represents the lateral acceleration of the aircraft : in flight it shows sideslip (as computed by the FAC). One centimeter of displacement indicates 0.2g. The sideslip index is against its stop at 0.3g.



In case of engine failure at takeoff or go around, the sideslip index changes from yellow to blue.

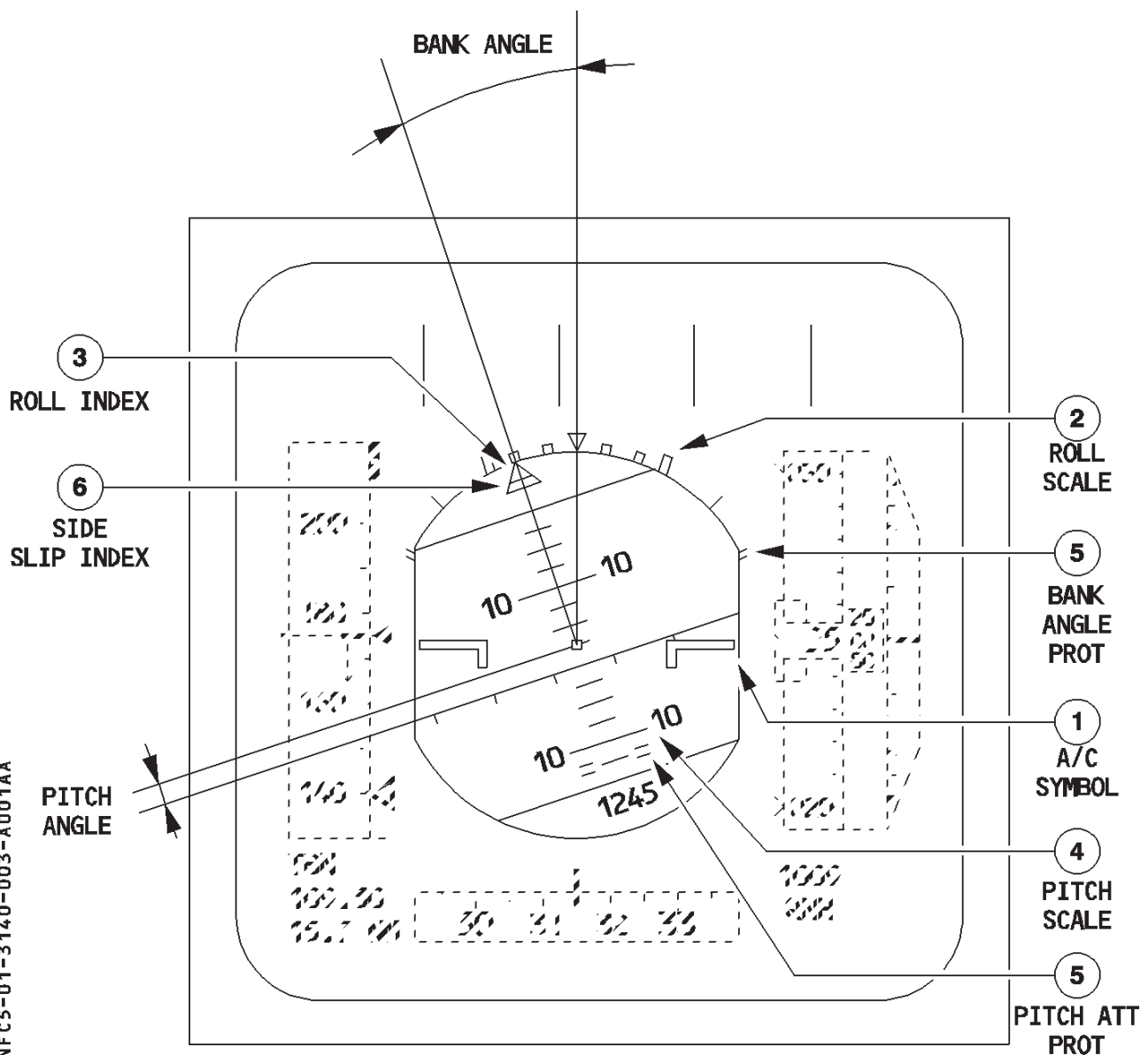
Note : The sideslip target is blue if :

- CONF 1, 2, or 3 is selected, and
- any ENG N1 > 80%, and
- the difference between the ENG N1's exceeds 35%.

In this case the sideslip index is called β target.

When this index is centered with the roll index, the sideslip equals the sideslip target for optimum aircraft performance.

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① Fixed Aircraft Symbol

R	This symbol is in black, and outlined in yellow. The yellow outline is dimmed if the crew
R	selects TRK-FPA, unless the FMA is in the TOGA or FLX mode.

② Roll Scale

This scale is in white, and has markers at 0,10, 20, 30, and 45 degrees of bank.

③ Roll Index (yellow)

This pointer indicates the bank angle. When the bank angle exceeds 45°, all the PFD symbols, except those for attitude, speed, heading, altitude, and vertical speed, disappear. The display returns to normal when the bank angle decreases below 40°.

④ Pitch Scale (white)

This scale has markers every ten degrees between 80° nose up and 80° nose down (every 2.5° between 10° nose down and 30° nose up). When pitch angle exceeds 25° nose up or 13° nose down, all the PFD displays except attitude, speed, speed trend, heading, altitude, and vertical speed disappear. Beyond 30°, large red arrowheads indicate that the attitude has become excessive and show the direction to move the nose in order to reduce it. The display returns to normal when pitch angle becomes less than 22° nose up or 10° nose down.

⑤ Flight Control Protection Symbols

The display shows these symbols in green :

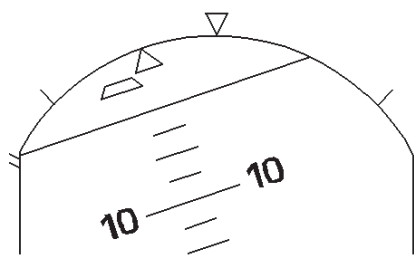
- on the roll scale at $\pm 67^\circ$ to mark the bank angle limits
- on the pitch scale at 15° nose down or 30° nose up to mark the pitch limits.

An amber $\times$ replaces these symbols if the corresponding protection is lost. (Refer to 1.27.30)

⑥ Sideslip Index (yellow)

R

This trapezoidal index moves beneath the roll index. On the ground it represents the lateral acceleration of the aircraft : in flight it shows sideslip (as computed by the FAC). One centimeter of displacement indicates 0.2g. The sideslip index is against its stop at 0.3g.



In case of engine failure at takeoff or go around, the sideslip index changes from yellow to blue.

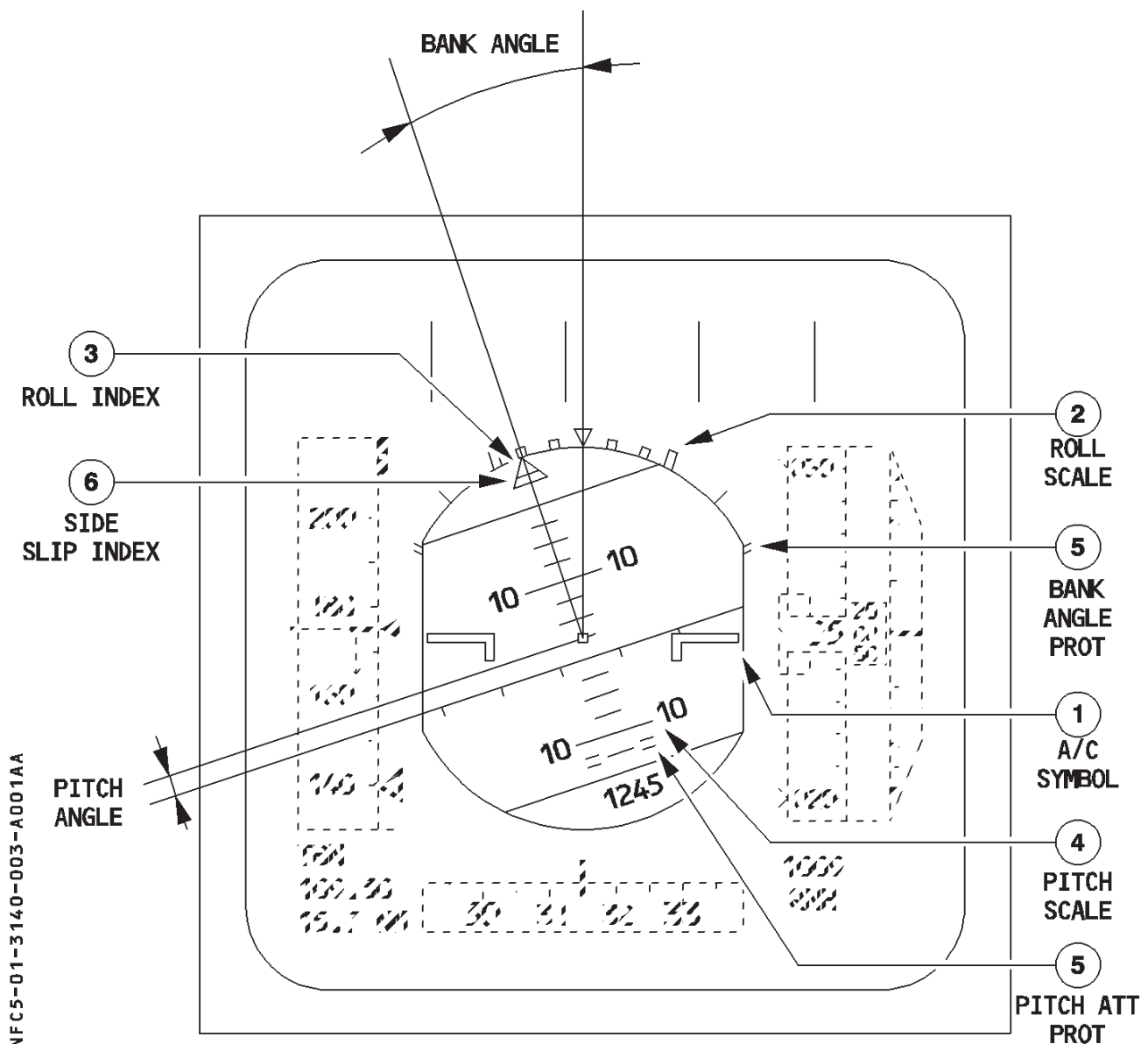
Note : The sideslip target is blue if :

- CONF 1, 2, or 3 is selected, and
- any ENG EPR > 1.25, and
- the difference between the ENG EPR's exceeds 0.25.

In this case the sideslip index is called β target.

When this index is centered with the roll index, the sideslip equals the sideslip target for optimum aircraft performance.

ATTITUDE DATA



① Fixed Aircraft Symbol

R This symbol is in black, and outlined in yellow. The yellow outline is dimmed if the crew
R selects TRK-FPA, unless the FMA is in the TOGA or FLX mode.

② Roll Scale

This scale is in white, and has markers at 0, 10, 20, 30, and 45 degrees of bank.

③ Roll Index (yellow)

This pointer indicates the bank angle. When the bank angle exceeds 45°, all the PFD symbols, except those for attitude, speed, heading, altitude, and vertical speed, disappear. The display returns to normal when the bank angle decreases below 40°.

④ Pitch Scale (white)

This scale has markers every ten degrees between 80° nose up and 80° nose down (every 2.5° between 10° nose down and 30° nose up). When pitch angle exceeds 25° nose up or 13° nose down, all the PFD displays except attitude, speed, speed trend, heading, altitude, and vertical speed disappear. Beyond 30°, large red arrowheads indicate that the attitude has become excessive and show the direction to move the nose in order to reduce it. The display returns to normal when pitch angle becomes less than 22° nose up or 10° nose down.

⑤ Flight Control Protection Symbols

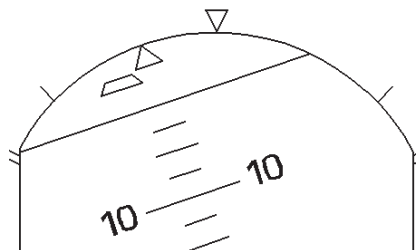
The display shows these symbols in green :

- On the roll scale at $\pm 67^\circ$ to mark the bank angle limits
- On the pitch scale at 15° nose down or 30° nose up to mark the pitch limits.

An amber $\times$ replaces these symbols if the corresponding protection is lost.
(Refer to 1.27.30)

⑥ Sideslip Index (yellow)

This trapezoidal index moves beneath the roll index. On ground, it represents the lateral acceleration of the aircraft. In flight, it shows sideslip (as computed by the FAC). One centimeter of displacement indicates 0.2g. The sideslip index is against its stop at 0.3g.



In case of engine failure at takeoff or go around, the sideslip index changes from yellow to blue.

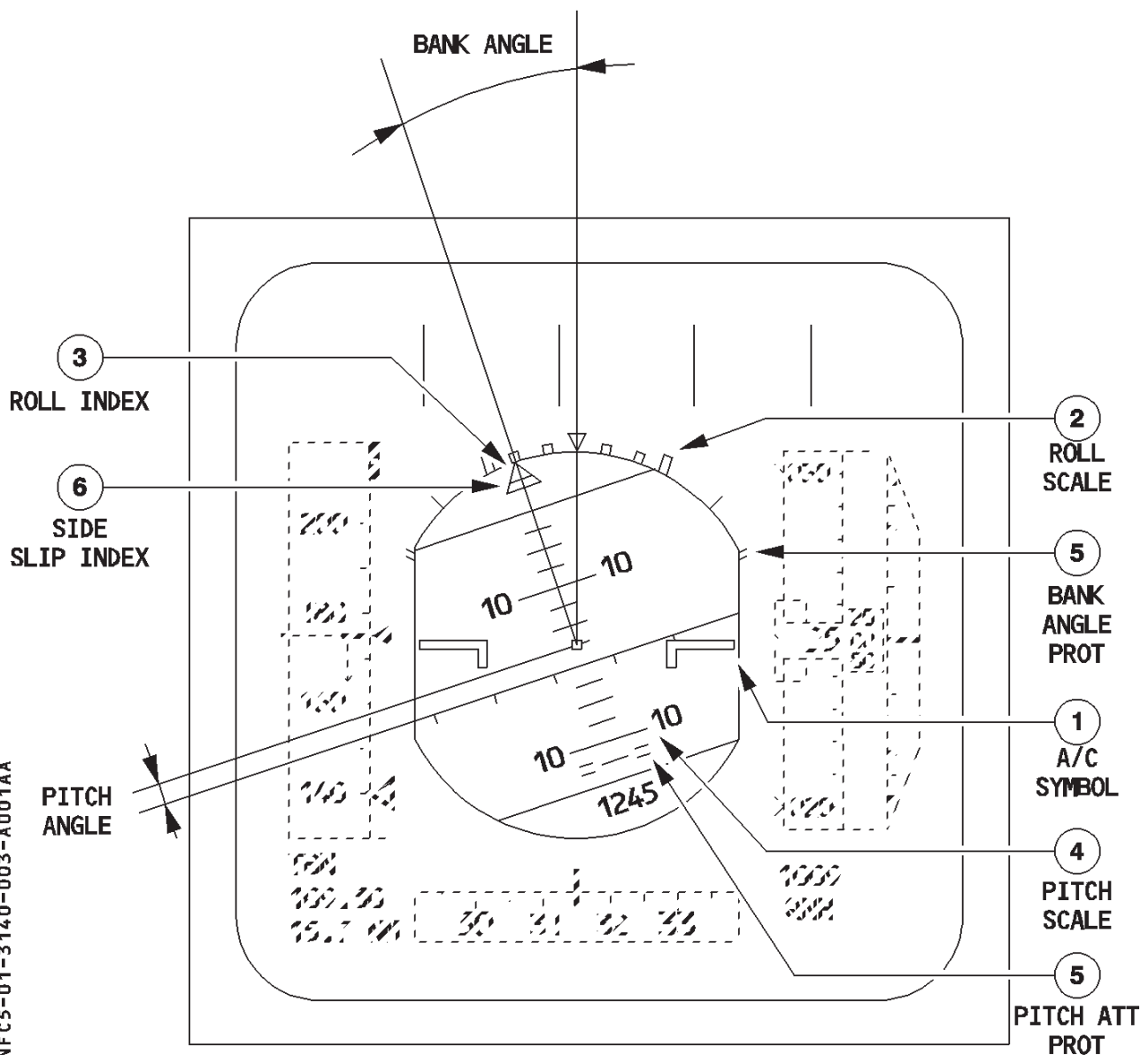
Note : The sideslip target is blue, if :

- CONF 1, 2, or 3 is selected, and
- Any ENG N1 > 80% or one Thrust Lever > MCT ($\geq$ FLX if FLX or DERATED TO), and
- The difference between the ENG N1's exceeds 35%.

In this case, the sideslip index is called β target.

When this index is centered with the roll index, the sideslip equals the sideslip target for optimum aircraft performance.

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This symbol is in black, and outlined in yellow. The yellow outline is dimmed if the crew selects TRK-FPA, unless the FMA is in the TOGA or FLX mode.

This scale is in white, and has markers at 0,10, 20, 30, and 45 degrees of bank.

This pointer indicates the bank angle. When the bank angle exceeds 45°, all the PFD symbols, except those for attitude, speed, heading, altitude, and vertical speed, disappear. The display returns to normal when the bank angle decreases below 40°.

④ Pitch Scale (white)

This scale has markers every ten degrees between 80° nose up and 80° nose down (every 2.5° between 10° nose down and 30° nose up). When pitch angle exceeds 25° nose up or 13° nose down, all the PFD displays except attitude, speed, speed trend, heading, altitude, and vertical speed disappear. Beyond 30°, large red arrowheads indicate that the attitude has become excessive and show the direction to move the nose in order to reduce it. The display returns to normal when pitch angle becomes less than 22° nose up or 10° nose down.

⑤ Flight Control Protection Symbols

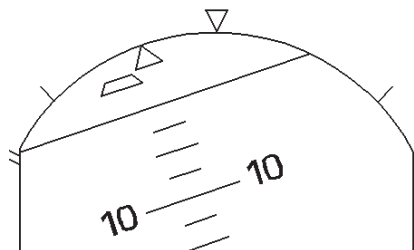
The display shows these symbols in green :

- On the roll scale at $\pm 67^\circ$ to mark the bank angle limits
- On the pitch scale at 15° nose down or 30° nose up to mark the pitch limits.

An amber $\times$ replaces these symbols if the corresponding protection is lost. (Refer to 1.27.30)

⑥ Sideslip Index (yellow)

This trapezoidal index moves beneath the roll index. On ground, it represents the lateral acceleration of the aircraft. In flight, it shows sideslip (as computed by the FAC). One centimeter of displacement indicates 0.2g. The sideslip index is against its stop at 0.3g.



In case of engine failure at takeoff or go around, the sideslip index changes from yellow to blue.

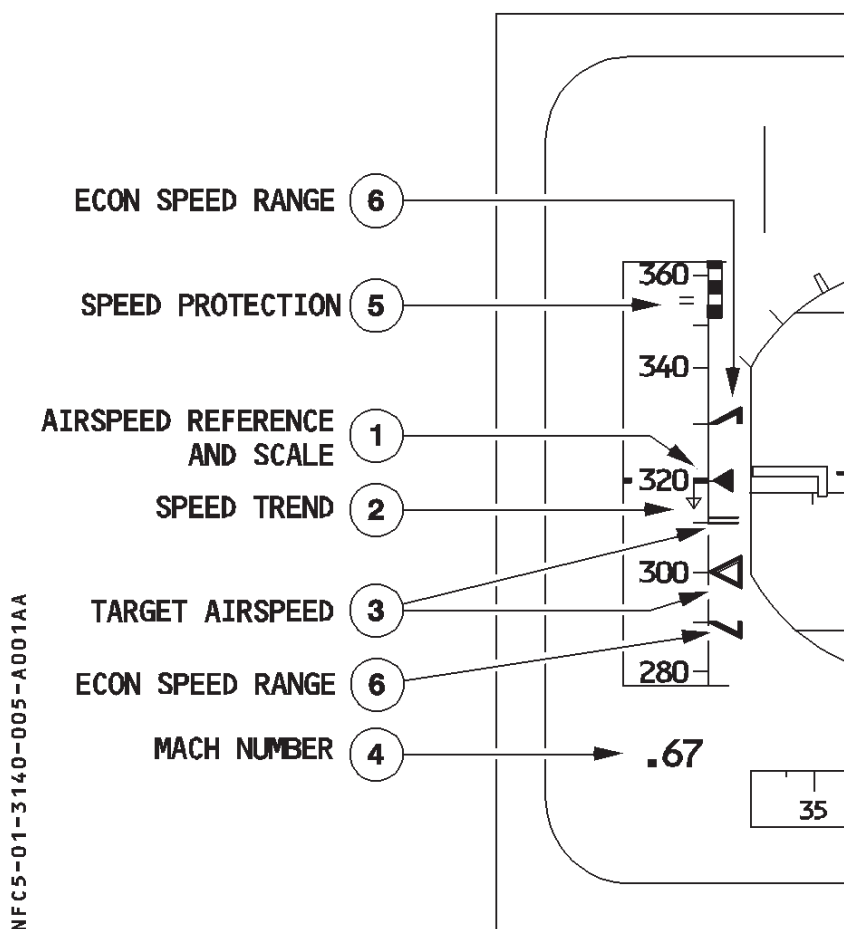
Note : The sideslip target is blue, if :

- CONF 1, 2, or 3 is selected, and
- Any ENG EPR > 1.25 or one Thrust Lever > MCT ($\geq$ FLX if FLX or DERATED TO), and
- The difference between the ENG EPR's exceeds 0.25.

In this case, the sideslip index is called β target.

When this index is centered with the roll index, the sideslip equals the sideslip target for optimum aircraft performance.

AIRSPEED



① Actual Airspeed Reference Line and Scale

R A white scale on a grey background moves in front of a fixed yellow reference line next to a yellow triangle to show airspeed. The minimum airspeed indication is 30 knots.

② Speed Trend (yellow)

This pointer starts at the speed symbol. The tip shows the speed the aircraft will reach in 10 seconds if its acceleration remains constant. The pointer appears only when it is greater than 2 knots and disappears when it is less than 1 knot. It also disappears if the FACs fail.

③ Target Airspeed (magenta or blue)

This symbol gives the target airspeed or the airspeed corresponding to the target Mach number.

The target airspeed is the airspeed computed by FMGC in managed speed mode (magenta) or entered manually on the FCU for selected speed mode (blue). The target speed is a magenta double bar (=) when associated with the ECON speed range. Otherwise it is a triangle (magenta or blue).

When the target speed is off the speed scale, its value is displayed as numbers below or above the speed scale.

④ Mach Number (green)

This is displayed when it is greater than 0.5.

⑤ Speed Protection (green)

R This symbol indicates the speed ($V_{MO} + 6 \text{ kt}$ or $MMO + 0.01$) at which overspeed protection becomes active. (Refer to 1.27.30)

R ⑥ ECON Speed Range (magenta)

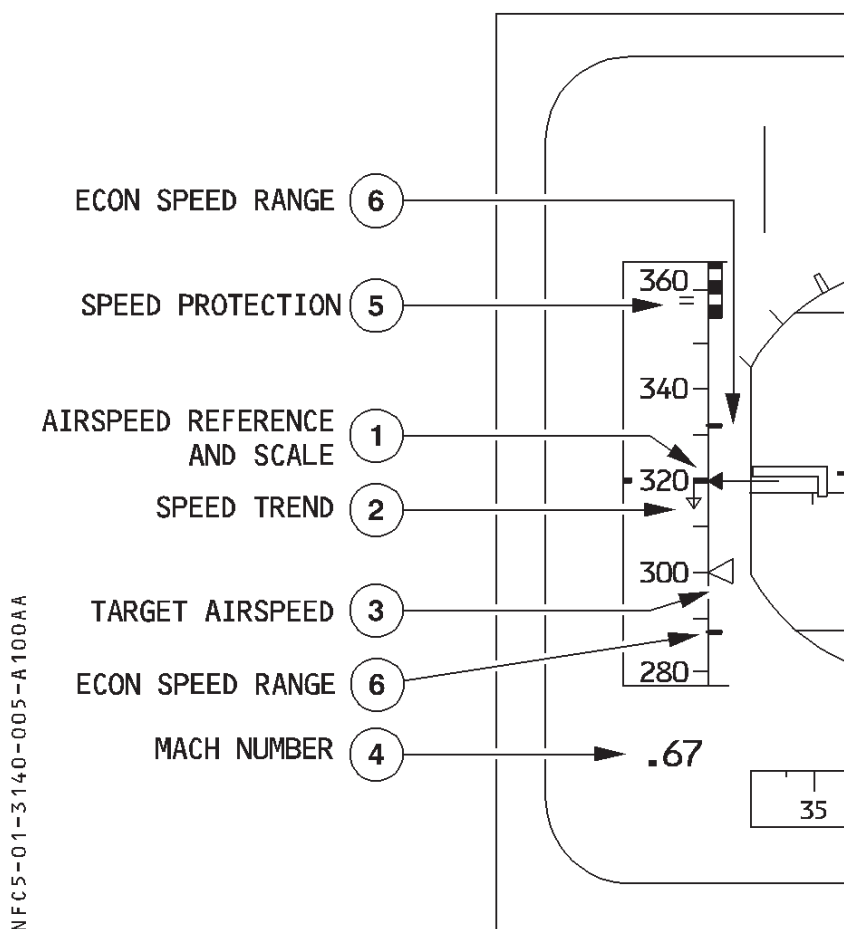
R In descent mode with ECON/AUTO SPD mode active, these two half triangles replace the selected speed symbol. It shows the upper and lower limits calculated by the FMGC.

- The upper speed is target speed + 20 knots, limited to V_{MAX} or $V_{MO} - 3 \text{ knots}$ or $MMO - 0.006$, whichever is lowest.

If a speed limit or a speed constraint applies, the upper margin is limited to ECON SPD + 5 knots.

- R – The lower speed margin is the target speed – 20 knots, limited to green dot, F, S, or VLS, whichever is higher.

AIRSPEED



① Actual Airspeed Reference Line and Scale

A white scale, on a grey background, moves in front of a fixed yellow reference line (next to a yellow triangle) to indicate airspeed. The minimum airspeed indication is 30 knots.

② Speed Trend (yellow)

This pointer starts at the speed symbol. The tip indicates the speed the aircraft will reach in 10 seconds, if its acceleration remains constant. The pointer only appears, when it is greater than 2 knots, and disappears when it is less than 1 knot. It also disappears, if the FACs fail.

③ Target Airspeed (magenta or blue)

This symbol gives the target airspeed, or the airspeed corresponding to the target Mach number.

The target airspeed is the airspeed computed by FMGC in managed speed mode (magenta), or the airspeed manually entered on the FCU in selected speed mode (blue). The target speed is indicated by a magenta or blue triangle.

When the target speed is off the speed scale, its value is displayed as numbers, either above or below the speed scale.

④ Mach Number (green)

It is displayed, when it is greater than 0.5.

⑤ Speed Protection (green)

This symbol indicates the speed ($V_{MO} + 6 \text{ kt}$ or $MMO + 0.01$) at which overspeed protection becomes active (Refer to 1.27.30).

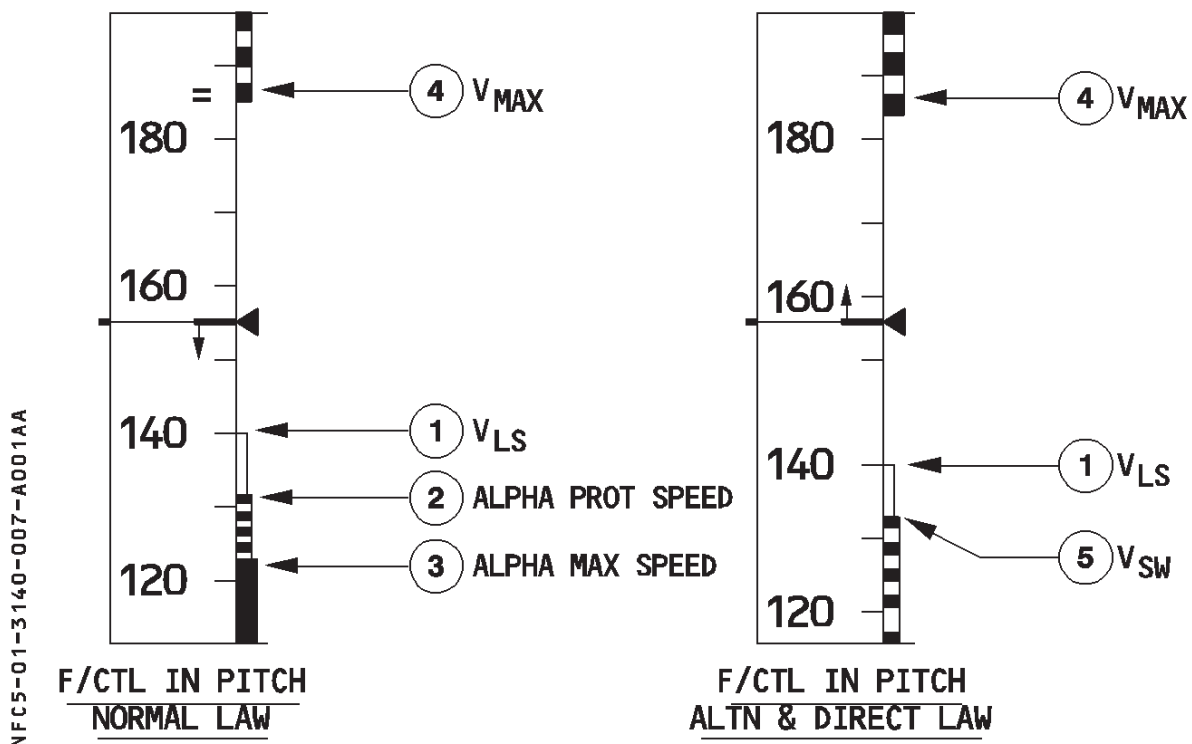
⑥ ECON Speed Range (magenta)

In descent mode with the ECON/AUTO SPD mode active, these two thick lines replace the selected speed symbol. It shows the upper and lower limits, calculated by the FMGC.

- The upper speed is target speed + 20 knots, limited to V_{MAX} or $V_{MO} - 3 \text{ knots}$ or $MMO - 0.006$, whichever is lowest.

If a speed limit or a speed constraint applies, the upper margin is limited to ECON SPD + 5 knots.

- The lower speed margin is the target speed – 20 knots, limited to green dot, F, S, or VLS, whichever is higher.



① Minimum Selectable Speed (VLS)

The top of the amber strip along the speed scale indicates this speed. It represents the lowest selectable speed providing an appropriate margin to the stall speed. (Refer to 3.04.10)

VLS information is inhibited from touchdown until 10 seconds after liftoff.

② Alpha Protection Speed

The top of a black and amber strip along the speed scale indicates this speed. It represents the speed corresponding to the angle of attack at which alpha protection becomes active (Refer to 1.27.20).

It is displayed when in pitch normal law.

③ Alpha Max Speed

The top of a red strip along the speed scale indicates this speed. It represents the speed corresponding to the maximum angle of attack that the aircraft can attain in pitch normal law (Refer to 1.27.20).

It is displayed when in pitch normal law.

④ VMAX

The lower end of a red and black strip along the speed scale defines this speed.

It is the lowest of the following :

- VMO or the speed corresponding to MMO
- VLE
- VFE

(Refer to 3.04.10)

⑤ Stall Warning Speed (VSW)

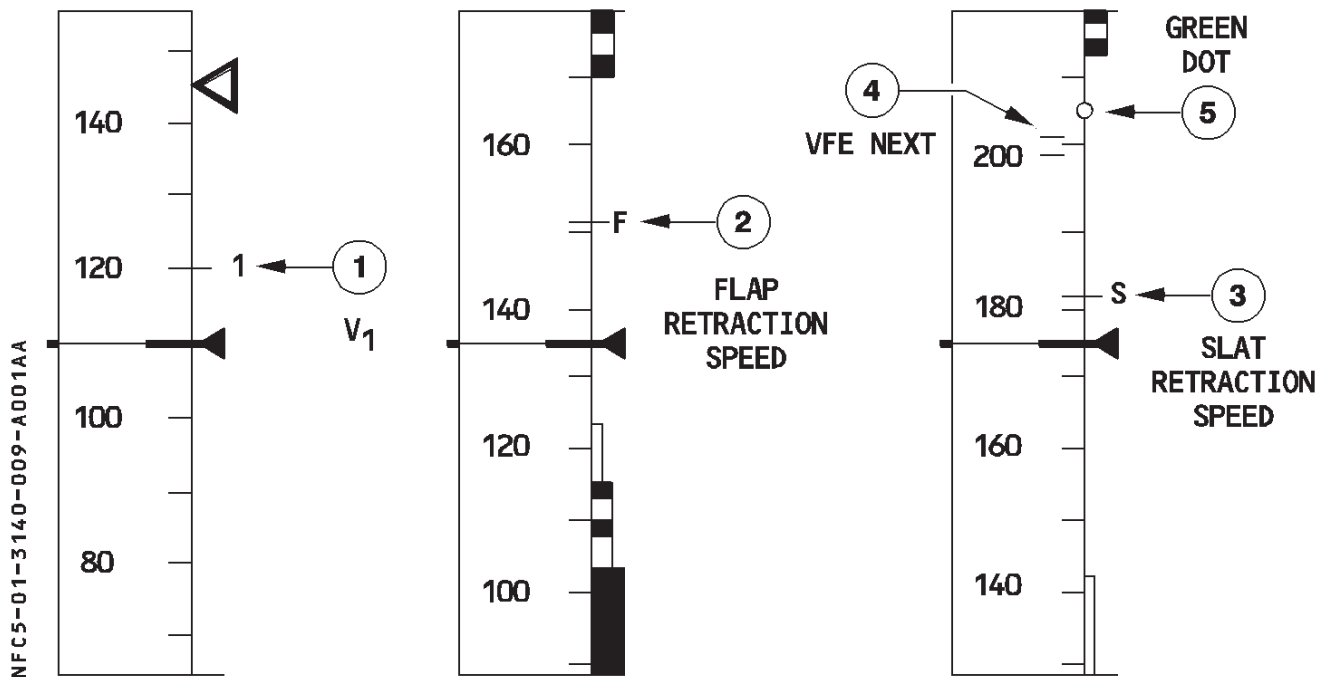
The top of a red and black strip along the speed scale defines this speed.

It is the speed corresponding to the stall warning. (Refer to 1.27.20).

VSW information is inhibited from touchdown until 5 seconds after liftoff.

It is displayed when operating in pitch alternate or pitch direct law.

R



① Decision Speed (V_1)

This is a blue symbol (numeral one) that the crew inserts manually through the MCDU. When it is off the scale, the upper part of the scale shows it in numbers. It disappears after liftoff. (Refer to 3.04.10)

② Minimum Flap Retraction Speed

This is a green symbol (letter F). It appears when the flap selector is in position 3 or 2. (Refer to 3.04.10)

③ Minimum Slat Retraction Speed

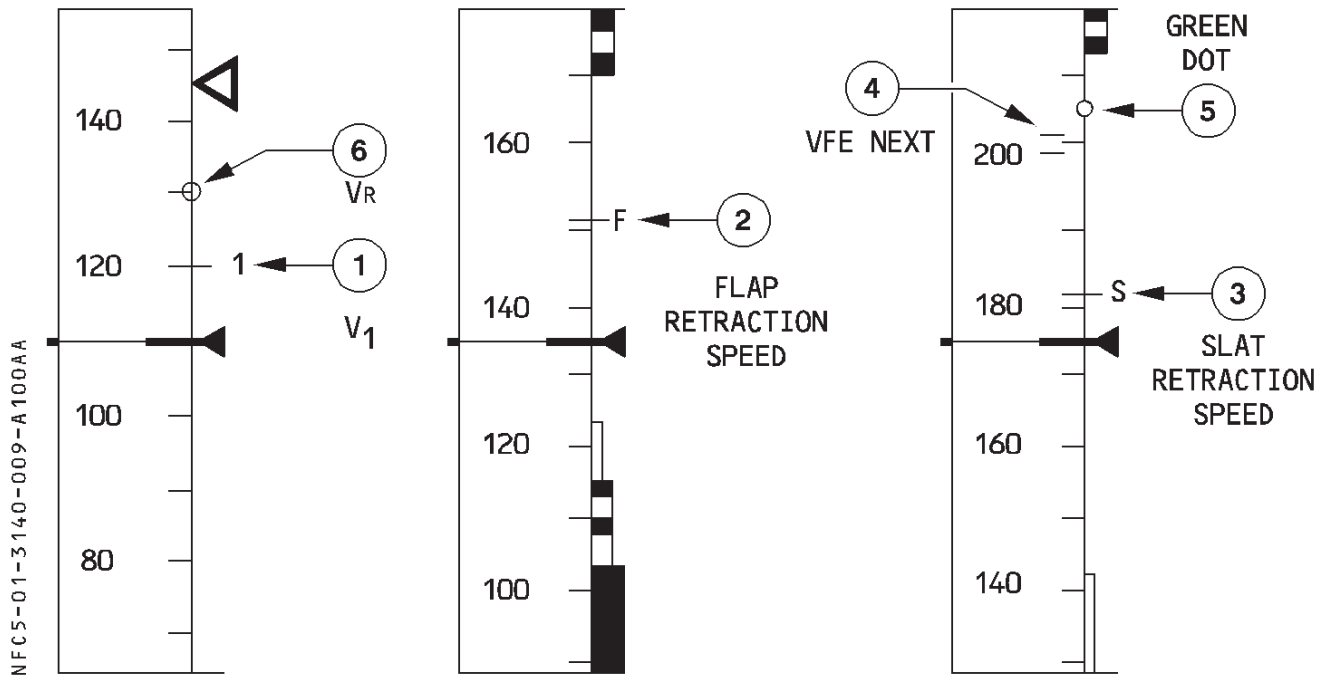
This is a green symbol (letter S). It appears when the flap selector is in position 1. (Refer to 3.04.10)

④ VFE NEXT

This symbol, an amber =, shows the VFE corresponding to the next flap lever position. It appears when the aircraft altitude is below 15,000 feet or 20 000 feet depending upon the FAC standard. (Refer to 3.04.10)

⑤ Green Dot (Engine-out operating speed in clean configuration)

This green dot appears when the aircraft is flying in the clean configuration. It shows the speed corresponding to the best lift-to-drag ratio.



① Decision Speed (V₁)

This is a blue symbol (numeral one) that the crew manually inserts via the MCDU. When it is off the scale, the upper part of the scale shows it in numbers. It disappears after liftoff. (Refer to 3.04.10).

② Minimum Flap Retraction Speed

This is a green symbol (letter F). It appears when the flap selector is in position 3 or 2. (Refer to 3.04.10).

③ Minimum Slat Retraction Speed

This is a green symbol (letter S). It appears when the flap selector is in position 1. (Refer to 3.04.10).

④ VFE NEXT

The VFE next symbol is an amber equal sign showing the VFE corresponding to the next flap lever position. It appears when the aircraft altitude is below 15,000 or 20,000 feet, depending upon the FAC standard. (Refer to 3.04.10).

⑤ Green Dot (Engine-out operating speed in clean configuration)

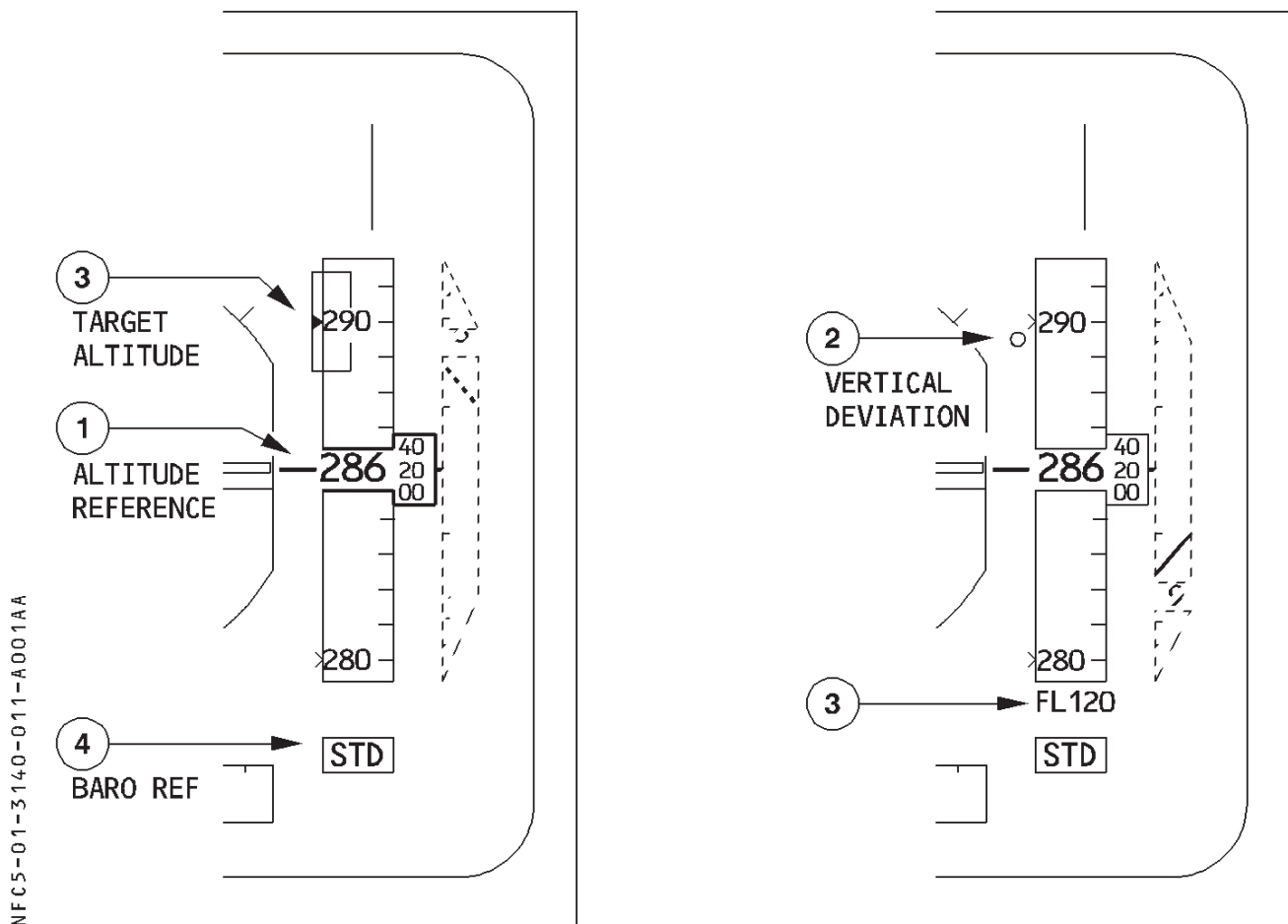
This green dot appears, when the aircraft is flying in the clean configuration. It shows the speed corresponding to the best lift-to-drag ratio.

⑥ Rotation speed : (VR)

R VR is entered on the PERF takeoff page of the MCDU, and is indicated by a cyan circle.
R This cyan circle is visible during takeoff.

Note : *V2 is represented by the target speed index during takeoff.
V2 is manually inserted by the crew via the MCDU.*

ALTITUDE



① Altitude Indication

R This appears both as a white moving scale, and as a green digital readout on a grey
 R background. Small white marks are positioned on the scale against the round values (e.g. 280, 290...). "NEG" appears in the window in white for negative values. The altitude window changes from yellow to amber, if the aircraft deviates from the FCU-selected altitude or flight level.

On any approach for which an MDA (MDH) is entered in the FMGS, the altitude numbers change from green to amber, when the aircraft goes below the MDA (MDH).

② Vertical Deviation (magenta)

R This symbol appears next to the altitude corresponding to the theoretical vertical profile
 R computed by the FMGC. It is displayed from the top of descent down to the MAP altitude.

The pilot can read the VDEV directly from the altitude scale. The range is ± 500 feet. When the VDEV value exceeds ± 500 feet, the symbol stays at the range limit and the PROG page displays the exact value.

R ③ Target Altitude or Selected Flight Level Symbol (blue)

This symbol shows the FCU selected altitude (if QNH baro reference is selected) or the selected flight level (if STD baro reference is selected.)

When the FMGC operates in the vertical managed mode, this symbol is magenta if it represents a flight plan altitude constraint that the FMGC will follow. If the target altitude or flight level is on the scale, the symbol is displayed and the numerical value appears inside the symbol.

If it is off the scale, the symbol is not displayed, and the numerical value appears above or underneath the scale.

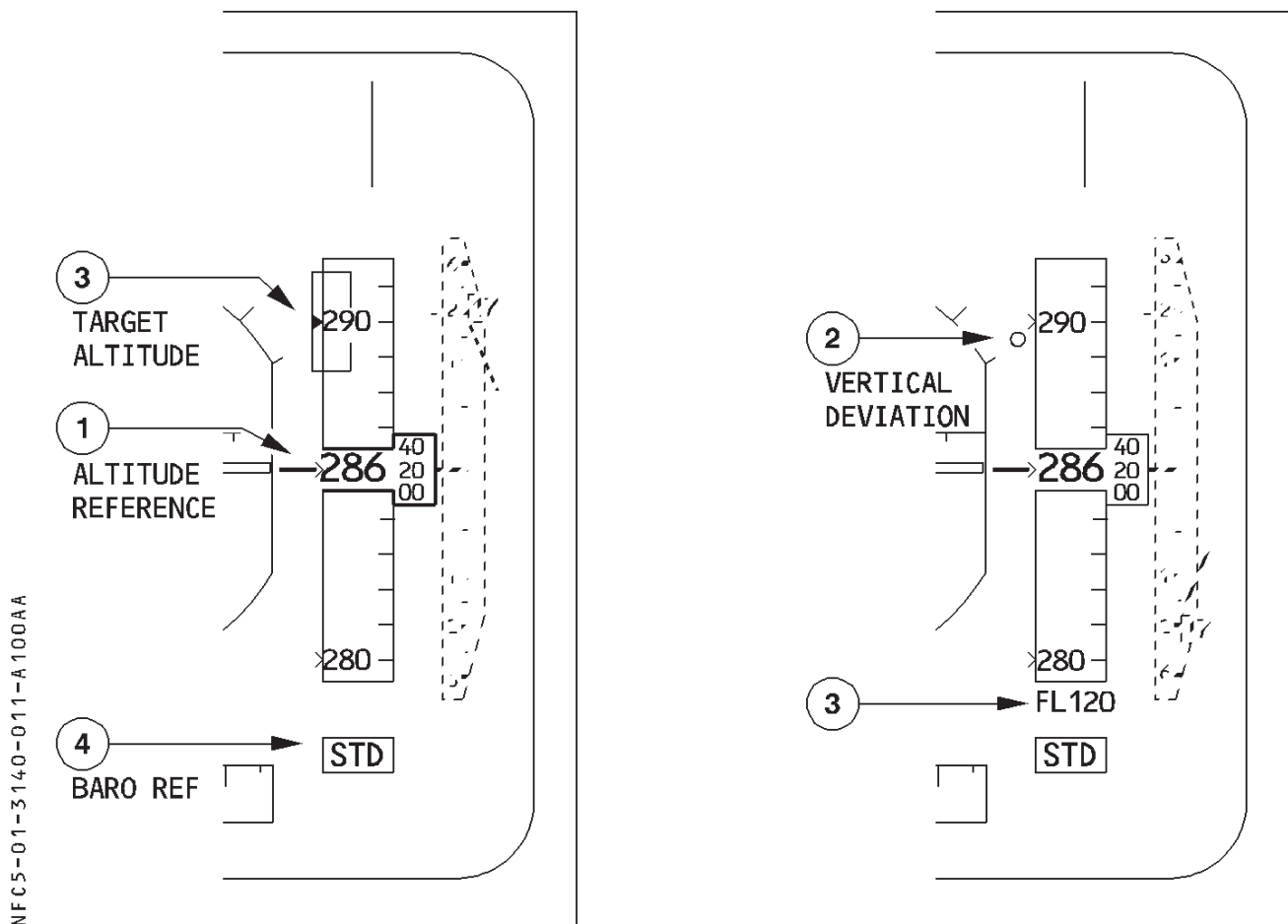
R ④ Barometric Reference

The display shows "STD" or it shows "QNH" and the numerical setting in hectoPascals or inches of mercury.

It pulses when the selection made by the pilot is not correct (STD not selected above transition altitude in climb or STD still selected in approach below transition level or 2500 feet radio height if transition level is not available).

R

ALTITUDE



① Altitude Indication

R This appears both as a white moving scale, and as a green digital readout on a grey background. Small white marks are positioned on the scale against the round values (e.g. 280, 290...). For negative altitudes, "NEG" appears in the window in white. The altitude window changes from yellow to amber, if the aircraft deviates from the FCU-selected altitude or flight level.

On any approach for which an MDA (MDH) is entered in the FMGS, the altitude numbers change from green to amber, when the aircraft goes below the MDA (MDH).

② Vertical Deviation (magenta)

R This symbol appears next to the altitude corresponding to the theoretical vertical profile computed by the FMGC. It is displayed from the top of descent down to the MAP altitude.

R The pilot can read the VDEV directly from the altitude scale. The range is ± 500 feet. When the VDEV value exceeds ± 500 feet, the symbol stays at the range limit and the PROG page displays the exact value.

R ③ Target Altitude or Selected Flight Level Symbol (blue)

This symbol shows the FCU selected altitude (if QNH baro reference is selected) or the selected flight level (if STD baro reference is selected.)

When the FMGC operates in the vertical managed mode, this symbol is magenta if it represents a flight plan altitude constraint that the FMGC will follow. If the target altitude or flight level is on the scale, the symbol is displayed and the numerical value appears inside the symbol.

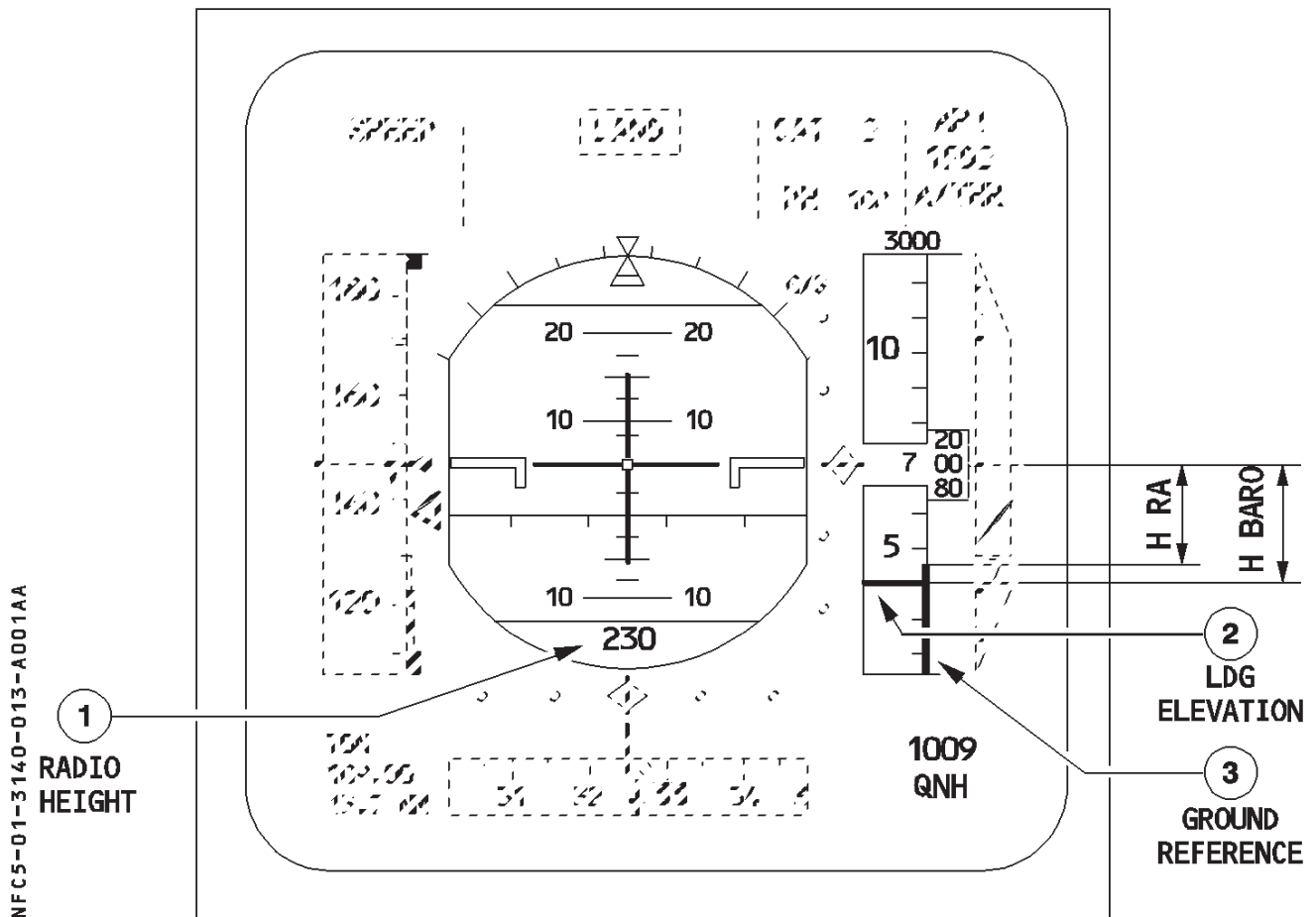
If it is off the scale, the symbol is not displayed, and the numerical value appears above or underneath the scale.

R ④ Barometric Reference

The display shows "STD" or it shows "QNH" and the numerical setting in hectoPascals or inches of mercury.

It pulses when the selection made by the pilot is not correct (STD not selected above transition altitude in climb or STD still selected in approach below transition level or 2500 feet radio height if transition level is not available).

R



① Radio Height

This quantity appears when it is less than 2500 feet.

- If a DH has been entered, the radio height appears :
 - in green when $DH + 100 \text{ feet} < RA < 2500 \text{ feet}$
 - in amber when $RA < DH + 100 \text{ feet}$

If "NO" is entered as the DH on the MCDU APPROACH page, 0 feet becomes a default value.

When the aircraft reaches the decision height selected on the MCDU, DH letters flash amber for three seconds, then stay in amber above the radio height indication.

- If no DH has been entered or if both FMGCs fail, the radio height appears :
 - in green when $400 \text{ feet} < RA < 2500 \text{ feet}$
 - in amber when $RA \leq 400 \text{ feet}$

The radio altitude indication changes every 10 feet down to 50 feet, then every 5 feet down to 10 feet, then every foot.

② Landing Elevation (blue)

The horizontal bar on the altitude scale shows the landing elevation at the flight-planned destination.

It is displayed :

- during flight phases 7 and 8 and
- if the QNH reference mode is selected.

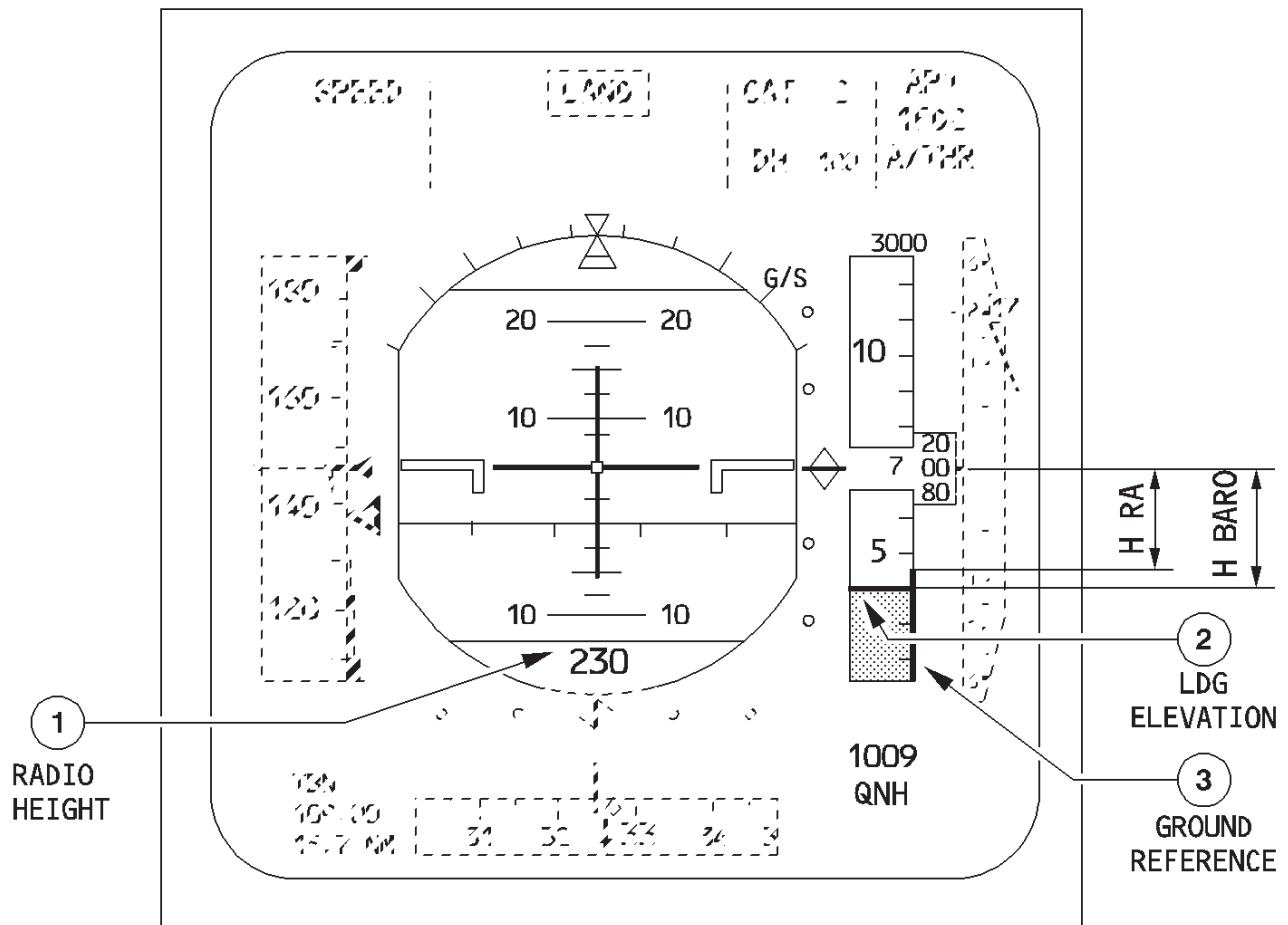
③ Ground reference

A red ribbon on the right of the altitude scale represents the field elevation. This ribbon, which is driven by the radio altimeter signal, is displayed below 570 feet.

It moves up, as does the lower line of the attitude sphere, with the altitude scale as the aircraft descends. When the aircraft has touched down, the top of this ribbon is at the middle of the altitude window.

R

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① Radio Height

This quantity appears, when it is less than 2500 feet.

- If a DH has been entered, the radio height appears :
 - In green, when $DH + 100 \text{ feet} < RA < 2500 \text{ feet}$
 - In amber, when $RA < DH + 100 \text{ feet}$.

If "NO" is entered as the DH on the MCDU APPROACH page, 0 feet becomes a default value.

When the aircraft reaches the decision height selected on the MCDU, DH letters flash amber for nine seconds, then remain amber above the radio height indication.

- If no DH has been entered, or if both FMGCs fail, the radio height appears :
 - In green, when $400 \text{ feet} < RA < 2500 \text{ feet}$
 - In amber, when $RA \leq 400 \text{ feet}$.

The radio altitude indication changes every 10 feet down to 50 feet, then every 5 feet down to 10 feet, then every foot.

R

② Landing Elevation (blue)

The top of the brown surface on the altitude scale represents the landing elevation at the flight-planned destination.

It is displayed during flight phases 7 and 8 and if the STD reference mode is not selected.

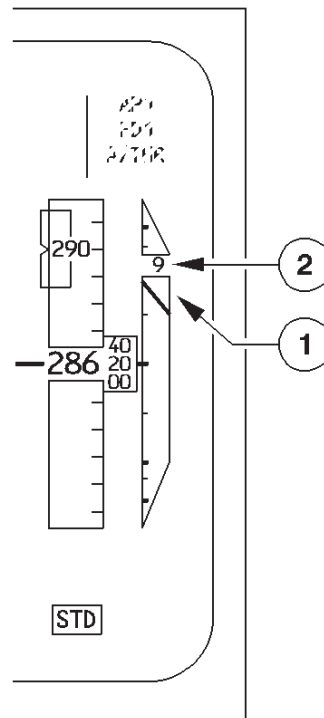
③ Ground reference

A red ribbon on the right of the altitude scale represents the field elevation. This ribbon, driven by the radio altimeter signal, is displayed below 570 feet.

It moves up, as does the lower line of the attitude sphere, with the altitude scale as the aircraft descends. When the aircraft has touched down, the top of this ribbon is at the middle of the altitude window.

VERTICAL SPEED

- R The displayed vertical speed information is normally based on both inertial and barometric data. If inertial data is not available, it is automatically replaced by barometric information.
- R In this case, the window around the numerical value becomes amber.
- R



① Analog pointer

This pointer, which is normally green, points to a white vertical speed scale displayed on a grey background and graduated at intervals of 500 feet/minute. If the V / S is greater than 6000 feet/minute, the pointer stays at the end of the scale.

② Digital indication

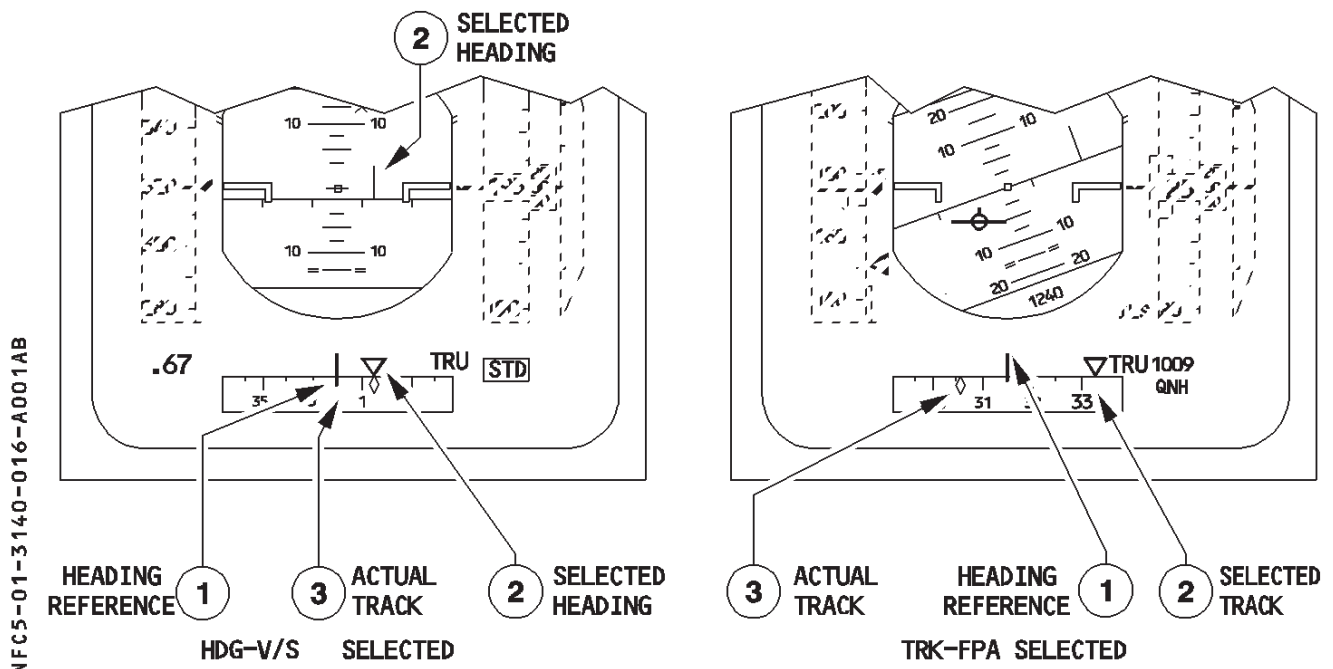
This number, normally green, is the vertical speed in hundreds of feet per minute. It disappears, if the vertical speed is less than 200 feet/minute.

The analog pointer and the digital indication become amber, if :

- V / S is greater than 6000 feet/minute, (climb or descent)
- V / S is greater than 2000 feet/minute, during descent when 1000 feet < RA < 2500 feet, or
- V / S is greater than 1200 feet/minute, during descent and RA < 1000 feet.

Note : For TCAS, refer to 1.34.80.

HEADING



① Heading Reference Line and Scale

A white scale on a grey background moves in front of a fixed yellow reference line to show the actual magnetic heading.

"TRU" appears when the display shows true heading, instead of magnetic heading (latitude above 73° North or below 60° South).

② Selected Heading or Track Index (blue)

R The pointer indicates the heading or track displayed on the FCU HDG-TRK window. The index is replaced by digits on the right or left side of the scale when the selected value is off the scale.

R If the FD pushbutton switch is OFF a second heading/track symbol appears on the horizon line and markers are displayed every 10°.

③ Actual Track Symbol

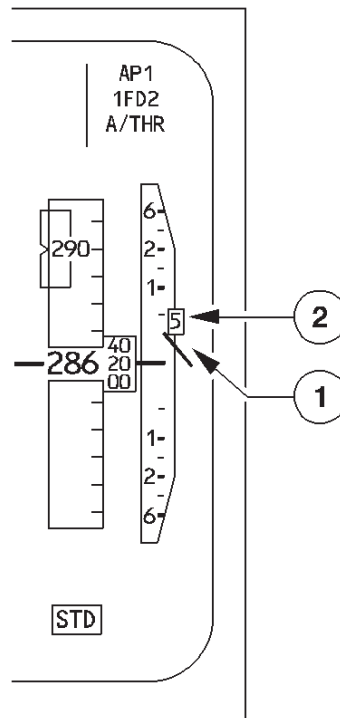
This symbol is a small green diamond.

VERTICAL SPEED

The displayed vertical speed information is normally based on both inertial and barometric data. If inertial data is not available, it is automatically replaced by barometric information. In this case, the window around the numerical value becomes amber.

R

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① Analog pointer

This pointer, which is normally in green, points to a white vertical speed scale, displayed on a grey background and graduated at intervals of 500 feet/minute.

If the V/S is greater than 6000 feet/minute, the pointer stays at the end of the scale.

② Digital indication

This number, normally in green, is the vertical speed in hundreds of feet per minute. It disappears, if the vertical speed is less than 200 feet/minute.

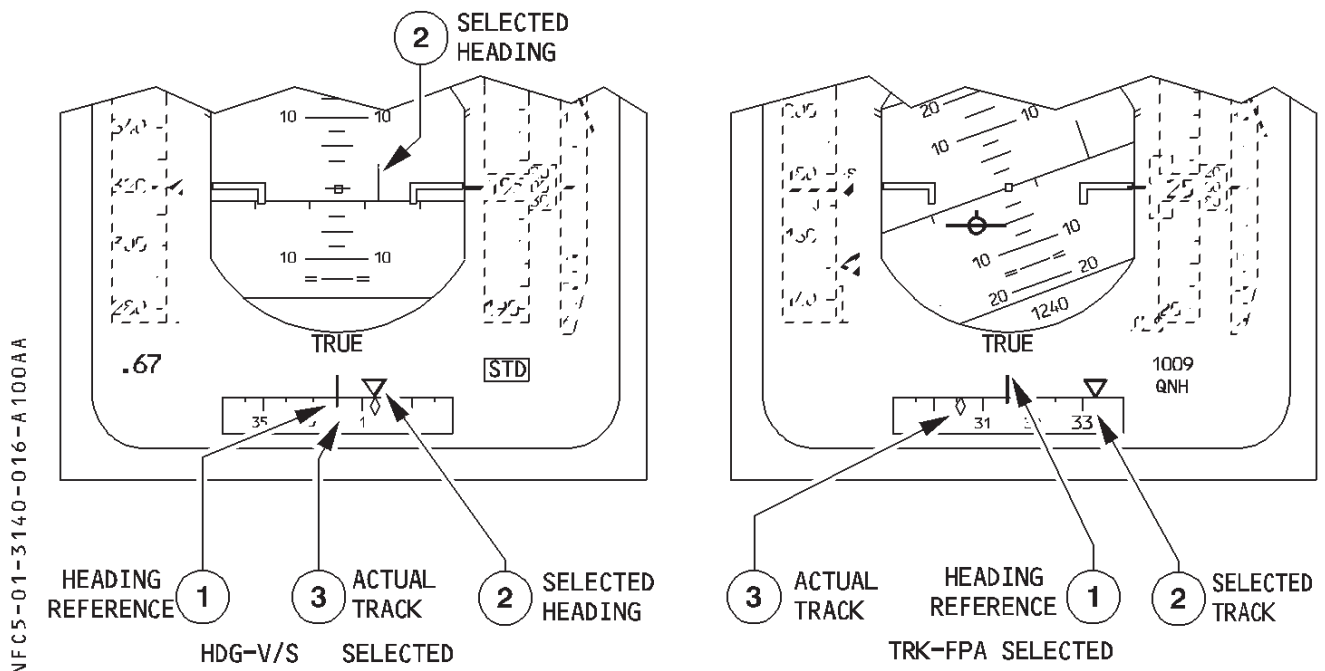
The analog pointer and the digital indication become amber, if :

- V/S is greater than 6000 feet/minute, (climb or descent)
- V/S is greater than 2000 feet/minute, during descent when $1000 \text{ feet} < RA < 2500 \text{ feet}$, or
- V/S is greater than 1200 feet/minute, during descent and $RA < 1000 \text{ feet}$.

Note : For TCAS, refer to 1.34.80.

HEADING

R



① Heading Reference Line and Scale

A white scale on a grey background moves in front of a fixed yellow reference line to show the actual magnetic heading.

"TRUE" appears, when the display shows true heading, instead of magnetic heading (latitude above 73° North or below 60° South).

② Selected Heading or Track Index (blue)

The pointer indicates the heading or track displayed on the FCU HDG-TRK window.

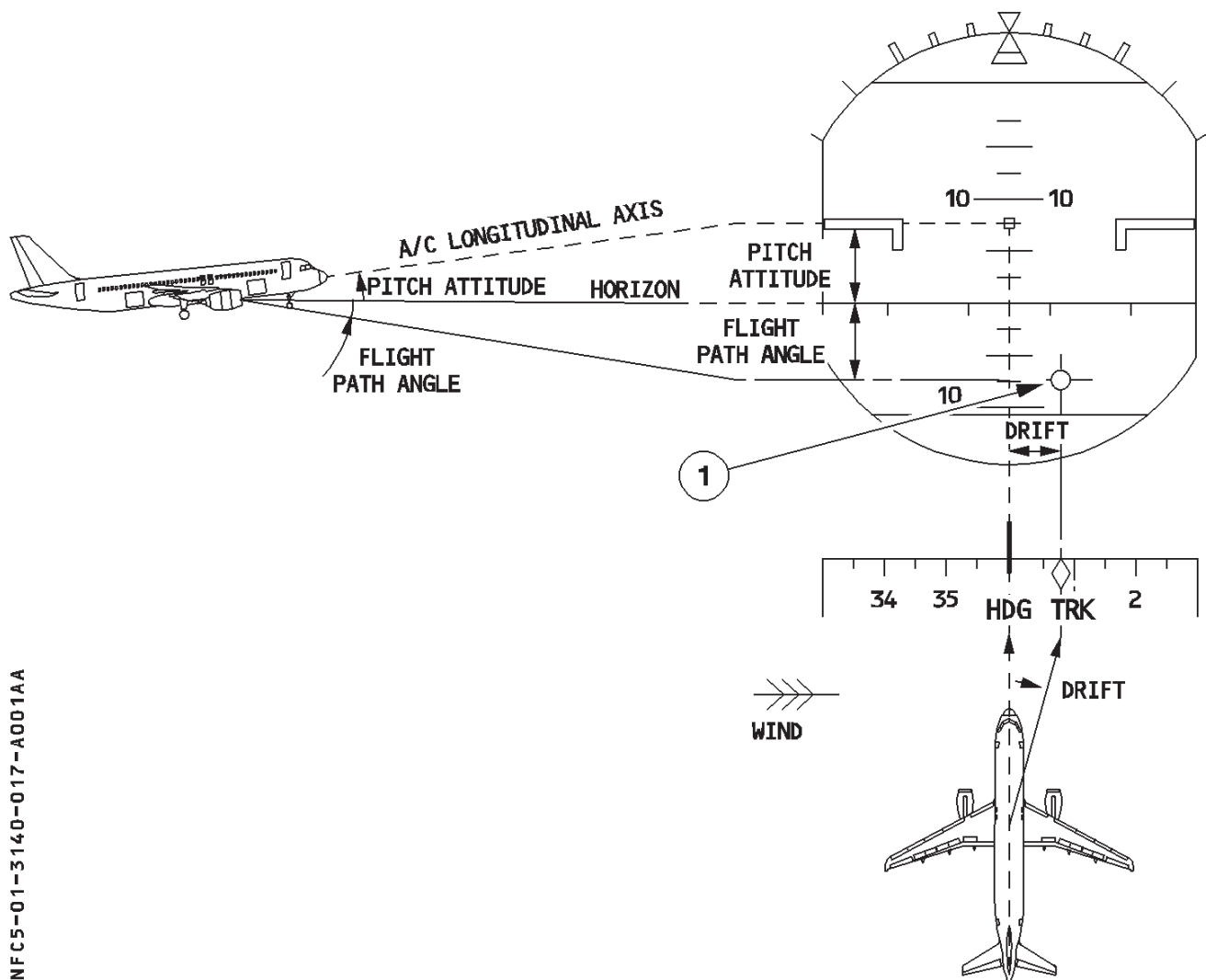
The index is replaced by digits on the right, or on the left, side of the scale when the selected value is off the scale.

If the FD pushbutton is OFF, a second heading/track symbol appears on the horizon line and markers are displayed every 10°.

③ Actual Track Symbol

This symbol is a small green diamond.

FLIGHT PATH VECTOR



① Flight Path Vector (FPV)

This symbol appears when the pilot selects TRK/FPA on the FCU.

The flight path vector represents the lateral and vertical trajectory of the aircraft with respect to the ground.

- On the lateral scale it indicates the aircraft's track.
- On the vertical scale it indicates the aircraft's flight path angle.

Example : The aircraft flies a track of 009 (heading 360°, wind from west) and descends with a flight path angle of -7.5° .

GUIDANCE

Two completely different flight director modes are available, each with its own characteristic symbols. The symbol displayed corresponds to the basic operating reference the pilot has selected – either HDG V/S or TRK FPA.

In normal operation PFD1 displays FD1 orders.

If FD1 fails, PFD1 automatically displays FD2 orders, and on the PFD1 the FD2 indication in the right column of the FMA flashes for a few seconds.

The same applies for FD2 orders normally displayed on PFD2.

IF THE CREW HAS SELECTED HDG V/S TO BE THE BASIC REFERENCE :

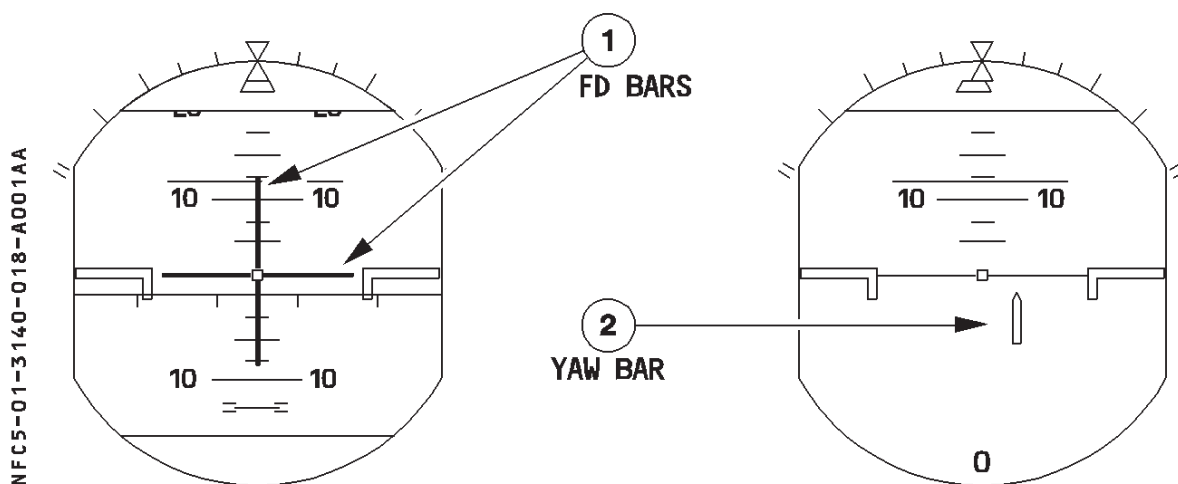
The PFD displays pitch and roll bars in green. They automatically move out of view at touchdown in ROLL OUT mode.

They flash for 10 seconds and then remain steady in the following conditions :

- reversion to the HDG V/S basic mode (manual or automatic), or
- change of selected flight level when ALT CAPTURE mode is engaged, or
- loss of LOC or G/S in LAND mode or loss of LAND mode, or
- at the first AP or FD engagement.

The PFD displays a yaw bar in green below 30 feet radio altitude if a localizer signal is available:

- during takeoff (in RWY mode)
- R – upon landing (in FLARE and ROLL OUT mode).



① FD Crossed Bars (green)

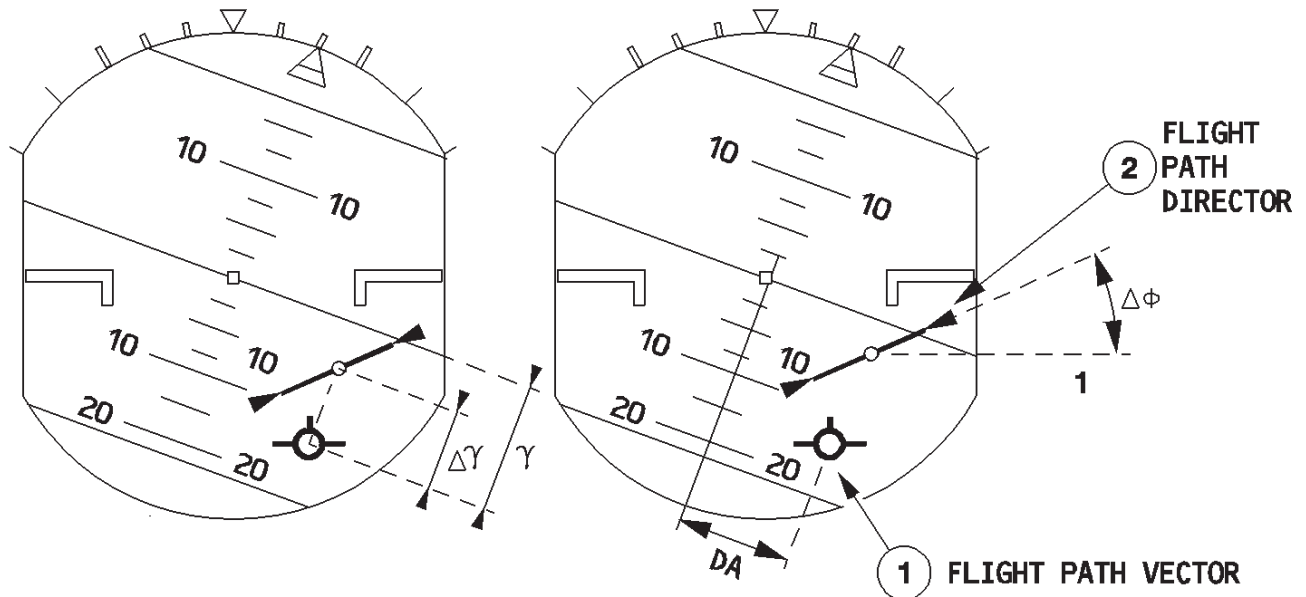
② Yaw Bar (green)

THE CREW HAS SELECTED TRK FPA AS THE BASIC REFERENCE :

An inertial flight path vector defines the aircraft's horizontal and vertical track, taking wind effect into account.

An associated flight path director symbol guides the flight crew onto the vertical and horizontal flight path targets.

R



γ REPRESENTS THE FLIGHT PATH ANGLE

DA REPRESENTS THE DRIFT ANGLE

$\Delta\gamma$ REPRESENTS THE DIFFERENCE BETWEEN THE ORDERED FLIGHT PATH ANGLE AND THE ACTUAL ONE

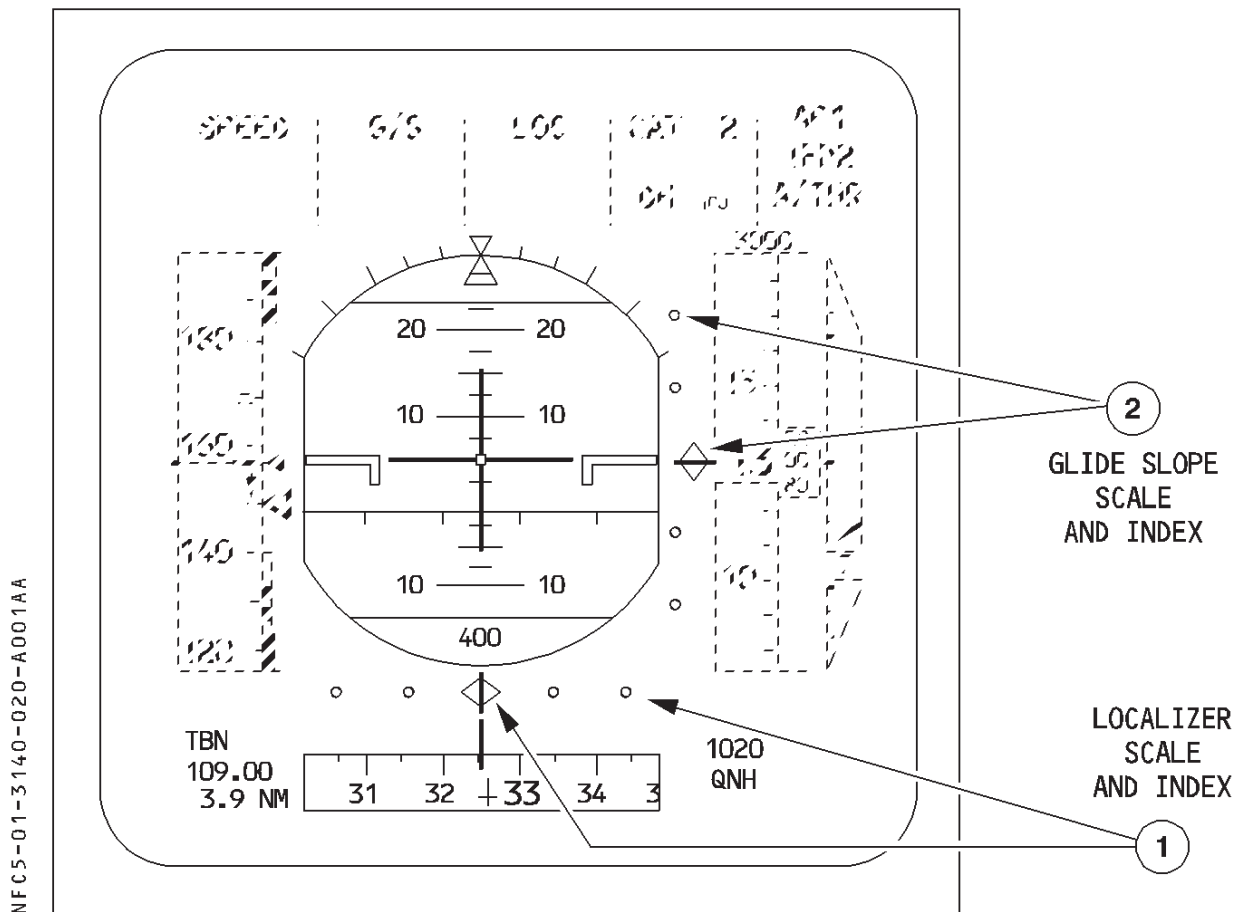
$\Delta\phi$ REPRESENTS THE DIFFERENCE BETWEEN THE ORDERED ROLL ANGLE AND THE ACTUAL ONE

① Flight Path Vector (green)

② Flight Path Director (green)

TRAJECTORY DEVIATION

ILS APPROACH



① Localizer Deviation Scale and Index

② Glide Slope Deviation Scale and Index

Deviation scales appear as soon as the flight crew pushes an ILS pushbutton switch on the EFIS control panel. Deviation indexes appear when the glide slope and localizer signals are valid if deviation scales are displayed.

When a deviation index is out of the displayed range, only half a symbol appears at the end of the scale.

The LOC scale flashes and continues to flash if the deviation exceeds 1/4 dot for two seconds (above 15 feet RA). The glide scale flashes and continues to flash if the deviation exceeds one dot for two seconds (above 100 feet RA).

"LOC" and the glide scale half index symbols flash and continue to flash when the deviation exceeds two dots for two seconds.

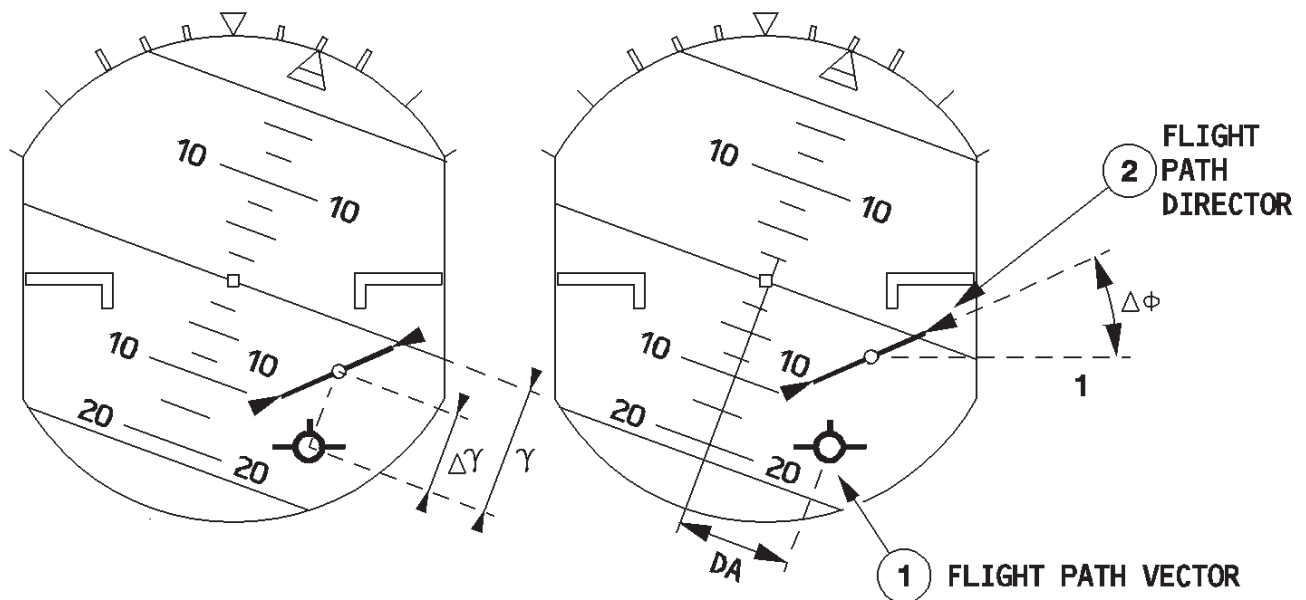
R One dot represents a deviation of $\pm 0.8^\circ$ on the localizer scale and $\pm 0.4^\circ$ on the glide
R slope scale.

THE CREW HAS SELECTED TRK FPA AS THE BASIC REFERENCE :

An inertial flight path vector defines the aircraft's horizontal and vertical track, taking wind effect into account.

An associated flight path director symbol guides the flight crew onto the vertical and horizontal flight path targets.

R



γ REPRESENTS THE FLIGHT PATH ANGLE

DA REPRESENTS THE DRIFT ANGLE

$\Delta\gamma$ REPRESENTS THE DIFFERENCE BETWEEN THE ORDERED FLIGHT PATH ANGLE AND THE ACTUAL ONE

$\Delta\phi$ REPRESENTS THE DIFFERENCE BETWEEN THE ORDERED ROLL ANGLE AND THE ACTUAL ONE

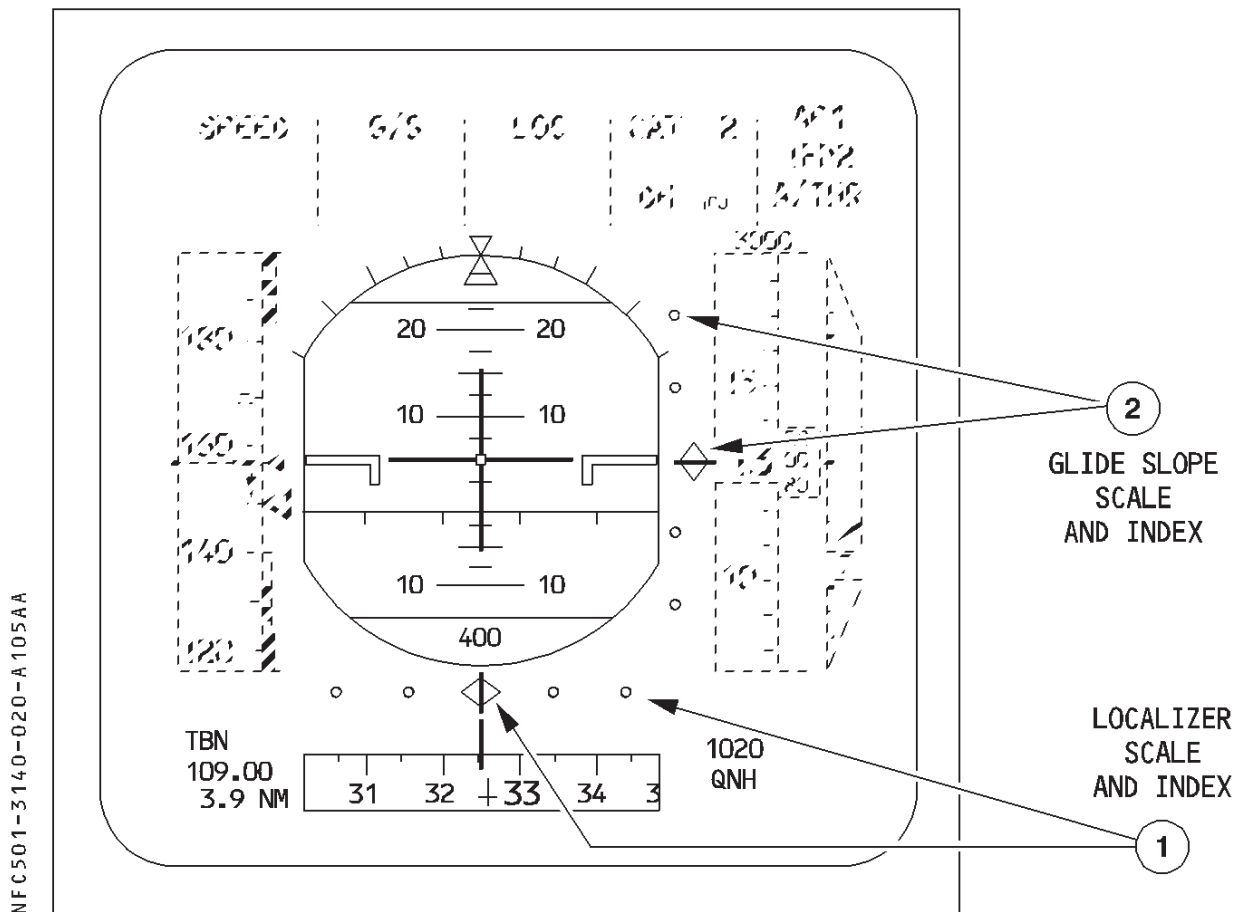
① Flight Path Vector (green)

② Flight Path Director (green)

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TRAJECTORY DEVIATION

ILS APPROACH



- ① Localizer Deviation Scale and Index
- ② Glide Slope Deviation Scale and Index

Deviation scales appear as soon as the flight crew pushes an LS pushbutton switch on the EFIS control panel. Deviation indexes appear when the glide slope and localizer signals are valid if deviation scales are displayed.

When a deviation index is out of the displayed range, only half a symbol appears at the end of the scale.

The LOC scale flashes and continues to flash if the deviation exceeds 1/4 dot for two seconds (above 15 feet RA). The glide scale flashes and continues to flash if the deviation exceeds one dot for two seconds (above 100 feet RA).

"LOC" and the glide scale half index symbols flash and continue to flash when the deviation exceeds two dots for two seconds.

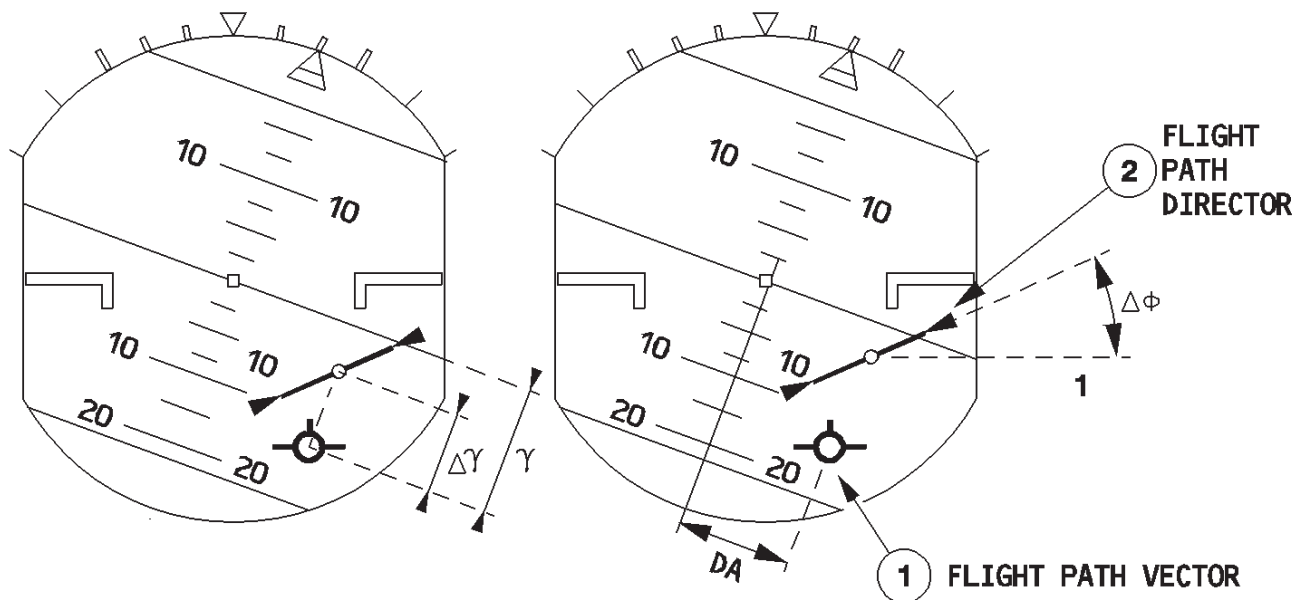
One dot represents a deviation of $\pm 0.8^\circ$ on the localizer scale and $\pm 0.4^\circ$ on the glide slope scale.

THE CREW HAS SELECTED TRK FPA AS THE BASIC REFERENCE :

An inertial flight path vector defines the aircraft's horizontal and vertical track, taking wind effect into account.

An associated flight path director symbol guides the flight crew onto the vertical and horizontal flight path targets.

R



γ REPRESENTS THE FLIGHT PATH ANGLE

DA REPRESENTS THE DRIFT ANGLE

$\Delta\gamma$ REPRESENTS THE DIFFERENCE BETWEEN THE ORDERED FLIGHT PATH ANGLE AND THE ACTUAL ONE

$\Delta\phi$ REPRESENTS THE DIFFERENCE BETWEEN THE ORDERED ROLL ANGLE AND THE ACTUAL ONE

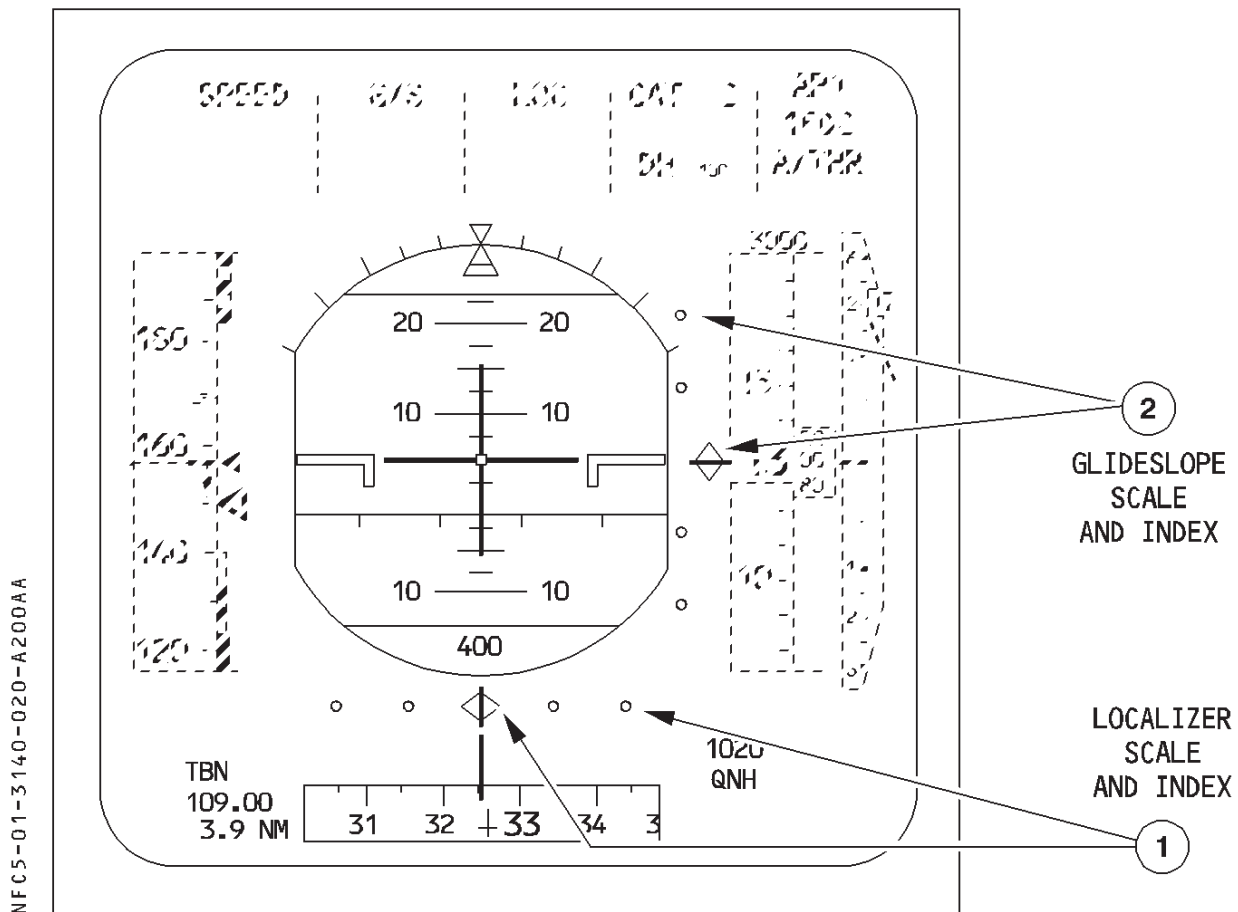
① Flight Path Vector (green)

② Flight Path Director (green)

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TRAJECTORY DEVIATION

ILS APPROACH



- ① Localizer Deviation Scale and Index
- ② Glideslope Deviation Scale and Index

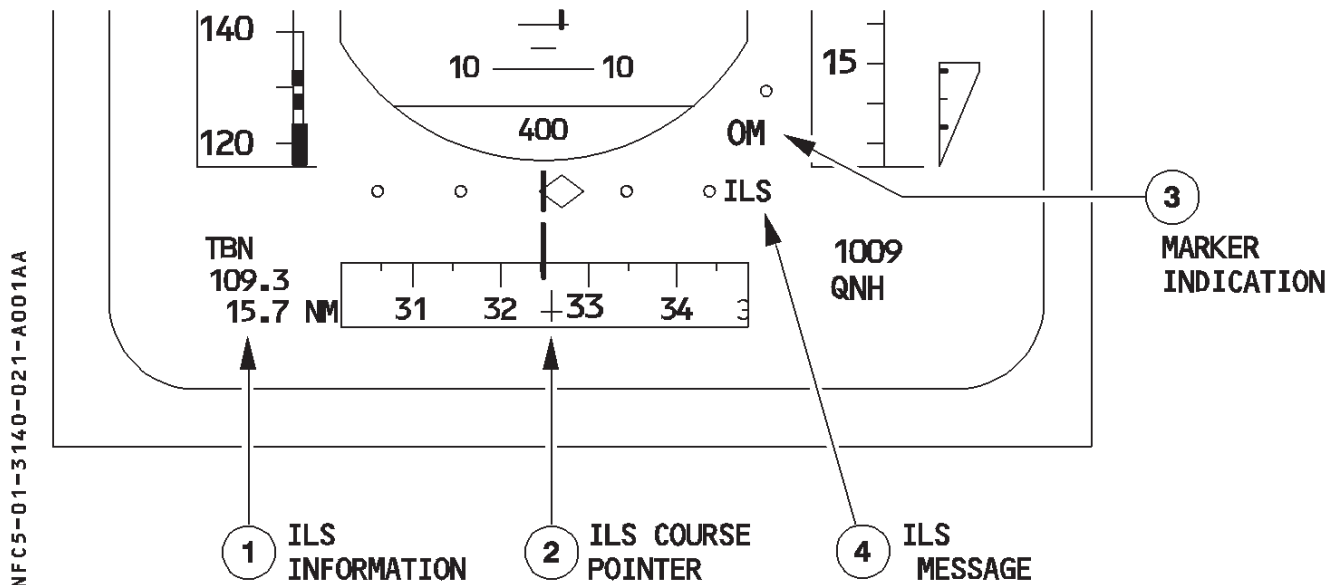
Deviation scales appear as soon as the flight crew pushes an LS pushbutton on the EFIS control panel. Deviation indexes appear, when the glideslope and localizer signals are valid, if deviation scales are displayed.

When a deviation index is out of the displayed range, only half a symbol appears at the end of the scale.

The LOC scale flashes and continues to flash, if the deviation exceeds 1/4 dot for two seconds (above 15 feet RA). The glide scale flashes and continues to flash, if the deviation exceeds one dot for two seconds (above 100 feet RA).

"LOC" and the glide scale half index symbols flash and continue to flash, when the deviation exceeds two dots for two seconds.

One dot represents a deviation of $\pm 0.8^\circ$ on the localizer scale and $\pm 0.4^\circ$ on the glideslope scale.



① ILS information (magenta)

The following information appears on the PFD, when the crew has selected an ILS frequency and course, and pushed the ILS pushbutton :

- ILS identification, as decoded by the ILS receiver ;
- ILS frequency ;
- DME distance, if the ILS has a DME

② ILS course Pointer (magenta)

This pointer appears on the PFD, when the crew has selected an ILS frequency and course, and pushed the ILS pushbutton.

It is a dagger-shaped symbol on the heading scale.

The ILS course (numerical) appears in magenta, on the right or the left side of the scale, when it is outside the displayed range of headings.

R

③ Marker Indications

OM appears in blue, when the aircraft flies over the outer marker.

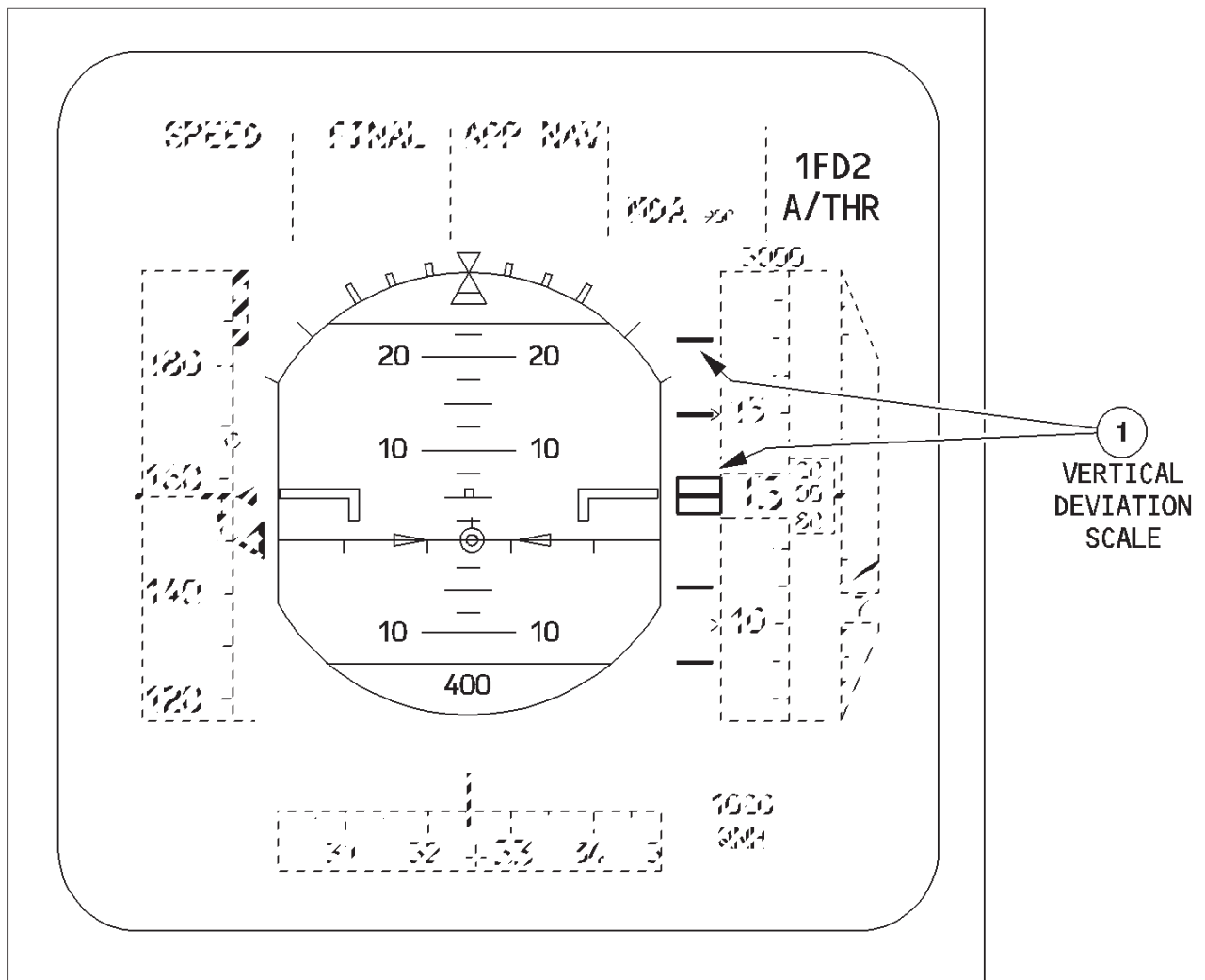
MM appears in amber, when it flies over the middle marker.

AWY appears in white, when it flies over an airways marker beacon or the ILS inner marker.

④ ILS Message

This flashes amber, when the APPR mode is armed and the ILS display is not selected.

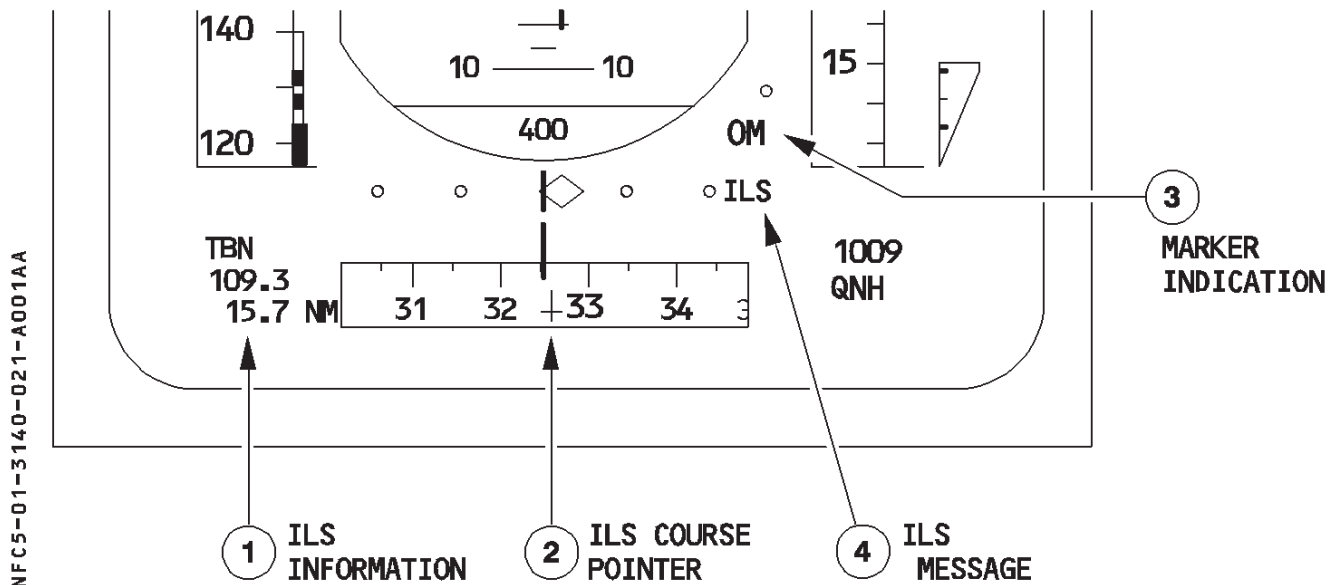
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R

Each index scale graduation represents 100 feet. The range is ± 200 feet.

Note : • If the ILS pushbutton is pressed, glide deviation has priority over vertical deviation information. As long as V/DEV display conditions are met, and the ILS pushbutton is selected, an amber V/DEV message flashes above the glide scale.



① ILS information (magenta)

The following information appears on the PFD, when the crew has selected an ILS frequency and course, and pushed the ILS pushbutton :

- ILS identification, as decoded by the ILS receiver ;
- ILS frequency ;
- DME distance, if the ILS has a DME

② ILS course Pointer (magenta)

This pointer appears on the PFD, when the crew has selected an ILS frequency and course, and pushed the ILS pushbutton.

It is a dagger-shaped symbol on the heading scale.

The ILS course (numerical) appears in magenta, on the right or the left side of the scale, when it is outside the displayed range of headings.

R

③ Marker Indications

OM appears in blue, when the aircraft flies over the outer marker.

MM appears in amber, when it flies over the middle marker.

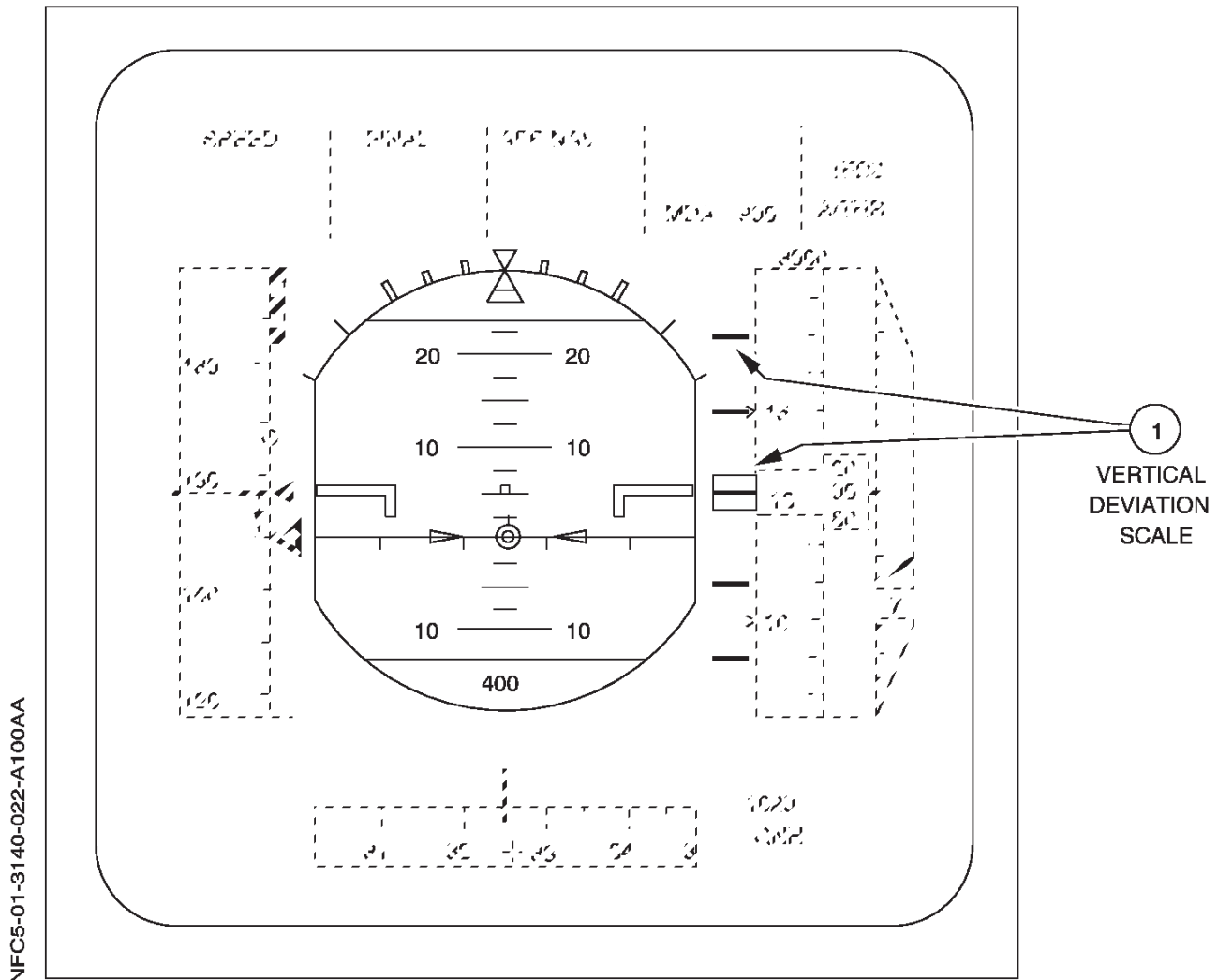
AWY appears in white, when it flies over an airways marker beacon or the ILS inner marker.

④ ILS Message

This flashes amber, when the APPR mode is armed and the ILS display is not selected.

NON PRECISION APPROACH

R



① Vertical Deviation Scale and Index

These symbols appear when in the approach phase and, when either FINAL is armed/engaged or a non-LS approach has been entered. They are displayed in the approach or go-around phase, until the MDA has been reached, or the MAP or the runway has been sequenced. They give the vertical deviation from the trajectory defined by the FMGC.

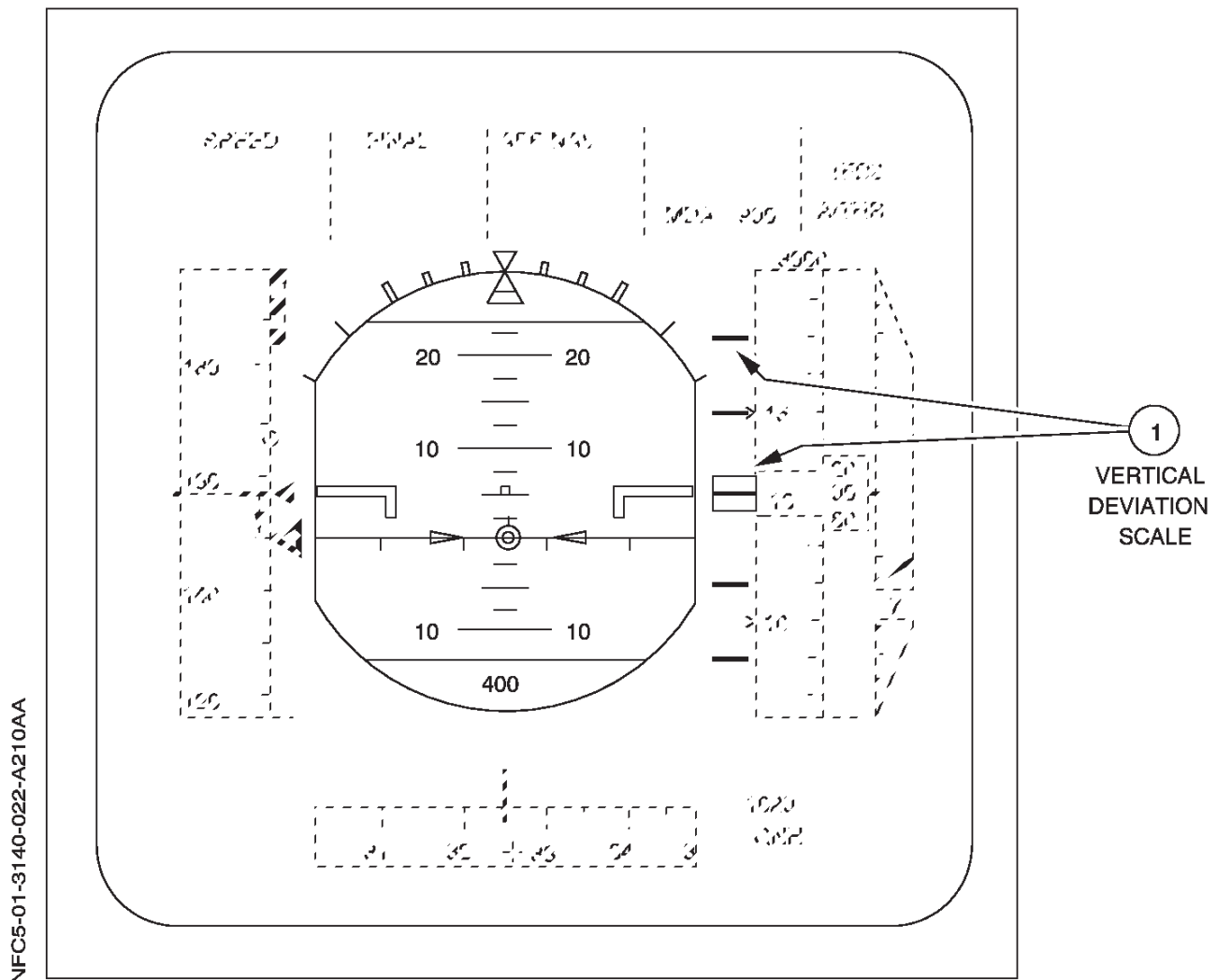
Each index scale graduation represents 100 feet. The range is ± 200 feet.

Note : If the ILS pushbutton is pressed, glide deviation has priority over vertical deviation information. As long as V/DEV display conditions are met, and the ILS pushbutton is selected, an amber V/DEV message flashes above the glide scale.



NON PRECISION APPROACH

R



① Vertical Deviation Scale and Index

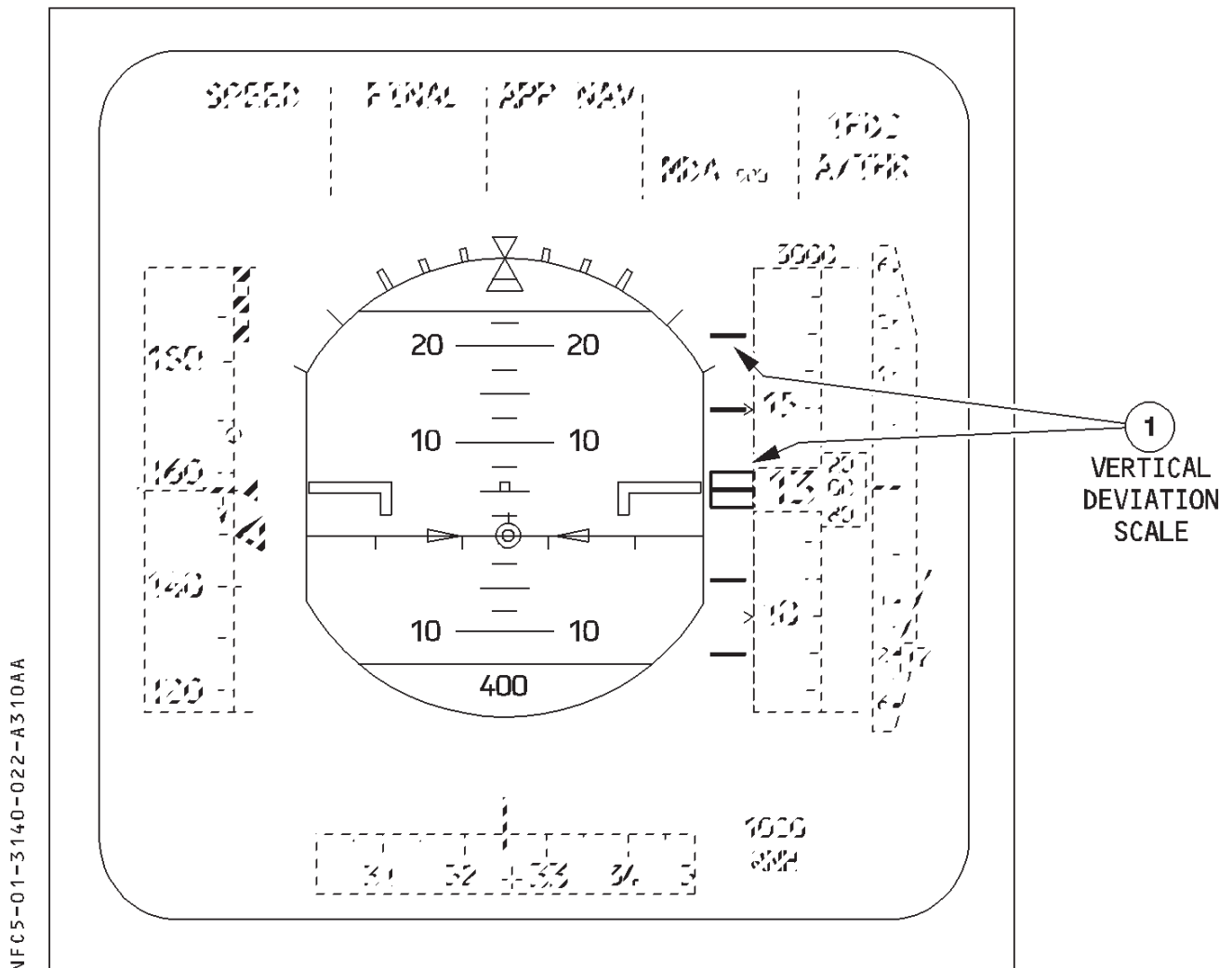
These symbols appear when in the approach phase and, when either FINAL is armed/engaged or a non-LS approach has been entered. They are displayed in the approach or go-around phase, until the MDA has been reached, or the MAP or the runway has been sequenced. They give the vertical deviation from the trajectory defined by the FMGC.

Each index scale graduation represents 100 feet. The range is ± 200 feet.

Note : · If the LS pushbutton is pressed, glide deviation has priority over vertical deviation information. As long as V/DEV display conditions are met, and the LS pushbutton is selected, an amber V/DEV message flashes above the glide scale.



NON PRECISION APPROACH



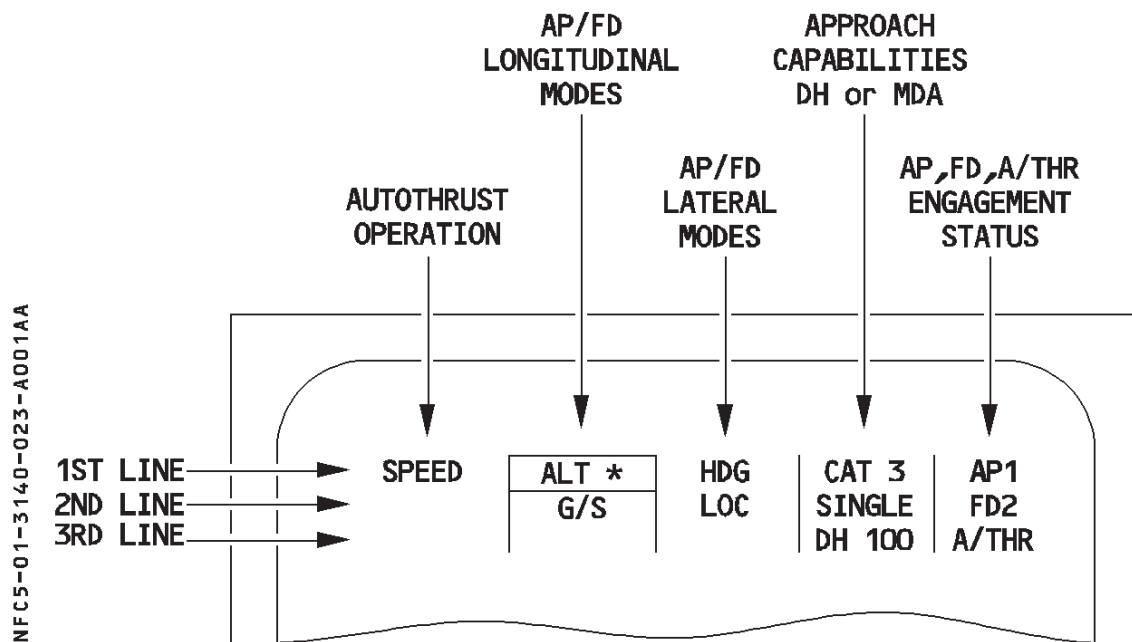
① Vertical Deviation Scale and Index

These symbols appear when in the approach phase and, when either FINAL is armed/engaged or a non-LS approach has been entered. They are displayed in the approach or go-around phase, until the MDA has been reached, or the MAP or the runway has been sequenced. They give the vertical deviation from the trajectory defined by the FMGC.

Each index scale graduation represents 100 feet. The range is ± 200 feet.

Note : If the LS pushbutton is pressed, glide deviation has priority over vertical deviation information. As long as V/DEV display conditions are met, and the LS pushbutton is selected, an amber V/DEV message flashes above the glide scale.

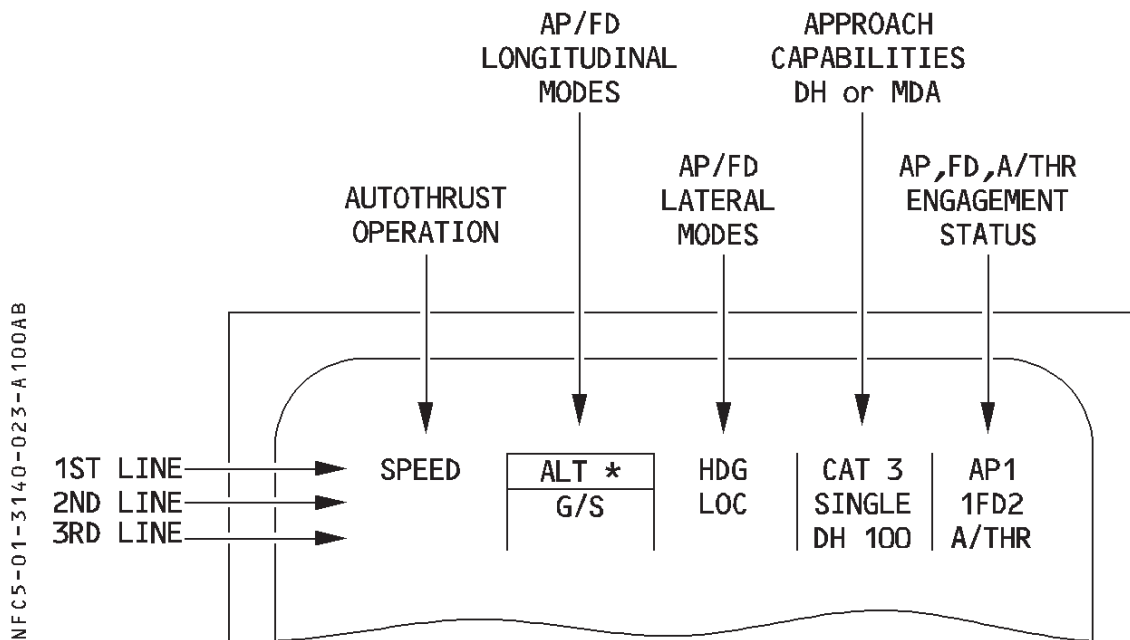
FLIGHT MODE ANNUNCIATOR



R For a detailed discussion of legends and messages that may appear during FMGS operations, see FLIGHT GUIDANCE chapter (Refer to 1.22.30).

INTENTIONALLY LEFT BLANK

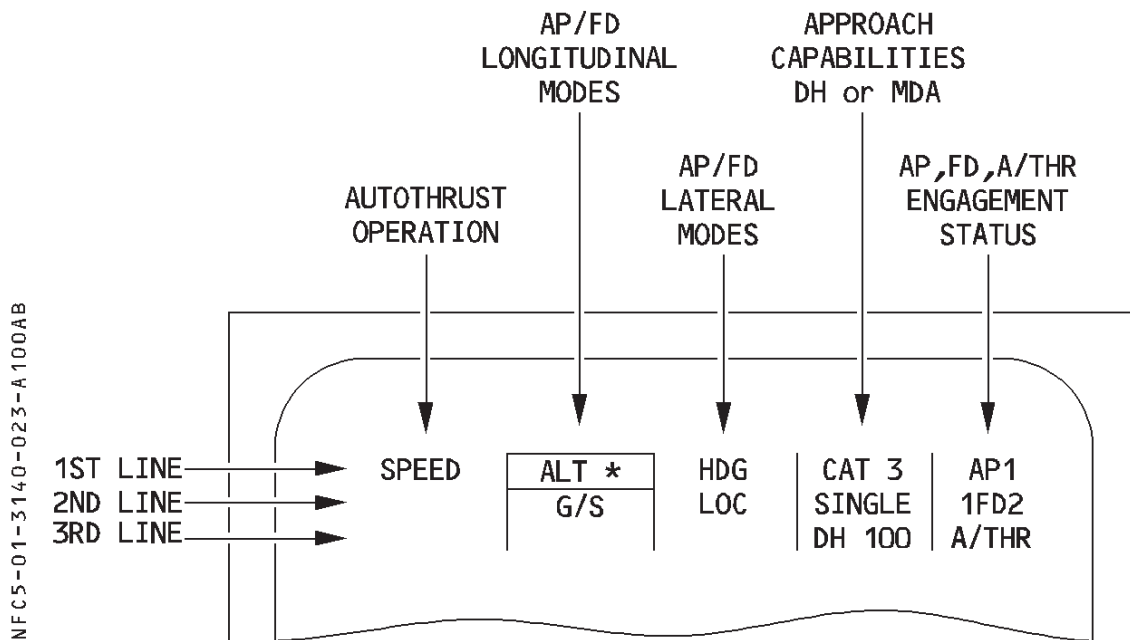
FLIGHT MODE ANNUNCIATOR



For a detailed discussion of legends and messages that may appear during FMGS operations, see FLIGHT GUIDANCE chapter (Refer to 1.22.30).

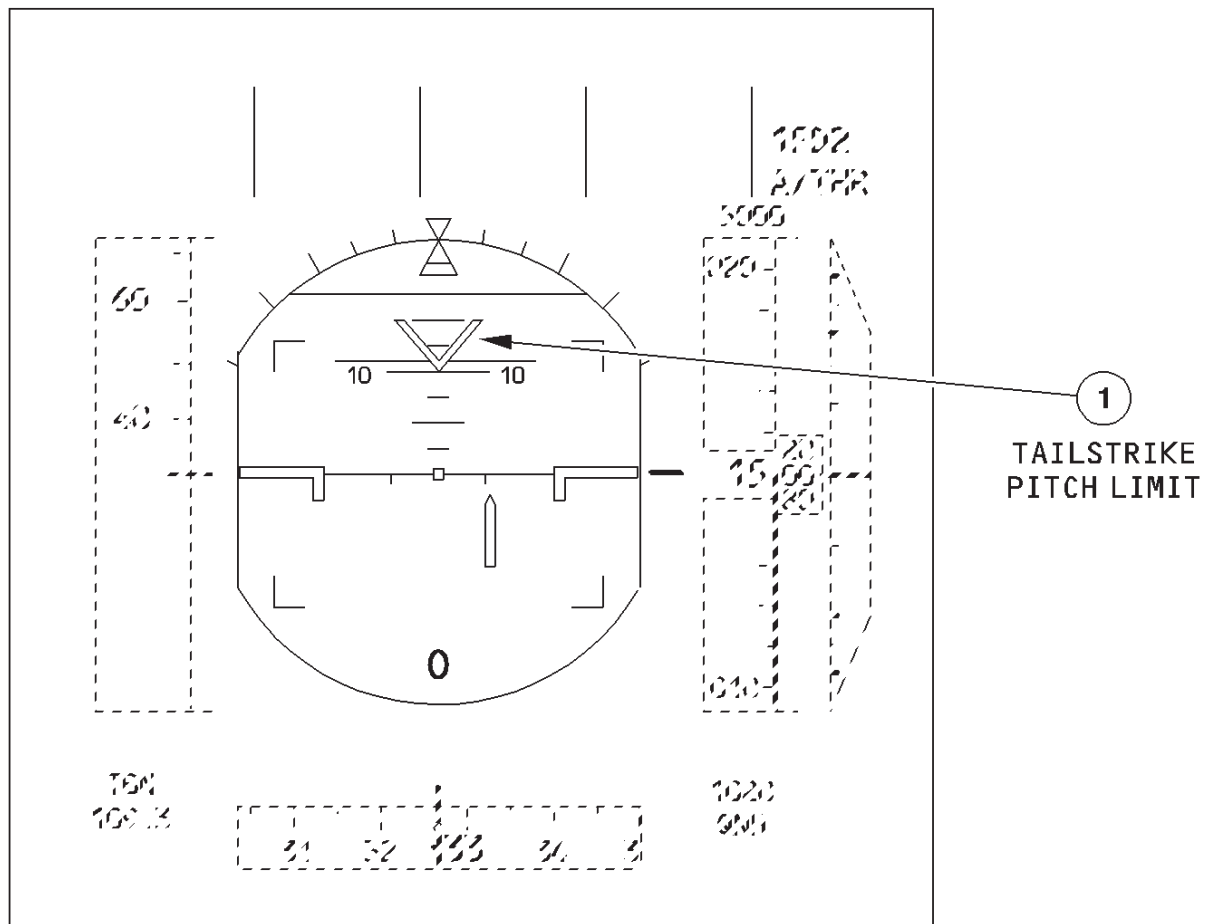
INTENTIONALLY LEFT BLANK

FLIGHT MODE ANNUNCIATOR



For a detailed discussion of legends and messages that may appear during FMGS operations, see FLIGHT GUIDANCE chapter (Refer to 1.22.30).

TAILSTRIKE PITCH LIMIT INDICATOR



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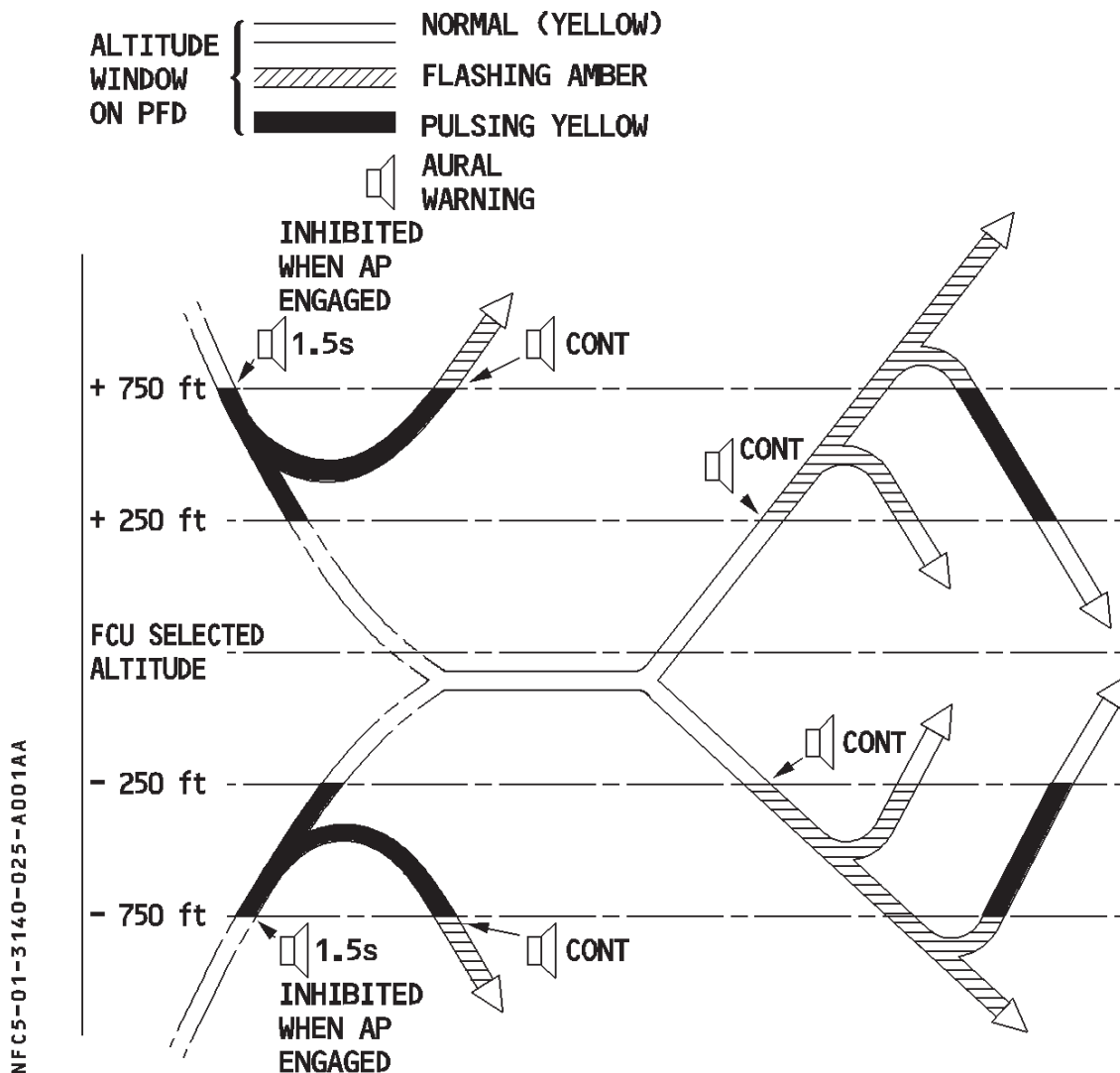
① Tailstrike Pitch Limit

The pitch limit indicates the maximum pitch attitude to avoid the tailstrike risk at landing. The indication is a fixed value corresponding to the main landing gear compressed. The indication appears at 400 feet radio height. The indication disappears, when there is no longer a risk of tailstrike.

ALTITUDE ALERT

The FWC generates an altitude warning (C chord sound and altitude window of PFD pulsing yellow or flashing amber) when the aircraft approaches a preselected altitude or flight level or when it deviates from its selected altitude or flight level.

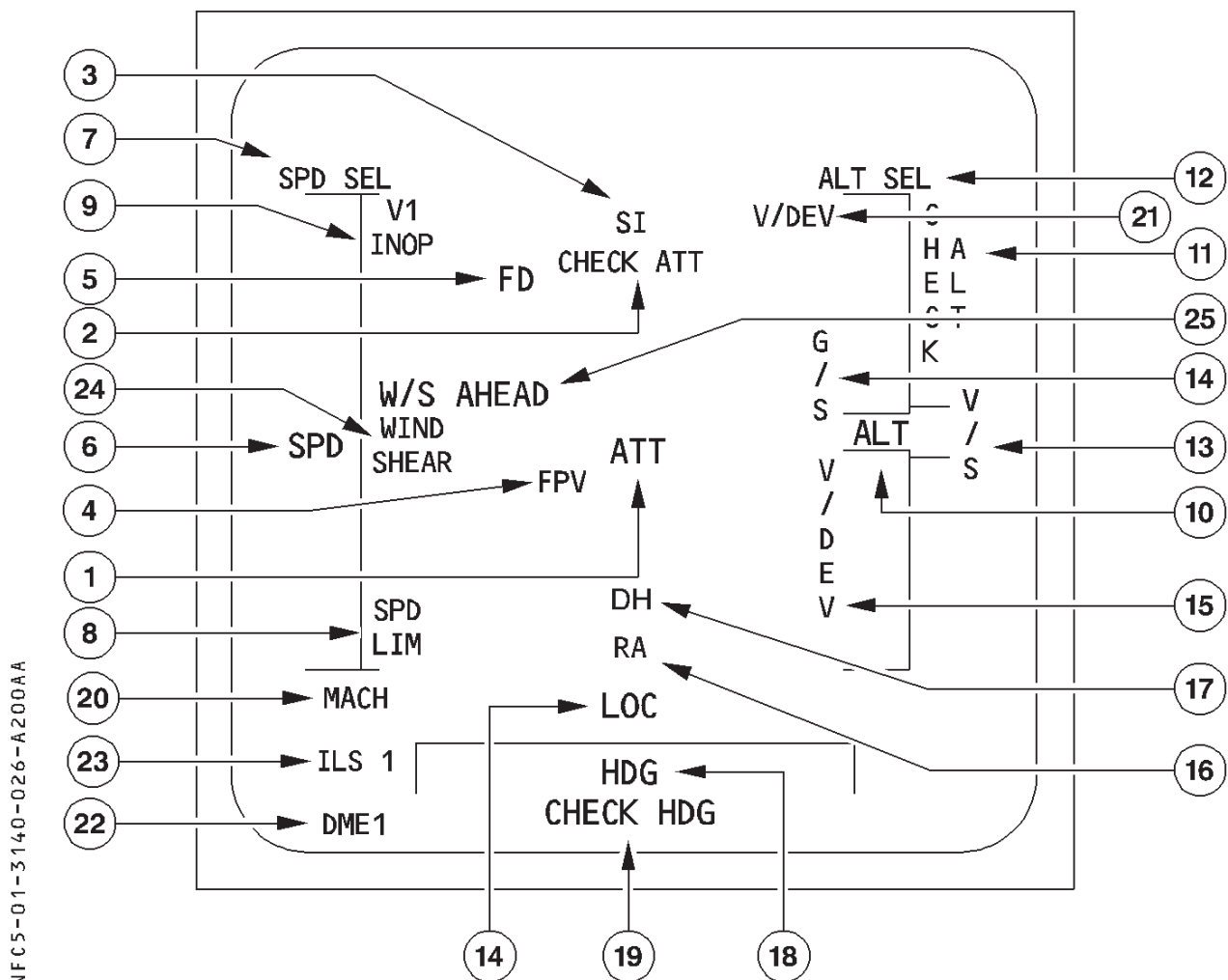
This warning results from a comparison between the altitude (ADIRS) and the preselected altitude displayed on FCU.



- The selection of a new altitude cancels the continuous C chord, as does the crew's pushing the EMER CANC pushbutton of the ECAM control panel or the pressing either MASTER WARN pushbutton.
- The selection of a new altitude stops the flashing of the altitude window.
- The altitude alert is inhibited :
 - when the slats are out with landing gear is selected down, or
 - in approach after the aircraft captures the glide slope, or
 - when the landing gear is locked down.

FLAGS AND MESSAGES DISPLAYED ON PFD

R



① ATT flag (red)

If the PFD loses all attitude data, its entire sphere is cleared to display the ATT flag.

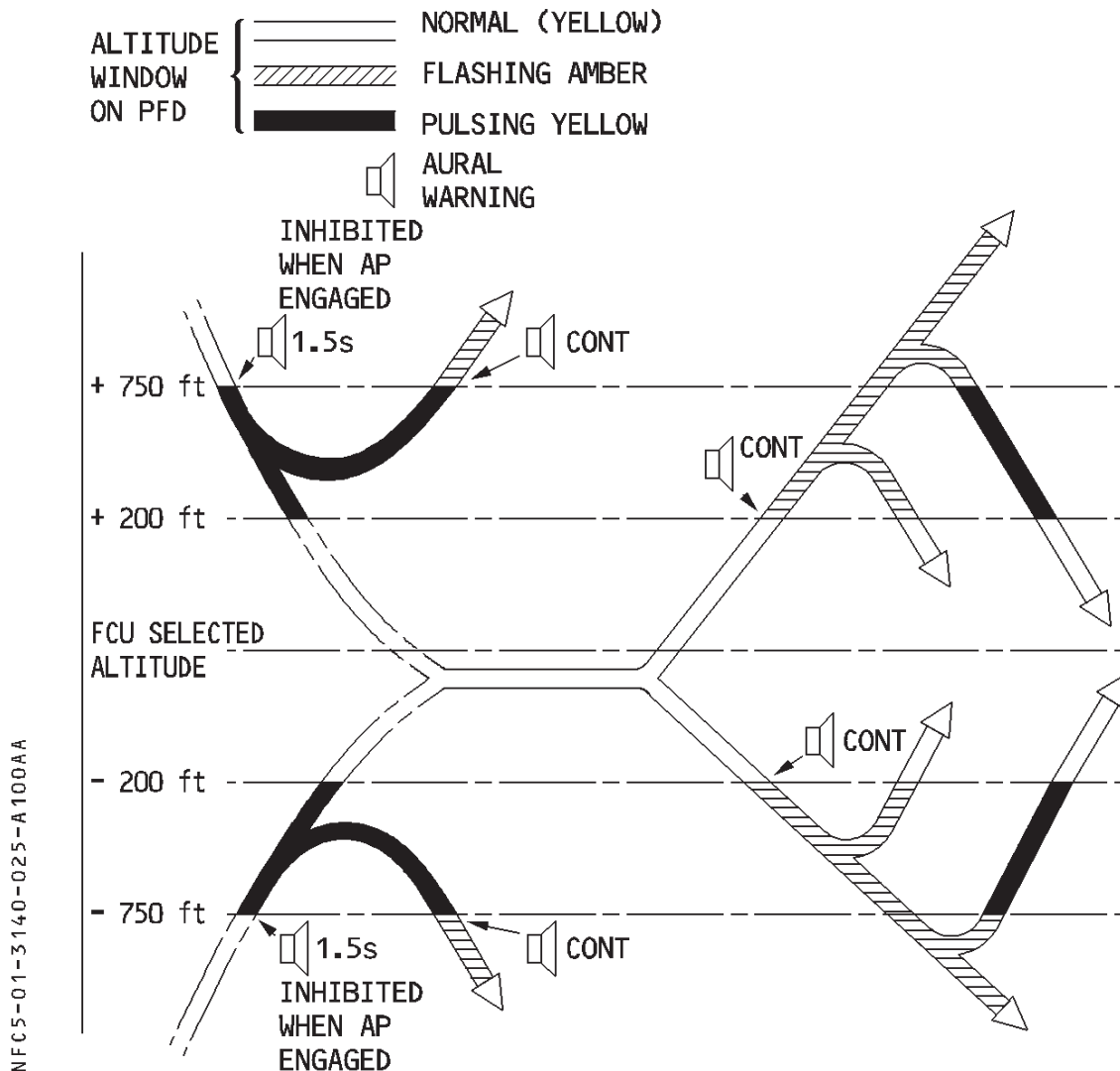
② CHECK ATT (amber)

“CHECK ATT” appears when there is a disagreement (of a least 5°) in the attitude information displayed by the two PFDs. The CHECK ATT flag appears on both PFDs, and a caution appears on the ECAM.

ALTITUDE ALERT

The FWC generates an altitude warning (C chord sound and PFD's altitude window pulses in yellow or flashes in amber), when the aircraft approaches a preselected altitude or flight level, or when it deviates from its selected altitude or flight level.

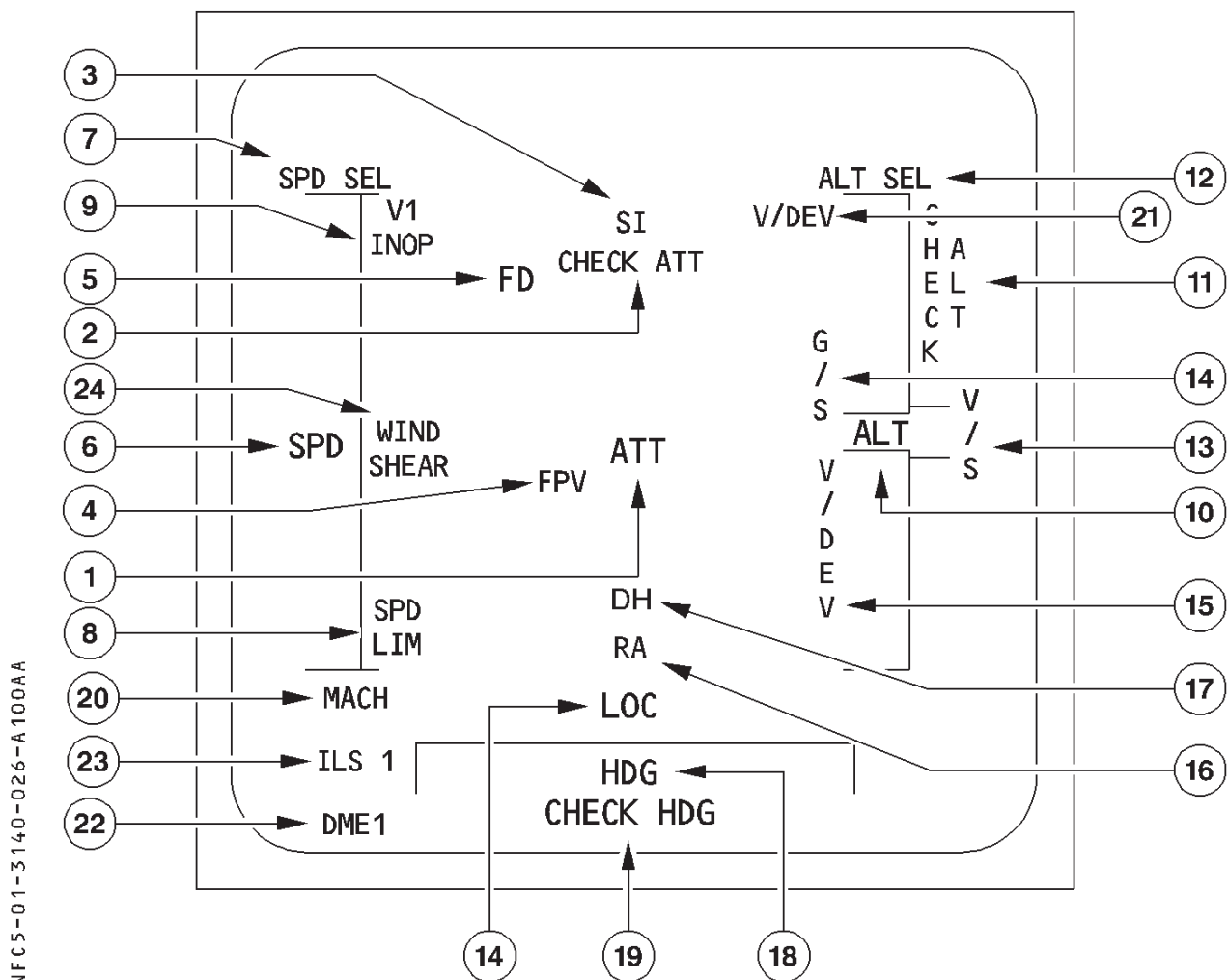
This warning results from a comparison between the altitude (ADIRS) and the preselected altitude displayed on FCU.



- Selecting a new altitude, or pushing the ECAM's EMER CANC pushbutton, or pressing either MASTER WARN pushbutton, cancels the continuous C chord.
- Selecting a new altitude stops the flashing of the altitude window.
- The altitude alert is inhibited :
 - When the slats are out, with the landing gear is selected down, or
 - In approach after the aircraft captures the glideslope, or
 - When the landing gear is locked down.

FLAGS AND MESSAGES DISPLAYED ON PFD

R



① ATT flag (red)

If the PFD loses all attitude data, its entire sphere is cleared to display the ATT flag.

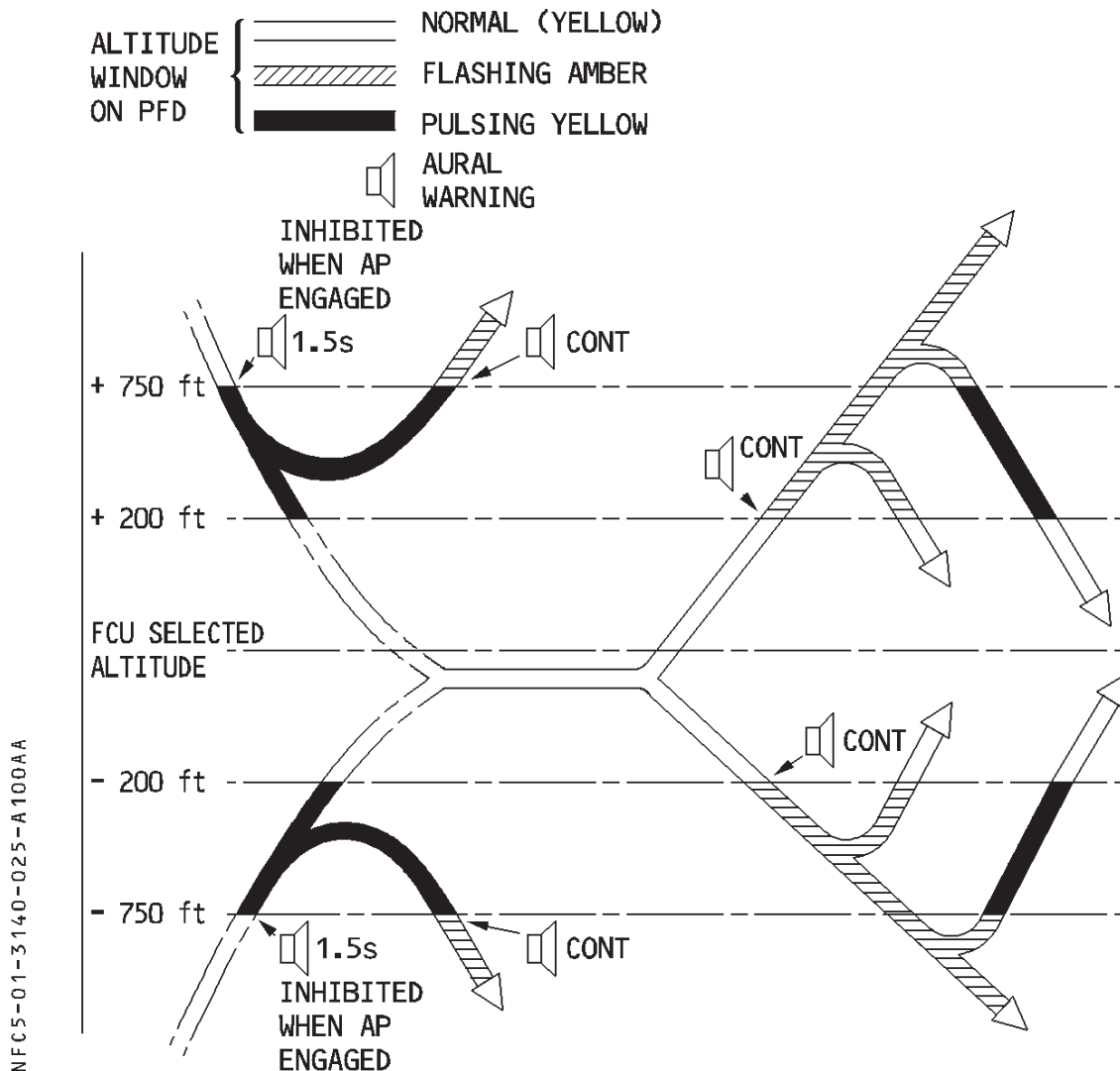
② CHECK ATT (amber)

“CHECK ATT” appears when there is a disagreement (of a least 5°) in the attitude information displayed by the two PFDs. The CHECK ATT flag appears on both PFDs, and a caution appears on the ECAM.

ALTITUDE ALERT

The FWC generates an altitude warning (C chord sound and PFD's altitude window pulses in yellow or flashes in amber), when the aircraft approaches a preselected altitude or flight level, or when it deviates from its selected altitude or flight level.

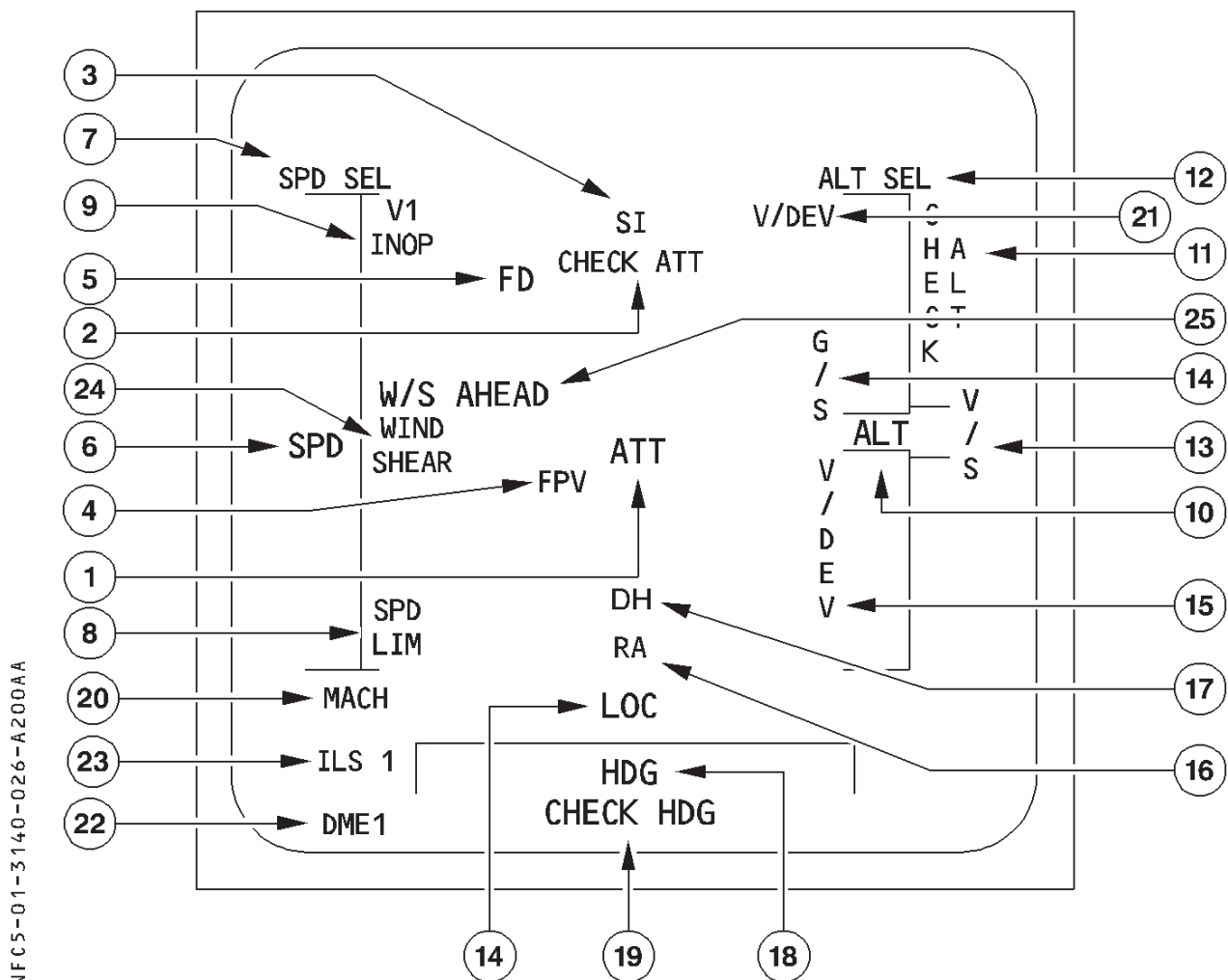
This warning results from a comparison between the altitude (ADIRS) and the preselected altitude displayed on FCU.



- Selecting a new altitude, or pushing the ECAM's EMER CANC pushbutton, or pressing either MASTER WARN pushbutton, cancels the continuous C chord.
- Selecting a new altitude stops the flashing of the altitude window.
- The altitude alert is inhibited :
 - When the slats are out, with the landing gear is selected down, or
 - In approach after the aircraft captures the glideslope, or
 - When the landing gear is locked down.

FLAGS AND MESSAGES DISPLAYED ON PFD

R



① ATT flag (red)

If the PFD loses all attitude data, its entire sphere is cleared to display the ATT flag.

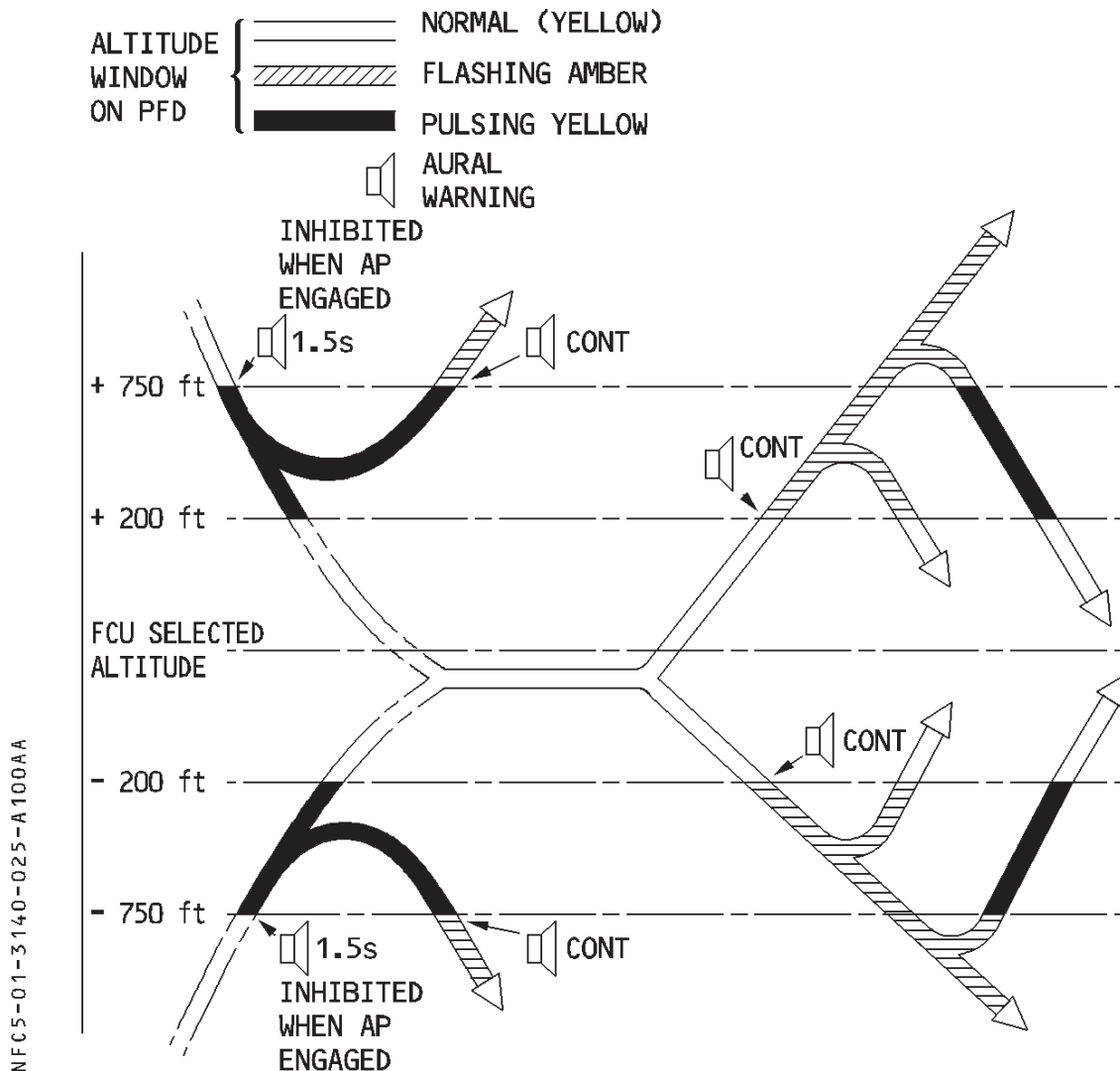
② CHECK ATT (amber)

“CHECK ATT” appears when there is a disagreement (of a least 5°) in the attitude information displayed by the two PFDs. The CHECK ATT flag appears on both PFDs, and a caution appears on the ECAM.

ALTITUDE ALERT

The FWC generates an altitude warning (C chord sound and PFD's altitude window pulses in yellow or flashes in amber), when the aircraft approaches a preselected altitude or flight level, or when it deviates from its selected altitude or flight level.

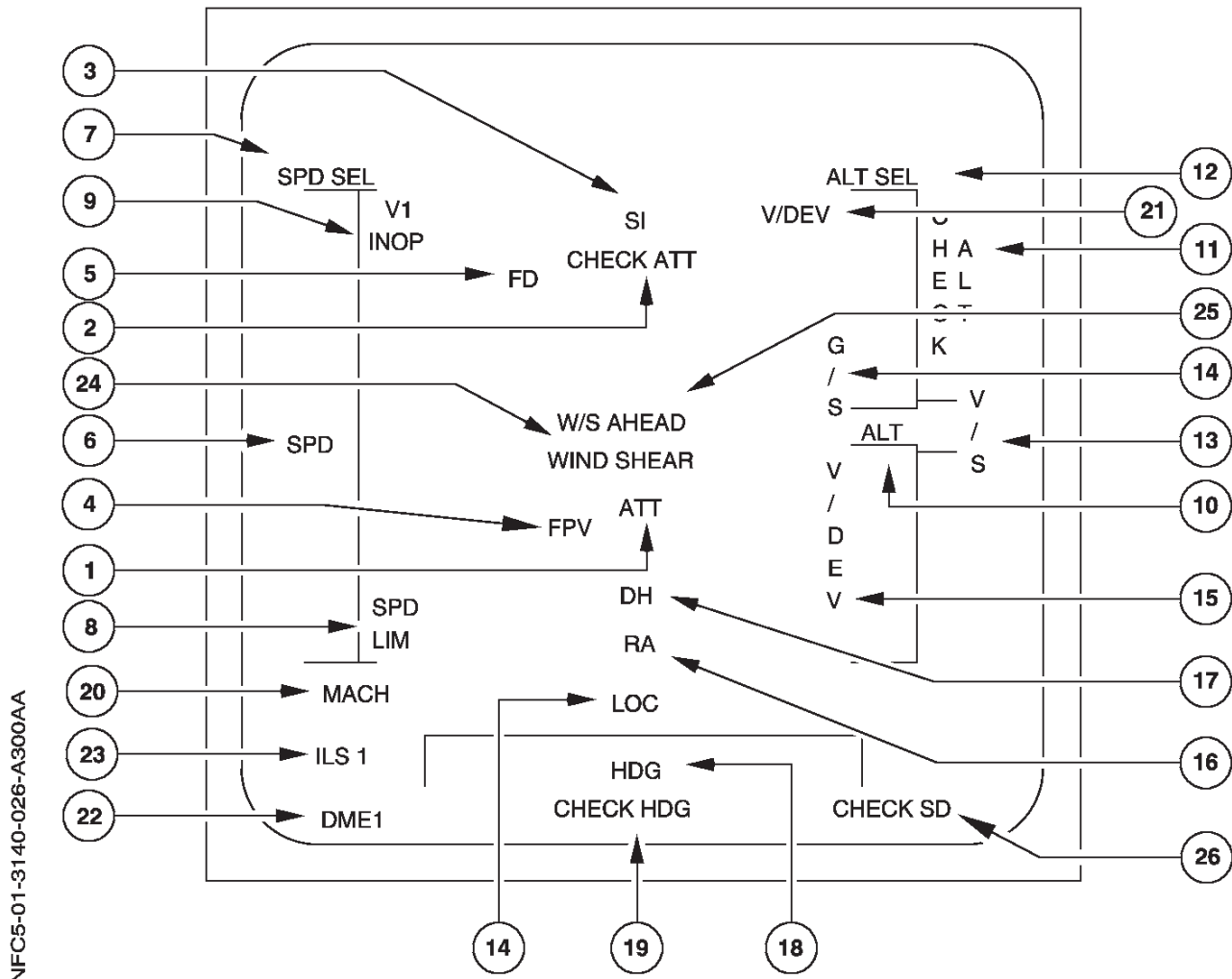
This warning results from a comparison between the altitude (ADIRS) and the preselected altitude displayed on FCU.



- Selecting a new altitude, or pushing the ECAM's EMER CANC pushbutton, or pressing either MASTER WARN pushbutton, cancels the continuous C chord.
- Selecting a new altitude stops the flashing of the altitude window.
- The altitude alert is inhibited :
 - When the slats are out, with the landing gear is selected down, or
 - In approach after the aircraft captures the glideslope, or
 - When the landing gear is locked down.

FLAGS AND MESSAGES DISPLAYED ON PFD

R



① ATT flag (red)

If the PFD loses all attitude data, its entire sphere is cleared to display the ATT flag.

R ② CHECK ATT, CHECK CAPT (F/O) PFD, CHECK EWD, DU NOT MONITORED (amber)

R For more information, refer to FCOM 1.31.05.

③ SI flag (red)

If the sideslip information is lost or any reverse is deployed in flight, the index disappears and a red SI flag appears.

④ FPV flag (red)

In the TRK FPA mode, when the drift angle or flight path angle is not valid, an FPV flag appears.

⑤ FD flag (red)

If both FMGCs fail, or if both FDs are disengaged and the FD pushbutton is on and the attitude is valid, a red FD flag appears.

⑥ SPD flag (red)

If the speed information fails, a SPD flag replaces the speed scale.

⑦ SPD SEL flag (red)

If the selected speed information fails, a SPD SEL flag appears.

⑧ SPD LIM flag (red)

This flag appears when both FACs are inoperative, or in case of SFCC dual flap/slat channel failure.

In this case, the following PFD information is lost : VLS, S, F, Green Dot, Vtrend, Vmax, VFE next, VSW.

⑨ V1 INOP flag (red)

When the V1 signal is not valid, a V1 INOP flag replaces the digital value.

⑩ ALT flag (red)

If the altitude information fails, the ALT flag replaces the altitude scale.

⑪ CHECK ALT flag (amber)

The CHECK ALT flag appears, as does an ECAM caution, if the disagree between the two PFD altitude indications is greater than 250 feet when QNH is selected, or 500 feet when STD is selected.

R The caution and the flag disappear, when the Pilot's and the Co-pilot's barometer references disagree.

⑫ ALT SEL flag (red)

If the selected altitude information fails, an ALT SEL flag appears.

⑬ V/S flag (red)

If the vertical speed information fails, the V/S flag replaces the vertical speed scale.

⑭ LOC and G/S flags (red)

If the localizer or glideslope receiver fails, a LOC or G/S flag appears on the deviation scale.

⑮ V/DEV flag (red)

If the vertical deviation information fails and the ILS pushbutton is not pressed, a V/DEV flag replaces the V/DEV scale.

⑯ RA flag (red)

If both radio altimeters fail, this flag appears in place of the radio height indication.

⑰ DH flag (amber)

A DH flag appears, when the aircraft reaches the selected DH.

⑱ HDG flag (red)

If the heading information fails, the HDG flag replaces the heading scale.

⑲ CHECK HDG flag (amber)

The CHECK HDG flag appears, as does an ECAM caution, if there is a discrepancy (5°) between pilots's and copilot's heading indications.

⑳ MACH flag (red)

This flag appears, if the Mach data fails.

㉑ V/DEV (amber)

At the top of the glide scale, this message flashes when FINAL mode is armed or engaged, and the ILS pushbutton is selected.

R

③ SI flag (red)

If the sideslip information is lost or any reverse is deployed in flight, the index disappears and a red SI flag appears.

④ FPV flag (red)

In the TRK FPA mode, when the drift angle or flight path angle is not valid, an FPV flag appears.

⑤ FD flag (red)

If both FMGCs fail, or if both FDs are disengaged and the FD pushbutton is on and the attitude is valid, a red FD flag appears.

⑥ SPD flag (red)

If the speed information fails, a SPD flag replaces the speed scale.

⑦ SPD SEL flag (red)

If the selected speed information fails, a SPD SEL flag appears.

⑧ SPD LIM flag (red)

This flag appears when both FACs are inoperative, or in case of SFCC dual flap/slat channel failure.

In this case, the following PFD information is lost : VLS, S, F, Green Dot, Vtrend, Vmax, VFE next, VSW.

⑨ V1 INOP flag (red)

When the V1 signal is not valid, a V1 INOP flag replaces the digital value.

⑩ ALT flag (red)

If the altitude information fails, the ALT flag replaces the altitude scale.

⑪ CHECK ALT flag (amber)

The CHECK ALT flag appears, as does an ECAM caution, if the disagree between the two PFD altitude indications is greater than 250 feet when QNH is selected, or 500 feet when STD is selected.

R The caution and the flag disappear, when the Pilot's and the Co-pilot's barometer references disagree.

⑫ ALT SEL flag (red)

If the selected altitude information fails, an ALT SEL flag appears.

⑬ V/S flag (red)

If the vertical speed information fails, the V/S flag is displayed instead of the vertical speed scale.

⑭ LOC and G/S flags (red)

If the localizer or glide slope receiver fails, a LOC or G/S flag appears on the deviation scale.

⑮ V/DEV flag (red)

If the vertical deviation information fails, and the flight crew does not press the ILS pushbutton, a V/DEV flag appears instead of the V/DEV scale.

⑯ RA flag (red)

If both radio altimeters fail, this flag appears instead of the radio height indication.

⑰ DH flag (amber)

A DH flag appears when the aircraft reaches the selected DH.

⑱ HDG flag (red)

If the heading information fails, the HDG flag appears instead of the heading scale.

⑲ CHECK HDG flag (amber)

The CHECK HDG flag appears, as does an ECAM caution, if there is a discrepancy (5°) between pilots's and copilot's heading indications.

⑳ MACH flag (red)

This flag appears if the Mach data fails.

㉑ V/DEV (amber)

At the top of the glide scale, this message flashes when the aircraft is in approach phase and, when either the FINAL mode is armed/engaged, or a non-LS approach is selected, and the ILS pushbutton is selected.

R

③ SI flag (red)

If the sideslip information is lost or any reverse is deployed in flight, the index disappears and a red SI flag appears.

④ FPV flag (red)

In the TRK FPA mode, when the drift angle or flight path angle is not valid, an FPV flag appears.

⑤ FD flag (red)

If both FMGCs fail, or if both FDs are disengaged and the FD pushbutton is on and the attitude is valid, a red FD flag appears.

⑥ SPD flag (red)

If the speed information fails, a SPD flag replaces the speed scale.

⑦ SPD SEL flag (red)

If the selected speed information fails, a SPD SEL flag appears.

⑧ SPD LIM flag (red)

This flag appears when both FACs are inoperative, or in case of SFCC dual flap/slat channel failure.

In this case, the following PFD information is lost : VLS, S, F, Green Dot, Vtrend, Vmax, VFE next, VSW.

⑨ V1 INOP flag (red)

When the V1 signal is not valid, a V1 INOP flag replaces the digital value.

⑩ ALT flag (red)

If the altitude information fails, the ALT flag replaces the altitude scale.

⑪ CHECK ALT flag (amber)

The CHECK ALT flag appears, as does an ECAM caution, if the disagree between the two PFD altitude indications is greater than 250 feet when QNH is selected, or 500 feet when STD is selected.

R The caution and the flag disappear, when the Pilot's and the Co-pilot's barometer references disagree.

⑫ ALT SEL flag (red)

If the selected altitude information fails, an ALT SEL flag appears.

⑬ V/S flag (red)

If the vertical speed information fails, the V/S flag replaces the vertical speed scale.

⑭ LOC and G/S flags (red)

If the localizer or glideslope receiver fails, a LOC or G/S flag appears on the deviation scale.

⑮ V/DEV flag (red)

If the vertical deviation information fails, and the LS pushbutton is not pressed, a V/DEV flag replaces the V/DEV scale.

⑯ RA flag (red)

If both radio altimeters fail, this flag appears in place of the radio height indication.

⑰ DH flag (amber)

A DH flag appears, when the aircraft reaches the selected DH.

⑱ HDG flag (red)

If the heading information fails, the HDG flag replaces the heading scale.

⑲ CHECK HDG flag (amber)

The CHECK HDG flag appears, as does an ECAM caution, if there is a discrepancy (5°) between pilots's and copilot's heading indications.

⑳ MACH flag (red)

This flag appears, if the Mach data fails.

㉑ V/DEV (amber)

At the top of the glide scale, this message flashes when in approach phase and, when either the FINAL mode is armed/engaged or a non-LS approach has been selected, and the LS pushbutton is selected.

③ SI flag (red)

If the sideslip information is lost or any reverse is deployed in flight, the index disappears and a red SI flag appears.

④ FPV flag (red)

In the TRK FPA mode, when the drift angle or flight path angle is not valid, an FPV flag appears.

⑤ FD flag (red)

If both FMGCs fail, or if both FDs are disengaged and the FD pushbutton is on and the attitude is valid, a red FD flag appears.

⑥ SPD flag (red)

If the speed information fails, a SPD flag replaces the speed scale.

⑦ SPD SEL flag (red)

If the selected speed information fails, a SPD SEL flag appears.

⑧ SPD LIM flag (red)

This flag appears when both FACs are inoperative, or in case of SFCC dual flap/slat channel failure.

In this case, the following PFD information is lost : VLS, S, F, Green Dot, Vtrend, Vmax, VFE next, VSW.

⑨ V1 INOP flag (red)

When the V1 signal is not valid, a V1 INOP flag replaces the digital value.

⑩ ALT flag (red)

If the altitude information fails, the ALT flag replaces the altitude scale.

⑪ CHECK ALT flag (amber)

The CHECK ALT flag appears, as does an ECAM caution, if the disagree between the two PFD altitude indications is greater than 250 feet when QNH is selected, or 500 feet when STD is selected.

R The caution and the flag disappear, when the Pilot's and the Co-pilot's barometer references disagree.

⑫ ALT SEL flag (red)

If the selected altitude information fails, an ALT SEL flag appears.

⑬ V/S flag (red)

If the vertical speed information fails, the V/S flag is displayed instead of the vertical speed scale.

⑭ LOC and G/S flags (red)

If the localizer or glide slope receiver fails, a LOC or G/S flag appears on the deviation scale.

⑮ V/DEV flag (red)

If the vertical deviation information fails, and the flight crew does not press the LS pushbutton, a V/DEV flag appears instead of the V/DEV scale.

⑯ RA flag (red)

If both radio altimeters fail, this flag appears instead of the radio height indication.

⑰ DH flag (amber)

A DH flag appears when the aircraft reaches the selected DH.

⑱ HDG flag (red)

If the heading information fails, the HDG flag appears instead of the heading scale.

⑲ CHECK HDG flag (amber)

For more information, refer to FCOM 1.31.05.

⑳ MACH flag (red)

This flag appears if the Mach data fails.

㉑ V/DEV (amber)

At the top of the glide scale, this message flashes when the aircraft is in approach phase and, when either the FINAL mode is armed/engaged, or a non-LS approach is selected, and the LS pushbutton is selected.

②② DME 1 flag (red)

When the DME distance is not valid, a DME1 (on PFD1) or DME2 (on PFD2) flag replaces the DME distance indication.

②③ ILS1 flag (red)

If an ILS frequency fails, or if either the LOC or G/S signals fail, an ILS1 (on PFD1) or ILS2 (on PFD2) flag replaces the ILS frequency indication.

②④ WINDSHEAR warning (red)

This message is displayed, when windshear is detected (reactive windshear warning) by the FAC.

R Refer to 1.22.40.

Note : 1. All flags, except SI, V1 INOP and DME 1 (which are steady), flash for 9 seconds, then remain steady.
The DH flag flashes for 3 seconds, then remains steady.
2. For information on the TCAS flag, refer to 1.34.80.

②② DME 1 flag (red)

When the DME distance is not valid, a DME1 (on PFD1) or DME2 (on PFD2) flag replaces the DME distance indication.

②③ ILS1 flag (red)

If an ILS frequency fails, or if either the LOC or G/S signals fail, an ILS1 (on PFD1) or ILS2 (on PFD2) flag replaces the ILS frequency indication.

②④ WINDSHEAR (red)

This message is displayed, when windshear is detected (reactive windshear warning) by the FAC.

R Refer to 1.22.40.

②⑤ W/S AHEAD

This message is displayed, when the predictive windshear system has detected windshear ahead of the aircraft.

The message is in amber or red, depending on the alert level.

Refer to 1.34.60

Note : 1. All flags, except SI, V1 INOP and DME 1 (which are steady), flash for 9 seconds, then remain steady.

The DH flag flashes for 3 seconds, then remains steady.

2. For information on the TCAS flag, refer to 1.34.80.

②② DME 1 flag (red)

When the DME distance is not valid, a DME1 (on PFD1) or DME2 (on PFD2) flag is displayed instead of the DME distance indication.

②③ ILS1 flag (red)

If an ILS frequency fails, or if either the LOC or the G/S signals fail, an ILS1 (on PFD1) or ILS2 (on PFD2) flag is displayed instead of the ILS frequency indication.

②④ WINDSHEAR warning (red)

This message is displayed, when the FAC detects windshear (reactive windshear warning).

For more information, refer to 1.22.40.

②⑤ W/S AHEAD

This message is displayed, when the predictive windshear system detects windshear ahead of the aircraft.

The message is displayed in amber or red, depending on the alert level.

For more information, refer to 1.34.60.

Note : 1. All flags except, V1 INOP which is steady, flash for 9 seconds, then remain steady.

2. For information on the TCAS flag, refer to FCOM 1.34.80.

R ②⑥ CHECK SD, CHECK CAPT (F/O) ND (amber)

R For more information, refer to FCOM 1.31.05.

GENERAL

There are five different displays (five modes to display navigation information) :

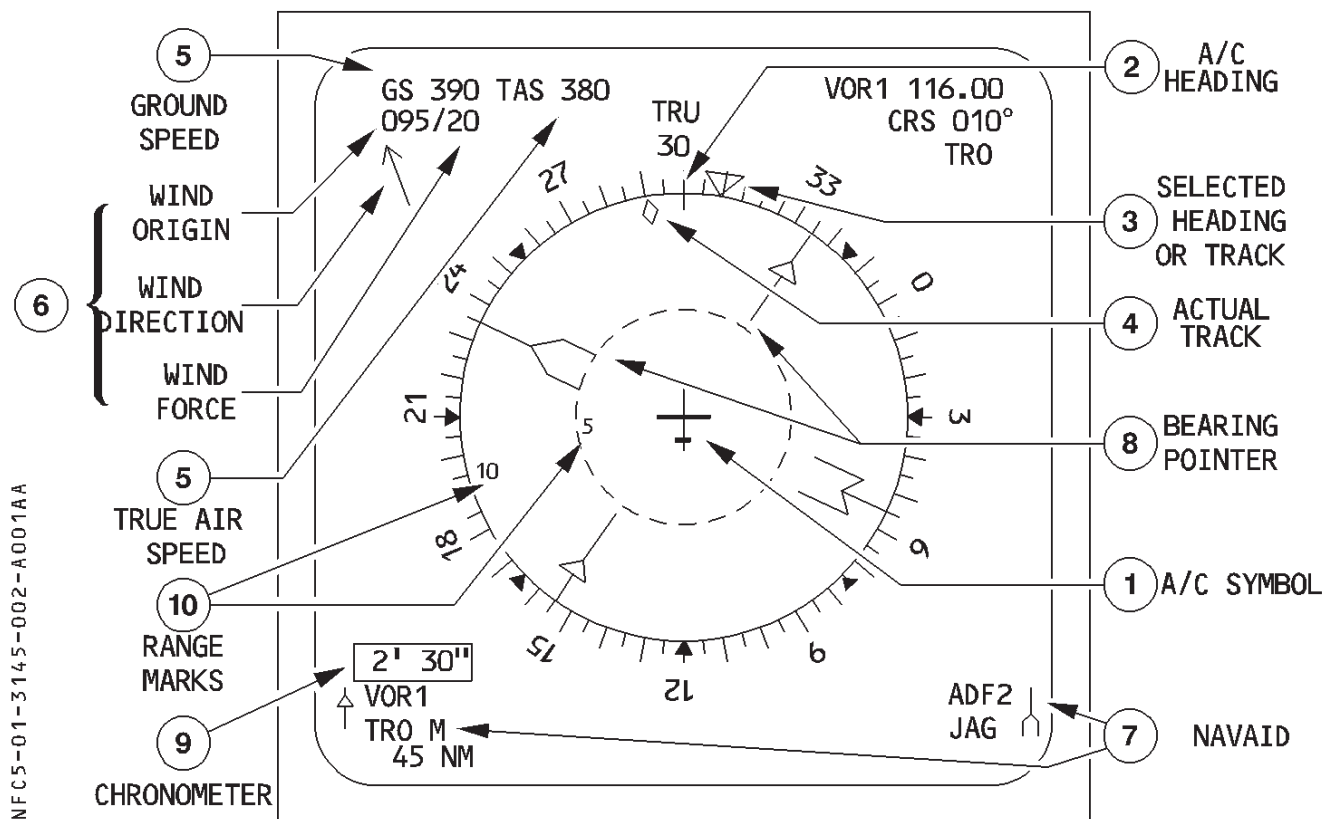
- ROSE ILS
- ROSE VOR
- ROSE NAV
- ARC
- PLAN

The Navigation Display (ND) can provide a weather radar image in all modes, except PLAN.

Note : *In case avionics ventilation is not sufficient (e.g. due to a blower and extract fan failure), and the Navigation Display (ND) Unit temperature exceeds a defined threshold, the ND will not display the weather radar image, in order to limit power consumption and prevent a DU overheat. Any additional increase in temperature will lead to a complete cut off of the power supply to this display unit.*

R
R

ROSE MODES



① Aircraft symbol (yellow)

Fixed and centered in the display, this symbol points to the yellow lubber line.

② Aircraft heading

The fixed yellow lubber line points to the aircraft magnetic heading on the moving white compass rose. Small white triangles are fixed at 45° intervals on the circumference of the compass rose.

"TRU" appears at the top of the compass rose when it is displaying true heading instead of magnetic heading (latitude above 73° North or 60° South).

③ Selected heading or track (blue)

This pointer shows the heading or track indicated on the HDG TRK counter of the FCU.

④ Actual aircraft track (green)

This symbol is a small green diamond.

⑤ Ground speed and true air speed (green)

ADIRS furnishes these speeds.

GENERAL

There are five different displays (five modes to display navigation information) :

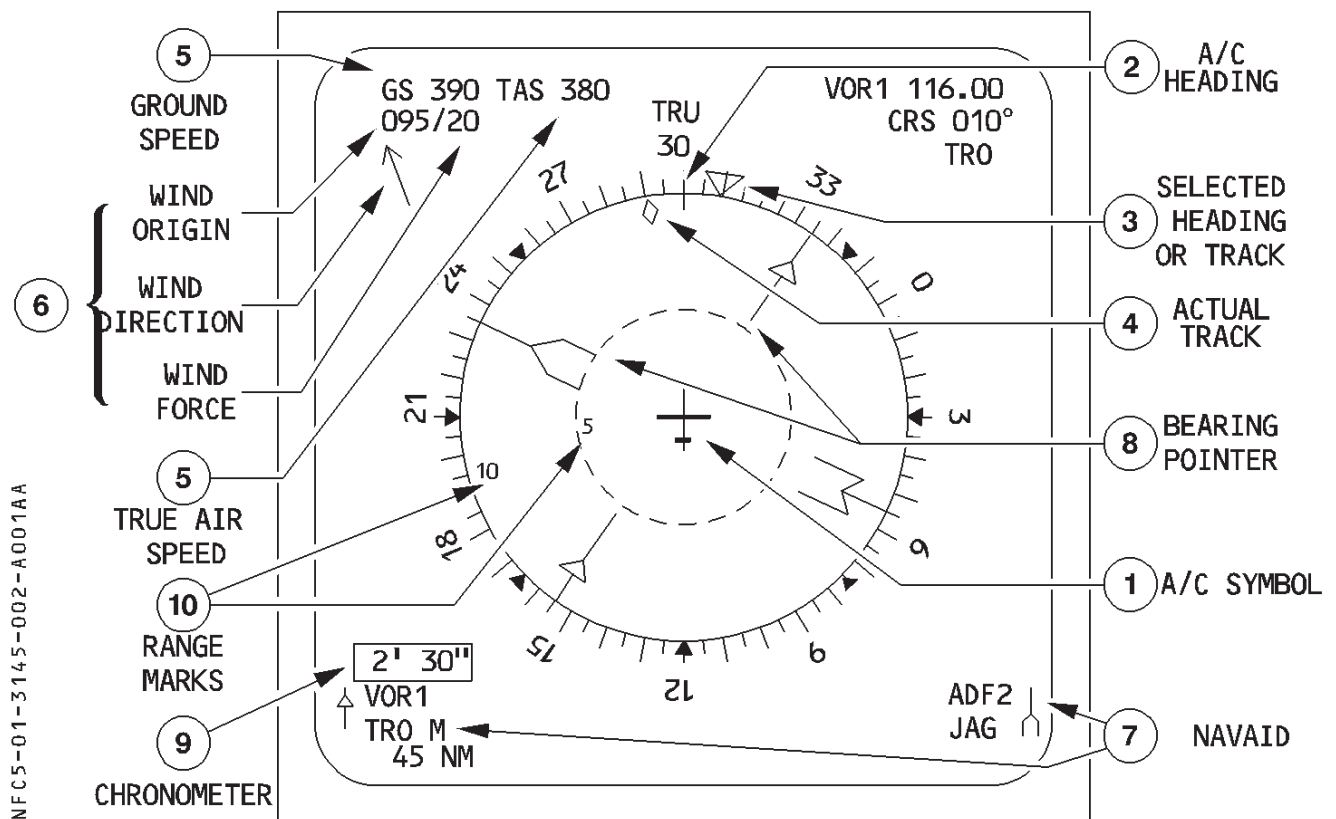
- ROSE LS
- ROSE VOR
- ROSE NAV
- ARC
- PLAN

The Navigation Display (ND) can provide a weather radar image in all modes, except PLAN.

Note : *In case avionics ventilation is not sufficient (e.g. due to a blower and extract fan failure), and the Navigation Display (ND) Unit temperature exceeds a defined threshold, the ND will not display the weather radar image, in order to limit power consumption and prevent a DU overheat. Any additional increase in temperature will lead to a complete cut off of the power supply to this display unit.*

R
R

ROSE MODES



① Aircraft symbol (yellow)

Fixed and centered in the display, this symbol points to the yellow lubber line.

② Aircraft heading

The fixed yellow lubber line points to the aircraft magnetic heading on the moving white compass rose. Small white triangles are fixed at 45° intervals on the circumference of the compass rose.

"TRU" appears at the top of the compass rose when it is displaying true heading instead of magnetic heading (latitude above 73° North or 60° South).

③ Selected heading or track (blue)

This pointer shows the heading or track indicated on the HDG TRK counter of the FCU.

④ Actual aircraft track (green)

This symbol is a small green diamond.

⑤ Ground speed and true air speed (green)

ADIRS furnishes these speeds.

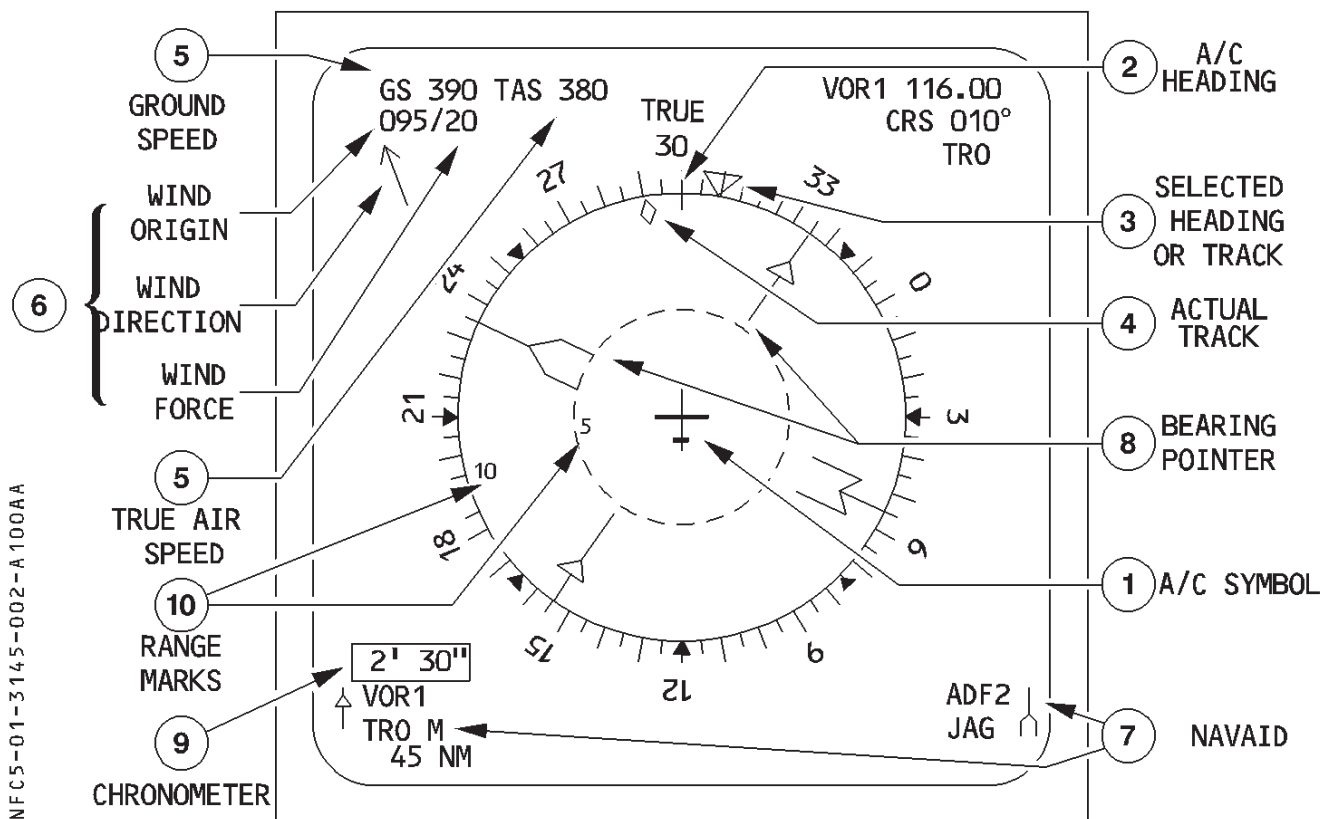
GENERAL

There are five different displays (that display navigation information) :

- ROSE LS
- ROSE VOR
- ROSE NAV
- ARC
- PLAN

The Navigation Display (ND) can provide a weather radar image in all modes, except PLAN.

ROSE MODES



① Aircraft symbol (yellow)

Fixed and centered in the display, this symbol points to the yellow lubber line.

② Aircraft heading

The fixed yellow lubber line points to the aircraft magnetic heading on the moving white compass rose. Small white triangles are fixed at 45° intervals on the circumference of the compass rose.

"TRUE" appears at the top of the compass rose, when it is displaying true heading instead of magnetic heading (latitude above 73° North or 60° South).

③ Selected heading or track (blue)

This pointer shows the heading or track indicated on the FCU's HDG TRK counter.

④ Actual aircraft track (green)

This symbol is a small green diamond.

⑤ Ground speed and true air speed (green)

ADIRS furnishes these speeds.

⑥ Wind direction and speed

ADIRS furnishes the wind direction and speed. The digital direction is with respect to true north, and the analog direction (green arrow) is with respect to magnetic north. The arrow appears only if the wind speed is greater than two knots.

If the display does not receive either wind speed or direction, dashes replace the numbers on the display.

⑦ Nav aids

When the ADF-OFF-VOR selector switch on either the pilot's or copilot's EFIS control panel is set to ADF or VOR, the outside ND displays the following characteristics of the corresponding nav aid in white for VOR or in green for ADF (left side for receiver 1 and right side for receiver 2) :

- Type of nav aid (ADF or VOR)
- Shape and color of the associated bearing pointer (if the bearing pointer is in view).
- Nav aid identification (or frequency by default)
- DME distance if a DME is collocated with the selected VOR.ADF and DME distance are never displayed at the same time.
- Mode of tuning
 - M for a nav aid tuned manually by the pilot through the MCDU (underlined and dimmed),
 - R for a nav aid tuned from an RMP (Radio Management Panel) (underlined and dimmed),
 - Nothing for a nav aid tuned automatically by the FMGC.

If reception fails, the ND stops displaying the associated data (except for the identification or frequency).

⑧ Bearing pointer (green for ADF, white for VOR)

This pointer appears when bearing data is available.

R If the aircraft is not receiving the beacon or if a receiver fails, the associated bearing pointer disappears.

⑨ Chronometer Indication (white)

These numbers appear when the outside chronometer is started.

They display the elapsed time.

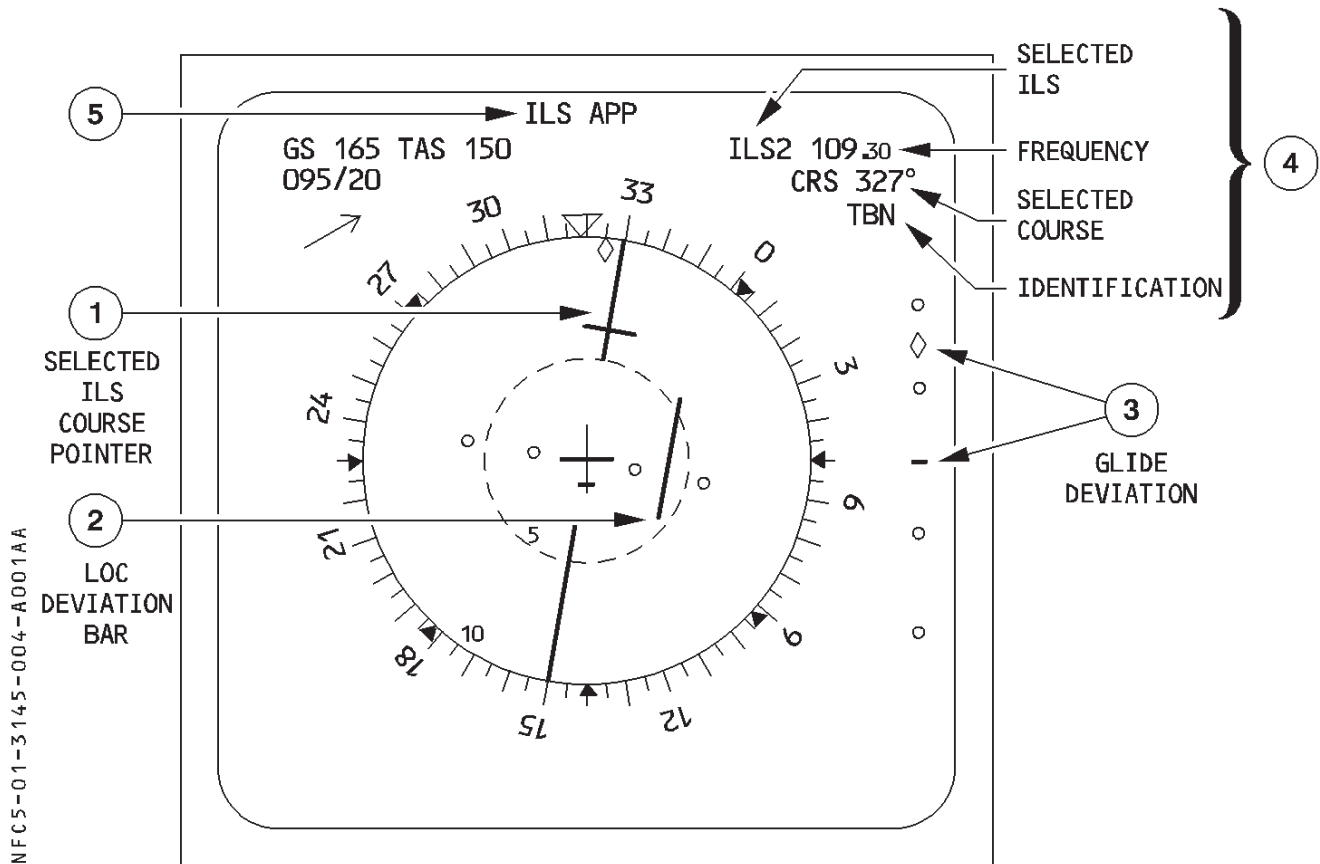
The indication is in minutes and seconds from 0 to 59' 59", and in hours and minutes from 1 H to 99 H 59'. (Seconds are not displayed beyond 59' 59").

R ⑩ Range marks

R The range scale value selected on the EFIS control panel (10 to 320 NM) governs the
R scale of the ND.

ROSE ILS MODE

R



① ILS course pointer (magenta)

This dagger-shaped symbol points to the selected ILS course.

The ILS is selected either by the FMGC (autotuned or manually) or through the RMP in backup mode. If no course has been entered, the value defaults to 360°.

② Localizer deviation bar (magenta)

This bar moves laterally with respect to the course pointer. Its scale consists of two dots on each side of zero deviation. Each dot represents a deviation of about $\pm 0.8^\circ$.

If the deviation becomes excessive (1/4 dot, 0.2°) above 15 feet RA, the bar and the scale pulse.

⑥ Wind direction and speed

ADIRS furnishes the wind direction and speed. The digital direction is with respect to true north, and the analog direction (green arrow) is with respect to magnetic north. The arrow appears only if the wind speed is greater than two knots.

If the display does not receive either wind speed or direction, dashes replace the numbers on the display.

⑦ Nav aids

When the ADF-OFF-VOR selector switch on either the pilot's or copilot's EFIS control panel is set to ADF or VOR, the outside ND displays the following characteristics of the corresponding nav aid in white for VOR or in green for ADF (left side for receiver 1 and right side for receiver 2) :

- Type of nav aid (ADF or VOR)
- Shape and color of the associated bearing pointer (if the bearing pointer is in view).
- Nav aid identification (or frequency by default)
- DME distance if a DME is collocated with the selected VOR.ADF and DME distance are never displayed at the same time.
- Mode of tuning
 - M for a nav aid tuned manually by the pilot through the MCDU (underlined and dimmed),
 - R for a nav aid tuned from an RMP (Radio Management Panel) (underlined and dimmed),
 - Nothing for a nav aid tuned automatically by the FMGC.

If reception fails, the ND stops displaying the associated data (except for the identification or frequency).

⑧ Bearing pointer (green for ADF, white for VOR)

This pointer appears when bearing data is available.

R If the aircraft is not receiving the beacon or if a receiver fails, the associated bearing pointer disappears.

⑨ Chronometer Indication (white)

These numbers appear when the outside chronometer is started.

They display the elapsed time.

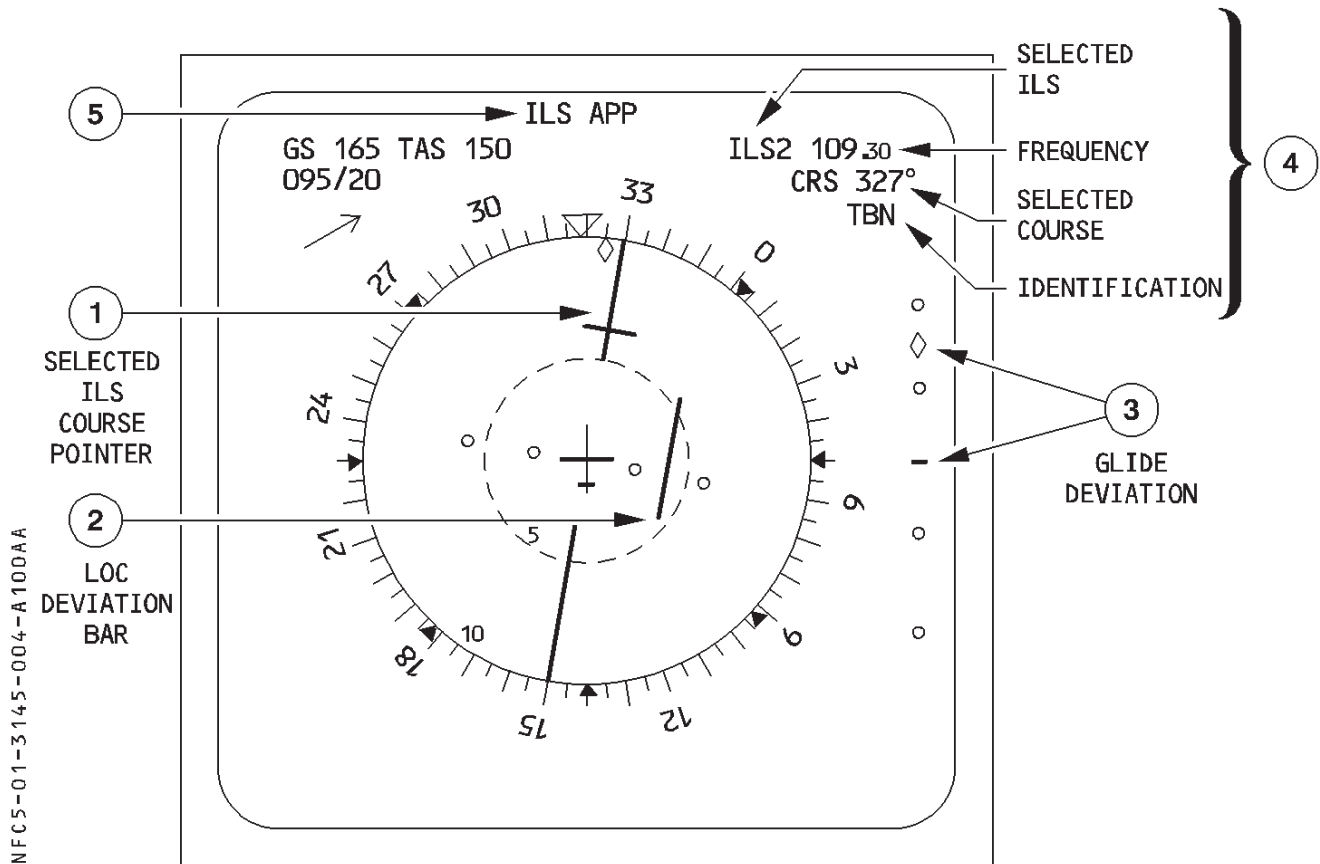
The indication is in minutes and seconds from 0 to 59' 59", and in hours and minutes from 1 H to 99 H 59'. (Seconds are not displayed beyond 59' 59").

R ⑩ Range marks

R The range scale value selected on the EFIS control panel (10 to 320 NM) governs the
R scale of the ND.

ROSE LS MODE

R



① ILS course pointer (magenta)

This dagger-shaped symbol points to the selected ILS course.

The ILS is selected either by the FMGC (autotuned or manually) or through the RMP in backup mode. If no course has been entered, the value defaults to 360°.

② Localizer deviation bar (magenta)

This bar moves laterally with respect to the course pointer. Its scale consists of two dots on each side of zero deviation. Each dot represents a deviation of about $\pm 0.8^\circ$.

If the deviation becomes excessive (1/4 dot, 0.2°) above 15 feet RA, the bar and the scale pulse.

⑥ Wind direction and speed

R ADIRS provides the wind direction and speed. The digital direction and the analog
R direction (indicated by the green arrow) reflect both the true north reference. The green
R arrow only appears, if the wind speed is above two knots.
If the display does not receive either wind speed or direction, dashes replace the numbers on the display.

⑦ Nav aids

When the ADF-OFF-VOR selector switch on either the pilot's or copilot's EFIS control panel is set to ADF or VOR, the outside ND displays the following characteristics of the corresponding nav aid in white for VOR or in green for ADF (left side for receiver 1 and right side for receiver 2) :

- Type of nav aid (ADF or VOR)
- Shape and color of the associated bearing pointer (if the bearing pointer is in view).
- Nav aid identification (or frequency by default)
- DME distance if a DME is collocated with the selected VOR. ADF and DME distance are never displayed at the same time.
- Mode of tuning
 - M for a nav aid tuned manually by the pilot through the MCDU (underlined and dimmed),
 - R for a nav aid tuned from an RMP (Radio Management Panel) (underlined and dimmed),
 - Nothing for a nav aid tuned automatically by the FMGC.

If reception fails, the ND stops displaying the associated data (except for the identification or frequency).

⑧ Bearing pointer (green for ADF, white for VOR)

This pointer appears when bearing data is available.

If the aircraft is not receiving the beacon or if a receiver fails, the associated bearing pointer disappears.

⑨ Chronometer Indication (white)

These numbers appear when the outside chronometer is started.

They display the elapsed time.

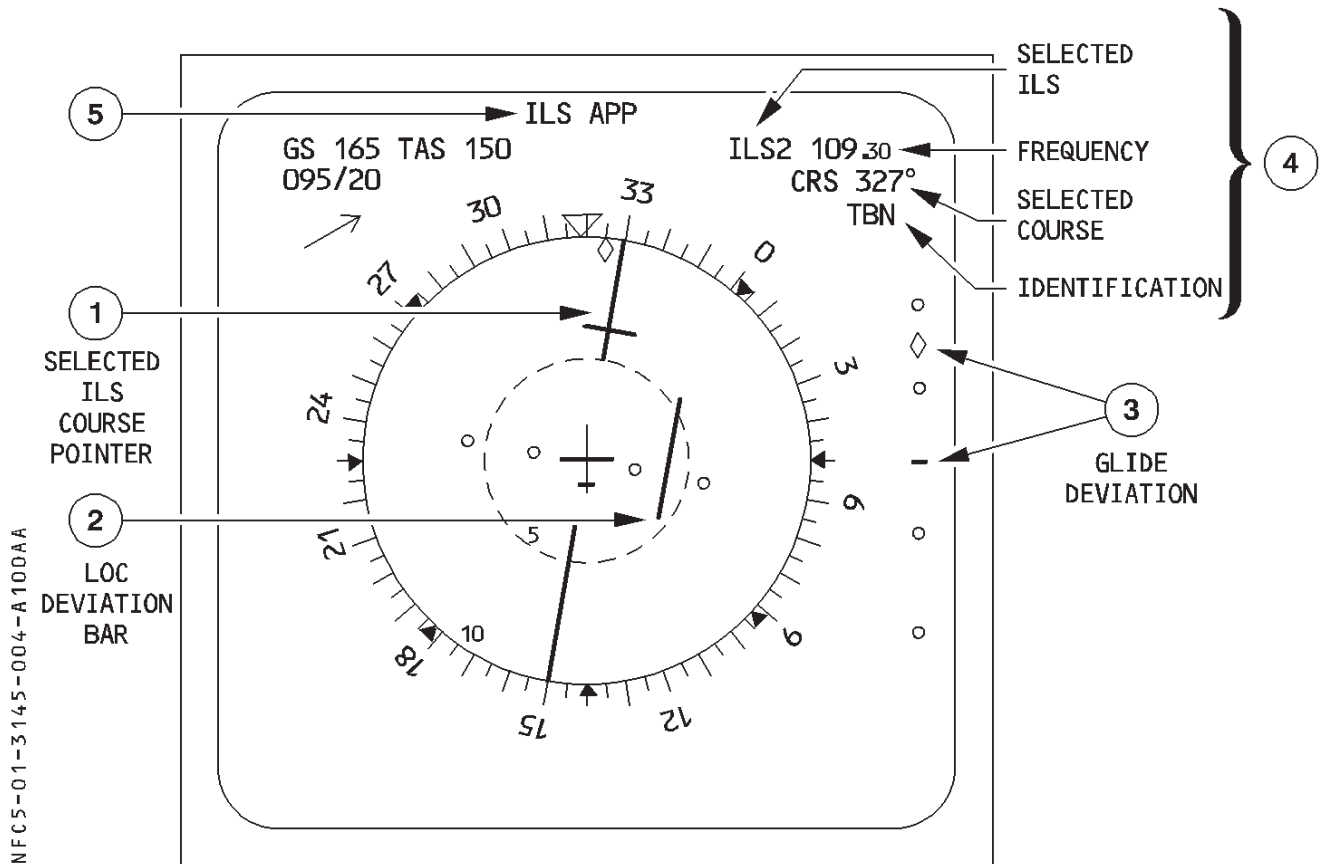
The indication is in minutes and seconds from 0 to 59' 59", and in hours and minutes from 1 H to 99 H 59'. (Seconds are not displayed beyond 59' 59").

⑩ Range marks

The range scale value selected on the EFIS control panel (10 to 320 NM) governs the scale of the ND.

ROSE LS MODE

R



① ILS course pointer (magenta)

This dagger-shaped symbol points to the selected ILS course. The ILS is selected either by the FMGC (autotuned or manually) or through the RMP in backup mode. If no course has been entered, the value defaults to 360°.

② Localizer deviation bar (magenta)

This bar moves laterally with respect to the course pointer. Its scale consists of two dots on each side of zero deviation. Each dot represents a deviation of about $\pm 0.8^\circ$. If the deviation becomes excessive (1/4 dot, 0.2°) above 15 feet RA, the bar and the scale pulse.

③ Glide deviation (magenta)

This diamond moves on a vertical scale consisting of two white dots on each side of the yellow reference line. Each dot represents a deviation of about $\pm 0.4^\circ$.

If the deviation becomes greater than one dot above 100 feet RA, the scale and the diamond flash.

④ Selected ILS information

This display shows the ILS frequency (magenta), selected course (blue), and identification (magenta).

⑤ ILS APP message (green)

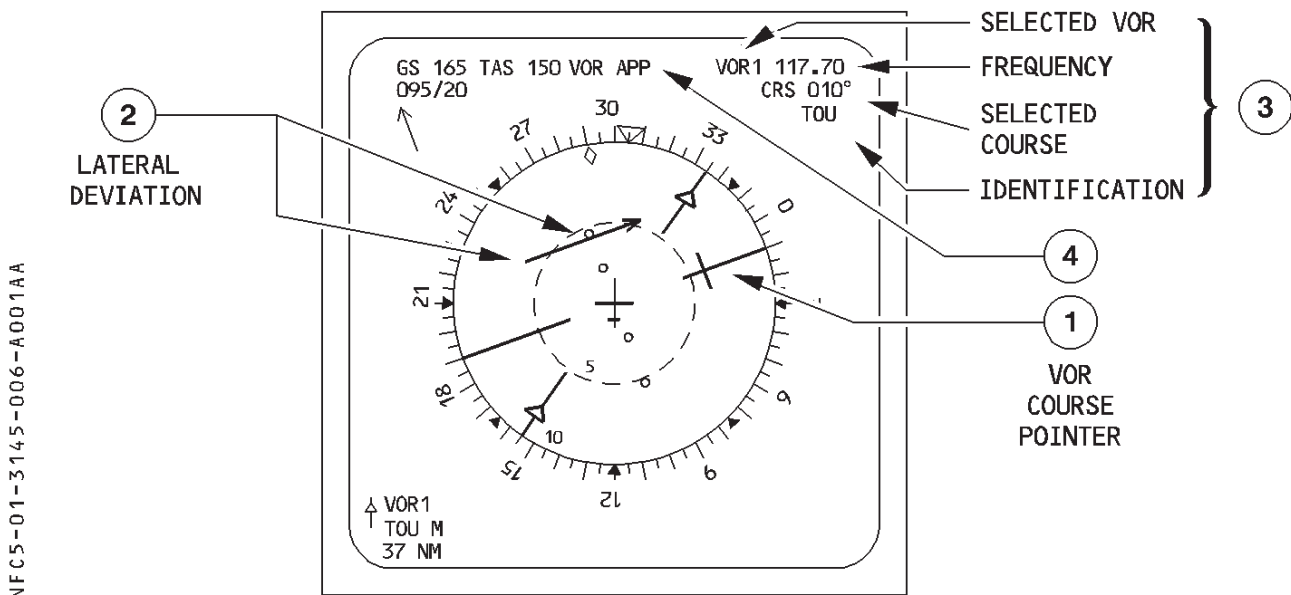
R This message appears when the flight crew has selected an ILS approach on the MCDU.

Note : *ILS1 information appears on PFD1 and ND2.*

ILS2 information appears on PFD2 and ND1.

ROSE VOR MODE

R



① VOR course pointer (blue)

This dagger-shaped symbol points to the selected VOR course.

The VOR course is automatically selected by the FMGC or manually by the crew using the MCDU pages or the RMP backup mode.

R

② Lateral deviation bar (blue)

This bar shows the VOR deviation on a lateral scale.

Each dot represents 5°. When the lateral deviation exceeds 10°, the bar remains displayed on the outer dot.

R

The arrow on the bar gives the TO/FROM indication.

③ VOR information (white)

This area displays the frequency of the selected VOR and its identification (if decoded by the receiver), the selected course.

④ VOR APP message (green)

This message appears when the flight crew has selected a VOR approach on the MCDU.

③ Glide deviation (magenta)

This diamond moves on a vertical scale consisting of two white dots on each side of the yellow reference line. Each dot represents a deviation of about $\pm 0.4^\circ$.

If the deviation becomes greater than one dot above 100 feet RA, the scale and the diamond flash.

④ Selected ILS information

This display shows the ILS frequency (magenta), selected course (blue), and identification (magenta).

⑤ ILS APP message (green)

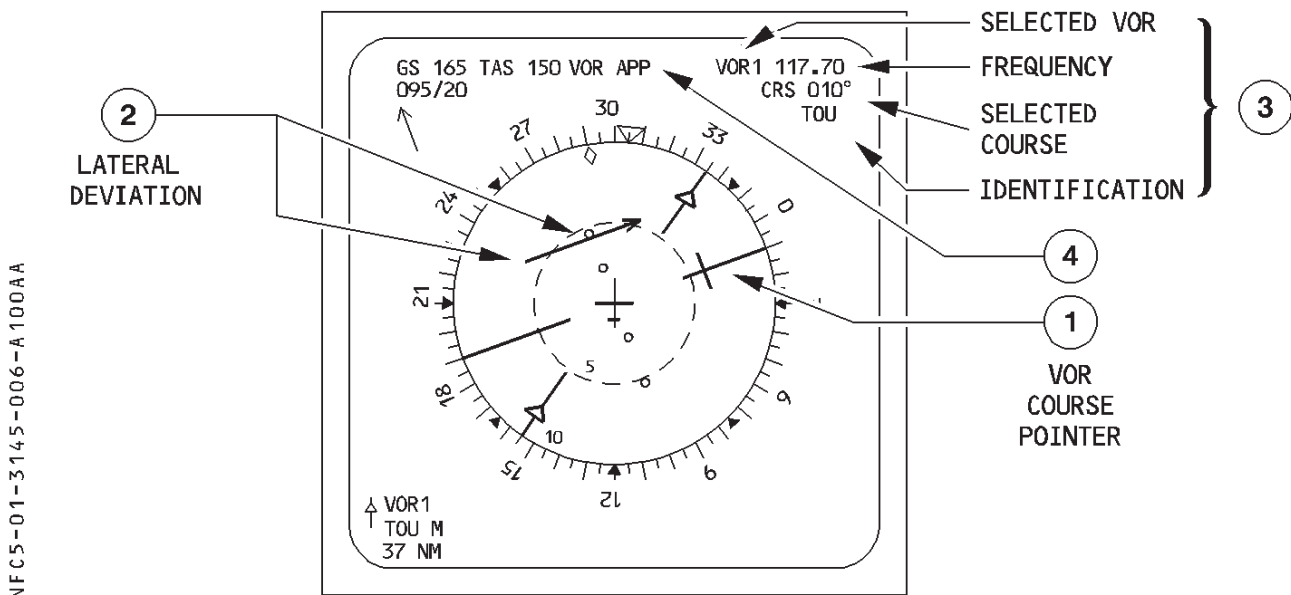
R This message appears when the flight crew has selected an ILS approach on the MCDU.

Note : *ILS1 information appears on PFD1 and ND2.*

ILS2 information appears on PFD2 and ND1.

ROSE VOR MODE

R



① VOR course pointer (blue)

This dagger-shaped symbol points to the selected VOR course.

The VOR course is automatically selected by the FMGC or manually by the crew using the MCDU pages or the RMP backup mode.

R

② Lateral deviation bar (blue)

This bar shows the VOR deviation on a lateral scale.

Each dot represents 5°. When the lateral deviation exceeds 10°, the bar remains displayed on the outer dot.

R

The arrow on the bar gives the TO/FROM indication.

③ VOR information (white)

This area displays the frequency of the selected VOR and its identification (if decoded by the receiver), the selected course.

④ VOR APP or GPS APP Messages (green)

VOR APPR appears when the flight crew has selected a VOR approach on the MCDU. GPS APP appears when the crew has selected a GPS approach.

③ Glide deviation (magenta)

This diamond moves on a vertical scale that has two white dots on each side of the yellow reference line. Each dot represents a deviation of approx. plus or minus 0.4 degrees.

If the deviation becomes greater than one dot above 100 feet RA, the scale and the diamond flash.

④ Selected ILS information

This display shows the ILS frequency (magenta), the selected course (blue), and the identification (magenta).

⑤ ILS APP message (green)

This message appears :

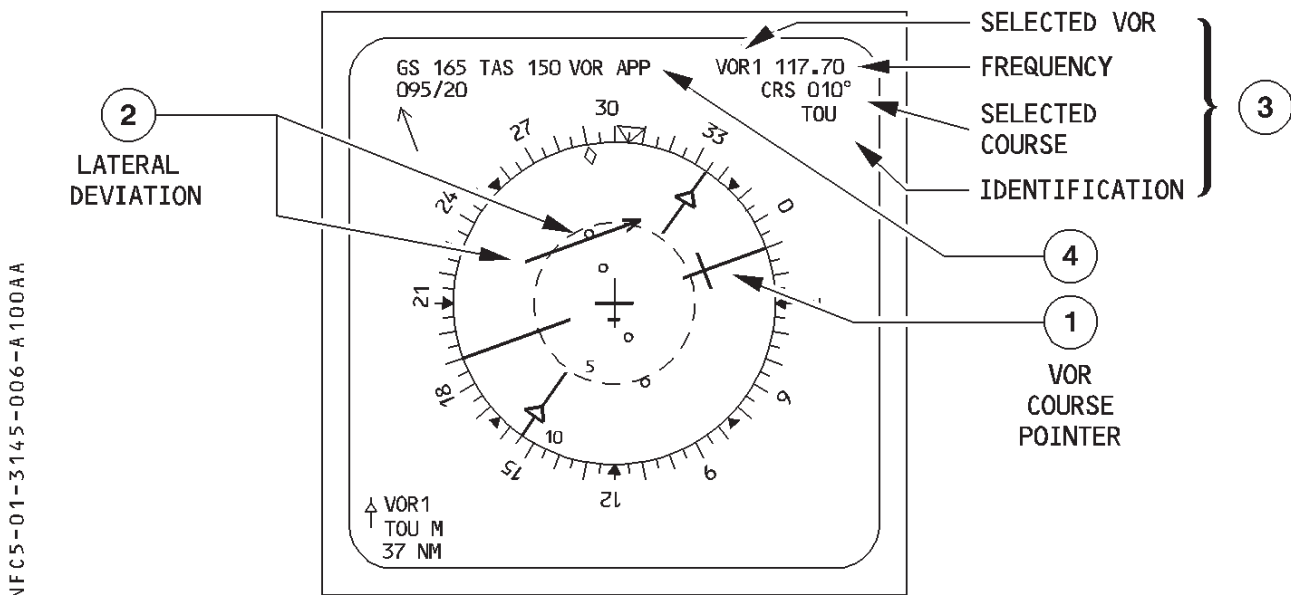
- When the flight crew selects an ILS approach on the MCDU, and
- When the FMS flight phase is DES, APP or GA, or the FMS phase is CRZ and the along track distance to destination is less than 250 NM.

Note : *ILS1 information appears on PFD1 and ND2.*

ILS2 information appears on PFD2 and ND1.

ROSE VOR MODE

R



① VOR course pointer (blue)

This dagger-shaped symbol points to the selected VOR course.

The VOR course is automatically selected by the FMGC or manually by the crew using the MCDU pages or the RMP backup mode.

R

② Lateral deviation bar (blue)

This bar shows the VOR deviation on a lateral scale.

Each dot represents 5°. When the lateral deviation exceeds 10°, the bar remains displayed on the outer dot.

R

The arrow on the bar gives the TO/FROM indication.

③ VOR information (white)

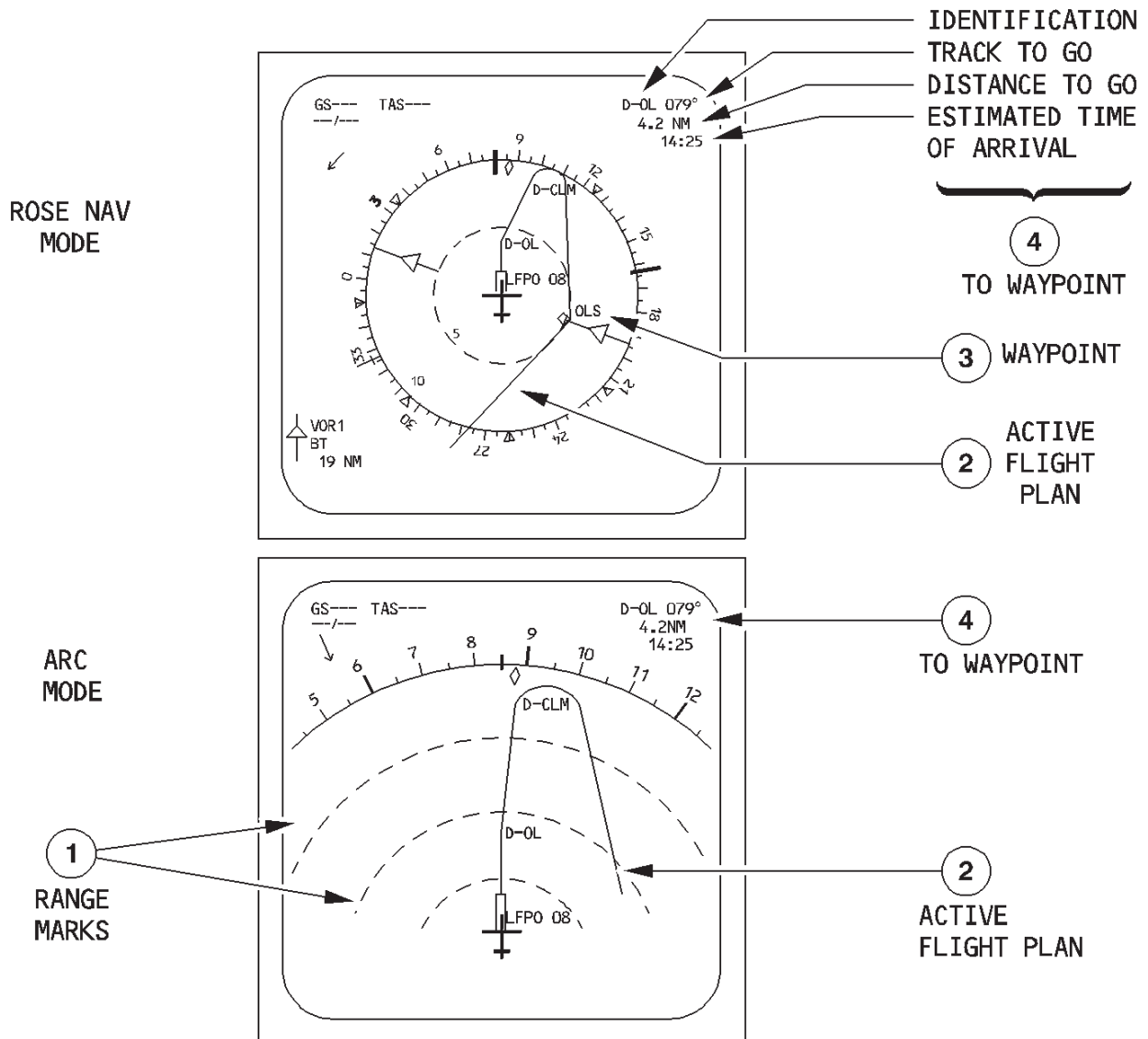
This area displays the frequency of the selected VOR and its identification (if decoded by the receiver), the selected course.

④ VOR APP or GPS APP Messages (green)

VOR APPR appears when the flight crew has selected a VOR approach on the MCDU. GPS APP appears when the crew has selected a GPS approach.

ROSE NAV MODE/ARC MODE

ROSE NAV and ARC modes give the pilot the same information, but ARC mode limits it to the forward 90° sector.



① Range Marks and Values

R

The values displayed on the ND are :

In ROSE NAV mode 1/4 of the selected range for the inner circle.

1/2 of the selected range for the heading scale circle.

In ARC mode 1/4 of the selected range for the first inner arc.

1/2 of the selected range for the second inner arc.

3/4 of the selected range for the third inner arc.

② Flight Plan

R

The crew can use the MCDU to select various types of flight plan :

- The active flight plan (the flight plan the aircraft is actually following when the NAV mode is engaged) is represented by a continuous green line. The ND shows only the part of the flight plan that is ahead of the aircraft, as well as the waypoints that are still to be overflown and the waypoint from which the aircraft is coming.

The ND does not show a SID or a STAR, except for the last waypoint of the SID and the first waypoint of the STAR, when the selected range is 160 or 320 NM.

If the primary flight plan is not active, it is represented by a dotted green line.

- A continuous blue line portrays the missed approach procedure, and a dashed blue line portrays the flight plan to the alternate.

The missed approach and the alternate flight plan are displayed when :

- In ARC or ROSE NAV mode, a missed approach waypoint or an alternate flight plan waypoint is displayed on the onside MCDU.
- In PLAN mode a missed approach or alternate waypoint is displayed in the 2L field of the onside MCDU.

- The secondary flight plan is represented by a continuous white line. The ND continues to display the active flight plan.

- Temporary flight plan

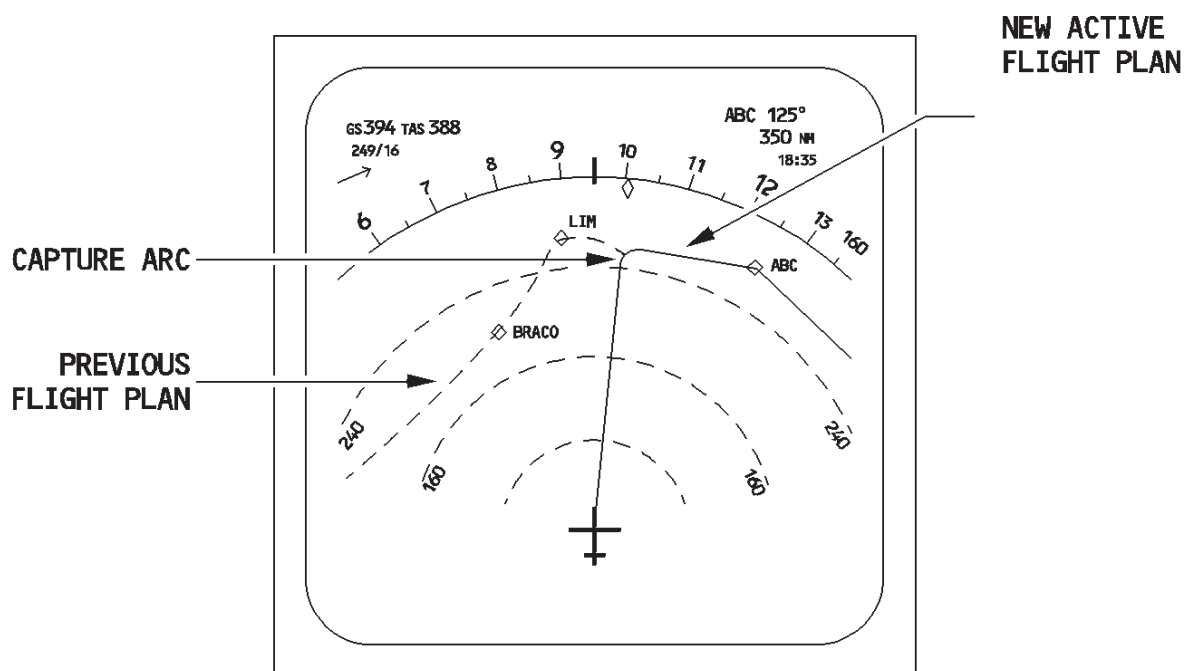
The revised portion of the flight plan is represented by a dotted yellow line.

- Flight plan capture

R

When the aircraft is off the primary flight plan and is flying toward it in HDG mode with the NAV mode armed, the ND shows the new active flight plan as a continuous green line if the FMGC has computed the intercept path.

The part of the flight plan before the interception point shows as a dotted green line.



NFC5-01-3145-008-A001AB

③ Waypoint

The ND can display various kinds of waypoints :

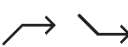
Flight plan waypoints

The ND displays these as green diamonds (white, for TO waypoints). When the flight crew selects the WPT option on his EFIS control panel, all waypoints other than flight plan waypoints are displayed in magenta.

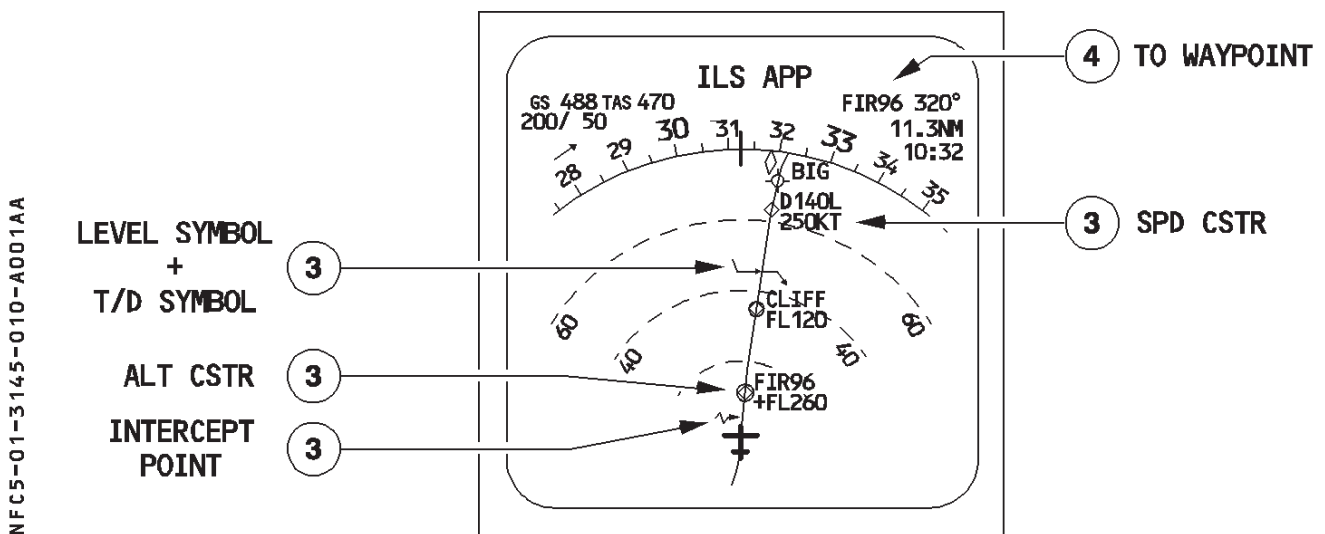
Pseudo waypoint

Point of the flight path where the aircraft is predicted to reach a selected altitude or speed.

R

Pseudo waypoint	Definition
	Level symbol (top of climb or level-off position), when the aircraft reaches : · The FCU-selected altitude (blue arrow), or · The constrained altitude, if it is more restrictive than the FCU altitude, and if appropriate modes are engaged (magenta) · It does not appear when the aircraft is within 100 feet above, or below, the selected altitude.
	Top of descent symbol, or continue descent symbol : · White, if DES is not armed · Blue, if DES is armed.
	Start of CLIMB symbol : · White, if CLB is not armed · Blue, if CLB is armed.
	Intercept point symbol : · White, if only the NAV mode is engaged · Blue, if DES mode is engaged · Indicates the point at which the aircraft is predicted to intercept the descent path, if there is any vertical deviation while the aircraft is in DES mode.
	Speed change symbol (magenta) : · Indicates the point at which the aircraft will start an automatic acceleration or deceleration from the current speed to a new computed speed for SPD LIM, SPD CSTR, or HOLDING SPD.
	Decelerate point symbol (magenta) : · Indicates where the aircraft will start an automatic deceleration toward VAPP (and thus switch to the approach phase). Although the symbol is always displayed, automatic deceleration only occurs if in managed speed, and NAV or approach mode is engaged.
	ALT CSTR symbol set around the constrained waypoint : · Magenta, when the ALT CSTR is predicted to be satisfied · Amber, when the ALT CSTR is predicted to be met · White, when the ALT CSTR is not taken into account by the FMGS, and NAV mode is engaged.
	Energy circle symbol (green arc) centered on the aircraft position and oriented to the current track line. Represents the Required Distance to Land. Only displayed if the lateral guidance mode is heading or track, and the current FMS flight phase is in cruise, descent or approach, and the aircraft is within 180 NM of the destination.

③ Waypoint (cont'd)

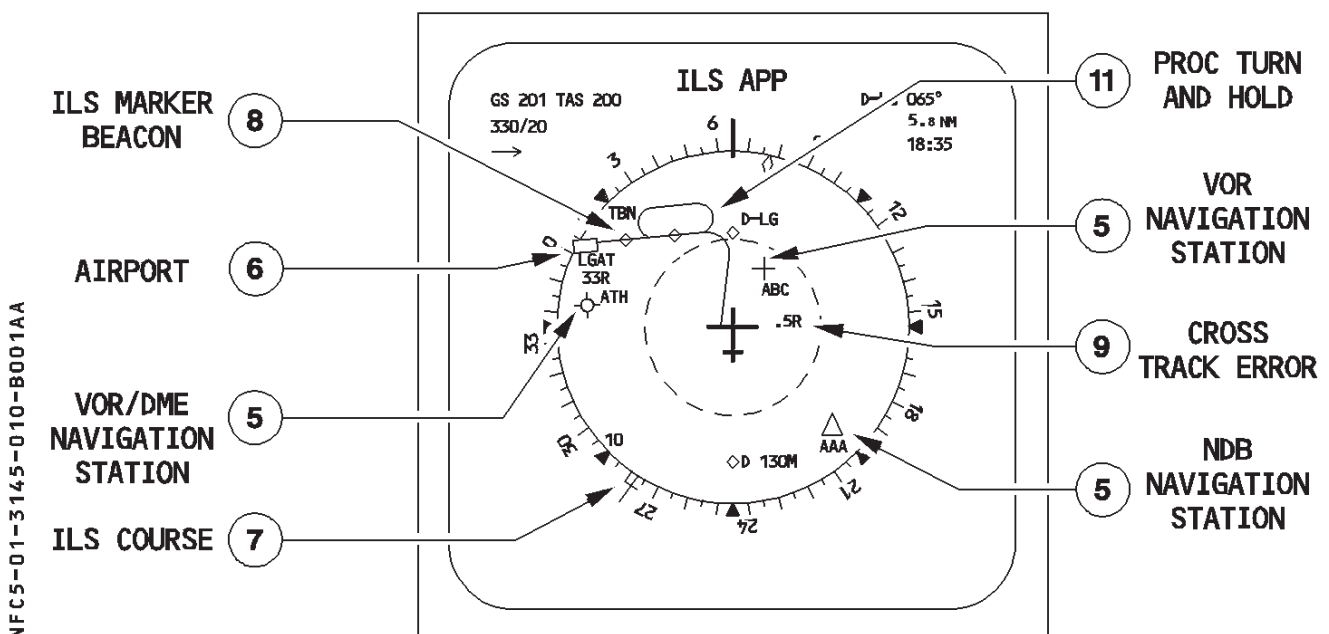


④ TO waypoint

This is the next waypoint to be overflown.

This area of the screen also shows :

- Waypoint identification (white).
- Track to go (green).
- Distance to go (green).
- Estimated time of arrival (green), assuming the aircraft will fly directly from its present position to the TO waypoint at the current ground speed.



③ Waypoint

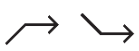
The ND can display various kinds of waypoints :

Flight plan waypoints

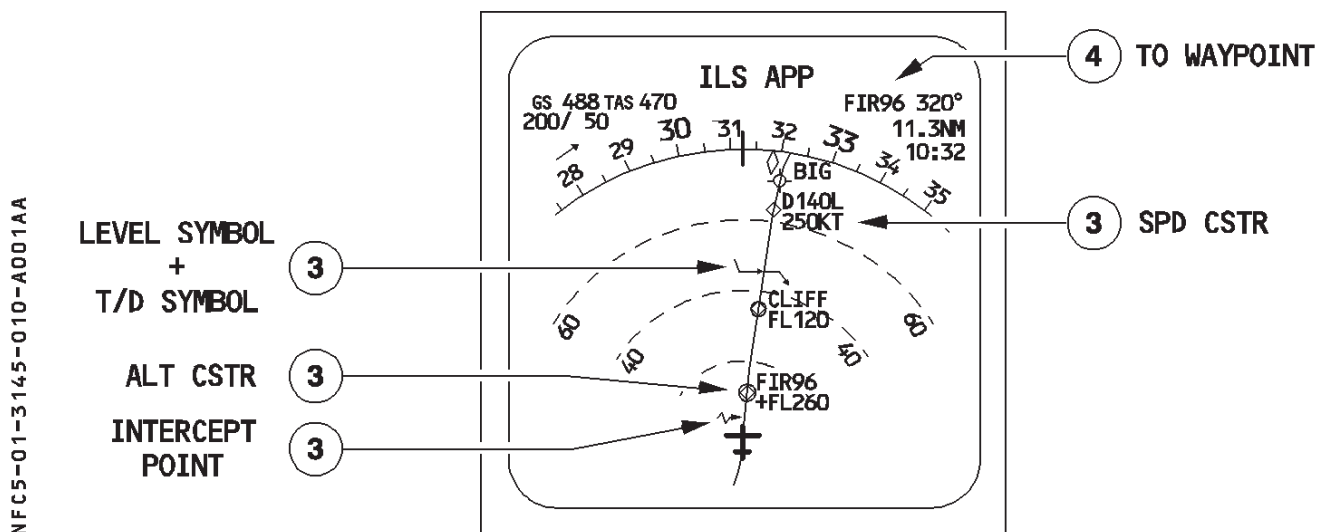
The ND displays these as green diamonds (white, for TO waypoints). When the flight crew selects the WPT option on his EFIS control panel, all waypoints other than flight plan waypoints are displayed in magenta.

Pseudo waypoint

Point of the flight path where the aircraft is predicted to reach a selected altitude or speed.

Pseudo waypoint	Definition
	Level symbol (top of climb or level-off position), when the aircraft reaches : · The FCU-selected altitude (blue arrow), or · The constrained altitude, if it is more restrictive than the FCU altitude, and if appropriate modes are engaged (magenta) · It does not appear when the aircraft is within 100 feet above, or below, the selected altitude.
	Top of descent symbol, or continue descent symbol : · White, if DES is not armed · Blue, if DES is armed.
	Start of CLIMB symbol : · White, if CLB is not armed · Blue, if CLB is armed.
	Intercept point symbol : · White, if only the NAV mode is engaged · Blue, if DES mode is engaged · Indicates the point at which the aircraft is predicted to intercept the descent path, if there is any vertical deviation while the aircraft is in DES mode.
	Speed change symbol (magenta) : · Indicates the point at which the aircraft will start an automatic acceleration or deceleration from the current speed to a new computed speed for SPD LIM, SPD CSTR, or HOLDING SPD.
	Decelerate point symbol : · Indicates the point at which the aircraft is predicted to decelerate for approach (and thus switch to the approach phase) · Magenta, if in managed speed and NAV or approach mode is engaged · White, if in selected speed or HDG/TRK mode · Automatic decelerations only occur when displayed in magenta.
	ALT CSTR symbol set around the constrained waypoint : · Magenta, when the ALT CSTR is predicted to be met · Amber, when the ALT CSTR is predicted to be missed · White, when the ALT CSTR is not taken into account by the FMGS, and NAV mode is engaged.
	Energy circle symbol (green arc) centered on the aircraft position and oriented to the current track line. Represents the Required Distance to Land. Only displayed if the lateral guidance mode is heading or track, and the current FMS flight phase is in cruise, descent or approach, and the aircraft is within 180 NM of the destination.

③ Waypoint (cont'd)

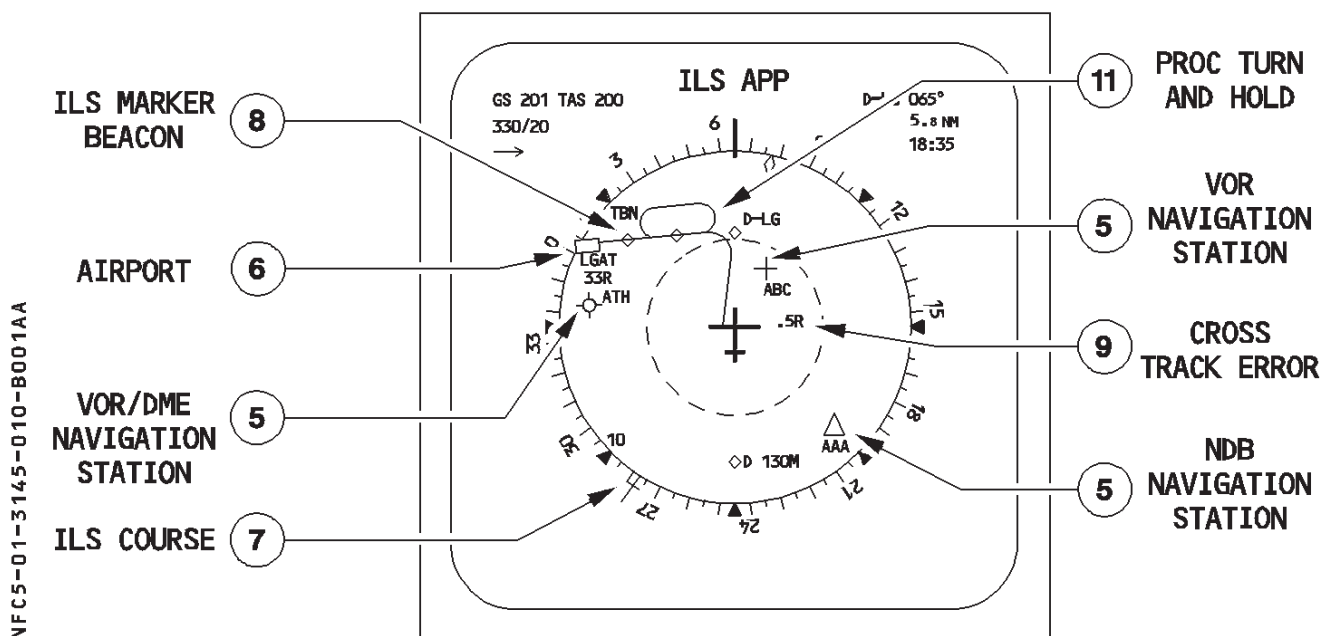


④ TO waypoint

This is the next waypoint to be overflown.

This area of the screen also shows :

- Waypoint identification (white).
- Track to go (green).
- Distance to go (green).
- Estimated time of arrival (green), assuming the aircraft will fly directly from its present position to the TO waypoint at the current ground speed.



⑤ Nav aids

The display uses specific symbols for nav aids :

○ DME or TACAN

+ VOR

⊙ VOR/DME

△ NDB

The symbol appears :

- In green if the nav aid is a current waypoint of the flight plan.
- In white if it is the TO waypoint.
- In blue when the nav aid is tuned for display either automatically by the FMGC or manually through the MCDU.
- In magenta when the nav aid is not part of the flight plan and is called for display as an option (corresponding option pushbutton pressed on the FCU EFIS control panel).

⑥ Airport

Airport included in the flight plan :

- If the runway is not specified, the airport is represented by a star and the identification is displayed in white.

Example : * LSGG

- If the runway is specified, it is represented by an oriented runway symbol in white.



The runway is drawn to scale (paved length) if the selected range is 10, 20 or 40 NM.

Optional airport information

The airports that are not displayed as part of the flight plan may be called for display (ARPT pushbutton on the EFIS control panel).

They are represented by a star and the identification in magenta.

⑦ ILS Course (Magenta)

When the pilot pushes the ILS pushbutton switch on the EFIS control panel, and if an ILS station has been selected, the display shows an ILS course symbol.

⑧ ILS Marker Beacons

The screen shows these as waypoints (diamonds).

When the aircraft overflies a marker beacon, the corresponding symbol flashes :

Blue for the outer marker.

Amber for the middle marker.

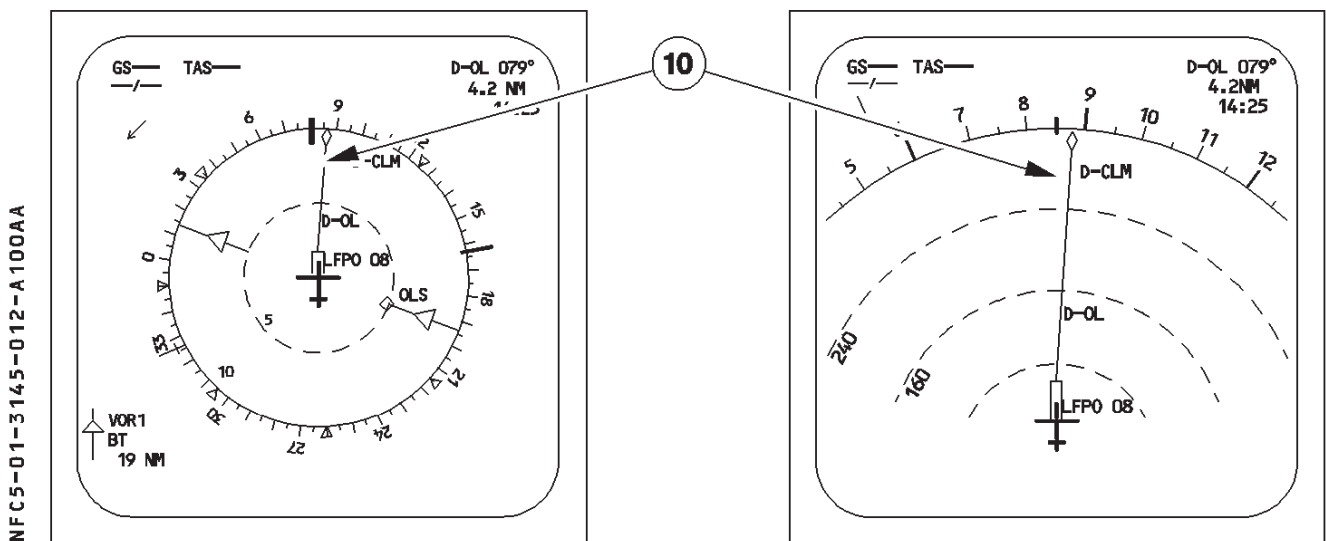
White for the inner marker.

9 Cross Track Error

This is the aircraft's lateral deviation from the active leg of the flight plan (related to the great circle route). It is indicated in nautical miles (NM), with the letter R (right) or L (left), according to the position of the aircraft with respect to the flight plan.

10 Track line

This line appears in green only in the ROSE NAV or ARC mode when HDG or TRK has been selected on the FCU.



⑤ Nav aids

The display uses specific symbols for nav aids :

○ DME or TACAN

+ VOR

⊙ VOR/DME

△ NDB

The symbol appears :

- In green if the nav aid is a current waypoint of the flight plan.
- In white if it is the TO waypoint.
- In blue when the nav aid is tuned for display either automatically by the FMGC or manually through the MCDU.
- In magenta when the nav aid is not part of the flight plan and is called for display as an option (corresponding option pushbutton pressed on the FCU EFIS control panel).

⑥ Airport

Airport included in the flight plan :

- If the runway is not specified, the airport is represented by a star and the identification is displayed in white.

Example : * LSGG

- If the runway is specified, it is represented by an oriented runway symbol in white.



The runway is drawn to scale (paved length) if the selected range is 10, 20 or 40 NM.

Optional airport information

The airports that are not displayed as part of the flight plan may be called for display (ARPT pushbutton on the EFIS control panel).

They are represented by a star and the identification in magenta.

⑦ ILS Course (Magenta)

When the pilot pushes the LS pushbutton switch on the EFIS control panel, and if an ILS station has been selected, the display shows an ILS course symbol.

⑧ ILS Marker Beacons

The screen shows these as waypoints (diamonds).

When the aircraft overflies a marker beacon, the corresponding symbol flashes :

Blue for the outer marker.

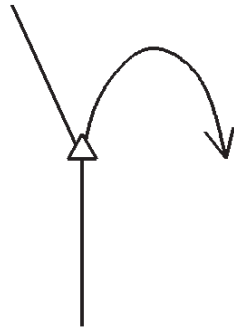
Amber for the middle marker.

White for the inner marker.

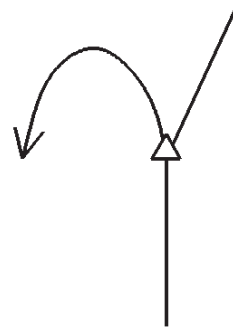
11 Procedure turns and holding patterns

These appear only when they are part of the flight plan. For the 160 and 320 NM range scales, each one is represented by a white arrow that originates at the associated fix and indicates the direction of the turn.

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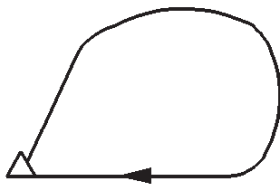


OR

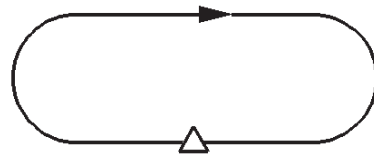


For shorter range scales and if the procedure turn or the holding pattern is in the next or the active leg, the display shows the full circuit or pattern.

NFC5-01-3145-013-B001AA



OR



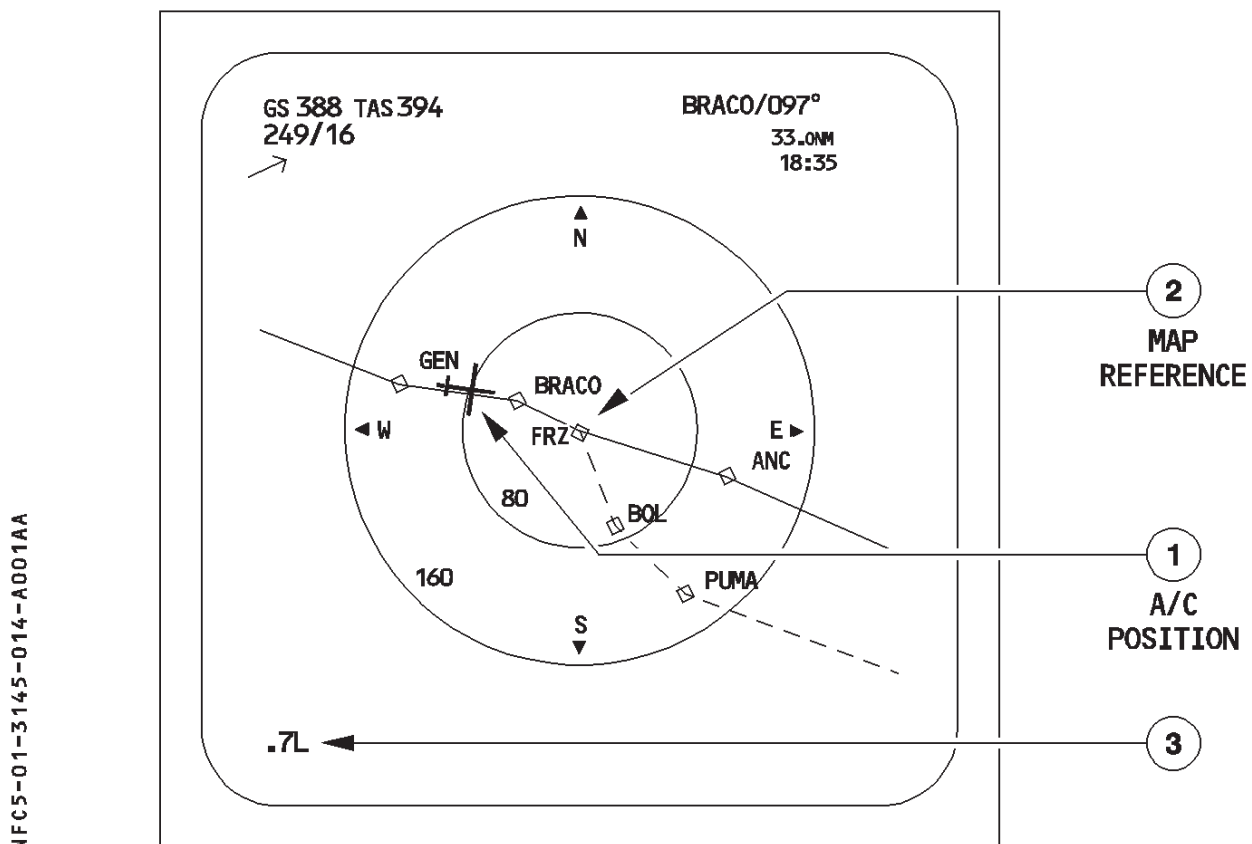
PLAN MODE

This mode displays statically the flight plan legs on a map oriented to true north. The map is centered on a map reference point that the pilot chooses by slewing to it on his MCDU. The map reference point is the waypoint displayed on the second line of the MCDU F-PLN page. It can be either the active waypoint (next waypoint to be overflown) or any other waypoint of the flight plan.

The pilot can slew the overall flight plan and display it in PLAN mode.

The pilot chooses the scale of the map with the range selector (the diameter of the outer circle corresponds to the selected range).

R Data on nav aids and their characteristics and associated bearing pointers are not available
R in this mode.



1 Aircraft Position and True Track

The orientation of the yellow aircraft symbol always indicates the true track of the aircraft. Its position represents the aircraft position given by the FMGS.

2 Map Reference Point

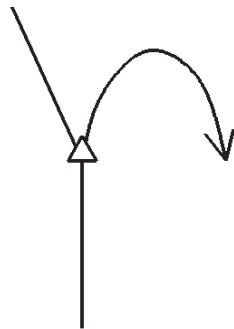
3 Cross Track Error

See ROSE NAV MODE/ARC MODE chapter.

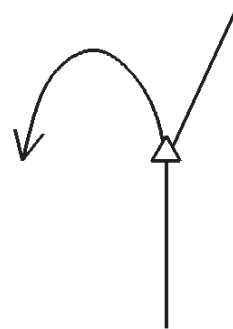
11 Procedure turns and holding patterns

These appear only when they are part of the flight plan. For the 160 and 320 NM range scales, each one is represented by a white arrow that originates at the associated fix and indicates the direction of the turn.

NFC5-01-3145-013-A001AA

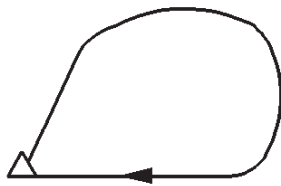


OR

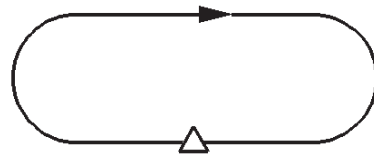


For shorter range scales and if the procedure turn or the holding pattern is in the next or the active leg, the display shows the full circuit or pattern.

NFC5-01-3145-013-B001AA

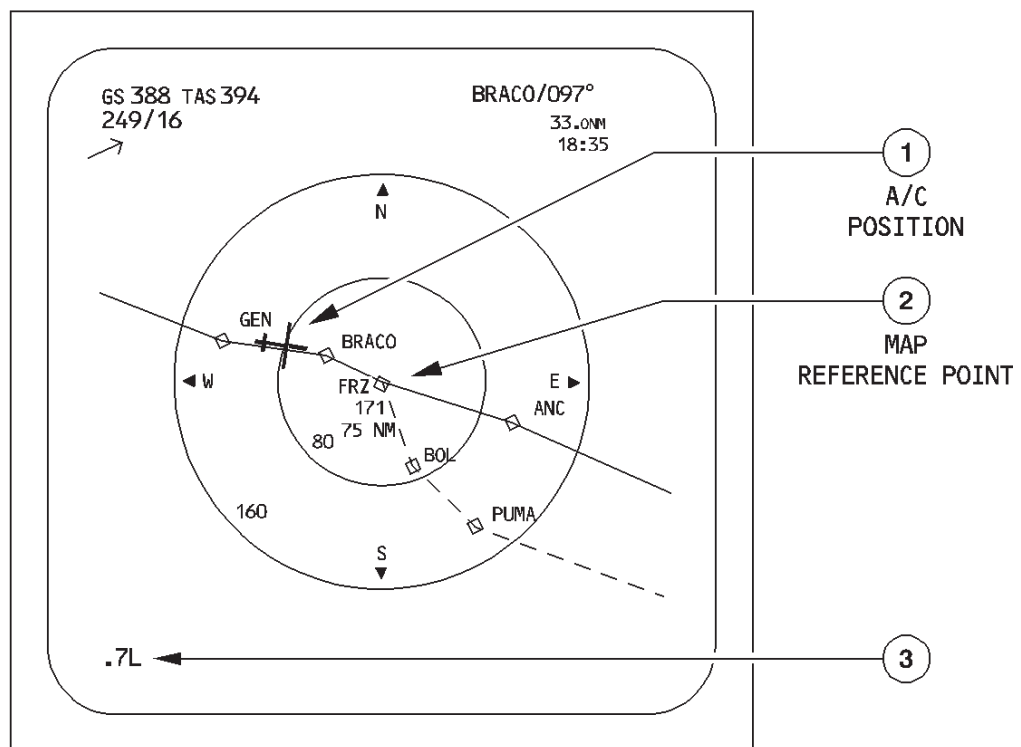


OR



NFC5-01-3145-014-A100AA

Data on nav aids and on their characteristics and associated bearing pointers are not available in this mode.



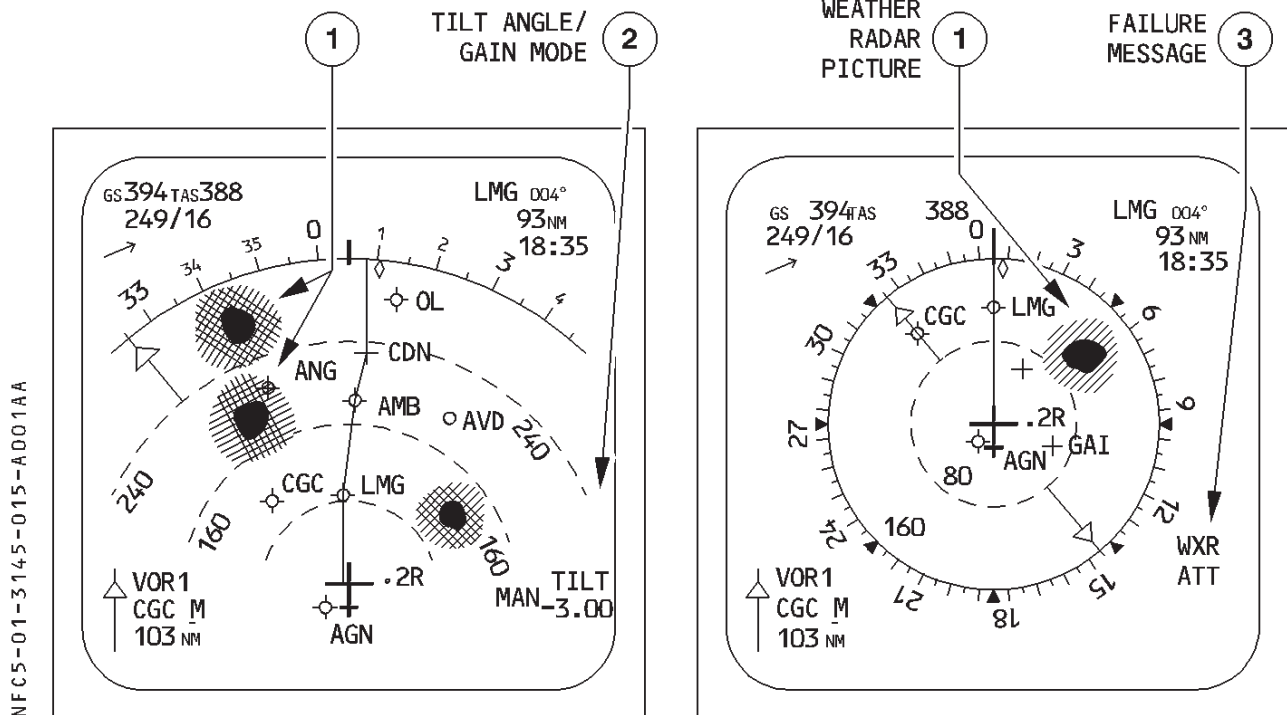
The orientation of the yellow aircraft symbol always indicates the true track of the aircraft. Its position represents the aircraft position given by the FMGS.

If the CSTR option is not selected, the track and distance from the map reference point to the next F-PLN waypoint is displayed in magenta.

Refer to the ROSE NAV MODE/ARC MODE chapter.

WEATHER RADAR

R



R ① Weather Radar Picture

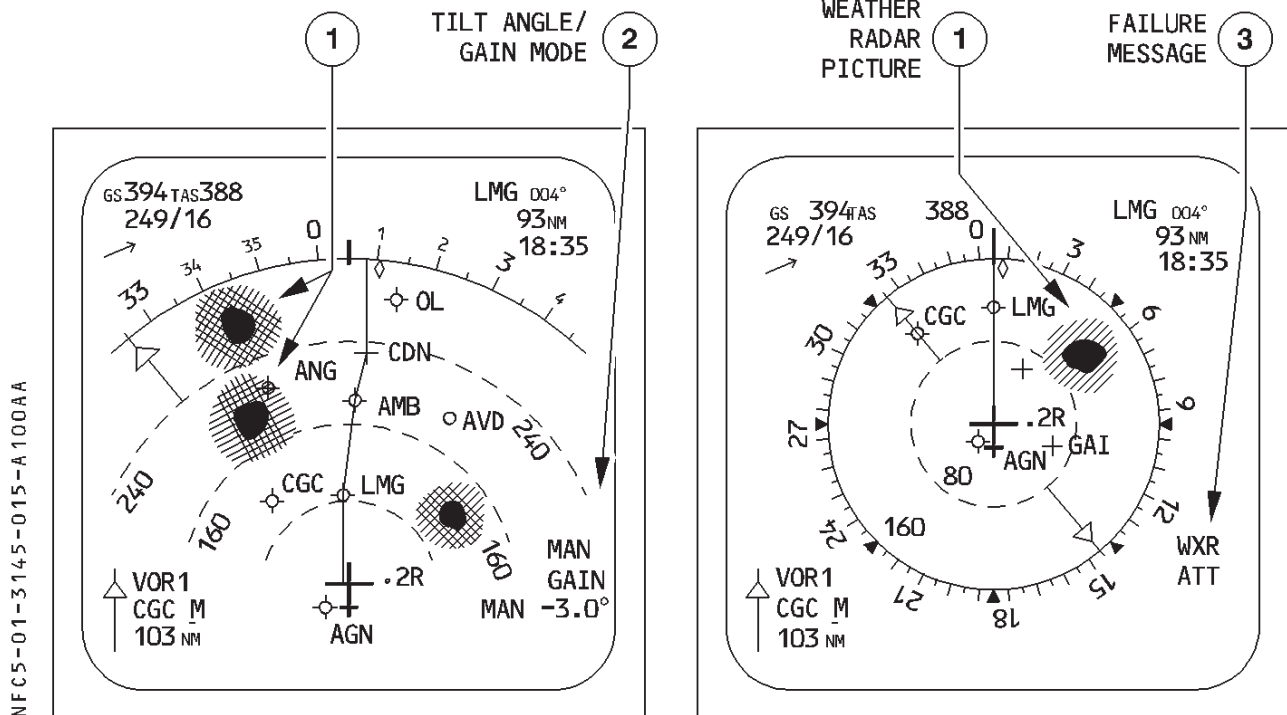
- R — When the radar is operating, and when the ND is not in PLAN mode, the ND displays the weather radar picture.
- R — The echoes appear in different colors, depending on the precipitation rates (black, green, yellow, red or magenta).
- R — The selected ND range will determine how often the image is refreshed.

R ② Tilt Angle and Gain Mode

- R — The value of the tilt angle is in degrees, and quarters of a degree. It appears in blue in the lower right-hand corner of the screen. This angle is the angle between the horizon and the radar beam axis.
- R — "MAN" appears in white, when the manual gain mode is selected.

WEATHER RADAR

R



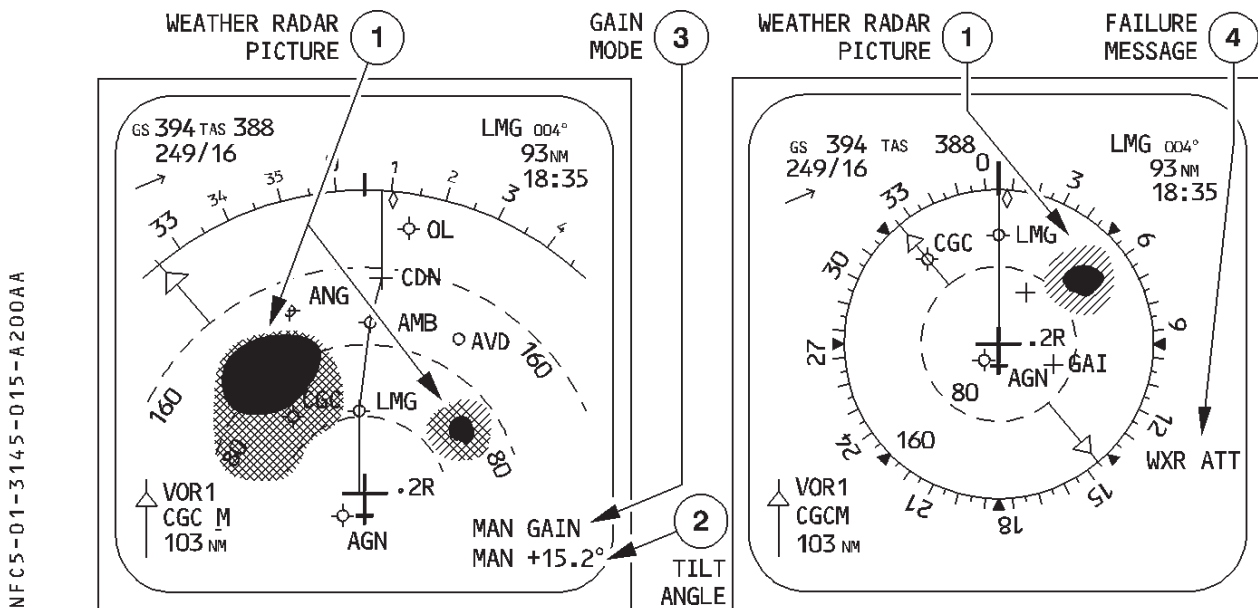
R ① Weather Radar Picture

- R — When the radar is operating, and when the ND is not in PLAN mode, the ND displays the weather radar picture.
- R — The echoes appear in different colors, depending on the precipitation rates (black, green, yellow, red or magenta).
- R — The selected ND range will determine how often the image is refreshed.

R ② Tilt Angle and Gain Mode

- R — The value of the tilt angle is in degrees, and quarters of a degree. It appears in blue in the lower right-hand corner of the screen along with MAN indication. This angle is the angle between the horizon and the radar beam axis.
- R — "MAN GAIN" appears in white, when the manual gain mode is selected.

WEATHER RADAR



① Weather Radar Picture

- When the radar is operating, and when the ND is not in PLAN mode, the ND displays the weather radar picture.
- The echoes appear in different colors, depending on the precipitation rates (black, green, yellow, red or magenta).
- The selected ND range will determine how often the image is refreshed.

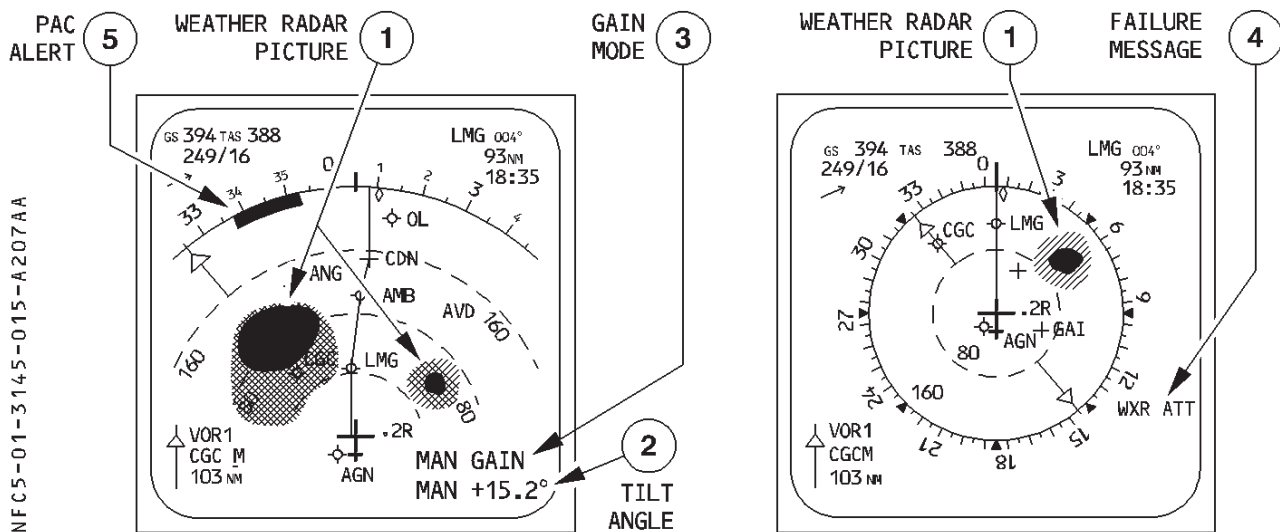
R ② Tilt Angle

- The auto-tilt function of the weather radar automatically sets the tilt angle to optimize weather radar detection.
- The value of the tilt angle is in degrees, and quarters of a degree. It appears in the lower right corner of the screen :
 - In green, when the flight crew sets the TILT toggle switch to AUTO
 - In blue, next to the blue "MAN" indication, when the flight crew sets the TILT toggle switch to MAN.
- The tilt angle is the angle between the horizon and the radar beam axis.
- When the auto-tilt function is lost, the tilt value is dashed, and the "NO AUTO TILT" message appears in amber on the ND, until the flight crew sets the TILT toggle switch to MAN.

R ③ Gain Mode

- "MAN GAIN" appears in white, when the manual gain mode is selected.

WEATHER RADAR



① Weather Radar Picture

- When the radar is operating, and when the ND is not in PLAN mode, the ND displays the weather radar picture.
- The echoes appear in different colors, depending on the precipitation rates (black, green, yellow, red or magenta).
- The selected ND range will determine how often the image is refreshed.

② Tilt Angle

- The multiscan function of the weather radar alternatively scans at low and high beam, and automatically sets the tilt of these beams to optimize the weather radar detection. The displayed weather radar picture is the result of the storing and merging of the information from each beam.
- The value of the tilt angle is in degrees, and quarters of a degree. It appears in the lower right corner of the screen :
 - In green, when the MULTISCAN toggle switch is set to AUTO. This value represents the average of the lower and the upper beam tilts.
 - In blue, next to the blue “MAN” indication, when the flight crew sets the MULTISCAN toggle switch to MAN.

The tilt angle is the angle between the horizon and the radar beam axis.
- When the multiscan function is lost, the tilt value is dashed and the “NO AUTO TILT” message appears in amber on the ND, until the flight crew sets the MULTISCAN toggle switch to MAN.

③ Gain Mode

- “MAN GAIN” appears in white, when the flight crew selects the manual gain mode.

R ③ Failure Messages

R The ND lists the detected failures.

R If the message is in “red”, the ND does not display a radar image.

R If the message is in “amber”, the image is not affected.

R WXR RT (red) : Radar transceiver failure.

R WXR ANT (red) : Radar antenna failure.

R WXR CTL (red) : Radar control unit failure.

R WXR RNG (red) : Range error.

R WXR WEAK (amber) : Calibration failure.

R WXR ATT (amber) : Attitude control failure.

R WXR STAB (amber) : Antenna stabilization failure.

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R ④ Failure Messages

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WXR RNG (red) : Range error.

WXR WEAK (amber) : Calibration failure.

WXR ATT (amber) : Attitude control failure.

WXR STAB (amber) : Antenna stabilization failure.

R ⑤ PAC alert

R When the flight crew sets the weather radar mode to WX or WX + T, and sets the
R gain to CAL, and when the aircraft is within 80 NM of a thunderstorm, the Path
R Attenuation Compensation (PAC) alert is available.

R The PAC alert displays a yellow arc on the outermost scale of the ND, when an
R intervening rainfall creates an attenuated area behind a thunderstorm (also called a
R radar shadow).

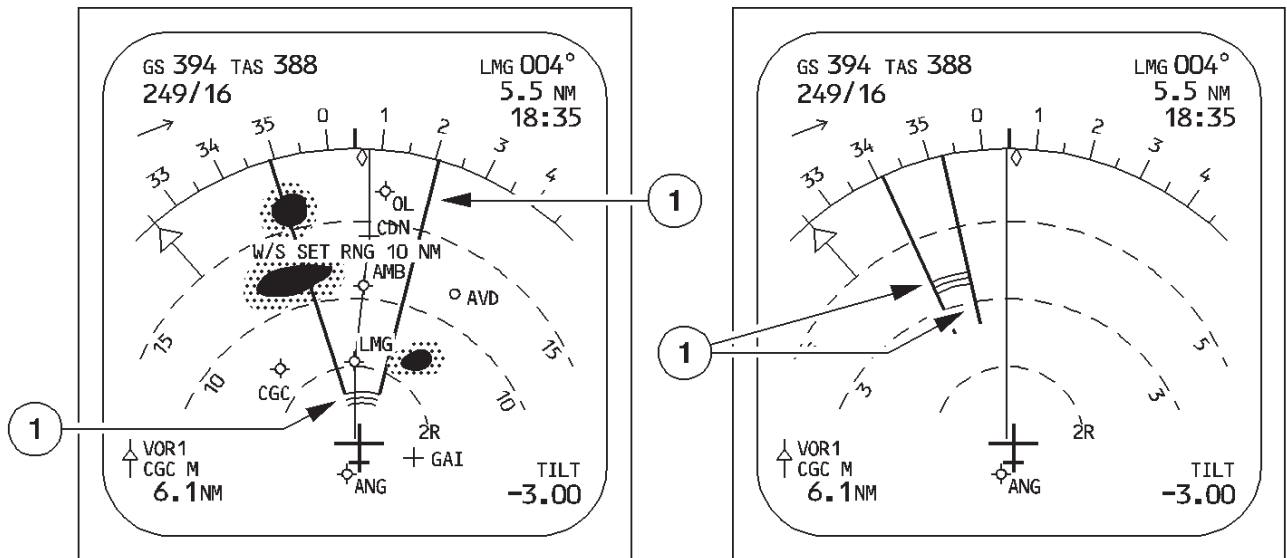
R Note : The PAC alert is available during both automatic and manual operation
R (MULTISCAN toggle switch either in AUTO or MAN position).

INTENTIONALLY LEFT BLANK

PREDICTIVE WINDSHEAR SYSTEM

R

NFC5-01-3145-016-A100AA



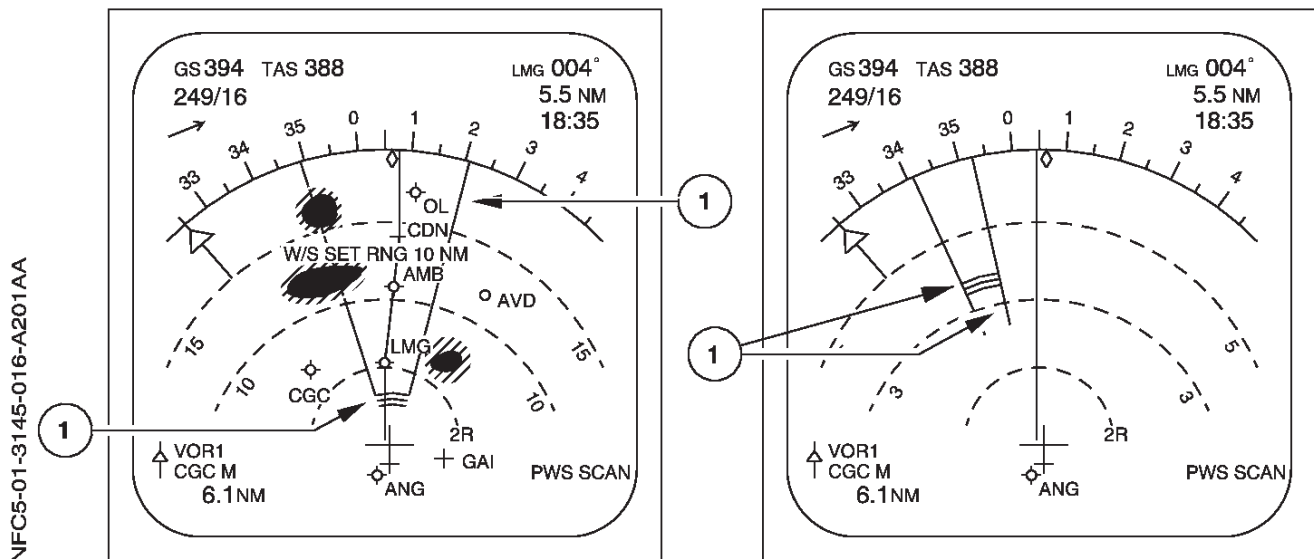
① Predictive windshear area indication

The predicted windshear area is indicated by a red and black icon and two yellow radial lines. Windshear information is available in ARC and ROSE ND modes.

When the ND range is set above 10 NM, a W/S SET RNG 10 NM (Windshear, set range 10 NM) message appears, requesting the crew to adjust the ND range. It is displayed even if the weather radar is switched off, provided the WINDSHEAR switch on the weather radar panel is set to AUTO.

Depending on the windshear alert level, ND indication may be completed with a PFD message (Refer to 1.31.40).

PREDICTIVE WINDSHEAR SYSTEM



1 Predictive windshear area indication

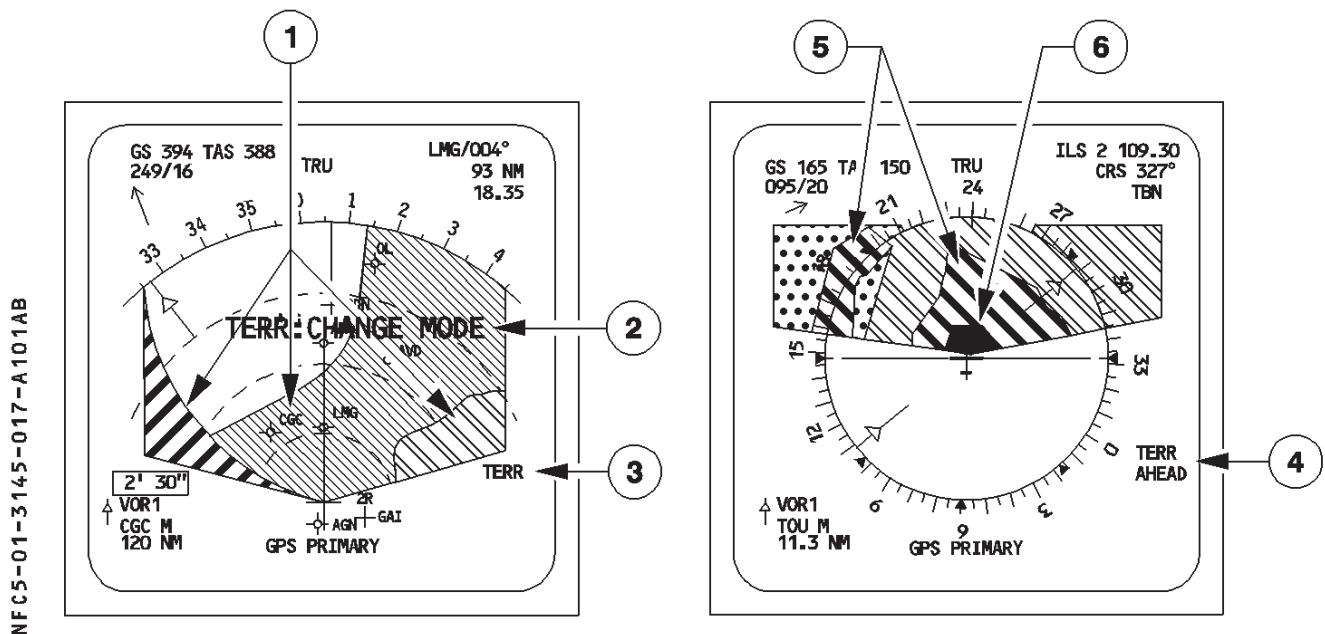
A PWS SCAN message is displayed when the Predictive Windshear system is active on the ND.

The predicted windshear area is indicated by a red and black icon and two yellow radial lines. Windshear information is available in ARC and ROSE ND modes.

When the ND range is set above 10 NM, a W/S SET RNG 10 NM (Windshear, set range 10 NM) message appears, requesting the crew to adjust the ND range. It is displayed even if the weather radar is switched off, provided the WINDSHEAR switch on the weather radar panel is set to AUTO.

Depending on the windshear alert level, ND indication may be completed with a PFD message (Refer to 1.31.40).

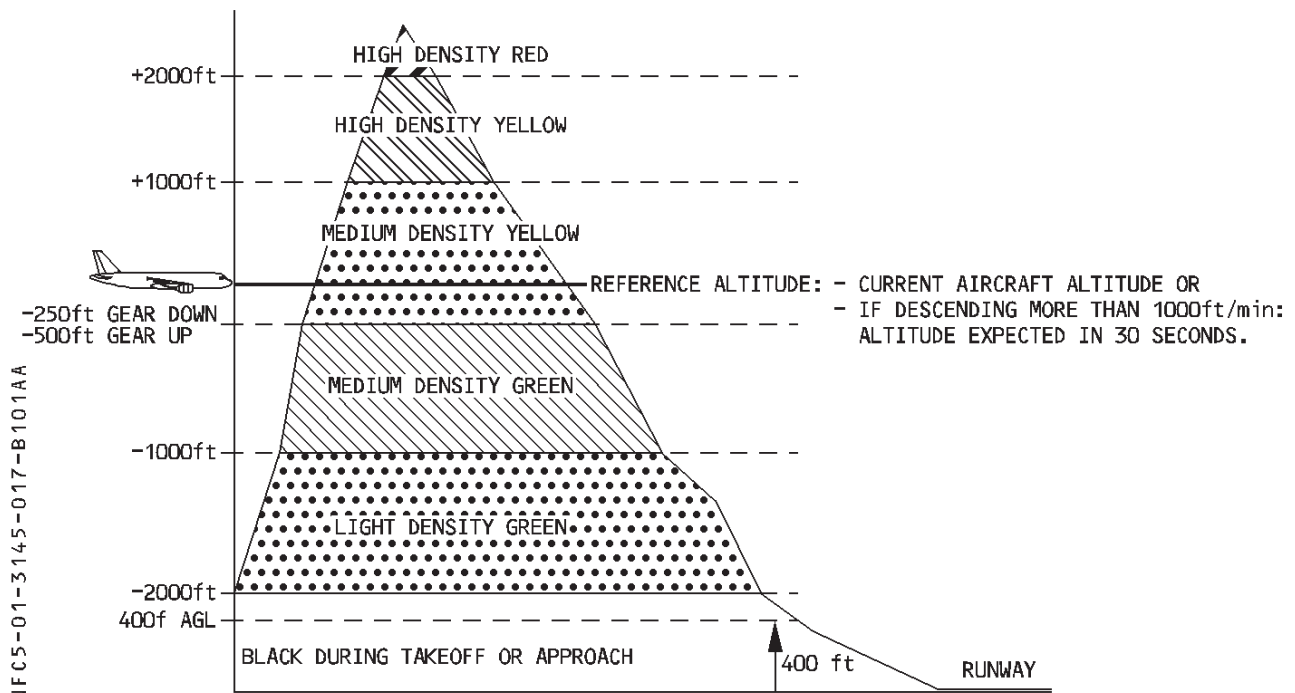
EGPWS



1 EGPWS terrain picture

The ND presents the EGPWS terrain picture, when the TERR ON ND switch is selected ON, and the ND is not in PLAN mode. The terrain picture replaces the weather radar image.

The terrain appears in different colors and densities, according to its relative height :



R Note : Areas without available terrain data in the EGPWS database appear in magenta.

R ② Center Part Messages

- R — The “TERR CHANGE MODE” indication is displayed in red (or amber), in the case of
- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the current selected
- R display mode is PLAN.
- R — The “TERR REDUCE RANGE” indication is displayed in red (or amber), in the case of
- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the selected range is
- R 160 NM or 320 NM.

③ TERR indication

To differentiate between the terrain and the weather display, the weather radar TILT is replaced by a blue TERR, and the terrain display sweeps from the center outward to both ND sides.

④ Warning and caution messages

TERR AHEAD (amber) : For a caution.

TERR AHEAD (red) : For a warning.

When triggered, these messages flash for 9 seconds, then remain steady until the caution or warning alert condition disappears.

TERR RNG (red) : For a RANGE error warning.

TERR TST (amber) : Appears during the EGPWS test, when the terrain pattern is displayed, and there is no failure.

⑤ Terrain caution alert

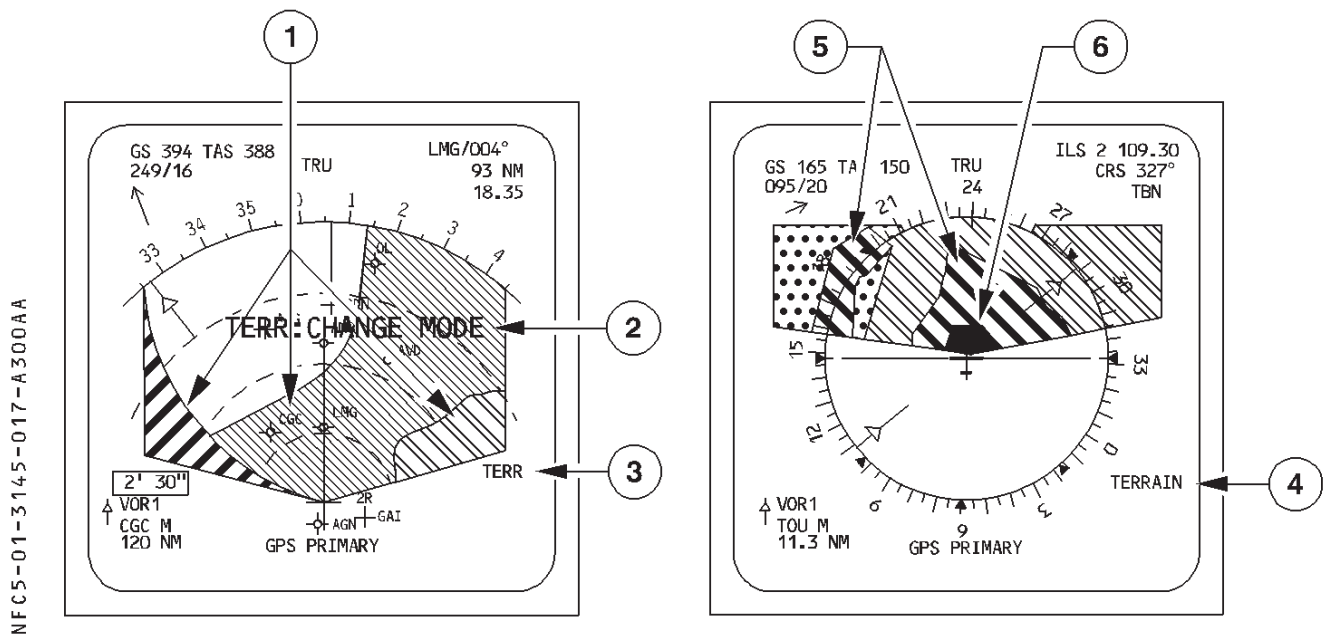
Generated when a conflict exists between the terrain caution envelope, ahead of the aircraft, and the terrain data stored in the database. The conflict area is shown in solid yellow.

⑥ Terrain warning alert

Generated when a conflict exists between the terrain warning envelope, ahead of the aircraft, and the terrain data stored in the database. The conflict area is shown in solid red.

Note : When an alert is generated (either caution or warning) and TERR ON ND is not selected, the terrain is automatically displayed and the TERR ON ND's pushbutton ON light comes on.

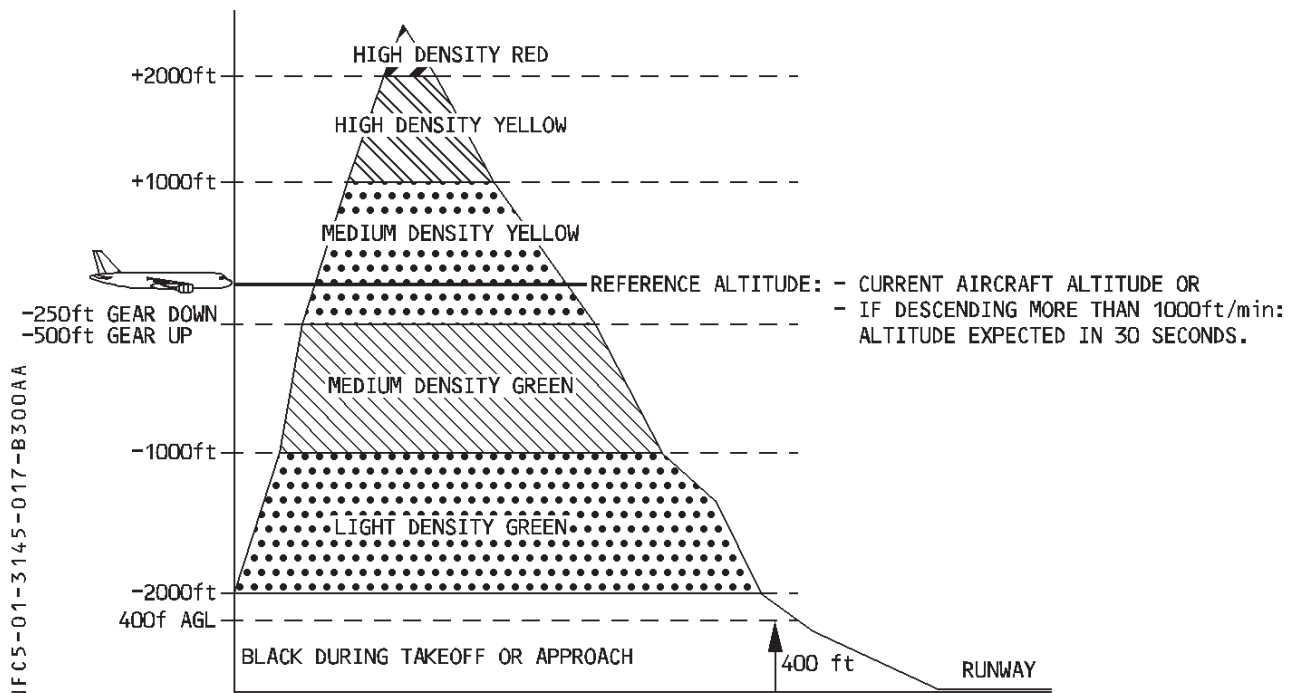
EGPWS



① EGPWS terrain picture

The ND presents the EGPWS terrain picture, when the TERR ON ND switch is selected ON, and the ND is not in PLAN mode. The terrain picture replaces the weather radar image.

The terrain appears in different colors and densities, depending on its relative height :



Note : Areas without available terrain data in the EGPWS database appear in magenta.

R ② Center Part Messages

- R — The “TERR CHANGE MODE” indication is displayed in red (or amber), in the case of
- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the current selected
- R display mode is PLAN.
- R — The “TERR REDUCE RANGE” indication is displayed in red (or amber), in the case of
- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the selected range is
- R 160 NM or 320 NM.

③ TERR indication

To differentiate between the terrain and weather display, the weather radar TILT is replaced by a blue TERR, and the terrain display sweeps from the center outward to both ND sides.

④ Warning and caution messages

TERRAIN (amber) : For a caution.

TERRAIN (red) : For a warning.

When triggered, these messages flash for 9 seconds, then remain steady until the caution or warning alert condition disappears.

TERR RNG (red) : For a RANGE error warning.

TERR TST (amber) : Appears during the EGPWS test when the terrain pattern is displayed and there is no failure.

⑤ Terrain caution alert

Generated when a conflict exists between the terrain caution envelope, ahead of the aircraft, and the terrain data stored in the database. The conflict area is shown in solid yellow.

⑥ Terrain warning alert

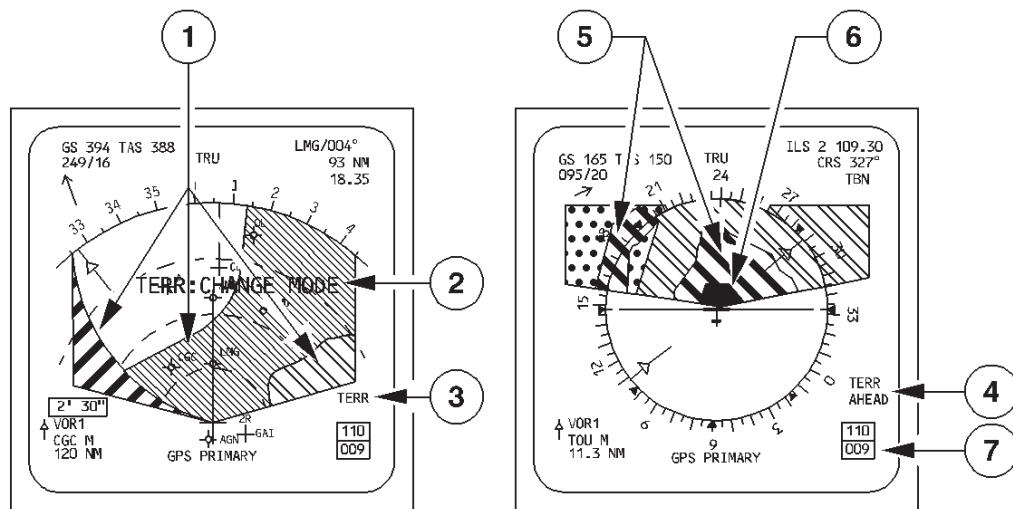
Generated when a conflict exists between the terrain warning envelope, ahead of the aircraft, and the terrain data stored in the database. The conflict area is shown in solid red.

Note : When an alert is generated (either caution or warning) and TERR ON ND is not selected, the terrain is automatically displayed and the TERR ON ND's pushbutton ON light comes on.

EGPWS

R

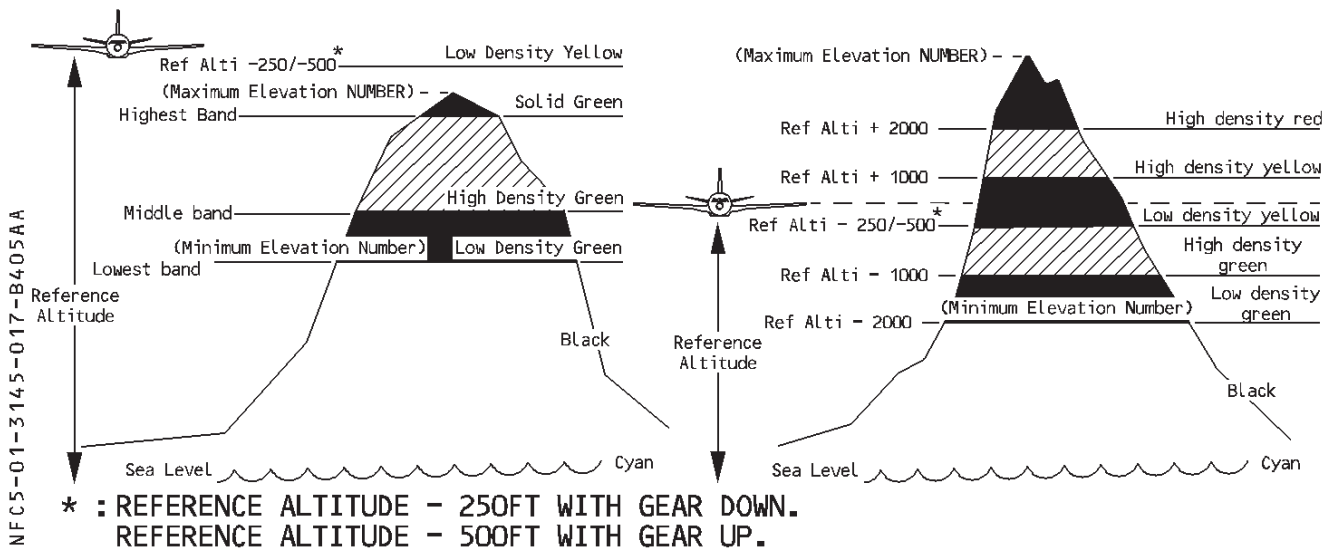
NFC5-01-3145-017-A405AA



1 EGPWS terrain picture

The ND displays the EGPWS terrain picture, when the TERR ON ND switch is selected ON, and the ND is not in PLAN mode. The terrain picture replaces the weather radar image. Terrain data is displayed independently of the aircraft relative altitude. The terrain appears in different colors and densities, according to its relative height :

High aircraft relative altitude (feet) Low aircraft relative altitude (feet)



Note : – Areas without available terrain data in the EGPWS database appear in magenta.

- The reference altitude is computed based on the current aircraft altitude or, if descending more than 1000 ft/min, the altitude expected in 30 seconds.
- In case of flight above the maximum elevation number, the relief between the minimum and maximum displayed elevations is displayed by using three different green levels.

R ② Center Part Messages

- R — The “TERR CHANGE MODE” indication is displayed in red (or amber), in the case of
- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the current selected
- R display mode is PLAN.
- R — The “TERR REDUCE RANGE” indication is displayed in red (or amber), in the case of
- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the selected range is
- R 160 NM or 320 NM.

③ TERR indication

To differentiate between the terrain and the weather display, the weather radar TILT is replaced by a blue TERR, and the terrain display sweeps from the center outward to both ND sides.

④ Warning and caution messages

TERR AHEAD (amber) : For a caution.

TERR AHEAD (red) : For a warning.

When triggered, these messages flash for 9 seconds, then remain steady until the caution or warning alert condition disappears.

TERR RNG (red) : For a RANGE error warning.

TERR TST (amber) : Appears during the EGPWS test, when the terrain pattern is displayed, and there is no failure.

⑤ Terrain caution alert

Generated when a conflict exists between the terrain caution envelope, ahead of the aircraft, and the terrain data stored in the database. The conflict area is shown in solid yellow.

⑥ Terrain warning alert

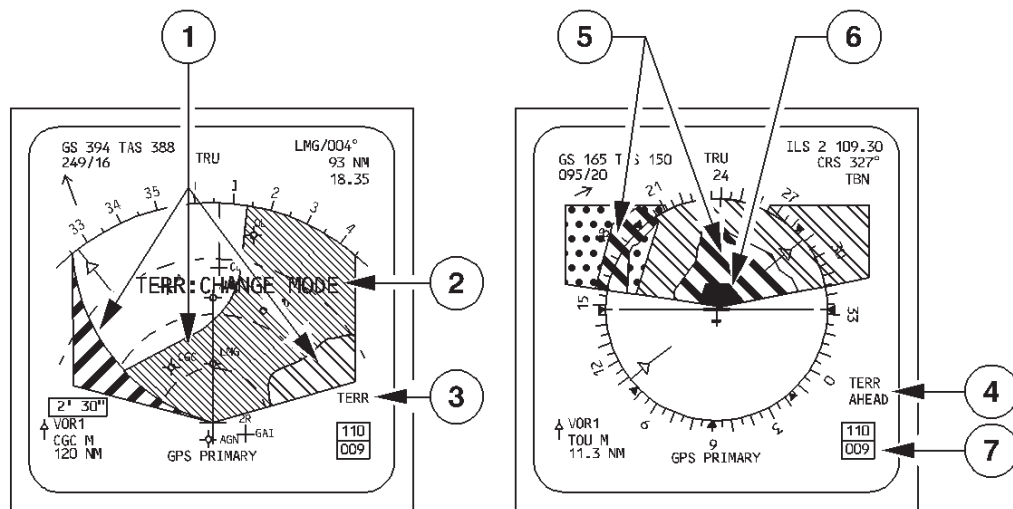
Generated when a conflict exists between the terrain warning envelope, ahead of the aircraft, and the terrain data stored in the database. The conflict area is shown in solid red.

Note : When an alert is generated (either caution or warning) and TERR ON ND is not selected, the terrain is automatically displayed and the TERR ON ND's pushbutton ON light comes on.

EGPWS

R

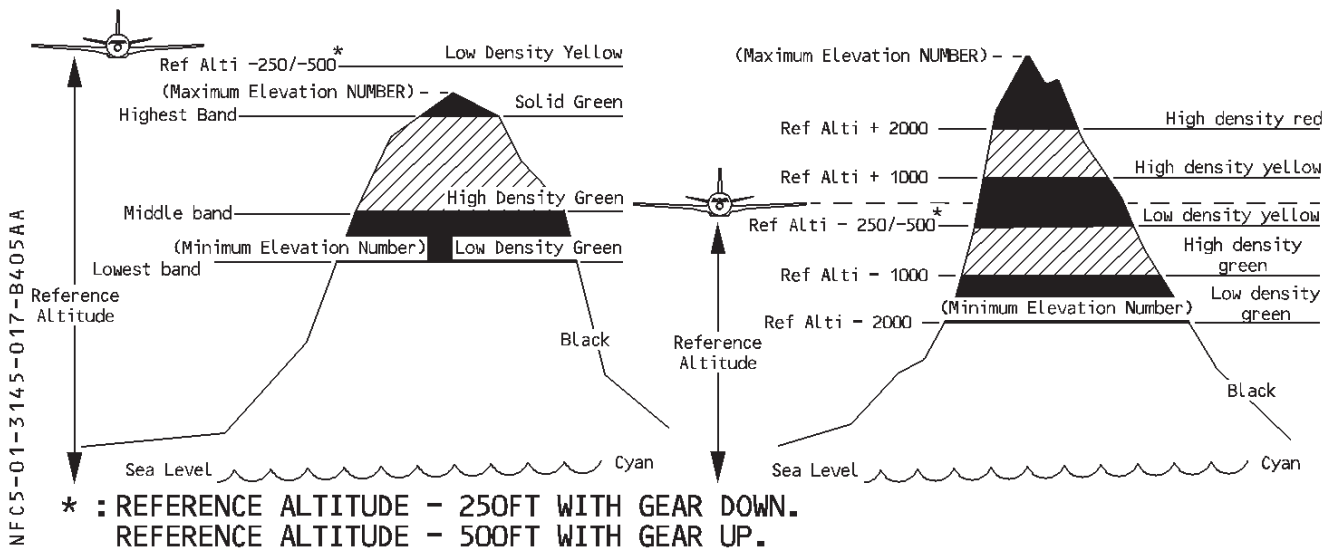
NFC5-01-3145-017-A405AA



1 EGPWS terrain picture

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High aircraft relative altitude (feet) Low aircraft relative altitude (feet)



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- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the selected range is
- R 160 NM or 320 NM.

③ TERR indication

To differentiate between the terrain and weather display, the weather radar TILT is replaced by a blue TERR, and the terrain display sweeps from the center outward to both NDs.

④ Warning and caution messages

TERR AHEAD or OBST AHEAD (amber) : For a caution.

TERR AHEAD or OBST AHEAD (red) : For a warning.

When triggered, these messages flash for 9 seconds, then remain steady until the caution or warning alert condition disappears.

TERR RNG (red) : For a RANGE error warning.

TERR TST (amber) : Appears during the EGPWS test, when the terrain pattern is displayed, and there is no failure.

⑤ Terrain or obstacle caution alert

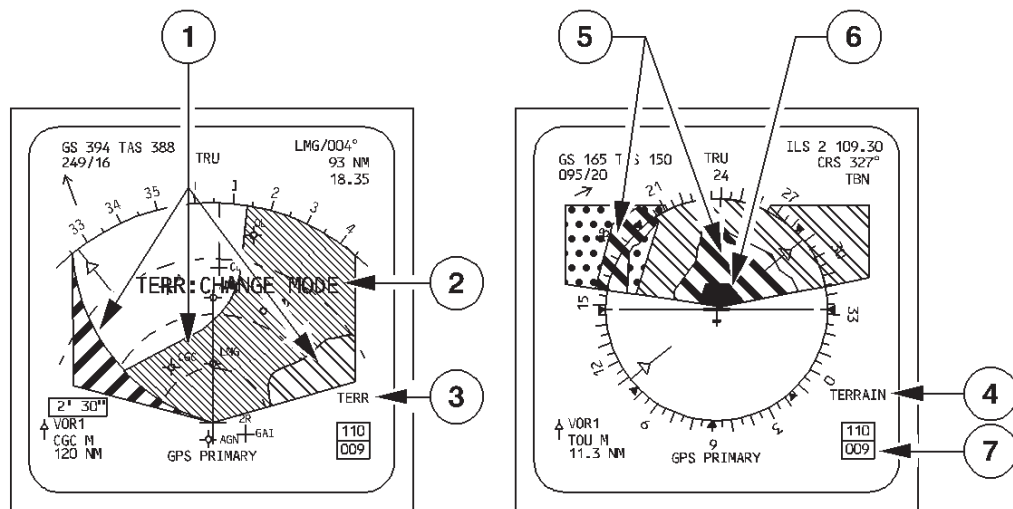
Generated when a conflict exists between the terrain caution envelope, ahead of the aircraft, and database-stored terrain/obstacles. The conflict area is displayed in solid yellow.

⑥ Terrain or obstacle warning alert

Generated when a conflict exists between the terrain warning envelope, ahead of the aircraft, and terrain/obstacles data stored in the database. The conflict area is displayed in solid red.

Note : When an alert is generated (either caution or warning) and TERR ON ND is not selected, the terrain is automatically displayed and the TERR ON ND pushbutton ON light comes on.

EGPWS

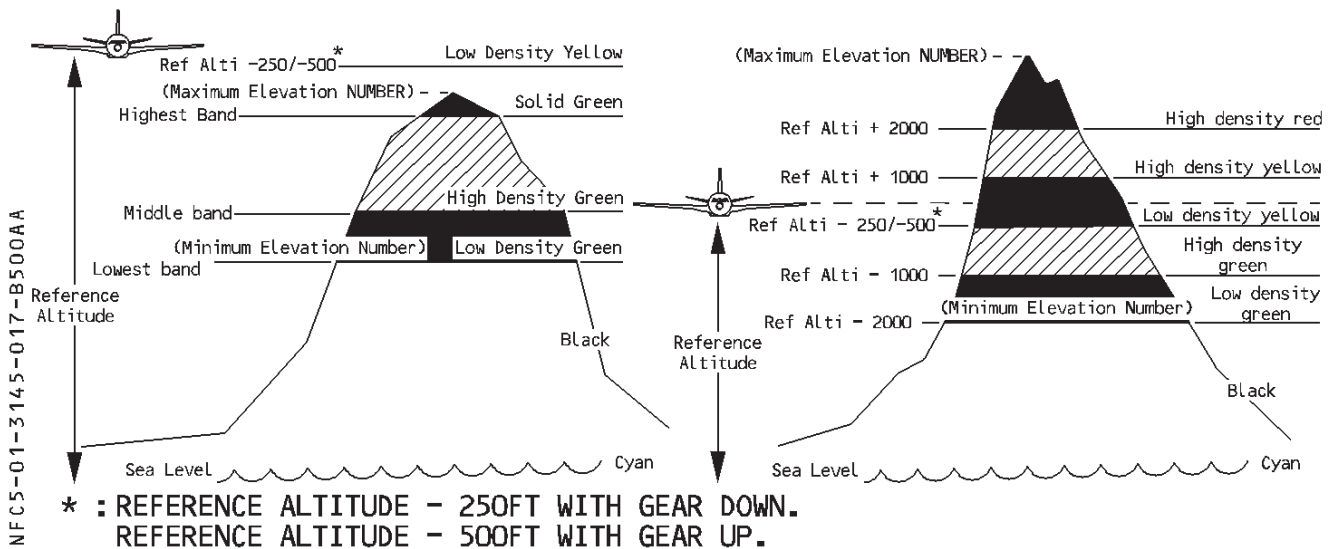


NFC5-01-3145-017-A500AA

① EGPWS terrain picture

The ND presents the EGPWS terrain picture, when the TERR ON ND switch is selected ON, and the ND is not in PLAN mode. The terrain picture replaces the weather radar image. Terrain data is displayed independently of the aircraft relative altitude. The terrain appears in different colors and densities, according to its relative height :

High aircraft relative altitude (feet) Low aircraft relative altitude (feet)



NFC5-01-3145-017-B500AA

Note : – Areas without available terrain data in the EGPWS database appear in magenta.

- The reference altitude is computed based on the current aircraft altitude or, if descending more than 1000 ft/min, the altitude expected in 30 seconds.
- In case of flight above the maximum elevation number, the relief between the minimum and maximum displayed elevations is displayed by using three different green levels.

R ② Center Part Messages

- R — The “TERR CHANGE MODE” indication is displayed in red (or amber), in the case of
- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the current selected
- R display mode is PLAN.
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- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the selected range is
- R 160 NM or 320 NM.

③ TERR indication

To differentiate between the terrain and weather display, weather radar TILT is replaced by a blue TERR, and the terrain display sweeps from the center outward to both NDs.

④ Warning and caution messages

TERRAIN or OBST (amber) : For a caution.

TERRAIN or OBST (red) : For a warning.

When triggered, these messages flash for 9 seconds, then remain steady until the caution or warning alert condition disappears.

TERR RNG (red) : For a RANGE error warning.

TERR TST (amber) : Appears during the EGPWS test, when the terrain pattern is displayed, and there is no failure.

⑤ Terrain or obstacle caution alert

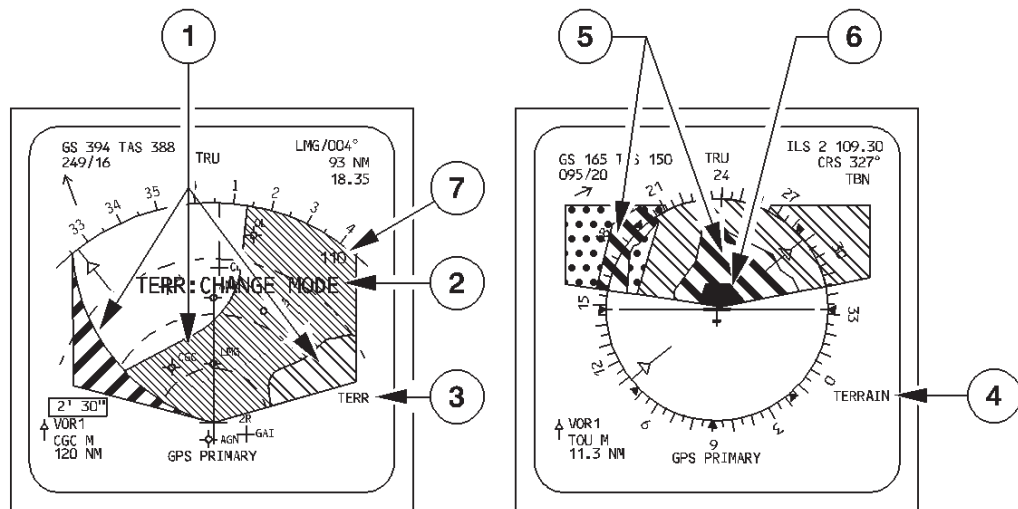
Generated when a conflict exists between the terrain caution envelope, ahead of the aircraft, and the database-stored terrain/obstacles. The area is shown in solid yellow.

⑥ Terrain or obstacle warning alert

Generated when a conflict exists between the terrain warning envelope, ahead of the aircraft, and the terrain/obstacles data stored in the database. The conflict area is displayed in solid red.

Note : When an alert is generated (either caution or warning) and TERR ON ND is not selected, the terrain is automatically displayed and the TERR ON ND's pushbutton ON light comes on.

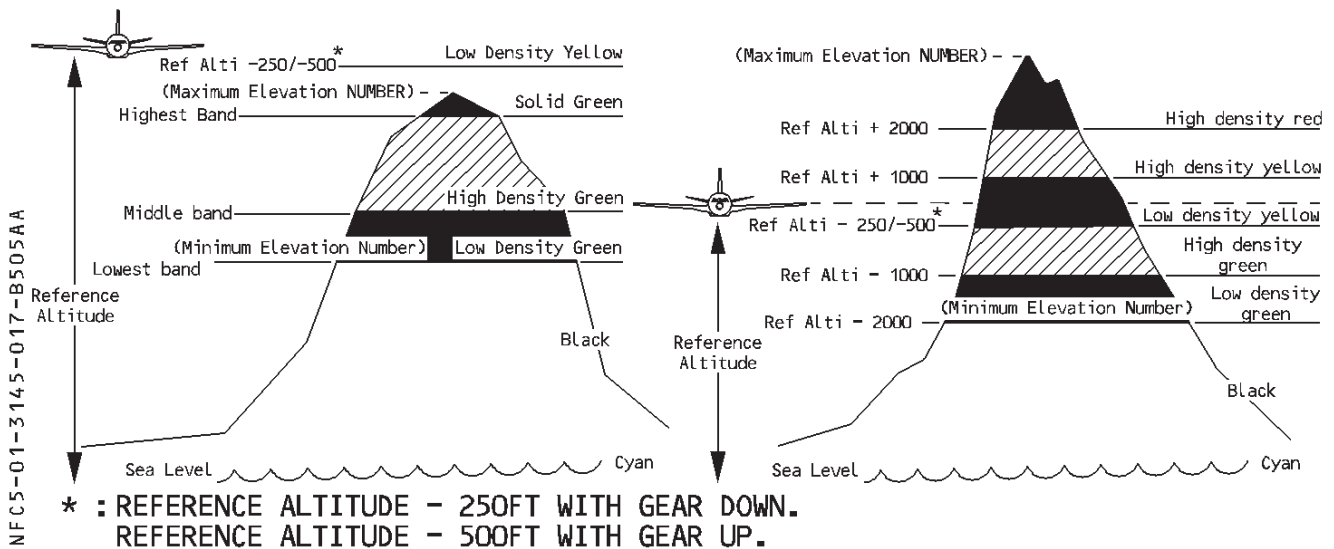
EGPWS



1 EGPWS terrain picture

The ND displays the EGPWS terrain picture, when the TERR ON ND switch is selected ON, and the ND is not in PLAN mode. The terrain picture replaces the weather radar image. Terrain data is displayed independently of the aircraft relative altitude. The terrain appears in different colors and densities, according to its relative height :

High aircraft relative altitude (feet) Low aircraft relative altitude (feet)



Note : – Areas without available terrain data in the EGPWS database appear in magenta.

- The reference altitude is computed based on the current aircraft altitude or, if descending more than 1000 ft/min, the altitude expected in 30 seconds.
- In case of flight above the maximum elevation number, the relief between the minimum and maximum displayed elevations is displayed by using three different green levels.

R ② Center Part Messages

- R — The “TERR CHANGE MODE” indication is displayed in red (or amber), in the case of
- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the current selected
- R display mode is PLAN.
- R — The “TERR REDUCE RANGE” indication is displayed in red (or amber), in the case of
- R a Terrain Awareness Display (TAD) warning (or caution) alert, if the selected range is
- R 160 NM or 320 NM.

③ TERR indication

To differentiate between the terrain and weather display, weather radar TILT is replaced by a blue TERR, and the terrain display sweeps from the center outward to both NDs.

④ Warning and caution messages

TERRAIN or OBST (amber) : For a caution.

TERRAIN or OBST (red) : For a warning.

When triggered, these messages flash for 9 seconds, then remain steady until the caution or warning alert condition disappears.

TERR RNG (red) : For a RANGE error warning.

TERR TST (amber) : Appears during the EGPWS test, when the terrain pattern is displayed, and there is no failure.

⑤ Terrain or obstacle caution alert

Generated when a conflict exists between the terrain caution envelope, ahead of the aircraft, and the database-stored terrain/obstacles. The area is shown in solid yellow.

⑥ Terrain or obstacle warning alert

Generated when a conflict exists between the terrain warning envelope, ahead of the aircraft, and the terrain/obstacles data stored in the database. The conflict area is displayed in solid red.

Note : When an alert is generated (either caution or warning) and TERR ON ND is not selected, the terrain is automatically displayed and the TERR ON ND's pushbutton ON light comes on.

⑦ Lowest and highest elevations

Minimum and maximum elevations encountered within the selected range, are respectively displayed in this square, using the color code of paragraph 1.

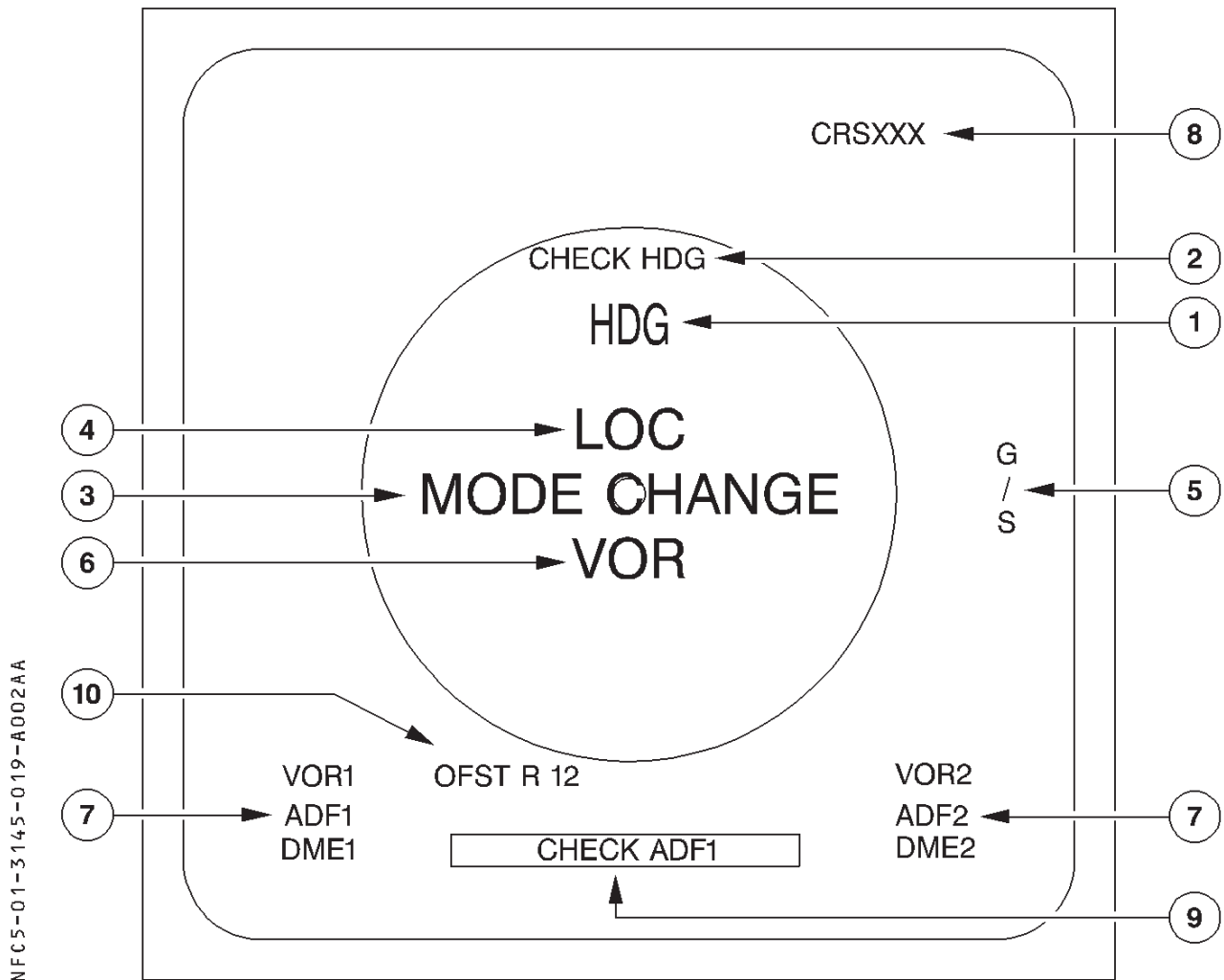
Note : *The elevations shown on the ND correspond to the terrain included in the selected ND range, ahead of the aircraft. In ARC mode, the elevations are linked with the terrain displayed on the ND. In ROSE mode, the elevations may not represent the lowest and highest terrain currently displayed on the ND.*

⑦ Highest Elevation

The highest elevation encountered within the selected range is only displayed in ARC mode, in the top righthand corner of the terrain frame, using the color code of paragraph 1.

FLAGS AND MESSAGES DISPLAYED ON ND

R


① HDG Flag (red)

If the heading data fails, the rose, arc and associated symbols disappear.

A HDG flag flashes for 9 seconds, then remains steady in the upper part of the ND.

② CHECK HDG Flag (amber)

When the flight warning computer detects a disagree (5°) between sides 1 and 2, a CHECK HDG flag appears on both NDs, and a caution appears on the ECAM.

③ Center Part Messages

- The screen displays a MODE CHANGE message in green if there is a discrepancy between the selected mode on the EFIS control panel and the mode sent from the onside FMGC, or while the DMC is preparing a new page for display.
- The screen displays a RANGE CHANGE message in green if there is a discrepancy between the range selected on the EFIS control panel and the range sent from the onside FMGC. A MODE CHANGE message has priority over a RANGE CHANGE message.
- The screen displays a MAP NOT AVAIL message in red for several reasons :
 - The MODE CHANGE or RANGE CHANGE message has been displayed more than six seconds or
 - The FMGC has failed or
 - The FMGC has delivered an invalid aircraft position.

④ LOC Flag (red)

If LOC data fails, this flag flashes for nine seconds, then remains steady.

⑤ G/S Flag (red)

If G/S data fails, this flag flashes for nine seconds, then remains steady.

⑥ VOR Flag (red)

In ROSE VOR mode, when the VOR bearing is not valid, this flag flashes for nine seconds, then remains steady.

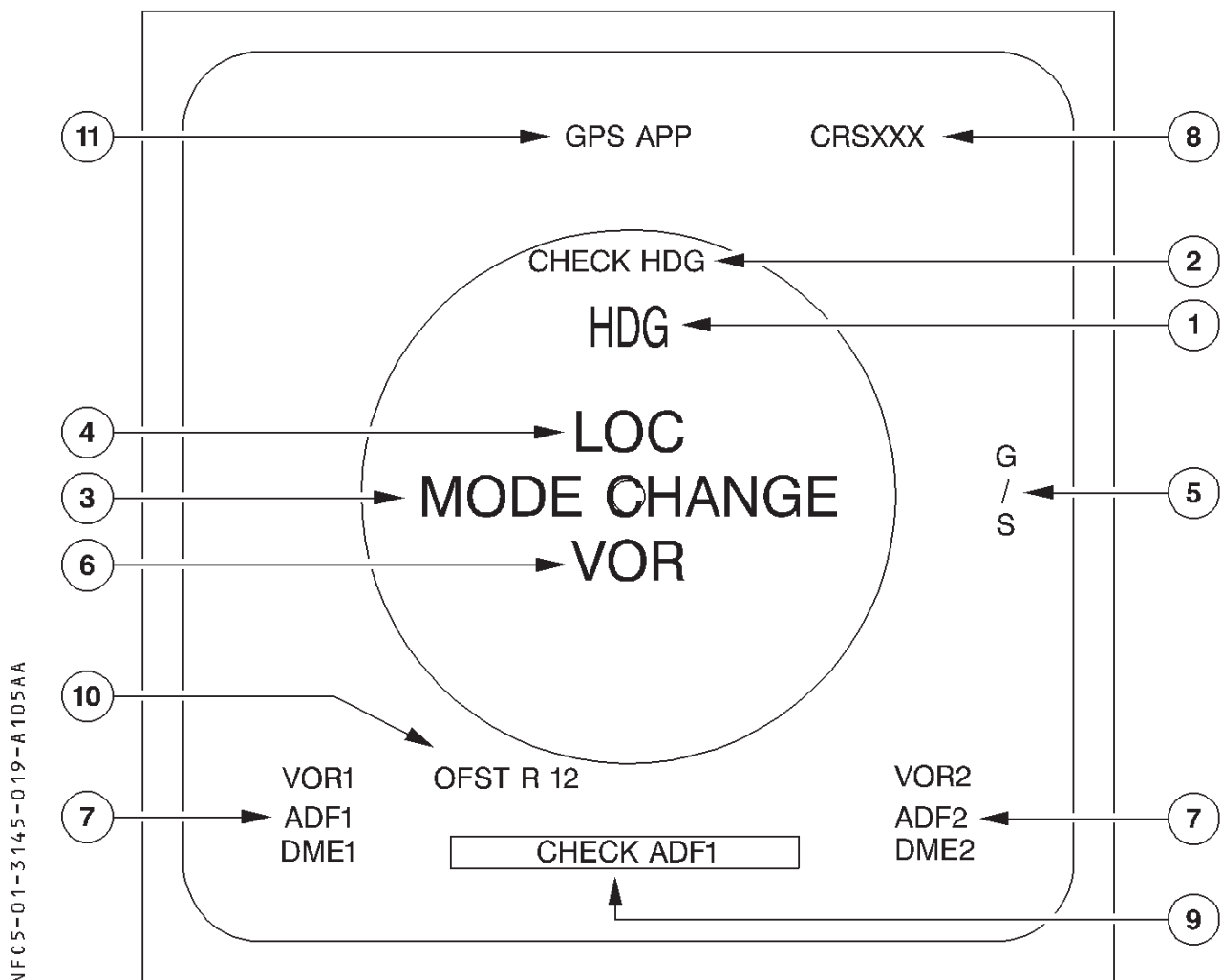
⑦ VOR1(2) or ADF1(2) or DME1 Flag (red)

If a navigation receiver fails, the appropriate one of these flags flashes for nine seconds, then remains steady.

⑧ VOR Course Flag

If the VOR course fails, a red CRSXXX flag appears.

If there is non-computed data (NCD), a blue CRS - - - flag appears.

FLAGS AND MESSAGES DISPLAYED ON ND

① HDG flag (red)

If the heading data fails, the rose, arc and associated symbols disappear.

A HDG flag on the upper part of the ND flashes for 9 seconds, then remains steady.

② CHECK HDG flag (amber)

When the flight warning computer detects a disagree (5 degrees) between sides 1 and 2, a CHECK HDG flag appears on both NDs, and a caution appears on the ECAM.

③ Center Part Messages

- The screen displays a MODE CHANGE message in green if there is a discrepancy between the selected mode on the EFIS control panel and the mode sent from the onside FMGC, or while the DMC is preparing a new page for display.
- The screen displays a RANGE CHANGE message in green if there is a discrepancy between the range selected on the EFIS control panel and the range sent from the onside FMGC. A MODE CHANGE message has priority over a RANGE CHANGE message.
- The screen displays a MAP NOT AVAIL message in red for several reasons :
 - The MODE CHANGE or RANGE CHANGE message has been displayed more than six seconds or
 - The FMGC has failed or
 - The FMGC has delivered an invalid aircraft position.

④ LOC Flag (red)

If LOC data fails, this flag flashes for nine seconds, then remains steady.

⑤ G/S Flag (red)

If G/S data fails, this flag flashes for nine seconds, then remains steady.

⑥ VOR Flag (red)

In ROSE VOR mode, when the VOR bearing is not valid, this flag flashes for nine seconds, then remains steady.

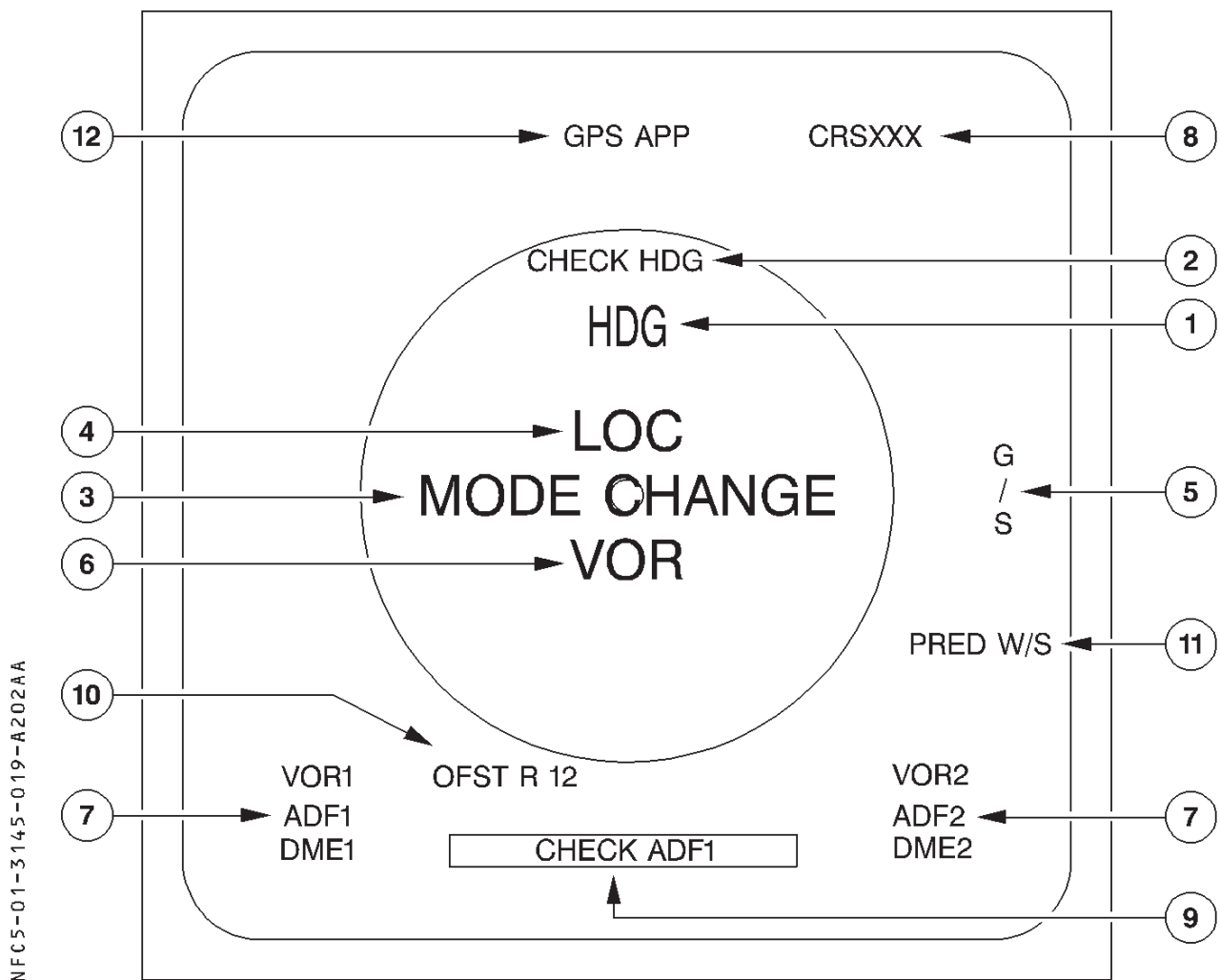
⑦ VOR1(2) or ADF1(2) or DME1 Flag (red)

If a navigation receiver fails, the appropriate one of these flags flashes for nine seconds, then remains steady.

⑧ VOR Course Flag

If the VOR course fails, a red CRSXXX flag appears.

If there is non-computed data (NCD), a blue CRS - - - flag appears.

FLAGS AND MESSAGES DISPLAYED ON ND

① HDG flag (red)

If the heading data fails, the rose, arc and associated symbols disappear.
A HDG flag on the upper part of the ND flashes for 9 seconds, then remains steady.

② CHECK HDG flag (amber)

When the flight warning computer detects a disagree (5°) between sides 1 and 2, a CHECK HDG flag appears on both NDs, and a caution appears on the ECAM.

③ Center Part Messages

- The screen displays a MODE CHANGE message in green if there is a discrepancy between the selected mode on the EFIS control panel and the mode sent from the onside FMGC, or while the DMC is preparing a new page for display.
- The screen displays a RANGE CHANGE message in green if there is a discrepancy between the range selected on the EFIS control panel and the range sent from the onside FMGC. A MODE CHANGE message has priority over a RANGE CHANGE message.
- The screen displays a MAP NOT AVAIL message in red for several reasons :
 - The MODE CHANGE or RANGE CHANGE message has been displayed more than six seconds or
 - The FMGC has failed or
 - The FMGC has delivered an invalid aircraft position.
- The screen displays a W/S SET RNG 10 NM message if a predictive windshear alert is triggered and the range is above 10 NM.
The message is displayed in the color corresponding to the windshear alert : red for a warning, amber for a caution.
- The screen displays a W/S CHANGE MODE message if a predictive windshear alert is triggered and the ND is not in ARC or ROSE mode. The message appears in red for a warning, or amber for a caution.

④ LOC Flag (red)

If LOC data fails, this flag flashes for nine seconds, then remains steady.

⑤ G/S Flag (red)

If G/S data fails, this flag flashes for nine seconds, then remains steady.

⑥ VOR Flag (red)

In ROSE VOR mode, when the VOR bearing is not valid, this flag flashes for nine seconds, then remains steady.

⑦ VOR1(2) or ADF1(2) or DME1 Flag (red)

If a navigation receiver fails, the appropriate one of these flags flashes for nine seconds, then remains steady.

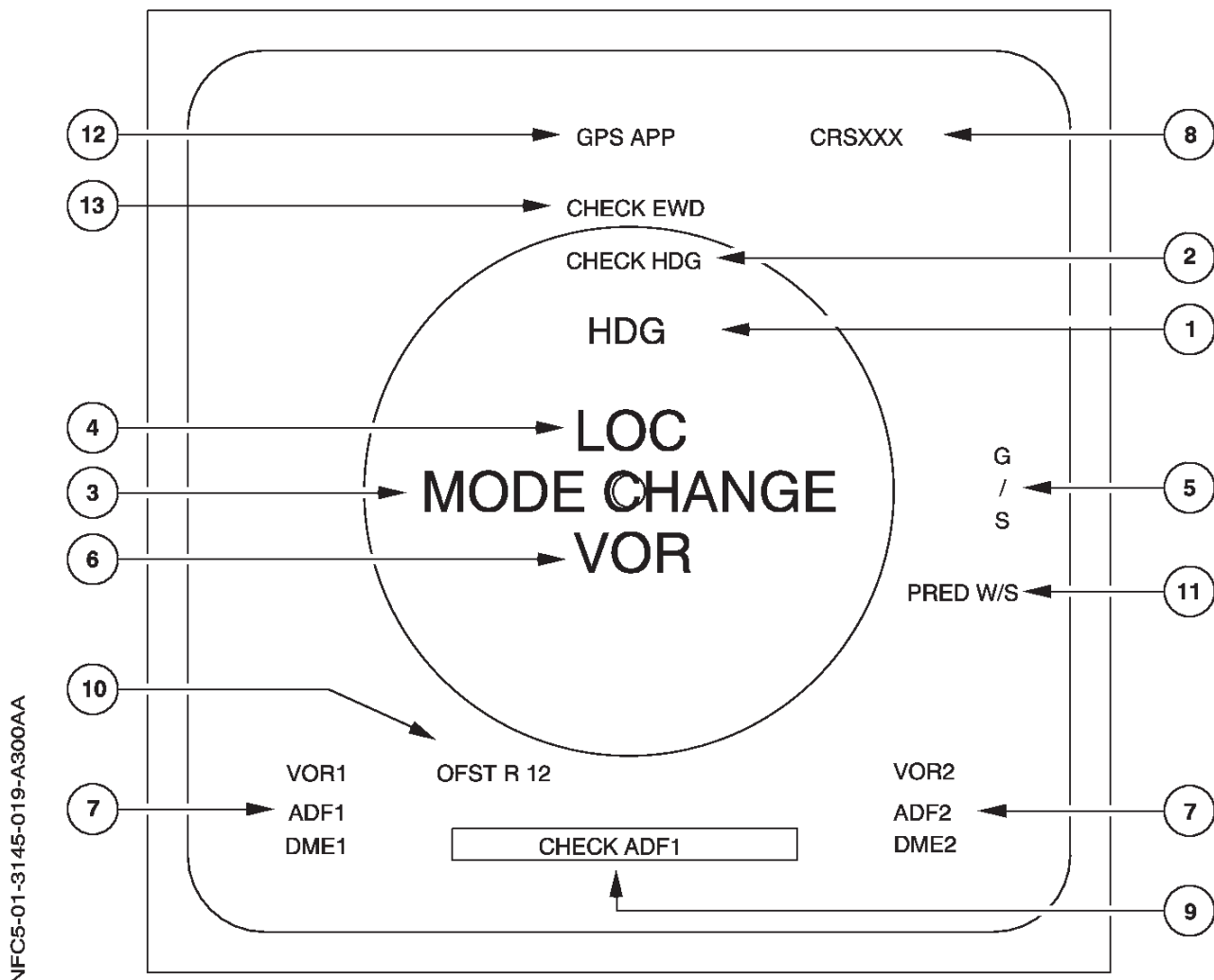
⑧ VOR Course Flag

If the VOR course fails, a red CRSXXX flag appears.

If there is non-computed data (NCD), a blue CRS - - - flag appears.

FLAGS AND MESSAGES DISPLAYED ON ND

R


① HDG flag (red)

If the heading data fails, the rose, arc and associated symbols disappear.

A HDG flag on the upper part of the ND flashes for 9 seconds, then remains steady.

② CHECK HDG flag (amber)

R

For more information, refer to FCOM 1.31.05.

③ Center Part Messages

- The screen displays a MODE CHANGE message in green if there is a discrepancy between the selected mode on the EFIS control panel and the mode sent from the onside FMGC, or while the DMC is preparing a new page for display.
- The screen displays a RANGE CHANGE message in green if there is a discrepancy between the range selected on the EFIS control panel and the range sent from the onside FMGC. A MODE CHANGE message has priority over a RANGE CHANGE message.
- The screen displays a MAP NOT AVAIL message in red for several reasons :
 - The MODE CHANGE or RANGE CHANGE message has been displayed more than six seconds or
 - The FMGC has failed or
 - The FMGC has delivered an invalid aircraft position.
- The screen displays a W/S SET RNG 10 NM message if a predictive windshear alert is triggered and the range is above 10 NM.
The message is displayed in the color corresponding to the windshear alert : red for a warning, amber for a caution.
- The screen displays a W/S CHANGE MODE message if a predictive windshear alert is triggered and the ND is not in ARC or ROSE mode. The message appears in red for a warning, or amber for a caution.

④ LOC Flag (red)

If LOC data fails, this flag flashes for nine seconds, then remains steady.

⑤ G/S Flag (red)

If G/S data fails, this flag flashes for nine seconds, then remains steady.

⑥ VOR Flag (red)

In ROSE VOR mode, when the VOR bearing is not valid, this flag flashes for nine seconds, then remains steady.

⑦ VOR1(2) or ADF1(2) or DME1 Flag (red)

If a navigation receiver fails, the appropriate one of these flags flashes for nine seconds, then remains steady.

⑧ VOR Course Flag

If the VOR course fails, a red CRSXXX flag appears.

If there is non-computed data (NCD), a blue CRS - - - flag appears.



⑨ Other messages

- MAP PARTLY DISPLAYED : In case of incomplete data transmission between the FMGC (priority criteria) and the DMC, or if the DMC cannot draw the complete MAP.
(amber)
This message is also displayed when a very long leg exists in the flight plan. A leg is considered as "very long" when the starting point (or endpoint) is located at more than 45° from the aircraft location (45° of longitude or latitude).
This DMC limitation results from a compromise between accurate drawing precision and maximum leg length that can be displayed.
- NAV ACCUR UPGRAD, or : Signals a change in navigation accuracy.
(white)
- NAV ACCUR DOWNGRAD
(amber)
- SPECIFIC VOR/D UNAVAIL: If the navaid, that is tuned for the selected approach or departure, is not available.
(amber)
- SET OFFSIDE RNG/MODE : Displayed on ND1(2), in case of an FMGC1(2) failure when the two ND ranges or modes selected on the EFIS control panels are different.
(amber)
- ↓ (green) : Overflow arrow, displayed when more than one of the following messages are present at the same time :
- NAV ACCUR DOWNGRAD
 - NAV ACCUR UPGRAD
 - SPECIF VOR/D UNAVAIL
 - MAP PARTLY DISPLAYED
 - SET OFFSIDE RNG/MODE

R

R

⑩ OFST R(L) XX message (yellow)

- R The screen displays this message, when a temporary or an offset flight plan is entered.
The offset value is given in NM.



⑨ Other messages

- MAP PARTLY DISPLAYED : In case of incomplete data transmission between the FMGC (priority criteria) and the DMC, or if the DMC cannot draw the complete MAP.
(amber)
This message is also displayed when a very long leg exists in the flight plan. A leg is considered as "very long" when the starting point (or endpoint) is located at more than 45° from the aircraft location (45° of longitude or latitude).
This DMC limitation results from a compromise between accurate drawing precision and maximum leg length that can be displayed.
- NAV ACCUR UPGRAD, or : Signals a change in navigation accuracy.
(white)
- NAV ACCUR DOWNGRAD
(amber)
- SPECIFIC VOR/D UNAVAIL: If the navaid, that is tuned for the selected approach or departure, is not available.
(amber)
- SET OFFSIDE RNG/MODE : Displayed on ND1(2), in case of an FMGC1(2) failure when the two ND ranges or modes selected on the EFIS control panels are different.
(amber)
- R ↓ (green) : Overflow arrow, displayed when more than one of the following messages are present at the same time :
- NAV ACCUR DOWNGRAD
 - NAV ACCUR UPGRAD
 - SPECIF VOR/D UNAVAIL
 - MAP PARTLY DISPLAYED
 - SET OFFSIDE RNG/MODE
- R

Note : For information about the TCAS messages, refer to 1.34.80.

⑩ OFST R(L) XX message (yellow)

The screen displays this message, when a temporary or an offset flight plan is entered.
The offset value is given in NM.

R Note : *For information about the TCAS messages, refer to 1.34.80.*

⑨ Other messages

- MAP PARTLY DISPLAYED** : In case of incomplete data transmission between the FMGC (priority criteria) and the DMC, or if the DMC cannot draw the complete MAP.
(amber)
This message is also displayed when a very long leg exists in the flight plan. A leg is considered as "very long" when the starting point (or endpoint) is located at more than 45° from the aircraft location (45° of longitude or latitude).
This DMC limitation results from a compromise between accurate drawing precision and maximum leg length that can be displayed.
- NAV ACCUR UPGRAD, or** : Signals a change in navigation accuracy.
(white)
- NAV ACCUR DOWNGRAD**
(amber)
- SPECIFIC VOR/D UNAVAIL** : If the navaid, that is tuned for the selected approach or departure, is not available.
(amber)
- SET OFFSIDE RNG/MODE** : Displayed on ND1(2), in case of an FMGC1(2) failure when the two ND ranges or modes selected on the EFIS control panels are different.
(amber)
- GPS PRIMARY** : This message appears when GPS PRIMARY mode is available, or has been recovered. The pilot can clear this message by pressing the CLR key on the MCDU.
(white, boxed white)
- GPS PRIMARY LOST** : This message appears when GPS PRIMARY is not available, and not clearable by pilot action.
(amber, boxed white)
- ↓ (green)** : Overflow arrow, displayed when more than one of the following messages are present at the same time :
- NAV ACCUR DOWNGRAD (inhibited when the navigation mode is IRS/GPS)
 - NAV ACCUR UPGRAD (inhibited when the navigation mode is IRS/GPS)
 - SPECIF VOR/D UNAVAIL
 - MAP PARTLY DISPLAYED
 - SET OFFSIDE RNG/MODE
 - GPS PRIMARY
 - GPS PRIMARY LOST

Note : For information about the TCAS messages, refer to 1.34.80.

R
R
R
R

⑩ OFST R(L) XX message (yellow)

The screen displays this message, when the flight crew enters a temporary, or an offset flight plan. The offset value is provided in NM.

Note : *For information about the TCAS messages, refer to 1.34.80.*

⑪ GPS APP (green)

This message is displayed, when the flight crew selects a GPS approach.

⑨ Other messages

- MAP PARTLY DISPLAYED : In case of incomplete data transmission between the FMGC (priority criteria) and the DMC, or if the DMC cannot draw the complete MAP.
(amber)
This message is also displayed when a very long leg exists in the flight plan. A leg is considered as "very long" when the starting point (or endpoint) is located at more than 45° from the aircraft location (45° of longitude or latitude).
This DMC limitation results from a compromise between accurate drawing precision and maximum leg length that can be displayed.
- NAV ACCUR UPGRAD, or : Signals a change in navigation accuracy.
(white)
- NAV ACCUR DOWNGRAD
(amber)
- SPECIFIC VOR/D UNAVAIL: If the navaid, that is tuned for the selected approach or departure, is not available.
(amber)
- SET OFFSIDE RNG/MODE : Displayed on ND1(2), in case of an FMGC1(2) failure when the two ND ranges or modes selected on the EFIS control panels are different.
(amber)
- GPS PRIMARY : This message appears when GPS PRIMARY mode is available, or has been recovered. The pilot can clear this message by pressing the CLR key on the MCDU.
(white, boxed white)
- GPS PRIMARY LOST : This message appears when GPS PRIMARY is not available, and not clearable by pilot action.
(amber, boxed white)
- ↓ (green) : Overflow arrow, displayed when more than one of the following messages are present at the same time :
- NAV ACCUR DOWNGRAD (inhibited when the navigation mode is IRS/GPS)
 - NAV ACCUR UPGRAD (inhibited when the navigation mode is IRS/GPS)
 - SPECIF VOR/D UNAVAIL
 - MAP PARTLY DISPLAYED
 - SET OFFSIDE RNG/MODE
 - GPS PRIMARY
 - GPS PRIMARY LOST

Note : For information about the TCAS messages, refer to 1.34.80.

R
R
R
R

⑩ OFST R(L) XX message (yellow)

The screen displays this message, when a temporary or an offset flight plan is entered. The offset value is given in NM.

Note : For information about the TCAS messages, refer to 1.34.80.

⑪ PRED W/S flag (amber)

The WINDSHEAR switch on the weather radar panel is set to AUTO, and a Predictive Windshear System fault is detected. This message appears on ground, or when flaps and slats are extended.

It is associated with a single chime. The radar image remains available, provided that the fault does not affect the radar mode.

⑫ GPS APP (green)

This message is displayed, when a GPS approach has been selected.

⑨ Other messages

- MAP PARTLY DISPLAYED : In case of incomplete data transmission between the FMGC (priority criteria) and the DMC, or if the DMC cannot draw the complete MAP.
(amber)
This message is also displayed when a very long leg exists in the flight plan. A leg is considered as "very long" when the starting point (or endpoint) is located at more than 45° from the aircraft location (45° of longitude or latitude).
This DMC limitation results from a compromise between accurate drawing precision and maximum leg length that can be displayed.
- NAV ACCUR UPGRAD, or : Signals a change in navigation accuracy.
(white)
- NAV ACCUR DOWNGRAD
(amber)
- SPECIFIC VOR/D UNAVAIL: If the navaid, that is tuned for the selected approach or departure, is not available.
(amber)
- SET OFFSIDE RNG/MODE : Displayed on ND1(2), in case of an FMGC1(2) failure when the two ND ranges or modes selected on the EFIS control panels are different.
(amber)
- GPS PRIMARY : This message appears when GPS PRIMARY mode is available, or has been recovered. The pilot can clear this message by pressing the CLR key on the MCDU.
(white, boxed white)
- GPS PRIMARY LOST : This message appears when GPS PRIMARY is not available, and not clearable by pilot action.
(amber, boxed white)
- ↓ (green) : Overflow arrow, displayed when more than one of the following messages are present at the same time :
- NAV ACCUR DOWNGRAD (inhibited when the navigation mode is IRS/GPS)
 - NAV ACCUR UPGRAD (inhibited when the navigation mode is IRS/GPS)
 - SPECIF VOR/D UNAVAIL
 - MAP PARTLY DISPLAYED
 - SET OFFSIDE RNG/MODE
 - GPS PRIMARY
 - GPS PRIMARY LOST

Note : For information about the TCAS messages, refer to 1.34.80.

R
R
R
R

⑩ OFST R(L) XX message (yellow)

The screen displays this message, when the flight crew enters a temporary or an offset flight plan. The offset value is provided in NM.

Note : For information about the TCAS messages, refer to 1.34.80.

⑪ PRED W/S flag (amber)

The WINDSHEAR switch on the weather radar panel is set to AUTO, and a Predictive Windshear System fault is detected. This message appears when the aircraft is on ground, or when flaps and slats are extended.

It is associated with a single chime. The radar image remains available, provided that the fault does not affect the radar mode.

⑫ GPS APP (green)

This message is displayed, when the flight crew selects a GPS approach.

⑬ CHECK EWD, CHECK CAPT (F/O) PFD, CHECK SD, CHECK CAPT (F/O) ND (amber)

For more information, refer to FCOM 1.31.05.

⑨ Other messages

- MAP PARTLY DISPLAYED** : In case of incomplete data transmission between the FMGC (priority criteria) and the DMC, or if the DMC cannot draw the complete MAP.
(amber)
This message is also displayed when a very long leg exists in the flight plan. A leg is considered as "very long" when the starting point (or endpoint) is located at more than 45° from the aircraft location (45° of longitude or latitude).
This DMC limitation results from a compromise between accurate drawing precision and maximum leg length that can be displayed.
- NAV ACCUR UPGRAD, or** : Signals a change in navigation accuracy.
(white)
- NAV ACCUR DOWNGRAD**
(amber)
- SPECIFIC VOR/D UNAVAIL** : If the navaid, that is tuned for the selected approach or departure, is not available.
(amber)
- SET OFFSIDE RNG/MODE** : Displayed on ND1(2), in case of an FMGC1(2) failure when the two ND ranges or modes selected on the EFIS control panels are different.
(amber)
- OFFSIDE FM CONTROL** : If the offside FM supplies the onside ND.
(amber)
- GPS PRIMARY** : This message appears when GPS PRIMARY mode is available, or has been recovered. The pilot can clear this message by pressing the CLR key on the MCDU.
(white, boxed white)
- GPS PRIMARY LOST** : This message appears when GPS PRIMARY is not available, and not clearable by pilot action.
(amber, boxed white)
- ↓ (green)** : Overflow arrow, displayed when more than one of the following messages are present at the same time :
 - NAV ACCUR DOWNGRAD
 - NAV ACCUR UPGRAD
 - SPECIF VOR/D UNAVAIL
 - MAP PARTLY DISPLAYED
 - SET OFFSIDE RNG/MODE
 - GPS PRIMARY
 - GPS PRIMARY LOST

Note : For information about the TCAS messages, refer to 1.34.80.

⑩ OFST R(L) XX message (yellow)

The screen displays this message, when the flight crew enters a temporary or an offset flight plan. The offset value is provided in NM.

Note : For information about the TCAS messages, refer to 1.34.80.

⑪ PRED W/S flag (amber)

The WINDSHEAR switch on the weather radar panel is set to AUTO, and a Predictive Windshear System fault is detected. This message appears when the aircraft is on ground, or when flaps and slats are extended.

It is associated with a single chime. The radar image remains available, provided that the fault does not affect the radar mode.

⑫ GPS APP (green)

This message is displayed, when the flight crew selects a GPS approach.

⑬ CHECK EWD, CHECK CAPT (F/O) PFD, CHECK SD, CHECK CAPT (F/O) ND (amber)

For more information, refer to FCOM 1.31.05.

⑨ Other messages

- MAP PARTLY DISPLAYED** : In case of incomplete data transmission between the FMGC (priority criteria) and the DMC, or if the DMC cannot draw the complete MAP.
(amber)
This message is also displayed when a very long leg exists in the flight plan. A leg is considered as "very long" when the starting point (or endpoint) is located at more than 45° from the aircraft location (45° of longitude or latitude).
This DMC limitation results from a compromise between accurate drawing precision and maximum leg length that can be displayed.
- NAV ACCUR UPGRAD, or** : Signals a change in navigation accuracy.
(white)
- NAV ACCUR DOWNGRAD**
(amber)
- SPECIFIC VOR/D UNAVAIL** : If the navaid, that is tuned for the selected approach or departure, is not available.
(amber)
- BACK-UP NAV** : If the MCDU back-up navigation mode is activated
(amber) (refer to 1.22).
- SET OFFSIDE RNG/MODE** : Displayed on ND1(2), in case of an FMGC1(2) failure when the two ND ranges or modes selected on the EFIS control panels are different.
(amber)
- OFFSIDE FM CONTROL** : If the offside FM supplies the onside ND.
(amber)
- GPS PRIMARY** : This message appears when GPS PRIMARY mode is available, or has been recovered. The pilot can clear this message by pressing the CLR key on the MCDU.
(white, boxed white)
- GPS PRIMARY LOST** : This message appears when GPS PRIMARY is not available, and not clearable by pilot action.
(amber, boxed white)
- ↓ (green)** : Overflow arrow, displayed when more than one of the following messages are present at the same time :
- NAV ACCUR DOWNGRAD
 - NAV ACCUR UPGRAD
 - SPECIF VOR/D UNAVAIL
 - MAP PARTLY DISPLAYED
 - SET OFFSIDE RNG/MODE
 - GPS PRIMARY
 - GPS PRIMARY LOST

Note : For information about the TCAS messages, refer to 1.34.80.

⑩ OFST R(L) XX message (yellow)

The screen displays this message, when the flight crew enters a temporary or an offset flight plan. The offset value is provided in NM.

Note : For information about the TCAS messages, refer to 1.34.80.

⑪ PRED W/S flag (amber)

The WINDSHEAR switch on the weather radar panel is set to AUTO, and a Predictive Windshear System fault is detected. This message appears when the aircraft is on ground, or when flaps and slats are extended.

It is associated with a single chime. The radar image remains available, provided that the fault does not affect the radar mode.

⑫ GPS APP (green)

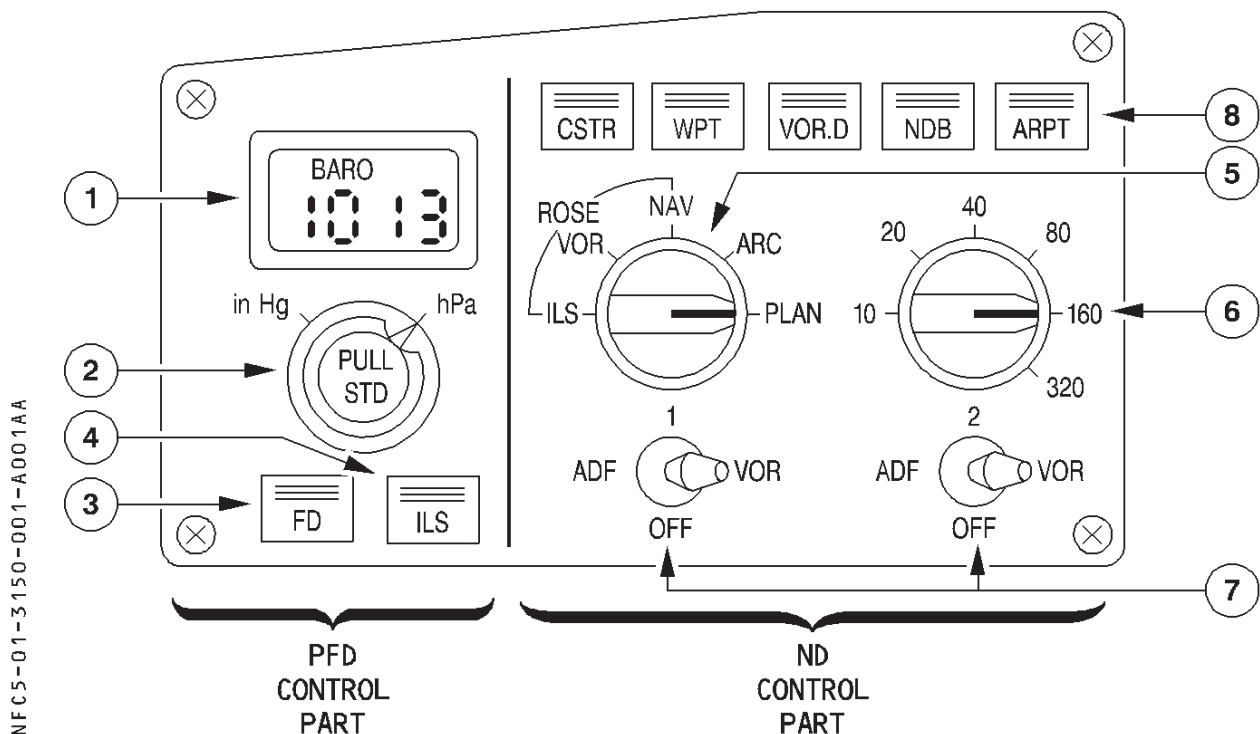
This message is displayed, when the flight crew selects a GPS approach.

⑬ CHECK EWD, CHECK CAPT (F/O) PFD, CHECK SD, CHECK CAPT (F/O) ND (amber)

For more information, refer to FCOM 1.31.05.

EFIS CONTROL PANEL

R



① Barometer Reference Display Window

Range : 745 hPa to 1050 hPa.

② Barometer Reference Selector

- a) Outer ring : For selection of the units for the barometer reference-either hectoPascals or inches of mercury.

Note : The unit selected does not appear on the PFD.

- b) Inner knob : For selection of the reference value displayed in the barometer reference display window and on the PFD below the altitude scale.
At FCU initialization, the window displays 1013 or 29.92, depending on the unit selected.
- Pulling the knob selects the standard baro reference setting. The PFD then displays "STD." (Rotating the knob has no effect.)
 - Pushing the knob from the STD position makes the last selected QNH baro setting displayed.

③ FD Pushbutton

Pushing this button removes the FD bars from the associated PFD (or removes the flight path director symbol if the TRK FPA reference is selected).

The pushbutton light goes out.

Pushing it again restores the FD bars (or the FPD symbol) and the green pushbutton light comes on.

④ ILS Pushbutton

- R Pushing this button displays the localizer and glide slope scales on the PFD. Deviation symbols appear if there is a valid ILS signal. The green pushbutton light comes on.

⑤ Mode Select Switch

This switch selects a navigation display for the onside ND.

⑥ Range Select Switch

This switch selects a range scale for the onside ND.

Note : If the mode or the range data fails, the default selection is the ROSE NAV mode and 80 NM range.

⑦ ADF-VOR Select Switches

These switches select ADF or VOR bearing pointers and DME distance on the onside ND, as well as the corresponding navaid data characteristics in any mode except PLAN mode.

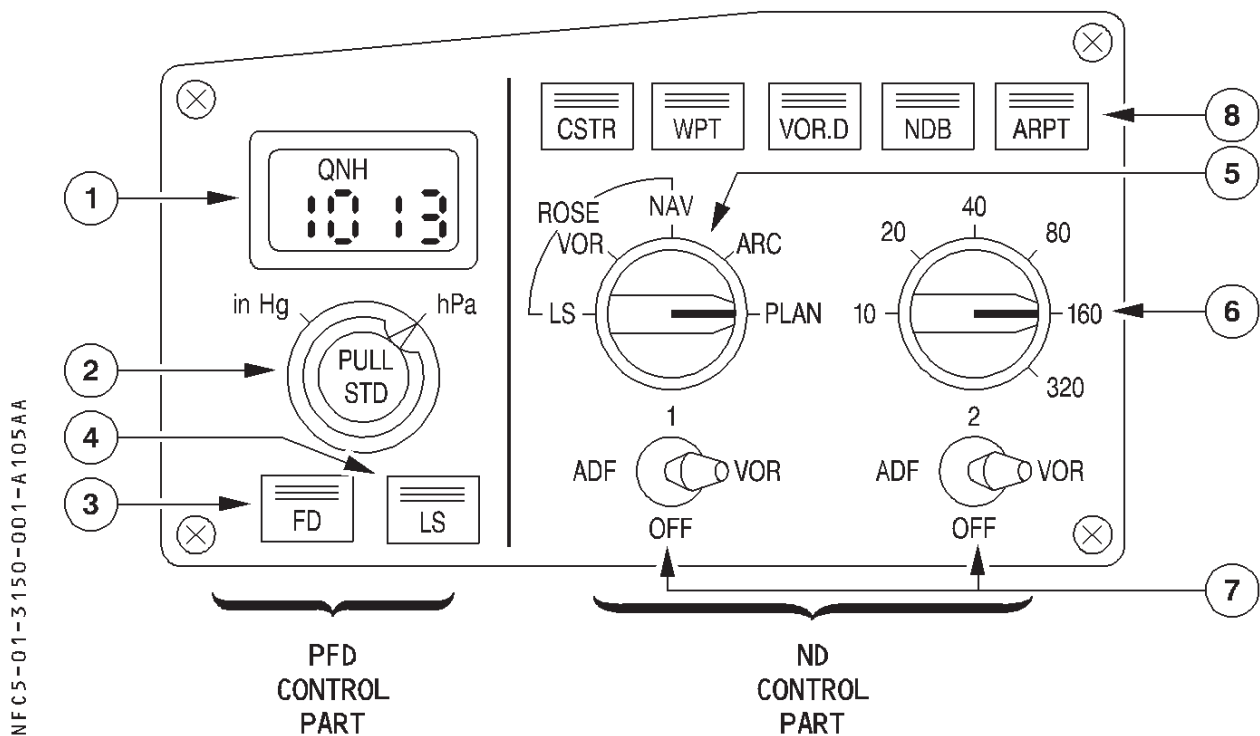
⑧ Optional Data Display Pushbutton

Pushing this button displays optional data in addition to the data permanently displayed in PLAN, ARC, or ROSE NAV modes. The green pushbutton light comes on.

Only one option can be activated at a time.

EFIS CONTROL PANEL

R



1 Barometer Reference Display Window

Range : 745 hPa to 1100 hPa.

2 Barometer Reference Selector

- a) Outer ring : For selection of the units for the barometer reference-either hectoPascals or inches of mercury.

Note : The unit selected does not appear on the PFD.

- b) Inner knob : For selection of the reference value displayed in the barometer reference display window and on the PFD below the altitude scale.
At FCU initialization, the window displays 1013 or 29.92, depending on the unit selected.
- Pulling the knob selects the standard baro reference setting. The PFD then displays "STD." (Rotating the knob has no effect.)
 - Pushing the knob from the STD position makes the last selected QNH baro setting available.

③ FD Pushbutton

Pushing this button removes the FD bars from the associated PFD (or removes the flight path director symbol if the TRK FPA reference is selected).

The pushbutton light goes out.

Pushing it again restores the FD bars (or the FPD symbol) and the green pushbutton light comes on.

R ④ LS Pushbutton

Pushing this button displays the localizer and glide slope scales on the PFD.

Deviation symbols appear if there is a valid ILS signal.

The green pushbutton light comes on.

⑤ Mode Select Switch

This switch selects a navigation display for the onside ND.

⑥ Range Select Switch

This switch selects a range scale for the onside ND.

Note : If the mode or the range data fails, the default selection is the ROSE NAV mode and 80 NM range.

⑦ ADF-VOR Select Switches

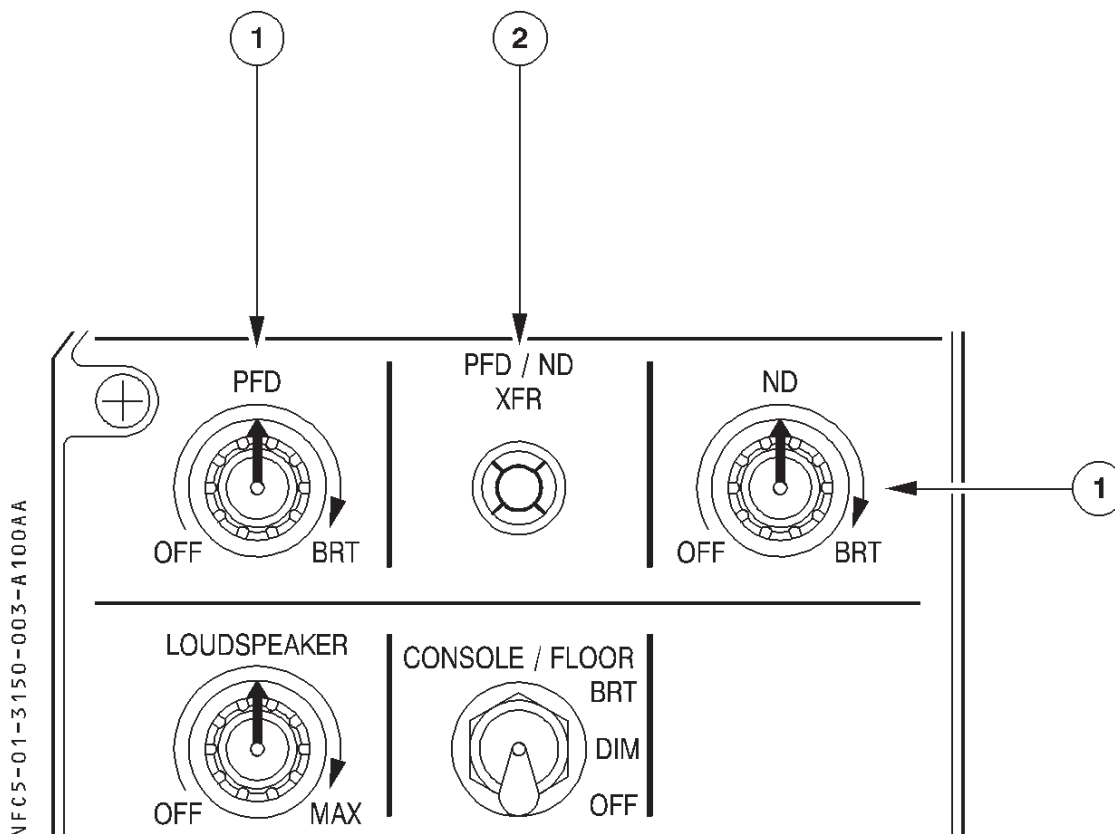
These switches select ADF or VOR bearing pointers and DME distance on the onside ND, as well as the corresponding navaid data characteristics in any mode except PLAN mode.

⑧ Optional Data Display Pushbutton

Pushing this button displays optional data in addition to the data permanently displayed in PLAN, ARC, or ROSE NAV modes. The green pushbutton light comes on.

Only one option can be activated at a time.

OTHER EFIS CONTROLS



① OFF/BRT knobs

- These knobs turn the PFD and ND display units on and off, and control their brightness.
- The display brightness automatically adjusts for changing light conditions. It may also be adjusted manually.

PFD Brightness Control Knob

Rotating this knob all the way counterclockwise switches off the PFD. In this case, the PFD image is automatically displayed on the NDU, but the pilot may recover the ND by means of the PFD-ND XFR pushbutton.

ND Brightness Control Knob

R The outer knob controls the brightness of both the weather radar image and the EGPWS
R terrain display.

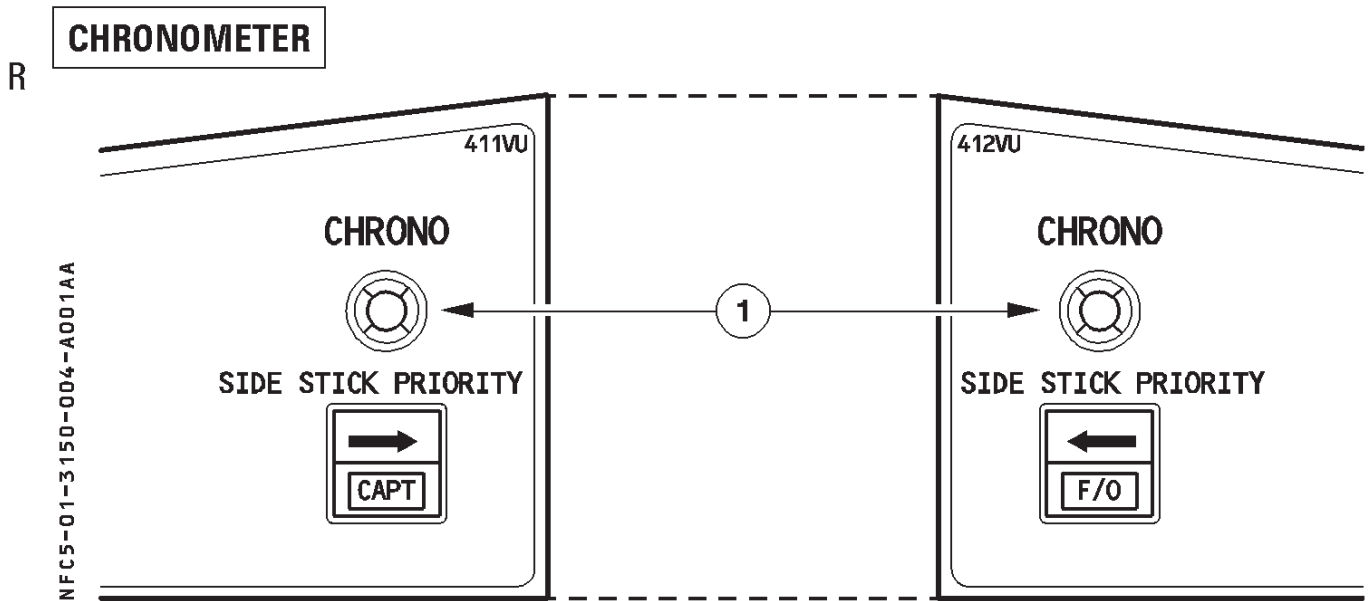
The inner knob controls the general brightness of the ND symbols.

Rotating this knob all the way counterclockwise switches off the NDU.

② PFD/ND XFR Pushbutton

Pushing this button interchanges the PFD and the ND.

If the PFDU fails, the PFD automatically transfers to the NDU.



① CHRONO Pushbutton

Pushing this button displays chronometer time on the onside ND.

Pushing it again freezes the displayed value.

Pushing it a third time resets the chronometer, and the chronometer time disappears from the display.

GENERAL

A fully independent clock is on the right side of the control panel.

It sends time to the centralized fault data interface unit, the flight data interface unit, and the flight management and guidance computer.

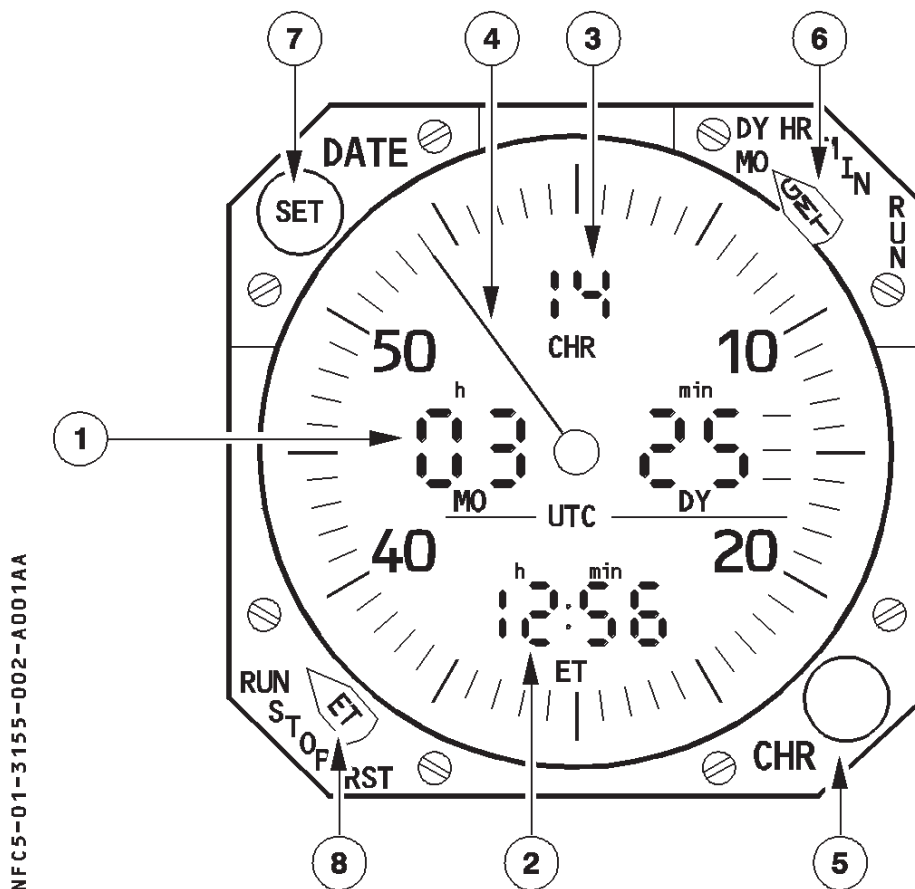
The clock has two electrical supplies, one of which is a direct connection to the aircraft battery hot bus.

The clock also has an internal battery that allows it to memorize time (for up to 50 hours) if the aircraft battery is removed.

The clock performs four functions :

- R — It displays "UTC" (GMT) time in hours and minutes on the center counter.
- It displays elapsed time (ET) (from engine startup) in hours and minutes on the lower counter.
- It drives the chronometer (CHR), which measures a time interval (from the pushing of the CHRONO button) in minutes and seconds and displays it with a needle and in numbers on the upper counter.
- It can replace the UTC with the date.

CONTROLS AND INDICATORS



① UTC (GMT) counter

This counter displays the present time in 24-hour format.

A trend indicator, which divides a minute into four 15-second periods, uses three horizontal marks :

From 0 to 14 seconds No mark is lighted.

From 15 to 29 seconds The highest mark is lighted.

From 30 to 44 seconds The two highest marks are lighted.

From 45 to 59 seconds All the three marks are lighted.

R ② Elapsed Time (ET)

This counter registers the elapsed time up to 99 hours and 59 minutes.

③ Chrono (CHR) counter

This Counter registers the elapsed time up to 59 minutes. It is controlled by the CHR pushbutton.

GENERAL

A fully independent clock is on the right side of the control panel.

It sends time to the centralized fault data interface unit, the flight data interface unit, and the flight management and guidance computer.

The clock has two electrical supplies, one of which is a direct connection to the aircraft battery hot bus.

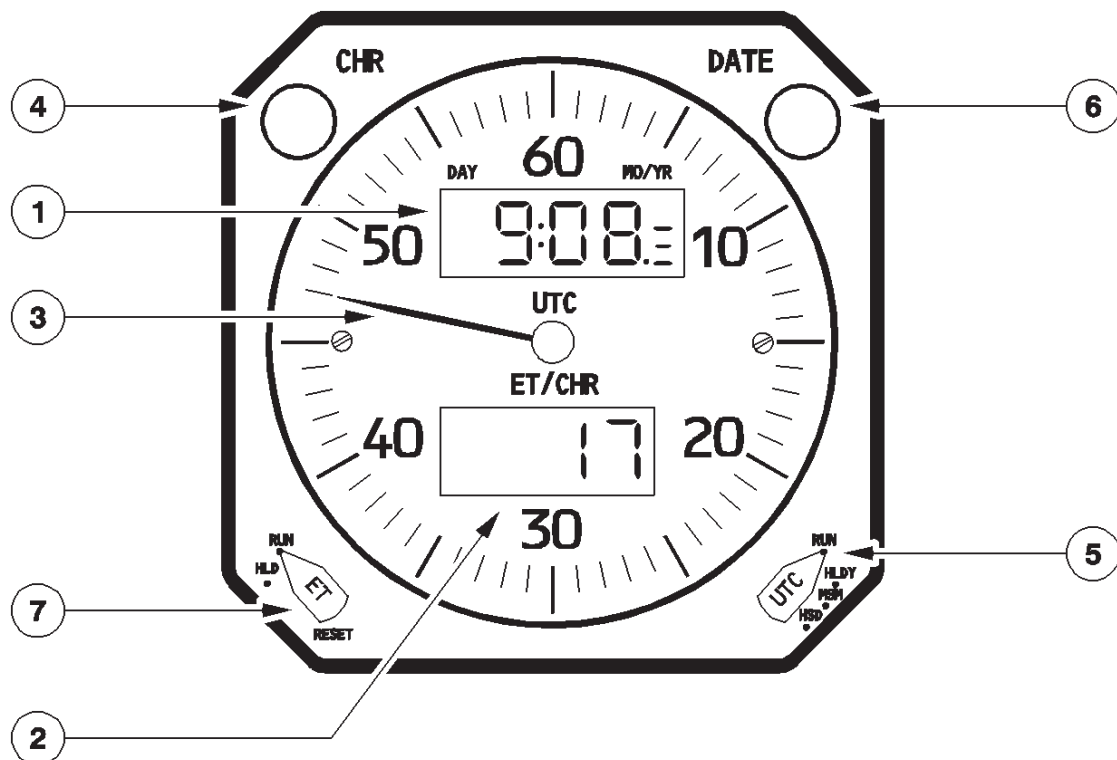
The clock also has an internal battery that allows it to memorize time (for up to 50 hours) if the aircraft battery is removed.

The clock performs four functions :

- R — It displays "UTC" (GMT) time in hours and minutes on the center counter.
- It displays elapsed time (ET) (from engine startup) in hours and minutes on the lower counter.
- It drives the chronometer (CHR), which measures a time interval (from the pushing of the CHRONO button) in minutes and seconds and displays it with a needle and in numbers on the upper counter.
- It can replace the UTC with the date.

CONTROLS AND INDICATORS

NFC5-01-3155-002-A100AB



① UTC (GMT) counter

This counter displays the present time in 24-hour format.

A trend indicator, which divides a minute into four 15-second periods, uses three horizontal marks :

From 0 to 14 seconds No mark is lighted.

From 15 to 29 seconds The highest mark is lighted.

From 30 to 44 seconds The two highest marks are lighted.

From 45 to 59 seconds All the three marks are lighted.

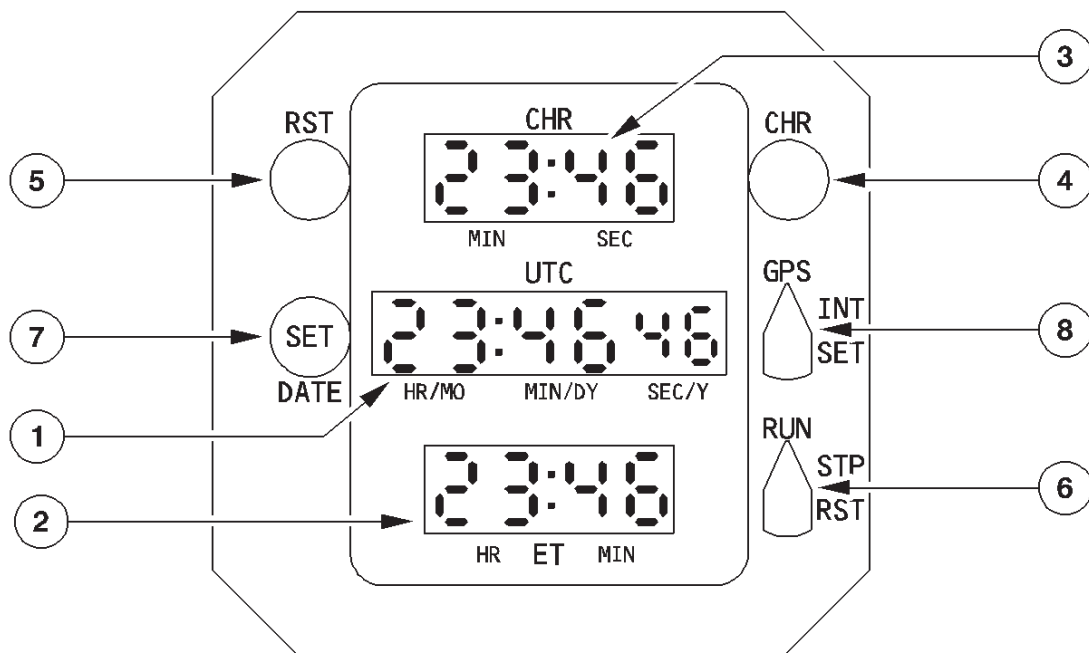
② Elapsed Time (ET) and chrono counter (CHR)

R This counter registers the elapsed time up to 99 hours and 59 minutes.

GENERAL

- R A fully independent clock is on the right side of the control panel.
It sends time to the centralized fault data interface unit, the flight data interface unit, and the flight management and guidance computer.
The clock has two electrical supplies, one of which is a direct connection to the aircraft battery hot bus.
The clock performs four functions :
- It displays “UTC” (GMT) time in hours, minutes and seconds on the center counter.
 - It displays elapsed time (ET) (from engine startup) in hours and minutes on the lower counter.
 - It drives the chronometer (CHR), which measures a time interval (from the pushing of the CHRONO button) in minutes and seconds.
 - It can replace the UTC with the date.

CONTROLS AND INDICATORS



NFC5-01-3155-002-A110AA

① UTC (GMT) counter

This counter displays the present time in 24-hour format from 0 to 23 hours 59 minutes 59 seconds.

② Elapsed Time (ET) counter

This counter registers the aircraft's flight time from 0 to 99 hours 59 minutes.

③ Chrono (CHR) counter

This counter registers elapsed time from 0 to 99 minutes 59 seconds. It is controlled by the CHR pushbutton.

④ CHR needle

This needle ticks off chronograph seconds.
It is controlled by the CHR pushbutton.

⑤ CHR pushbutton

First push : starts the second needle and the minute display.

Second push : stops the second needle and sums up the minutes in the minute display.

Third push : resets the second needle to zero and erases the minute display.

⑥ UTC (GMT) selector

"MO" : sets months and years

"DY" : sets days

"HR" : sets hours

"MIN" : sets minutes

"RUN" : starts the UTC counter

Note : Push and turn the selector to switch from MIN to RUN.

⑦ SET pushbutton

Sets the MO, DY, HR, MIN as selected by the UTC selector.

To increase the quantity, turn the button either way.

To decrease the quantity, push the button lightly.

⑧ ET (elapsed time) selector

"RUN" position : starts the ET counter.

"STOP" position : stops the counter, keeps the display at its last indication.

"RST" position : resets counter to zero (transient position).

R UPDATING MONTH, YEAR

- R Set the UTC selector on MO. The month and the year appear.
- R — To advance the month, turn the SET pushbutton until the proper month appears.
- R — To decrement the month, press the SET pushbutton.
- R — The year changes automatically when the month changes between 12 and 1.

UPDATING DAY, HOUR, MINUTE

Use the procedure for changing day, hour and minute after selecting the appropriate position for the UTC selector.

Note : *To return to normal function, switch the UTC selector to "RUN."*

③ CHR needle

This needle ticks off chronograph seconds.
It is controlled by the CHR pushbutton.

④ CHR pushbutton

- . First push : starts the second needle and the ET/CHR display.
- . Second push : stops the second needle and sums up the minutes in the ET/CHR display.
- . Third push : resets the second needle to zero and erases the ET/CHR display.

⑤ UTC (GMT) selector

«RUN» : STARTS THE UTC COUNTER	
TIME MODE	DATE * MODE
«HLDY» : STOPS UTC COUNTER	TO ADVANCE YEARS FOR SETTING
«MSM» : TO ADVANCE MINUTES : FOR SETTING	TO ADVANCE MONTHS FOR SETTING
«HSD» : TO ADVANCE HOURS FOR SETTING	TO ADVANCE DAYS FOR SETTING

* Press the date pushbutton to switch the clock to date mode.

⑥ DATE pushbutton

- R 1st push : sets the clock to date mode. The date (day, month, year) replaces the
R UTC time display.
- 2nd push : sets the clock to time mode. The date display disappears.

Note : In order to select the date mode, the UTC selector must be set to RUN.

⑦ ET (elapsed time) selector

- . "RUN" position : starts the ET counter.
- . "HLD" position : stops the counter, keeps the display at its last indication.
- . "RESET" position : resets counter to zero (transient position).

UPDATING OF DAY, MONTH AND YEAR

Press the DATE pushbutton to set the date mode. The day, month and year appear.

- To advance the day, turn and maintain the UTC selector at the MSD position until the proper day appears.
- For month and year proceed in the same way, using MSM to advance month or HLDY to advance years.

Note : To return to normal functioning, switch the UTC selector to RUN position.

UPDATING OF HOUR AND MINUTE

Use the same procedure in time mode for changing hour and minute. To advance the minute turn the UTC selector to MSM and to HSD for the hour.

Note : To return to normal functioning, switch the UTC selector to RUN position.

④ CHR pushbutton

First push : starts the CHR counter

Second push : stops the CHR counter, keeps the display at its last indication.

⑤ Reset (RST) pushbutton

When pressed, the CHR counter restarts from 0 if the chrono is running.

⑥ ET selector

"RUN" : the ET counter starts

"STP" : the ET counter stops counting

spring loaded "RST": the ET counter is blanked. The selector returns to its STP position when the selector is released.

Note : *A cumulative elapsed time can be realized by alternatively setting this switch in "RUN" and "STP" position.*

⑦ DATE/SET button

First push : sets the clock to date mode. The UTC time display is replaced by the date (day month year).

Second push : sets the clock to time mode. The date display disappears.

Note : *in order to select the date mode, the UTC selector must be set on "GPS" or "INT" position.*

⑧ UTC selector

“GPS”: Time (or date, if selected) is displayed, and this data is synchronized on GPS information.

Note : – If the signal between the GPS and the clock is not detected, dashes are displayed. Only the “INT” and “SET” positions are then available.

– If the signal is detected, but GPS data is invalid, the clock automatically runs on its internal time.

R – The clock will automatically resynchronize on the GPS information, as soon as
R the GPS data becomes available.

“INT”: Internal time (or date, if selected) is displayed.

R Note : – The clock’s internal time is always synchronized with the latest valid GPS
R information.

R – If there is no valid GPS information at power up, the internal time will be
R 00:00:00, until the clock is initialized, or until valid GPS information is present.

R “SET”: Allows the internal time and date to be initialized.

R INTERNAL TIME AND DATE INITIALIZATION

Set the UTC selector on “SET”. The minute digits flash, and the seconds’ digits are blank.

To increase data, turn the DATE/SET button clockwise.

To decrease data, turn the DATE/SET button counterclockwise.

– First, push on DATE/SET : To set the hour.

– Second, push on DATE/SET : To set the year.

– Third, push on DATE/SET : To set the month.

– Fourth, push on DATE/SET : To set the day.

Switch the UTC selector to the “INT” position, and the clock starts with the seconds’ digits at 00.

Note : This process must be completed in less than one minute. Otherwise, it will be necessary to reset the CFDS in order to synchronize the lower ECAM time display with the cockpit clock display. Resetting the CFDS is a maintenance operation.

R FLIGHT DATA RECORDING SYSTEM

R DESCRIPTION

R The Flight Data recording System, which records the mandatory parameters, consists of the following components :

- R — A Flight Data Interface Unit (FDIU)
- R — A Digital Flight Data Recorder (DFDR)
- R — A three-axis Linear Accelerometer (LA)

R The FDIU collects and processes parameters from the SDACs, DMCs, FWCs, FCDCs, BSCU, the DFDR event pushbutton, the GND CTL pushbutton and the Clock.

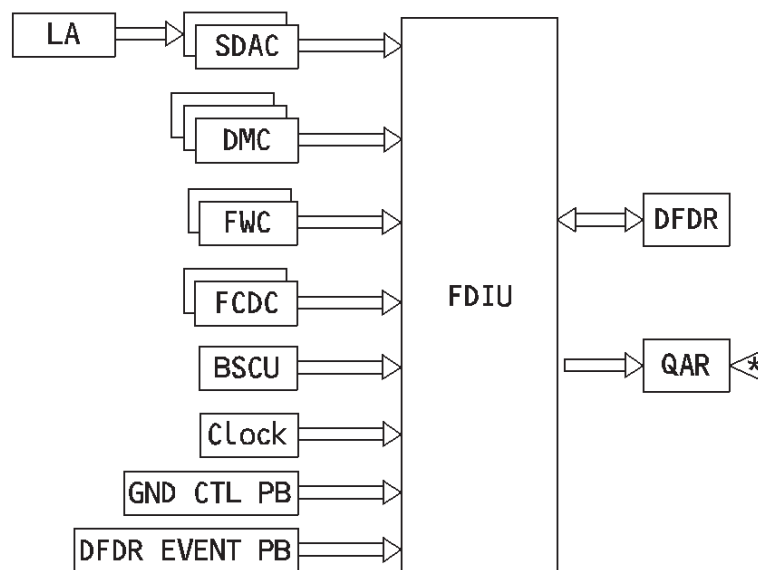
R It stores the mandatory flight parameters in the DFDR.

R The DFDR can store the last 25 hours data, at least. It stores this data on a fireproof and shockproof device. An underwater locator beacon is attached to the DFDR.

R The linear accelerometer measures the acceleration of the aircraft along each of the three axes.

R The QAR is an operational recorder that stores the same data as the DFDR. However the QAR is more accessible for the maintenance crew.

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R The recording system is automatically active :

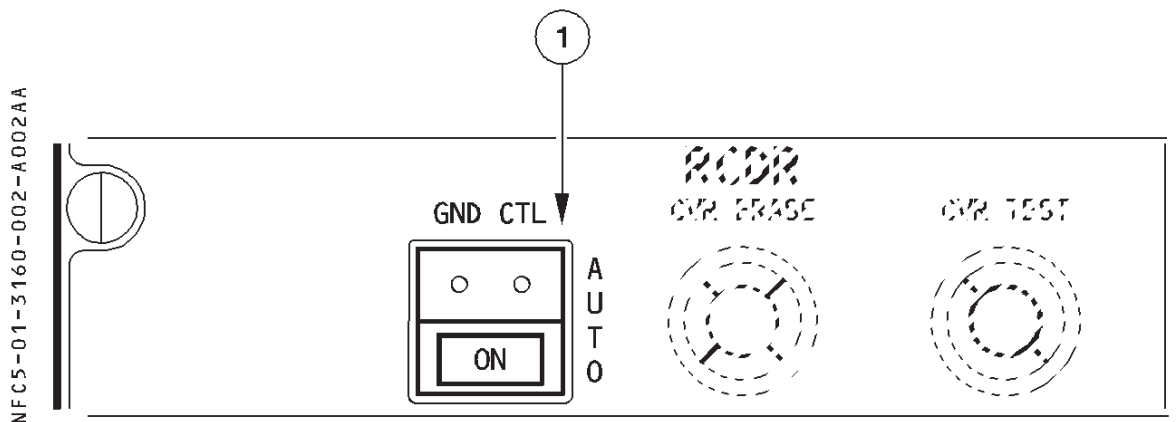
- R — On the ground, during the first five minutes after the aircraft electric network is energized.
- R — On the ground, after the first engine start.
- R — In flight (whether the engines are running or not).

R On the ground, the recording system stops automatically five minutes after the second engine shuts down.

R On the ground, the crew can start the recording system manually by pressing the GND CTL pushbutton.

R CONTROLS AND INDICATORS

OVERHEAD PANEL

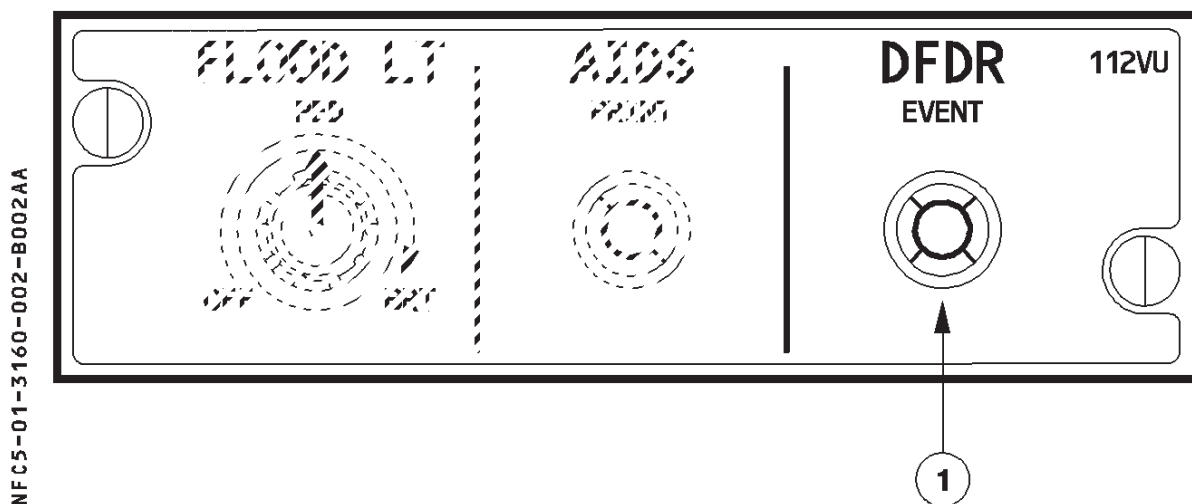


① GND CTL pushbutton (springloaded)

- R – ON The Cockpit Voice Recorder (CVR) and the Flight Data Recorders are active. The ON light is on.
- R – AUTO The Cockpit Voice Recorder (CVR) and the Flight Data Recorders are active, according to the logic.

The system automatically switches from ON to AUTO at the first engine start, and also in case of an electrical transient.

PEDESTAL



① DFDR EVENT pushbutton

- R – Pressing this button (briefly) sets an event mark on the Flight Data records.

FLIGHT DATA RECORDING SYSTEM

DESCRIPTION

The Flight Data Recording System, which records the mandatory parameters, consists of the following components :

- A Flight Data Interface and Management Unit (FDIMU)
- A Digital Flight Data Recorder (DFDR)
- A three-axis Linear Accelerometer (LA)

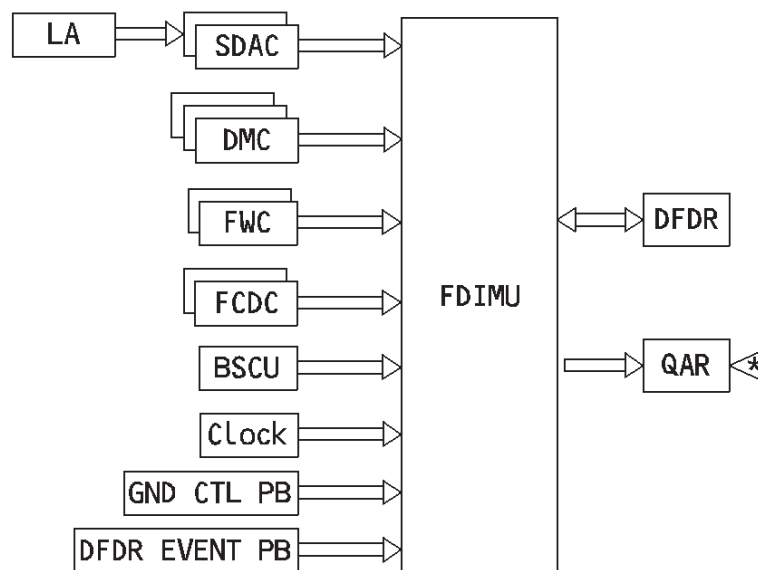
The FDIMU collects and processes parameters from the SDACs, DMCs, FWCs, FCDCs, BSCU, the DFDR event pushbutton, the GND CTL pushbutton and the Clock.

It stores the mandatory flight parameters in the DFDR.

The DFDR can store the last 25 hours, at least. It stores this data on a fireproof and shockproof device. An underwater locator beacon is attached to the DFDR.

The linear accelerometer measures the acceleration of the aircraft along each of the three axes.

The QAR is an optional recorder that stores the same data as the DFDR. However the QAR is more accessible for the maintenance crew.



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The recording system is automatically active :

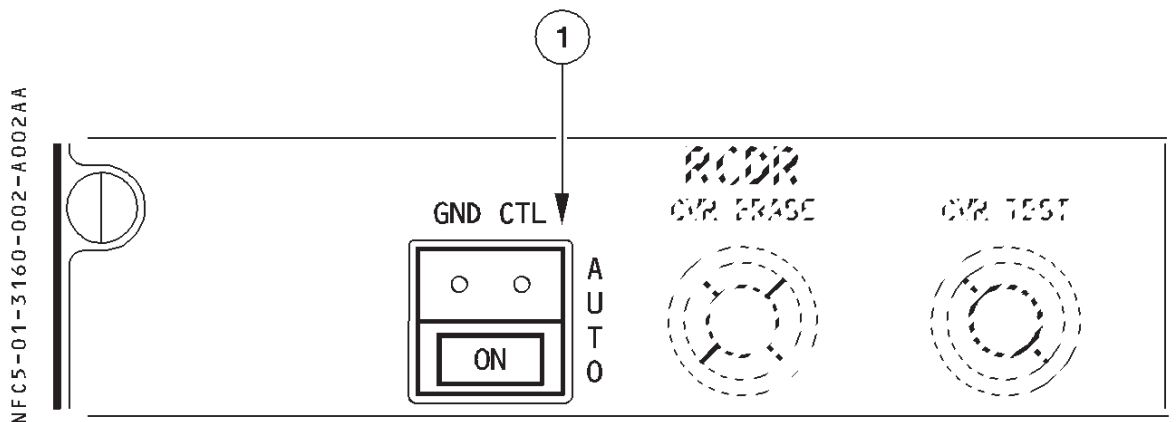
- On the ground, during the first five minutes after the aircraft electric network is energized.
- On the ground, after the first engine start.
- In flight (whether the engines are running or not).

On the ground, the recording system stops automatically five minutes after the second engine shuts down.

On the ground, the crew can start the recording system manually by pressing the GND CTL pushbutton.

R CONTROLS AND INDICATORS

OVERHEAD PANEL

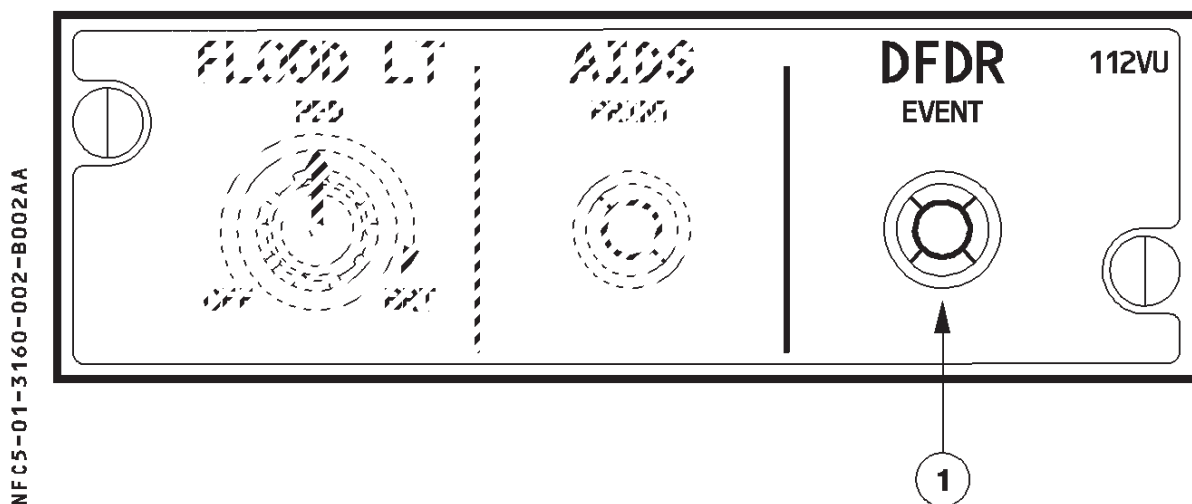


① GND CTL pushbutton (springloaded)

- R – ON The Cockpit Voice Recorder (CVR) and the Flight Data Recorders are active. The ON light is on.
- R – AUTO The Cockpit Voice Recorder (CVR) and the Flight Data Recorders are active, according to the logic.

The system automatically switches from ON to AUTO at the first engine start, and also in case of an electrical transient.

PEDESTAL



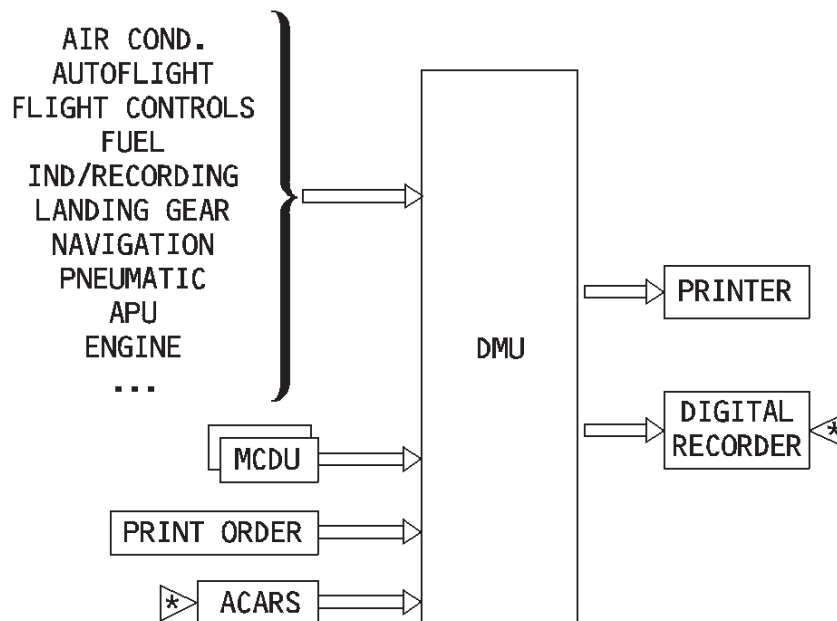
① DFDR EVENT pushbutton

- R – Pressing this button (briefly) sets an event mark on the Flight Data records.

AIRCRAFT INTEGRATED DATA SYSTEM (AIDS)

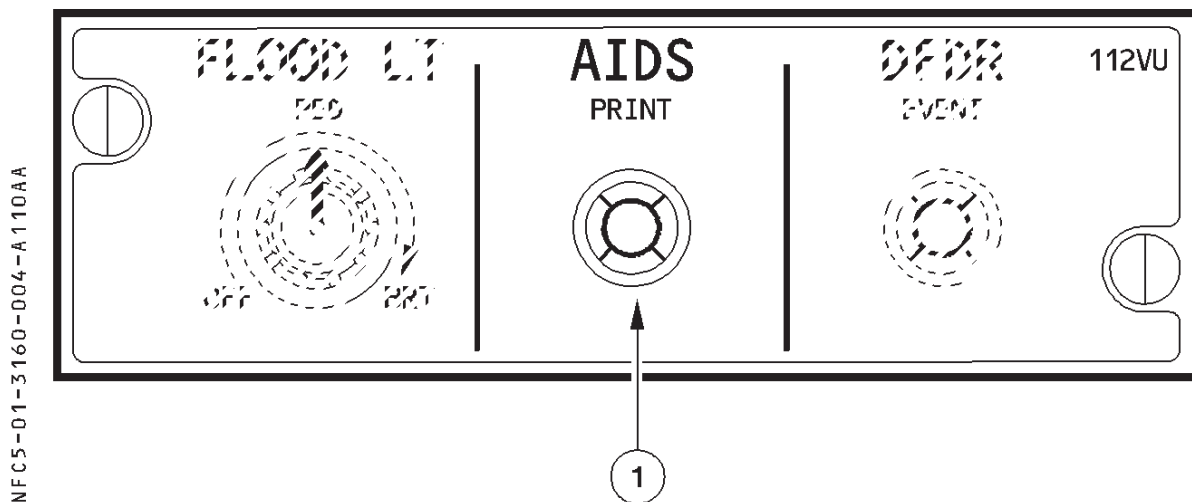
DESCRIPTION

- R The AIDS is used to monitor various aircraft system parameters in order to make maintenance easier and to allow formulating operational recommendations.
- R The AIDS can generate system reports. The Airbus Standard Reports are preprogrammed reports available at aircraft delivery. The operator can create its own reports.
- R The AIDS consists of a Data Managements Unit (DMU) connected as shown below.
- R The system may be programmed using the MCDUs. The crew can select any report to be displayed on the MCDUs
- R The Printer prints the flight phase programmed reports or any report selected on the MCDU. This printing may be automatic or in response to the AIDS PRINT pushbutton.
- R The AIDS may send automatic reports via ACARS (✱).
- R An optional Digital Recorder may be installed to extend the recording capacity.



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CONTROLS ON PEDESTAL



① AIDS PRINT pushbutton

Pushing this pushbutton causes the immediate printing of a specific report, depending on the flight phase. The crew may then use the MCDU to select another report for immediate printing.

AIRCRAFT INTEGRATED DATA SYSTEM (AIDS)

DESCRIPTION

The AIDS is used to monitor various aircraft system parameters in order to make maintenance easier and to allow formulating operational recommendations.

The AIDS can generate system reports. The Airbus Standard Reports are preprogrammed reports available at aircraft delivery. The operator can create its own reports.

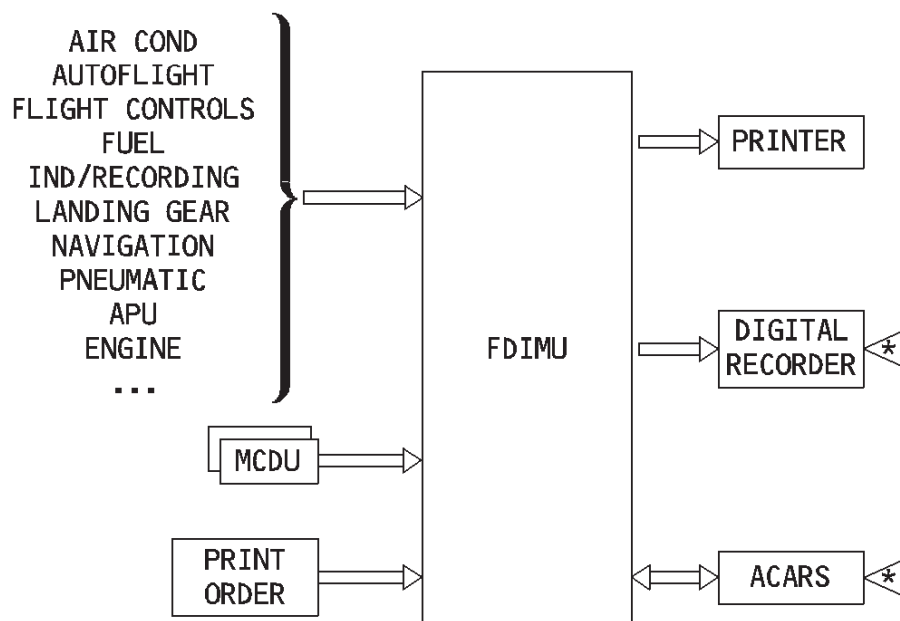
The AIDS uses the Flight Data Interface and Management Unit (FDIMU) to acquire the relevant aircraft system parameters. The FDIMU is connected to the rest of the AIDS as shown below.

The system may be programmed using the MCDUs. The crew can select any report to be displayed on the MCDUs

The Printer prints the flight phase programmed reports or any report selected on the MCDU. This printing may be automatic or in response to the AIDS PRINT pushbutton.

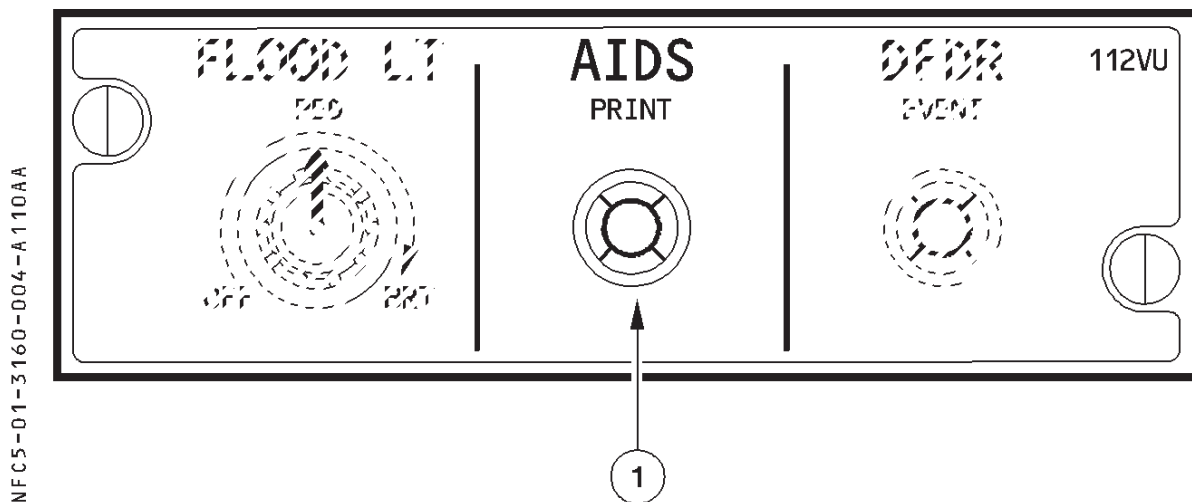
The AIDS may send automatic reports via ACARS (*).

An optional Digital Recorder may be installed to extend the recording capacity.



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CONTROLS ON PEDESTAL

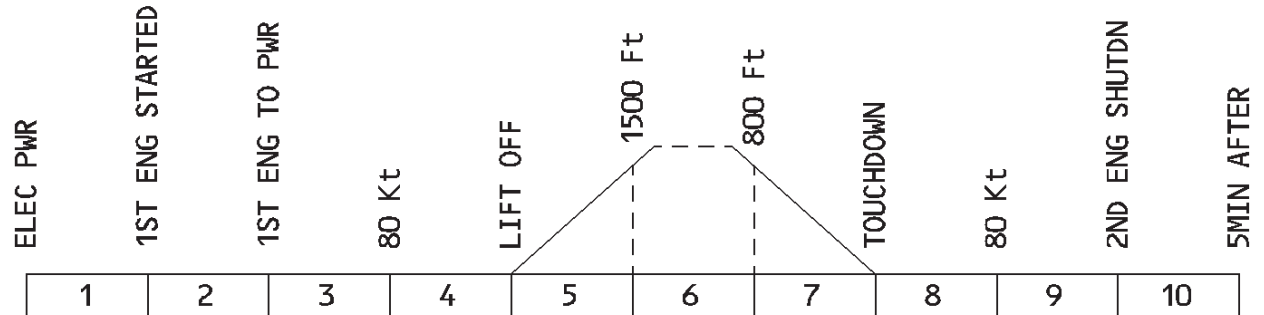


① AIDS PRINT pushbutton

Pushing this pushbutton causes the immediate printing of a specific report, depending on the flight phase. The crew may then use the MCDU to select another report for immediate printing.

WARNINGS AND CAUTIONS

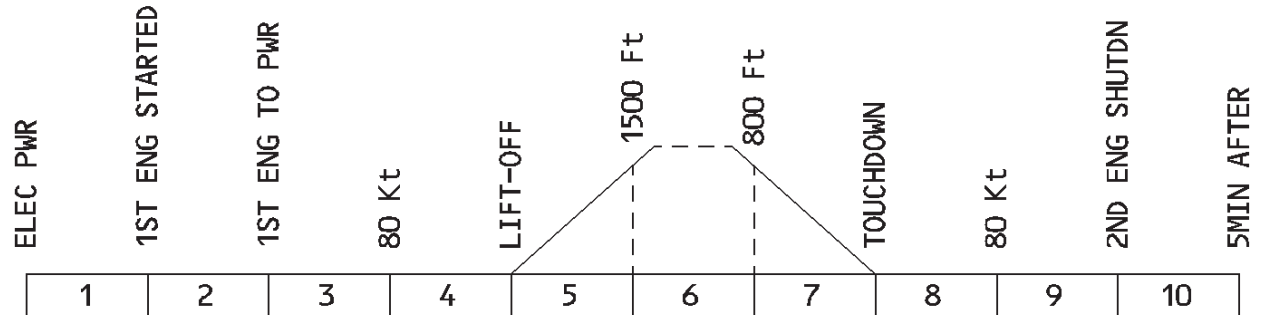
NFC5-01-3170-001-A100AA



E / WD: FAILURE TITLE conditions	AURAL WARNING	MASTER LIGHT	SD PAGE CALLED	LOCAL WARNINGS	FLT PHASE INHIB	
SDAC 1 + 2 FAULT	SINGLE CHIME	MASTER CAUT	NIL	NIL	4, 5, 7, 8	
DMC 1 (2) FAULT						
DMC 3 FAULT						
SDAC 1 (2) FAULT	NIL	NIL			NIL	3, 4, 5, 7, 8
FWC 1 (2) FAULT						NIL
FWC 1 + 2 FAULT						3, 4, 5, 7, 8
DFDR FAULT						
FDIU FAULT						
OEB/FWC DISCREPANCY FWC1 and FWC2 do not have the same OEBs listed in their OEB reminder database.	SINGLE CHIME	MASTER CAUT			3 to 8	

WARNINGS AND CAUTIONS

NFC5-01-3170-001-A205AA



E / WD: FAILURE TITLE conditions	AURAL WARNING	MASTER LIGHT	SD PAGE CALLED	LOCAL WARNINGS	FLT PHASE INHIB	
SDAC 1 + 2 FAULT	SINGLE CHIME	MASTER CAUT	NIL	NIL	4, 5, 7, 8	
DMC 1 (2) FAULT						
DMC 3 FAULT						
SDAC 1 (2) FAULT	NIL	NIL			NIL	3, 4, 5, 7, 8
FWC 1 (2) FAULT						NIL
FWC 1 + 2 FAULT						3, 4, 5, 7, 8
DFDR FAULT						
RECORDER SYS FAULT						
OEB/FWC DISCREPANCY FWC1 and FWC2 do not have the same OEBs listed in their OEB reminder database.	SINGLE CHIME	MASTER CAUT			3 to 8	

BUS EQUIPMENT LIST

R

		NORM		EMER ELEC		
		AC	DC	AC ESS	DC ESS	HOT
DU	CAPT PFD			X		
	CAPT ND			SHED		
	F/O PFD	AC2				
	F/O ND	AC2				
	UPPER ECAM DU			X		
	LOWER ECAM DU	AC2				
DMC	DMC 1			X		
	DMC 2	AC2				
	DMC 3	AC1		X *		
FWC	FWC 1			X		
	FWC 2	AC2				
SDAC	SDAC 1			X		
	SDAC 2	AC2				
ECP	ECP				X	
FDIU/QAR <4		AC2				
CLOCK					X	X

* in case of EIS DMC switching to CAPT/3, with AC BUS 1 failed.